INDUSTRY STARTER PACKS FOR ENTERPRISE ARCHITECTS

Healthcare

The architecture workbook — the TOGAF ADM against a real repository, phase by phase

Volume 2. Not a book to read front to back: open it at the phase you are in and work the numbered steps. Every instruction names a real sheet and real columns of the repository in front of you, and every finding was checked cell by cell — including the eight relationship sheets that resolve perfectly and mean nothing, the fourteen entities mastered where the first system was integrated, and the seven work packages marked delivered before any of them starts. It creates the six sheets a healthcare provider needs and the file does not have: consent, the device register, the code system, the measure definition, the disclosure event and the retention clock.

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Contents

BEFORE YOU START

Item 1 · How to use this workbook 3

PRELIMINARY PHASE

Item 2 · Standing the practice up where nothing is specified nationally 12

PHASE A

Item 3 · Architecture Vision 25

PHASE B

Item 4 · Business Architecture 34

PHASE C

Item 5 · Data Architecture 45

Item 6 · Application Architecture 59

PHASE D

Item 7 · Technology Architecture, devices and the downtime path 70

PHASE E

Item 8 · Opportunities and Solutions 83

PHASE F

Item 9 · Migration Planning 92

PHASE G

Item 10 · Implementation Governance 102

PHASE H

Item 11 · Architecture Change Management and the retention clock 113

REQUIREMENTS MANAGEMENT

Item 12 · Requirements Management 123

APPENDICES

Appendix A · The sheet-by-sheet reference 134

Appendix B · The report catalogue 145

Appendix C · Workshop packs, checklists and the ninety-day calendar 151

Appendix D · The repository in ArchiMate 4 163

Appendix E · Where to look in volume 1 175

Appendix F · The forms 181

Appendix G · The principles and standards starter set 216

Appendix H · Running the maturity assessment 229

ITEM 1

How to use this workbook

Eight of the twenty-five relationship sheets in the supplied repository were built by pairing two lists in row order; every row resolves, and almost none of them means anything.

Start with eight sheets that are perfect and worthless

Open the supplied workbook. Go to the tab named `R22 Org Unit owns Capability`. Row 4 gives you four columns — `Org unit`, `Unit name`, `Capability`, `Capability name` — and rows 5 to 14 give you ten rows of ownership. Read them in order.

`ORG-01` Executive owns `C01` Patient Access & Scheduling. `ORG-02` Finance owns `C02` Clinical Care Delivery. `ORG-03` Risk & Compliance owns `C03` Diagnostics. `ORG-04` Human Resources owns `C04` Medicines Management. `ORG-05` Technology owns `C05` Care Coordination. And so on to `ORG-10` Supply & Logistics, which owns `C10` Population Health & Analytics.

Human Resources owns prescribing, dispensing and medicines administration. Supply and logistics owns population health analytics. Commercial owns clinical rostering. Not one of the ten is right. `ORG-11` Internal Audit and `ORG-12` Legal own no capability at all, and the two capabilities a hospital would hand to a clinical function — `C02` and `C06` Clinical Governance & Safety — go to finance and to data and analytics.

Now check the sheet the way a repository tool checks it. Every `Org unit` value resolves to a row on `O09 Organization Unit`. Every `Capability` value resolves to a row on `O13 Business Capability`. Every name in column 2 matches the name that `Ref` carries on `O09`, and every name in column 4 matches the name that `Ref` carries on `O13`. There is not one dangling reference and not one name disagreement. The sheet is clean.

It is clean across the whole file. All 430 relationship rows on all 25 relationship sheets resolve on both ends, by `Ref` and by name, with no exceptions. `00 Index` row 8 claims exactly that — "Referential integrity ~ Every relationship row names both ends by reference and by name, so a broken link is visible without a lookup" — and the claim is true. It is the one promise the Index makes and keeps.

Figure 1.1 Referential integrity is perfect and the sheet is nonsense. R22 pairs the twelve organisation units against the ten level-1 capabilities in row order. Every Ref resolves on both ends and every name matches the Ref that carries it, which is the one promise 00 Index row 8 makes and keeps. Nothing on ZZ Checks or on any report tests the second question.

Seven more sheets were built the same way, and each one is a different variety of the same mistake.

- `R04`, ten rows, pairs `BSV-0n` to `A0n`: `BSV-04` Medicines Management Service is supported by `A04` RIS / PACS, the imaging system; `BSV-06` Clinical Governance & Safety Service by `A06` Theatre Management, while `A09` Incident & Risk — described on `O23` as "Safety incidents, investigation and action" — supports `BSV-09` Estates & Medical Devices Service. Six of ten are plainly wrong.
- `R07 Application deployed on Tec`, twelve rows, pairs `A0n` to `PTC-0n`. The imaging system runs on the image store; the billing system runs on the medical device gateways; the patient portal runs on the backup and archive tier. The component actually named "Integration engine servers" hosts RIS / PACS, while the integration engine sits on the cloud landing zone. Ten of twelve are wrong.
- `R09 Requirement satisfied by Bu`, thirty-two rows, lays the requirement list against the 37 rows of `O55 Building Block` in index order with a seven-row offset. Not one of the 32 reaches a block whose subject it names. `DBB-01 Patient mastering` exists; `REQ-008 Patient mastered once` reaches `BBB-08 Workforce & Rostering capability uplift`. `ABB-01 EHR / EPR target` exists; `REQ-016 EHR / EPR integration managed` reaches `DBB-06 Medication Order mastering`.
- `R13 Work Package delivers Build`, thirty-seven rows, cycles twelve work packages through the same 37 building blocks, so `WP-02 Data ownership and mastering` delivers one of the ten data blocks and `WP-04 Technology currency` delivers no technology block at all.
- `R20 Application located at Loca`, twelve rows, pairs `A0n` to `LOC-0n`, and contradicts `R07` plus `O27` on where eleven of twelve applications sit; the two agree on `A12` alone.

- `R23 Increment delivered by Proj`, six rows, pairs increment *n* to the odd-numbered projects, so `INC-5` "Analytics and evidence" is delivered by the resilience project while `PRJ-06 Analytics and reporting` delivers nothing.
- `R25 Application supports Produc`, five rows, strides `A01`, `A04`, `A07`, `A10`, `A01`. Emergency care is supported by the billing system; diagnostic services by the patient portal, while the laboratory and imaging systems support no product at all.

Then look at `ZZ Checks`, which is the file's own quality report: a short sheet of checks, each with a result and a threshold, and a closing note. Object types populated. Records. Relationship types. Relationship instances. Interfaces unmanaged. Direct database reads. Platform components past support. Security controls absent. Vacant architecture roles. Work packages past end date.

Nothing on that sheet, and nothing on any of the thirteen `RPT` sheets, tests whether the two ends of a resolved link belong together. There is no such check, because there is no such query. Belonging together is not a property of the data; it is a judgement somebody has to make and record.

That is the whole argument for this volume, and it is worth putting bluntly.

> **A REPOSITORY CAN PASS EVERY INTEGRITY CHECK AND BE WORTHLESS**
>
> Referential integrity is perfect in this file and eight relationship sheets are nonsense. Integrity is a property a script can assert. Meaning is a property a named person asserts, on a date, against something they know. The repository has nowhere to record that assertion, which is why nobody made it.
>
> So the work this volume teaches is not loading. `00 Index` row 11 tells the reader to "Load the structure as-is. Replace the records with the client's own during Phases B to D." Loading is a morning. **The work is the reconciliation**, and every phase item in this volume is organised around it: what was reconciled, by whom, against what, and where that is written down.

One note on how to read this with a client in the room. Per volume 1, chapter 21, each finding in the supplied file is a caricature of a documented real pattern rather than an invented embarrassment. The pattern here is a repository populated by somebody who had two lists and not the knowledge that joins them, under time pressure, before a gate. It is common and survivable, and only fatal when the resulting sheet is quoted at an executive as a statement about the organisation.

1.1 — Test one relationship sheet for meaning rather than resolution

Sheet `R22` · **Input** `O09` and `O13` open beside it · **Output** a verdict per row and a named reconciler · **Signed by** the Chief Enterprise Architect · **Time** two hours

1. Print `R22`'s ten rows. Print `O09`'s twelve rows and `O13`'s ten level-1 rows next to them.
2. For each row, ask one question of one person who does the work: does this unit decide how this capability is run, fund it, and answer for it when it fails? Record the answer and who gave it.
3. Mark each row `Agreed`, `Wrong` or `No owner`. A row nobody will claim is `No owner`, which is a finding and not a blank.
4. Count the `Agreed` rows. That count, over ten, is the sheet's meaning rate. Resolution was 10 of 10 before you started.
5. Repeat on one more of the eight — `R04` is the quickest, at ten rows — and stop. Two sheets are enough to establish the pattern and to price the rest.
6. In a client file: run the same test on the relationship sheet that carries ownership, whatever it is called. If the client's repository has no ownership relationship at all, that is the same finding reached faster.

Form 1-A — Relationship meaning test

Feeds `R22`, plus the reconciliation note that `ZZ Checks` has no row for · **Rows for a first pass** 10, one per `R22` row · **Signed by** the Chief Enterprise Architect · **Ratified by** `GB-01`

Columns 1 to 4 are `R22`'s own columns and are typed straight back into Excel. Columns 5 and 6 are working columns; the repository has nowhere to put them, which is the point of the exercise.

Org unit	Unit name	Capability	Capability name	Verdict	Who said so, and when
ORG-04	Human Resources	C04	Medicines Management	Wrong — example row	Chief Pharmacist, 4 Mar

What you were handed, and what each of the three files is for

Three artefacts arrive together, and they are not equally useful.

The metamodel workbook. 105 sheets: `00 Index`, 64 object sheets numbered `O01` to `O66` with no `O34` and no `O51`, `EXT Industry Extension`, 25 relationship sheets `R01` to `R25`, thirteen reports `RPT01` to `RPT13`, and `ZZ Checks`. This is the thing you work in. Every instruction in this volume names a sheet in it by its real tab name and its real columns.

The views deck. 27 slides: a title, a contents page and 25 views, one per ArchiMate 4 viewpoint. Item 16 takes it. For now, know that four of its captions were written for a different industry's repository and were never rewritten, so a caption that names an organisation you have not heard of is a copy that escaped a review rather than a part of the model.

Volume 1. The industry. This volume does not re-teach it. Where a step needs a definition, a number or a legal position, it cites the chapter and moves on, and item 17 is the index of those pointers.

The shape, in numbers you can check before lunch

Four conventions hold on every object sheet, and a step that assumes otherwise will mis-read the file:

- Row 1 is the sheet title. Row 2 is the organisation, the domain and a record count. Row 3 is blank.
- **Row 4 is the header row. Data starts at row 5.** A reader who treats row 1 as the header gets a one-row sheet whose single column is a title.
- The first data column is `Ref` on every object sheet.
- Each relationship sheet's row 2 banner carries its own instance count, and all 25 are correct.

Every object sheet has the same five-row anatomy

Row 4 is the header row and data starts at row 5. O20 Data Entity shown.

row 1 title	**Data Entity**							
row 2 banner	Ravensmere Health Network · Data domain · 14 records							
row 3 blank	*(blank)*							
row 4 HEADERS	**Ref**	**Data entity**	**Class**	**Definition**	**Classification**	**Owner**	**Master system**	**Lineage**
row 5 data	E01	Patient	Master	An identified…	Restricted	Head of Patient Access	ePrescribing	Partial
row 6 data	E02	Encounter	Transactional	A contact…	Public	Data Steward	Clinical Data Wareh.	Full
row 7 data	E13	Consent	Master	Recorded permission…	Confidential	Head of Revenue Cycle	Integration Engine	Full

The three traps

22 tab names are cut at exactly 31 characters, and the cut propagated into row 1: O26 row 1 reads "Logical Technology Componen".

R04's tab name was cut after a word boundary, so it ends in a trailing space. A lookup without it will not find the sheet.

O46 has two columns headed Ref. The second carries T01 to T07, which exist on no sheet, in no object type and in no other row.

Spelling is exact throughout this volume. A step that names a column the file does not have is a defect.

Figure 1.2 Row 4 is the header row and data starts at row 5. The same five-row anatomy holds on all 64 object sheets plus EXT. Below it, the three traps to know before writing a formula against any of them.

Three traps are worth knowing before you write a formula against any of it.

Twenty-two tab names are cut at exactly 31 characters, Excel's limit, and the cut landed mid-word: `O26 Logical Technology Componen` is missing its "t", and so are twenty of the twenty-five relationship sheets. The truncation propagated into the sheets themselves, because the title cell was written from the tab name rather than from the object type's real name — `O26` row 1 reads "Logical Technology Componen" and `O27` row 1 reads "Physical Technology Compone", and `00 Index` rows 40 and 41 reproduce both.

One tab name ends in a trailing space. `R04`'s name was cut after a word boundary, so the thirty-first character is a space. A formula, a lookup or a script written against the name without it will not find the sheet, and no amount of squinting at the tab will show you why. Match on the `R04` prefix.

`O46` has two columns headed `Ref` — the only duplicate header anywhere in the 105 sheets — and the second one carries values `T01` to `T07` that exist on no sheet, in no object type and in no other row. Two readings are available and both should be carried: either the column is a mis-keyed pointer to `O26`'s `LTC-01` to `LTC-07`, which match name for name and in order, or an object sheet numbered `T01` to `T07` was dropped and the numbering was never closed. The cells support the first. Do not pick one in a client deliverable; say there are two and that the file cannot distinguish them.

The number on the front page is wrong

`00 Index` row 2, `RPT01 Dashboard` row 6 and `ZZ Checks` row 6 all say 843 records. The 65 object sheets hold **820**.

The Index contradicts itself inside one sheet: rows 15 to 79 give a count for each of the 65 object types, every one of those figures matches its sheet exactly, and they sum to 820, under a banner that says 843. The difference, 23, is exactly the row count of `R06`, the interface sheet — so the likely mechanism is that interface rows were added into the object total. Treat the mechanism as inference and the arithmetic as fact, and correct it on day one, because every figure you quote afterwards inherits it.

1.2 — Count the repository before you quote anything from it

Sheet `00 Index` , `RPT01` , `ZZ Checks` · **Input** the workbook, read access only · **Output** a one-page count sheet with three corrections on it · **Signed by** the Chief Enterprise Architect · **Time** half a day

1. Count data rows from row 5 down on all 64 `O` sheets plus `EXT` . Do not trust row 2.
2. Compare your total to `00 Index` row 2, `RPT01` row 6 and `ZZ Checks` row 6. All three say 843. Yours says 820.
3. Sum the 65 per-sheet figures in `00 Index` rows 15 to 79. They also say 820, which is your corroboration and makes the finding unarguable.
4. Test two more `ZZ Checks` rows against the sheets. Row 18 says three work packages are past their end date; `O30` 's earliest end date is 2029-02 and the correct figure today is 0. Row 13 says two platform components are past support; `PTC-04` 's support end is 2026-05 and `PTC-06` 's is 2026-12, so today the figure is 1.
5. Write the three corrections on one page with the cell references, and send it to whoever owns the file before you send anything else.
6. In a client file: run the same count against whatever the client's own tool reports on its landing page. The question is not whether the total is right. It is whether anybody would have noticed.

Form 1-B — Repository count sheet

Feeds `00 Index` row 2, `RPT01` row 6 and `ZZ Checks` row 6 · **Rows for a first pass** 16, one per `ZZ Checks` row · **Signed by** the Chief Enterprise Architect · **Ratified by** `GB-01`

Columns 1 to 3 are `ZZ Checks` ' own columns and are typed straight back into Excel. Columns 4 to 6 are working columns.

Check	Result	Threshold	Counted	Correct figure	Where you counted it
Records	843	—	820 — example row	820	rows 5 down, 65 sheets

How to read a step

Every numbered instruction in this volume has the same head, and the head is the undertaking.

Sheet names the real tab you will be typing into, spelled as the file spells it. Where the sheet does not exist yet, the head says so and the item that creates it is named.

Input names what you must already have. A step whose input you do not have is a step you are not ready for, and running it anyway is how a repository fills with plausible rows.

Output names rows, not understanding. If a step ends without something written in a named place, it has not been done.

Signed by names one role. Not a committee, not a function, not "the business" — one role, held by one person, who will be asked about this row in a year. Where the signatory is a post the supplied file does not define, the step says so, and item 2 creates the post.

Time is an estimate for one architect with the file open and access to the people named. It assumes no workshop. Where a step needs a workshop, the workshop pack is in item 15.

Steps are numbered by item and then sequentially — `1.1` , `1.2` , `2.1` — and the numbering is continuous within an item. A cross-reference to `5.4` means item 5, step 4, and item 5 is the Data Architecture item on the contents page.

The two things you write in

Forms are bordered worksheet cards. Each one carries a number, the sheet it feeds, how many rows a first pass needs, who signs it and which body ratifies it. Four rules make them usable in print:

1. A form's first columns are the real columns of the real sheet, in sheet order, spelled exactly as the file spells them. Where a form adds working columns, it says which columns are typed straight into Excel and which are not.

2. Six columns maximum. Where a sheet has more than six, the form splits into part one, identify, and part two, assess — and the two parts carry consecutive letters.
3. One example row, marked as an example on its face, then eight to twelve blank rows.
4. **A form that feeds a sheet the repository does not have says so on its front.** Six do: the six sheets this volume tells you to create.

Form numbers run `<item>-<letter>`, so `1-A` is the first form in item 1 and `2-C` the third in item

1. Item 18 reprints all of them on their own, in order, for photocopying.

Ask blocks are questions with ruled space under them. They are not a questionnaire to send out. They are the questions to put to a named person, in a room, with the file open, and to write the answer down in front of them. An ask block with blank lines in a finished workbook is the honest record that the question was not asked.

Who signs what

The supplied file names five governance bodies on `057 Governance Body` and four gates on `058 Governance Gate`, and connects none of them to anything: `057`, `058`, `060` and `061` appear in no relationship sheet at all. The whole governance layer hangs off nothing, and item 2 is where you attach it.

For the rest of this volume, six signatories and four gates. Learn them once.

Who signs	Where the file defines them	What they sign
Chief Enterprise Architect	`ACT-08` and `ROL-08`	Every sheet definition, every repository correction, every step whose output is a repository row
Chief Data Officer	`ACT-05` and `ROL-05`	Classification, mastering, lineage, and the designation of data owners
Chief Compliance Officer	`ACT-07` and `ROL-07`	Obligations, instruments, retention periods and destruction authority
Clinical Safety Officer	Nowhere — item 2 creates the post	Anything whose failure mode is clinical: recovery objectives, device risk class, the downtime path
Head of Health Records	Nowhere — item 2 creates the post	Record classes, measured retrieval time, disposal evidence
The named data owner	`PRN-13` requires a person and not a committee	The rows of the entity they own, including every code system bound to it

The bodies ratify rather than sign. `GB-01` Architecture Board holds gates `GT-1` and `GT-2` and grants dispensations with an expiry. `GB-02` Investment Committee holds `GT-0` funding and `GT-3` benefit confirmation. `GB-03` Data Governance Council designates data owners and resolves disputes. `GB-04` Executive Committee is the escalation point. `GB-05` Architecture Function publishes criteria, assesses compliance and maintains the repository — which is you.

Two of the six exist in no row of the supplied file, and that is a statement about the organisation rather than an oversight. `009`'s twelve organisation units hold no medical directorate, no nursing, no pharmacy, no clinical governance unit and no diagnostics unit; `010`'s eighteen actors hold no clinician, no pharmacist, no clinical safety officer and no records manager; `011` mirrors `010` one for one. That absence is why `R22` cannot find a plausible owner for a clinical capability, and why eight of the nine owner titles on `020 Data Entity` exist nowhere else in the 105 sheets — leaving ten of fourteen entities owned by a title the repository does not define, while `ZZ Checks` row 17 reports "Vacant architecture roles ~ 0". Item 2 fills both posts.

Running this against a client's own repository

Every sheet reference, every defect, every "go and look" in this volume is Ravensmere Health Network, the supplied repository. That is deliberate: a worked example you can open and check beats an abstraction you cannot. But you are not being paid to repair Ravensmere.

The convention this volume uses is mechanical. A step is written against Ravensmere. Where running it against a client's own repository changes the step, the step carries a final instruction beginning **"In a client file:"** — and that instruction is the one you actually follow. Where no such instruction appears, the step transfers unchanged.

Three situations, and what to do in each.

A populated repository in a modelling tool. Map their object types onto these sheet names once, in writing, and keep the map. Most tools hold the same content metamodel under different labels, so the mapping is 50 to 60 rows of synonym and four or five genuine mismatches — and the mismatches are the interesting part. A tool with no equivalent of `020`'s `Master system` column cannot answer the mastering question; one with no equivalent of `049`'s `Tested` column cannot answer the resilience one.

A spreadsheet nobody maintains. The supplied workbook is better than what they have. Adopt its shape and migrate their content sheet by sheet, as each phase item reaches that sheet. Do not load everything first; that is how the file in front of you was made.

Nothing at all. Start with the six sheets this volume creates plus `020`, `023`, `037` and `R06`. That is enough to answer a regulator, and four weeks of work rather than a quarter.

1.3 — Map the client's repository onto this workbook's sheet names

Sheet `00 Index` as the index of targets · **Input** an export or a screen walk-through of the client's tool · **Output** one row per object type, mapped or marked absent · **Signed by** the Chief Enterprise Architect · **Time** a day

1. Take `00 Index` rows 15 to 79 — the 65 object types and their domains — as your target list.

2. For each, name the client's equivalent object type. Where there is none, write `Absent` rather than forcing a near-match; a forced match is finding 1 in a different file.
3. For each mapped type, name the client's equivalent of the columns this volume's steps actually use. The columns that matter most are `O20` 's `Classification` , `Owner` , `Master system` and `Lineage` ; `O23` 's `Disposition` , `Criticality` and `Health` ; `O37` 's `Classification` ; `O49` 's `Tested` ; and `O27` 's `Support end` .
4. Mark the six sheets of this volume's own that have no equivalent anywhere. They will not, because no tool ships them.
5. Circulate the map with a date on it, and version it from the first week.

Form 1-C — Repository mapping

Feeds the repository mapping note in the engagement pack; no sheet in this workbook holds it · **Rows for a first pass** 65, one per object type on `00 Index` · **Signed by** the Chief Enterprise Architect · **Ratified by** `GB-01`

Sheet	Records	Client object type	Columns this volume needs	Present	Note
O20 Data Entity	14	Data Object — example row	Classification, Owner, Master system, Lineage	3 of 4	no master system field

The six sheets you will create, and the numbering is fixed

The supplied file declares six industry extension types on `EXT Industry Extension` and populates none of them. Each of the six appears as one row of `O20 Data Entity` and nowhere else. Two of the six — Consent and Medical Device — have no attributes, no mastering rule, no building block and no relationship, so there is nowhere in the file to record what a consent covers or when a device was calibrated, both of which `EXT` itself says the type holds. The word "calibration" appears twice in the whole workbook and "risk class" once.

So this volume tells you to create six sheets. **The numbering is fixed before anybody types**, because the alternative is two architects numbering the same sheet differently in week three and a day lost reconciling it.

Sheet	Prefix	Defined in	Why the file needs it
`O67 Consent`	`CNS-`	Item 5	Consent to treatment and consent to disclosure are different objects with different lifecycles, and the file has one attributeless row for both
`O68 Medical Device`	`DEV-`	Item 7	`EXT` says the type holds calibration and risk class; the file holds neither
`O69 Code System`	`CSY-`	Item 2	Code validity is effective-dated and nothing in the file records an edition
`O70 Measure Definition`	`MDF-`	Item 5	Four defensible figures come off one list and the file has nowhere to say which was used
`O71 Disclosure Event`	`DSC-`	Item 5	A withdrawal of consent is not actionable without a record of what was disclosed to whom
`O72 Retention Clock`	`RTN-`	Item 2	The period, the instrument and the destruction authority are three different facts and the file holds none of them

There is no seventh, and nothing else may claim `O67` to `O72` . The sheets take the same shape as every other object sheet: row 1 title, row 2 organisation and count, row 4 headers, data from row 5, first column `Ref` . Put them in the workbook immediately after `O66 Architecture Increment` and before `EXT Industry Extension` , and spell the tab names at full length — all six are well inside 31 characters, so none of them will truncate the way twenty-two of the existing tabs did.

1.4 — Agree the six new sheets and their numbers before anybody types

Sheet `00 Index` , plus six new tabs · **Input** this item's table · **Output** six empty sheets with row 1, row 2 and row 4 populated, and six new rows on `00 Index` · **Signed by** the Chief Enterprise Architect · **Time** two hours

1. Create the six tabs in number order after `O66` , with the exact names in the table above.

2. Write row 1 as the sheet title, row 2 as the organisation and a record count of 0, and row 4 as the column list the owning item fixes. Leave row 3 blank.
3. Add six rows to `00 Index` after row 78, in the Index's own `<n> records · <domain>` format. Use the domain the owning item states.
4. Correct `00 Index` row 2's record count to 820 in the same edit, so the count and the six new rows are one change with one signature rather than two arguments.
5. Record in the change note that `O67` to `O72` are reserved and that no other sheet may claim them.

The two overlays you carry into every phase

Two things run the length of this volume and reappear in every phase item. Know them now; the items that own them argue them properly.

Overlay 1, retrievability and the clock. Every class of record has a retention period, a destruction authority and a retrieval time, and the third is the only one that can be measured. South Africa has no retention instrument: every period in circulation traces to professional guidance that disagrees with itself across its own versions [1][2], and the National Health Act makes the record-keeping duty expressly subject to the National Archives and Record Service of South Africa Act 43 of 1996 and to the Promotion of Access to Information Act 2 of 2000, which is where the authority to destroy actually sits [3]. The Auditor-General's audit of 27 facilities across nine provinces, reported to Parliament in 2026, found 15 reporting missing or lost records, 17 not tracking records after issue and 25 with inadequate storage — and in three of nine paid medico-legal claims reviewed, departments conceded because the record could not be found [4]. Item 2 sets the clock on `O72 Retention Clock`. Item 5 records it against the entities. Item 7 measures retrieval. Item 10 gates on it. Item 11 re-runs it.

Overlay 2, classification inherits downward. An attribute may be classified more tightly than the entity it belongs to. It may never be classified more loosely. The supplied file breaks that rule in 26 of 60 rows on `O37 Data Attribute`, including `ATR-001`, the patient identifier, at `Public` beneath an entity classified `Restricted` — against the file's own principle `PRN-14` at `O05` row 18, "Classification Travels with Data ~ Classification is a property of the data, not its location". Nothing would catch it: `O32`'s `STD-03 Data classification standard` is `Current` with conformance `Not assessed`, and `O65`'s `EP-04 Data platform` records "Classification not applied at read". Every item that touches `O20`, `O37`, `O42`, `O64` or `O65` carries the rule and the check.

Week one

Volume 1, chapter 22 gives the ninety-day decision order and the documents to ask for on the first day; item 15 turns it into dated weeks. This is the narrower thing: the five working days that make the rest of this workbook usable.

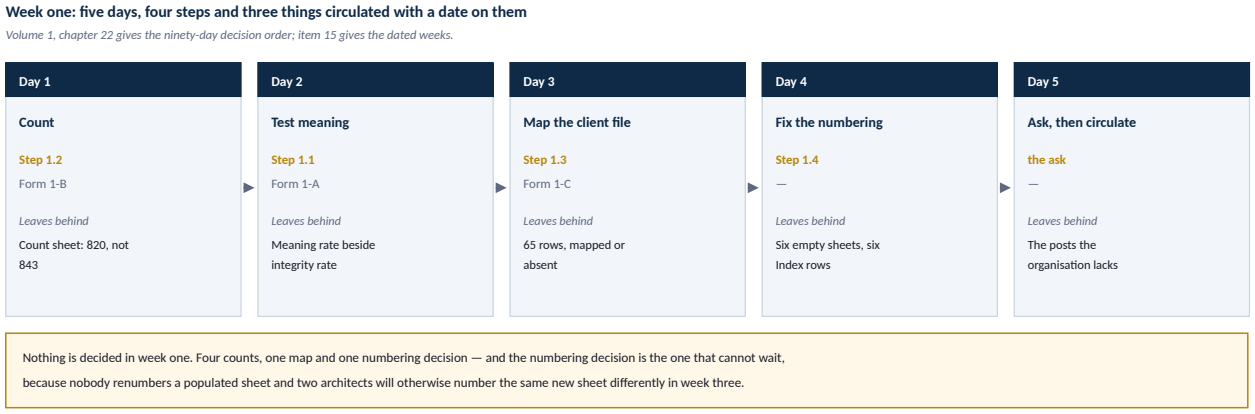


Figure 1.3 Week one, which decides nothing and corrects three things. Volume 1, chapter 22 gives the ninety-day decision order and the documents to ask for; item 15 turns it into dated weeks with named artefacts. This is the narrower list that makes the rest of the workbook usable.

Day one — get read access and count. Step 1.2, and nothing else. A corrected count sheet on the first afternoon changes how every later finding is received, because it establishes that you check.

Day two — test two relationship sheets. Step 1.1 on `R22`, then on `R04`. You now have a meaning rate to put beside the integrity rate, and you have met four people.

Day three — map the client's repository. Step 1.3. Where the client has none, spend the day on the documents volume 1, chapter 22 tells you to ask for, and record which do not exist. A request for a document that does not exist is a finding with a date on it.

Day four — fix the numbering. Step 1.4. Six empty sheets, six Index rows, one signature. Twenty minutes of typing, and the highest-value hour of the week, because nobody renumbers a populated sheet.

Day five — ask the four questions. Then circulate three things with a date on them: the count sheet, the meaning rate, and the list of posts the organisation does not have.

FOUR QUESTIONS TO ANSWER BEFORE YOU WORK ANY STEP IN THIS VOLUME

Put these to the person in charge of the establishment, or to the accounting officer's delegate, with the file open. Write the answers here, in front of them.

Who owns this repository, by name, and when did they last open it?

Which of the five bodies on `057` has met in the last quarter, and is there a minute?

Who is the Clinical Safety Officer and who is the Head of Health Records? If the posts do not exist, who does the work?

Which decisions in this estate are currently blocked on something nobody will sign?

Three organisations, kept apart

Three names recur and they do different jobs. Mixing them is the fastest way to make a page unusable.

Ravensmere Health Network is the supplied repository — fictional, as `00 Index` row 10 says, and the thing you open. Every sheet reference and every defect in this volume is Ravensmere.

Driehoek Health Group is a private hospital group: nine acute hospitals, two day clinics, about 1 750 registered beds. It carries the private-sector worked answers and the example rows in the forms.

Rietvlei District Health Service is one public health district: one district hospital, three community health centres, 22 fixed clinics and a mobile service. It carries the public-sector worked answers, and because the destruction authority runs differently for a public record it carries most of overlay 1.

Driehoek and Rietvlei never appear as a sheet reference, because they have no repository. Ravensmere never appears in a worked answer, because it has no people.

What this volume does not do

It does not teach the industry; volume 1 does, and item 17 is the map. It does not define terms; volume 1, chapter 4 defines sixty-two of them and this volume uses them without restating — admission is an event and not a stay, a funder claim and a medico-legal claim are never the bare word, and members pay a contribution. It reproduces no specification: no code list, no value set, no message layout, no framework decomposition. Where a specification governs a seam, the item names the seam and says what to ask the vendor.

And it does not promise that a repaired repository is a good architecture. It promises something narrower: that at the end of each phase you can say, in writing, with a signature and a date, what was reconciled, by whom, and against what. The supplied file cannot say that about one of its 430 relationship rows, and every one of them resolves.

THE CHAPTER IN TEN LINES

- Eight relationship sheets - R04, R07, R09, R13, R20, R22, R23 and R25 - were built by pairing two lists in row order.
- All 430 relationship rows resolve on both ends by Ref and by name, with no exceptions, exactly as 00 Index row 8 claims.
- Nothing on ZZ Checks or on any of the thirteen reports tests whether the two ends of a resolved link belong together.
- So a repository can pass every integrity check and be worthless, and the work is the reconciliation rather than the loading.
- Row 1 is the title, row 2 the banner, row 4 the headers and row 5 the first data row, on every object sheet.
- Twenty-two tab names are cut at 31 characters, R04 ends in a trailing space, and O46 has two columns headed Ref.
- The record count is 820, not the 843 printed on 00 Index, RPT01 and ZZ Checks; the Index own per-sheet figures sum to 820.
- Every step names one sheet, one input, one output, one signatory and a time, and a step whose input you lack is a step you are not ready for.
- Six signatories carry this volume, and two of them - Clinical Safety Officer and Head of Health Records - exist in no row of the file.
- Ravensmere is the repository you open; Driehoek and Rietvlei carry the worked answers and never appear as a sheet reference.
- Each step is written against Ravensmere, and where a client file changes the step the step says so in a final instruction.
- Six new sheets, O67 to O72, and the numbering is fixed before anybody types in one of them.
- Two overlays run the length of the volume: the retention clock, and classification that inherits downward and never loosens.

SOURCES FOR THIS CHAPTER

- **1.** HPCSA Booklet 9, Guidelines on Keeping Patient Records, September 2016 (retention at least six years from dormancy; minors to the 21st birthday; mentally incapacitated for the patient lifetime; occupational health 20 years; long-latency 25 years; cl. 9.5 Deputy Director-General authorisation; cl. 9.8 statutory precedence; para 13.7 billing records separate from care records) https://www.hpcsa-blogs.co.za/wp-content/uploads/2022/08/Booklet-9-Guidelines-on-Patient-Records.pdf
- **2.** HPCSA Booklet 9, Keeping of Patient Records, revision marked September 2022 (general records retained ideally indefinitely, particularly if electronic; which edition is in force could not be established) https://www.hpcsa-blogs.co.za/wp-content/uploads/2022/11/Booklet-9-Keeping-of-Patient-Records_Review-Draft_-FINAL_Sept-2022.pdf
- **3.** National Health Act 61 of 2003, consolidated text (s13 record-keeping duty, expressly subject to the National Archives and Record Service of South Africa Act 43 of 1996 and the Promotion of Access to Information Act 2 of 2000; s17 access control and the s17(2) offences) https://www.saflii.org/za/legis/consol_act/nha2003147.pdf
- **4.** Auditor-General findings on provincial health, reported via MedicalBrief, 14 October 2026 (27 facilities across nine provinces: 15 reporting missing or lost records, 17 not tracking records after issue, 25 with inadequate storage; three of nine paid medico-legal claims conceded for want of the record; claims balance R50,2bn at 31 March 2026 and R1,57bn paid in 2025/26; the recommendation to secure retrievable records) - headline source, secondary https://www.medicalbrief.co.za/health-sector-crippled-by-recurring-systemic-weaknesses-ag-warns/

ITEM 2

Standing the practice up where nothing is specified nationally

The file states the obligation, reports it unmet, and has no column in which the answer could be written - twice, which is why this phase creates two sheets.

The Preliminary question, in a country that has not answered it

The Preliminary Phase is normally the thinnest part of an engagement: confirm the framework, the principles and who governs, then move on. Here it is the thickest, and the reason is in volume 1, chapter 12. South Africa specifies almost nothing an architect needs — no national procedure classification in routine use, no national tariff since 28 July 2010 [1], no national medicines dictionary with a prescribable level, no case-mix grouping in force, no present-on-admission flag, no national planned-readmission algorithm, no national observation code subset, no national patient-identity merge policy, and no retention instrument at all.

So this phase is not a formality. It is where the organisation takes, locally and explicitly, a set of decisions nobody has taken nationally, and writes them down in a form that survives the architect leaving. Everything downstream dates from here.

The supplied repository shows what happens when the phase is skipped. Four sheets say something about one obligation, retention, and all four say it is unmanaged:

Sheet, row	What it says
`001` row 6	`DRV-02 Clinical records retention ~ Regulatory ~ Statutory retention periods by record type`, owner Chief Compliance Officer, pressure High
`063` row 6	`CO-02 Clinical records retention`, evidence `No mechanism`
`032` rows 8 and 19	`STD-04 Data retention standard ~ Emerging ~ Not assessed`; `STD-15 Records management standard ~ Deprecated ~ Breach` — a breach of a withdrawn standard, which nobody can act on
`042` row 10	`MTD-06 Retention rule ~ How long it is kept and why ~ Absent`

Now look for somewhere to fix it. `020 Data Entity`'s eight columns — `Ref`, `Data entity`, `Class`, `Definition`, `Classification`, `Owner`, `Master system`, `Lineage` — hold no period. `037 Data Attribute` gives all ten of its entities the same six attributes, Identifier, Name, Status, Effective date, Owner and Classification, and none is a retention attribute. `DRV-02` says "by record type"; `020`'s fourteen rows have nowhere to put a type's period.

That is the pattern this item fixes twice. The file states an obligation, reports it unmet, and holds no column in which the answer could be written. A repository that cannot hold the answer has not failed to record it; it has made recording it impossible.

Establishing the architecture function, which already exists on paper

Start with the good news, because there is some. `057 Governance Body` holds five rows and they are sensible:

Ref	Governance body	Cadence	Mandate, as the sheet words it
GB-01	Architecture Board	Monthly	Holds G1 and G2; approves dispensations with an expiry
GB-02	Investment Committee	Quarterly	Holds G0 funding and G3 benefit confirmation
GB-03	Data Governance Council	Monthly	Designates data owners and resolves disputes
GB-04	Executive Committee	Monthly	Escalation point when bodies cannot agree
GB-05	Architecture Function	Continuous	Publishes criteria, assesses compliance, maintains the repository

`058 Governance Gate` holds the four gates those bodies hold: `GT-0 G0 Charter` before work starts, `GT-1 G1 Design` before build, `GT-2 G2 Build complete` before release, `GT-3 G3 Benefit` after operation. `GB-05`'s mandate is a job description for the reader of this volume.

Now the bad news, in three parts.

The governance layer hangs off nothing. `057`, `058`, `060 Governance Decision` and `061 Governance Change Request` appear in no relationship sheet at all — along with sixteen other object sheets, 206 records in total, including every principle, constraint, business process, deliverable, gate, decision, change request and delivery risk. `RPT13 Orphan Analysis`, which `00 Index` row 12 calls the sheet that "finds what nothing references", reports two orphans, both applications.

One gate of four has ever been used. `059 Compliance Assessment` holds three rows, all testing a building block at `G2`, all `Non-compliant`, all with evidence `No evidence produced`. Thirty-four of `055`'s 37 building blocks are never assessed, and `GT-0`, `GT-1` and `GT-3` are exercised by nothing.

The decisions the bodies own are outstanding. Four of `060`'s seven are `Outstanding`: adopt the recommended alternative, select the integration product, fund compliance mechanisms, designate data owners. Four of `061`'s five change requests have no `Expires` value, including `CR-G05 Refuse or re-baseline the integration package` at status `Board decision required` with no date by which the board must decide. And `CR-G03 Raise unfunded building blocks as gaps` is raised against `Repository` — a change request against the file itself, still only `Raised`.

So the architecture function does not need establishing. It needs connecting, staffing and minuting.

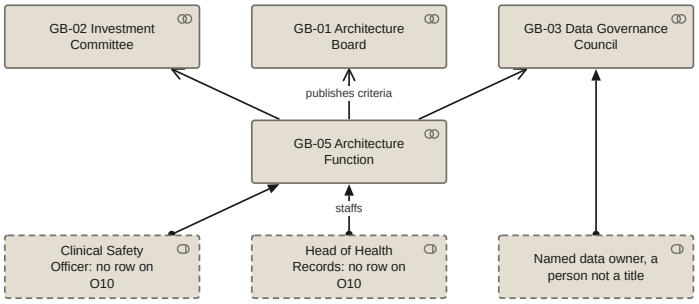


Figure 2.1 The governance layer the file names and does not connect. O57 holds all five bodies, including GB-04 Executive Committee as the escalation point, and O58 the four gates they hold. O57, O58, O60 and O61 appear in no relationship sheet at all. The three dashed roles are the posts this item creates: two have no row on O10 or O11, and the third must be a named person rather than the job title O20 carries.

2.1 — Staff the five bodies the file names and does not staff

Sheet `O57` · **Input** the current organogram and the last four quarters of minutes · **Output** one named person per seat, per body, with a first meeting date · **Signed by** the Chief Enterprise Architect · **Time** two days

1. For each of `GB-01` to `GB-05`, read the `Membership` cell. It names titles, not people.
2. Replace each title with a named person. Where the title has no holder, write the title, `vacant` and the date you established it. A vacancy with a date is a finding; a title is not.
3. For `GB-03`, note that `DEC-04 Designate data owners` is `Outstanding` while `O20` already names an owner for all fourteen entities, eight of those titles existing nowhere else in the file. Put it on the first agenda rather than resolving it here; item 5 owns it.
4. Fix the cadence to one the organisation will hold, because the gate dates are set from it, and record each body's first meeting date. That date, not the sheet, is the evidence of existence.
5. In a client file: where there is no body equivalent to `GB-03`, do not invent one. Record that data-ownership disputes escalate to the executive committee, and price that.

Form 2-A — Governance body membership

Feeds `O57` · **Rows for a first pass** 5, one per governance body · **Signed by** the Chief Enterprise Architect · **Ratified by** `GB-04`

Columns 1 to 4 are `O57`'s own columns and are typed straight back into Excel. Columns 5 and 6 are working columns.

Ref	Governance body	Cadence	Membership	Named holders	First meeting
GB-03	Data Governance Council	Monthly	CDO, business data owners, stewards	example row — 1 named, 4 vacant	18 Mar

The two posts the file has no row for

`O09 Organization Unit` holds twelve units: Executive, Finance, Risk & Compliance, Human Resources, Technology, Data & Analytics, Operations, Commercial, Customer Service, Supply & Logistics, Internal Audit, Legal. There is no medical directorate, no nursing, no pharmacy, no allied health, no clinical governance unit and no diagnostics unit — in a repository whose `00 Index` row 6 describes its own scope as "Provider healthcare — patients, encounters, clinical pathways and the records that follow them".

`O10 Actor` holds eighteen actors, all `Filled`: seven executives, five architects, three delivery and operations heads, a data steward, a security architect and an internal auditor. No clinician, no pharmacist, no clinical safety officer, no records manager. `O11 Role` mirrors it one for one, every `Performed by` reading `Person`, and `R21` reports all eighteen `Filled`.

That absence causes a surprising amount of the rest of the file: why `R22` cannot put a clinical capability with a plausible owner, why `O23`'s clinical applications are owned by human resources, internal audit and customer service, and why `O62 Delivery Risk` holds eight programme-generic risks and not one clinical one — with `RSK-06 Key supplier withdraws capability` owned by "Head of Procurement", a title appearing nowhere as an actor or a role.

Two posts have to exist before the rest of this volume can be worked, because its steps name them as signatories.

Clinical Safety Officer, who signs anything whose failure mode is clinical. `DRV-04 Clinical safety standards ~ Hazard assessment and clinical safety case for IT` is a stated obligation and `CO-04`'s evidence is `In breach`, while there is no hazard and no safety case anywhere in the 105 sheets — `O31 Deliverable`'s twenty artefacts run Architecture Vision to Requirements Catalogue and include none. The post needs no new unit; it needs a named clinician with time allocated and a reporting line into `GB-01`.

Head of Health Records, who signs record classes, measured retrieval time and disposal evidence. Volume 1, chapter 5 locates the statutory duty: the person in charge of the establishment must ensure a record is created and maintained for every patient, and that duty is expressly subject to the National Archives and Record Service of South Africa Act 43 of 1996 and to the Promotion of Access to Information Act 2 of 2000 [2]. Asking who holds it, by name, is a Preliminary question.

2.2 — Add the two roles the file has no row for, and say what they sign

Sheet `O10`, `O11`, `R21` · **Input** step 2.1's membership · **Output** two new rows on each of `O10` and `O11`, two on `R21`, and a one-line statement of what each signs · **Signed by** the Chief Enterprise Architect · **Time** half a day

1. Add `ACT-19 Clinical Safety Officer ~ Clinical` and `ACT-20 Head of Health Records ~ Records` to `O10`, with the real status. `Clinical` and `Records` are new values in the `Type` column, whose existing values are Executive, Architecture, Delivery, Operations, Data, Security and Assurance.
2. Add the matching `ROL-19` and `ROL-20` to `O11` with `Performed by ~ Person`, and two rows to `R21` with the real status rather than `Filled` by default.
3. Write one line per post naming what it signs in this volume. The signature list is the mandate; a job description is not needed.
4. Note on the change record that `O09` has no unit for either post, reported and not fixed. Creating a clinical directorate is not an architecture decision.
5. In a client file: both posts usually exist under other names — clinical risk lead, information governance manager, records manager. Use the client's name and map it once.

FOUR QUESTIONS TO PUT TO THE PERSON IN CHARGE OF THE ESTABLISHMENT

Ask these in a room, with `O57` and `O10` on the screen, and write the answers here in front of them.

Who signs off a design that carries a clinical risk, by name, today?

Who may authorise the destruction of a patient record here, and what do they sign?

Which of the five bodies on O57 has met in the last quarter, and where is the minute?

If a record cannot be found for a medico-legal claim, who finds that out, and when?

The principles position: decide the status, not the set

The temptation in Preliminary is to rewrite the principles. Do not. **Item 19 owns the starter set**, including the healthcare principles the generic twenty-four lack. Preliminary's job is narrower and takes an afternoon.

`O05 Principle` holds 24 rows under `Ref`, `Principle`, `Category`, `Statement`, `Owner`, `Status`. The set is the generic one — Primacy of Principles, Maximise Benefit to the Enterprise, Common Use Applications, Technology Independence, Ease of Use, Service Orientation and the rest. Not one of the 24 is about clinical safety, the clinical record, consent or the patient, in a provider repository. Three facts matter more than the wording.

All 24 are owned by "Architecture Board", which is `GB-01` on `O57` and appears on neither `O10` nor `O11`. Every principle in this file is owned by a committee, against the file's own `PRN-13 Data Ownership is Individual ~ An owner is a person, not a committee`.

Four are `Draft` and nothing turns on the difference. `PRN-05`, `PRN-10`, `PRN-15` and `PRN-20` are `Draft` — `PRN-20` being Least Privilege, in a file whose four security policies on `O64` are all `Breached`. No sheet records what a `Draft` principle does not bind, or when it will be decided.

All 24 are orphans. `O05` appears in no relationship sheet, so no requirement derives from a principle and `EP-01 Design review gate` has nothing to review against but prose.

2.3 — Re-own and re-status the twenty-four principles, and record the gap

Sheet `O05` · **Input** `O57`, `O10`, `O11`, and item 19's starter set to point at · **Output** 24 rows with a person in `Owner` and a decided `Status` · **Signed by** the Chief Enterprise Architect · **Time** half a day

1. Replace "Architecture Board" in `Owner` with a named person for each of the 24. One person may own several; a committee may own none.
2. Decide each `Draft`. A principle is `Approved` or removed; one that is permanently `Draft` is a preference, and a design cannot be refused against a preference.
3. Record, as a single finding, that none of the 24 addresses clinical safety, consent, the clinical record or the patient, and point at item 19 rather than drafting replacements. Preliminary fixes ownership and status; Phase A adopts the set.
4. Leave `O05`'s orphan status alone. Item 12 closes it, by tracing requirements to principles rather than only to drivers.

The four-way sourcing question, decided here and not in Phase E

Volume 1, chapter 15 opens on four ways to get a capability, and in South African healthcare all four are in use at once: build, buy, adopt open source, and consume a state service. The chapter's point is that each route commits the organisation to holding a different capability permanently, and the fourth is the one architects from other industries have no instinct for — there is no agreement to negotiate, no service level to agree and no escalation that works on a project timescale, so the only lever is the degraded-mode design on your own side.

That is why the question belongs here and not in Phase E. A dependency on a state service sets a recovery objective Phase D must then design to, and volume 1, chapter 1 gives the reason it cannot wait: downtime in this industry is clinical, the planning assumption is weeks rather than hours, and the dominant failure mode is a shared service the organisation does not own.

The supplied file has nowhere to record a route. `O23 Application Component`'s `Type` column holds Package, Custom and Platform and its `Disposition` column holds Invest, Retain, Replace, Retire and Consolidate; neither is a sourcing route. `O19`'s `Counterparty` column holds the system's own name in all ten rows — `CTR-01 EHR / EPR vendor agreement` has counterparty "EHR / EPR" — so the column that should name who the organisation is bound to names the thing it bought. `A11 Integration Engine` and `A12 Clinical Data Warehouse` have no `O19` row at all, which means the two platform components the rest of the estate depends on are, as far as this repository knows, supplied by nobody.

2.4 — Record the sourcing route per application and the state dependencies

Sheet `O23` and `O19` to read; the sourcing register in the Preliminary pack to write · **Input** the agreement file and one conversation with procurement · **Output** one route per application and one row per shared service not owned · **Signed by** the Chief Enterprise Architect · **Time** two days

1. For each of `O23`'s twelve applications, record one of four routes: built, bought, open source adopted, or a state service consumed. Where two apply, record both and say which carries the support obligation.
2. For each bought application, name the counterparty from the agreement rather than from `O19`, and record the expiry. `A11` and `A12` have no `O19` row; establish whether that is a missing row or a missing agreement.
3. List the shared services the organisation depends on and does not own. Volume 1, chapter 1 names the four that stop clinical work in South Africa: public-sector pathology, the private claims switching network, the national identity service and chronic medicines dispensing.
4. For each, record three things and nothing else: what stops when it stops, how long it stopped last time, and what the organisation did instead. The third is the degraded-mode design, usually undocumented rather than absent. Carry the result into Phase A as a constraint rather than a risk; `O07 Constraint` holds eight rows and a shared service you do not own belongs there.

Form 2-B — Sourcing route and shared-service dependency

Feeds the sourcing register in the Preliminary pack; no sheet holds a route · **Rows for a first pass** 12 applications plus the shared services · **Signed by** the Chief Enterprise Architect · **Ratified by** `GB-02`

Ref	Application	Type	Disposition	Sourcing route	What stops when it stops
A11	Integration Engine	Platform	Consolidate	bought — example row; no O19 row	19 of 23 interfaces bypass it already

O69 Code System, argued from the file

Search the supplied workbook for a code system. You will find one string, five times: "SNOMED CT, LOINC and dm+d coding at the point of use" — as `REQ-029`'s statement on `O06`, `LTC-02`'s rationale on `O26`, `TBB-02`'s requirement on `O46`, `RME-02`'s description on `O47`, and `TBB-02` again on `O55`. One sentence, copied five times, under the name "Terminology services". That is a **component name**, not a code system, and a component name has no edition, no version, no effective date and no licence position. The string "ICD" appears nowhere in the 105 sheets. Neither does DICOM.

Now look at the two entities defined by coding. `O20` row 8: `E04 Diagnosis ~ Transactional ~ A coded condition attributed to a patient`. Row 9: `E05 Procedure ~ Transactional ~ A coded intervention performed`. Both turn on the word "coded". Read their attributes on `O37 Data Attribute`: Identifier, Name, Status, Effective date, Owner, Classification — the same six every one of `O37`'s ten entities gets. **There is no code attribute, no code-system attribute and no code-version attribute anywhere in the repository.** A diagnosis defined as a coded condition has nowhere to record the code or the system it was coded against, while `DRV-06 Funder and coding rules ~ Accurate coding and claim substantiation` is a stated regulatory driver at pressure `High` whose `CO-06` evidence reads `In breach`.

One further piece of the file shows that a component name cannot carry the job. `O26` and `O47` give different statuses to the same seven items in four cases: terminology services is `Current` and `Candidate`, clinical decision support `Emerging` and `Adopted`, zero-downtime clinical systems `Target` and `Candidate`, de-identification and secure enclave `Target` and `Adopted`. A status two sheets disagree on is not a status — and four requirements, `REQ-028`, `REQ-029`, `REQ-030` and `REQ-032`, propose adopting components `O26` already records as `Current`, at priority `Could` and status `Proposed`.

So the argument for the sheet is not that code systems matter. It is that **code validity is effective-dated and nothing in this file records an edition.** Volume 1, chapter 12 gives the mechanism: the national reference table for diagnosis coding carries South African start and end dates per code, a status of added, modified or discontinued, and a revision history, and updates biennially on 1 July with three months allowed for operational change and six for system change [3]. A record from 2019 is validated against the table in force in 2019, and a warehouse joining historical coded data to the current table silently drops codes discontinued since.

And the licence position is a separate fact from the version, which is why the sheet needs both columns. Volume 1, chapter 12 carries three positions to know before filling it. The private-sector procedure code set is licensed, its local copyright sits with one body and the underlying American content with another, and its own standards document positions it for data rather than for billing while the industry standards body declines to recommend it nationally [4][5]. The observation naming scheme is royalty-free, which makes it the lowest-friction change available [6]. And the clinical terminology became deployable in-country when South Africa joined SNOMED International on or about 14 May 2026, because a member licence covers use within the member country — while no national release centre is operating, so there is no national extension, no national reference sets and no published national map [7][8]. The gazetted interoperability framework of 21 October 2022 names neither that terminology nor the observation naming scheme, and the classification appears in it only in an acronym list [9].

Four different answers in one column, and nowhere in the supplied file to put any of them.

> **`O69 Code System — new; the supplied repository does not have it`**
>
> **Prefix** `CSY-` · **Domain** Data · **Place** after `O68 Medical Device`, before `O70 Measure Definition` · **Rows for a first pass** 8 to 14 · **Signatory** the named data owner of the entity the system is bound to
>
> Columns, in sheet order: `Ref` · `Code system` · `Edition or version` · `Effective from` · `Effective to` · `Licence position` · `Where bound` · `Owner`
>
> One row per code system per edition, in use anywhere in the estate — including the ones nobody calls a code system: a laboratory's local test catalogue, a theatre picklist, a ward's reason-not-given list. `Effective to` is blank for the edition in force and dated for every superseded one, which is what makes a point-in-time join possible. `Where bound` names the entity and attribute, which is how item 5 joins `O69` to `O20` and `O37`.

2.5 — Define O69 Code System and fix its shape before anybody types in it

Sheet `O69`, new · **Input** step 1.4's six empty tabs · **Output** rows 1, 2 and 4 populated and one row on `00 Index` · **Signed by** the Chief Enterprise Architect · **Time** an hour

1. Write row 1 as `Code System`, row 2 as the organisation, `Data domain` and `0 records`, row 3 blank, row 4 as the eight column names above, spelled exactly.
2. Add `O69 Code System ~ 0 records · Data` to `00 Index`, in the Index's own format.
3. Record the prefix `CSY-` and the rule that one row is one code system at one edition. Two editions of the same system are two rows with the same `Code system` value and different `Effective from` dates. This is the decision that makes or breaks the sheet.
4. Record the rule that `Effective to` is blank for exactly one row per `Code system`. A system with two open-ended rows is a defect the sweep in item 20 will catch.

2.6 — Populate O69 from the coded fields and the interfaces

Sheet `O69` · **Input** `O37`, `O20`, `R06`, and one hour each with the coding lead, the laboratory and the pharmacy · **Output** one row per code system per edition · **Signed by** the named data owner, per entity · **Time** three days

1. Start with the coded fields rather than the standards. For each of `O20`'s fourteen entities, ask what in it is a code: `E04` and `E05` are coded by definition, and `E03`, `E06` and `E07` carry codes in practice.

2. Name the system and the edition in force for each. Where nobody can name the edition, write `unknown` and the date you asked. An `unknown` here is the finding.
3. Work `R06` 's 23 interfaces next. An interface carrying a code carries a code system, and the two ends may be on different editions.
4. Add the local code sets nobody calls code systems. They are the majority by row count and the source of most cross-site comparison failures.
5. Fill `Licence position` with permitted uses, permitted recipients and expiry, not a vendor name — this is the one column where an extract to a research partner may be a licensing breach rather than a privacy question — and `Where bound` with the entity and attribute. A code system bound to nothing is a licence somebody pays for and does not use.
6. In a client file: the fastest route in is the funder's rejection report. Every rejected code is a binding somebody got wrong, named with its system, edition and seam.

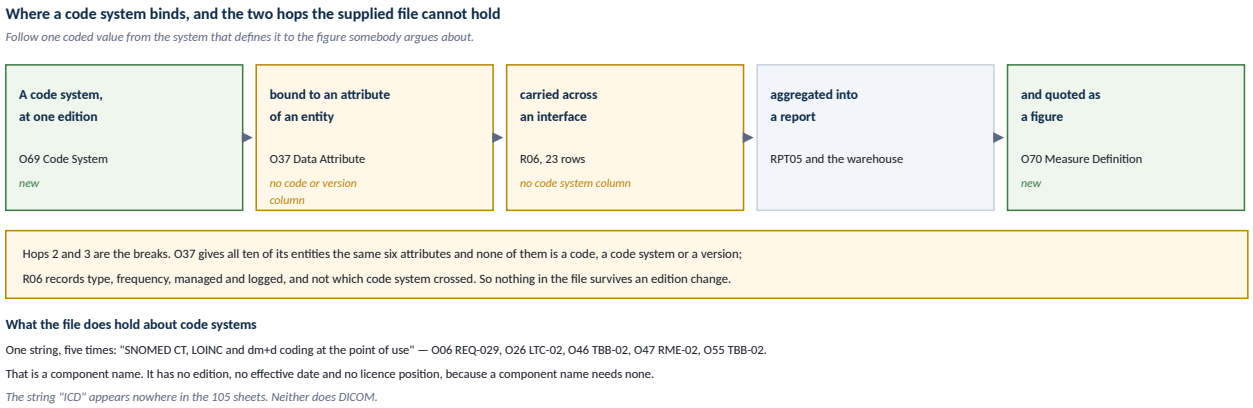


Figure 2.2 One coded value, five hops, and two of them have nowhere to live. O69 Code System and O70 Measure Definition are new sheets. The breaks in the middle are why a code system cannot be recorded in the supplied file even when the organisation knows which one it uses.

Form 2-C — O69 Code System, part one: identify

Feeds `O69` , which the supplied repository does not have · **Rows for a first pass** 8 to 14, one per code system per edition · **Signed by** the named data owner · **Ratified by** `GB-03`

All five columns go straight into Excel; form 2-D carries the other four, keyed on `Ref` .

Ref	Code system	Edition or version	Effective from	Effective to
CSY-03	Laboratory local test catalogue — example row	as at 2026-07 release	2026-07-01	

Form 2-D — O69 Code System, part two: assess

Feeds `O69`, which the supplied repository does not have · **Rows for a first pass** the same 8 to 14 rows as form 2-C · **Signed by** the named data owner · **Ratified by** `GB-03`

`Ref` and `Code system` repeat from form 2-C as the key; the last three go straight into Excel.

Ref	Code system	Licence position	Where bound	Owner
CSY-03	Laboratory local test catalogue	none needed — example row	E07, result code	Chief Pathologist

O72 Retention Clock, argued from the file and from the instruments

This is the sheet overlay 1 runs on, and it is set here. Every later phase references it: Phase C records the clock against the entities, Phase D measures retrieval, Phase G gates on the measurement, Phase H re-runs it. Nothing downstream works if these rows are wrong.

The case from the file was made at the top of this item. The case for **three columns rather than one** comes from the instruments, and it is the most useful thing an architect can know about South African health records.

There is no South African health-record retention instrument. Every period in circulation traces to professional guidance made under the Health Professions Act 56 of 1974 — not a statute and not a regulation — and that guidance disagrees with itself across its own versions [10][11].

Record class	September 2016 text	September 2022 revision
General adult records	At least 6 years from dormancy	Ideally indefinite, particularly if electronic; otherwise 6 years from dormancy
Minors, including obstetric	Until the 21st birthday	At least until the 21st birthday
Mentally incapacitated patients	The lifetime of the patient	For the patient's lifetime
Occupational health records	20 years after treatment	20 years after treatment
Long-latency exposure conditions	Minimum 25 years recommended	Minimum of 25 years recommended
Closed or deceased private practices	At least 12 months	At least 12 months in safe keeping

Three provisions travel with those periods, and each is a column on this sheet: statutory periods take precedence; records in provincial hospitals and clinics may be destroyed only with the authorisation of the relevant Deputy Director-General; and retention decisions should weigh storage cost against the value of records for defence, research and teaching [10].

The destruction authority is a different fact from the period, and it sits somewhere else. The National Health Act places the duty to create and maintain a record on the person in charge of the establishment, and makes that duty expressly subject to the National Archives and Record Service of South Africa Act 43 of 1996 and to the Promotion of Access to Information Act 2 of 2000 [2]. For a provincial facility that subordination is where the decision actually sits, which is why a district hospital cannot set its own schedule — and why a private group can. The same Act makes creating, changing or destroying a record without authority an offence, alongside failing to create or change one when properly required [2]. Under-documentation is capable of being an offence here, which is an unusually strong footing for a completeness requirement.

And "six years" is wrong three times over. It is not a statute; the same guidance's later edition says ideally indefinite; and six years does not guarantee the record exists when the medico-legal claim arrives, because the Prescription Act 68 of 1969's three-year period for an adult runs from when the claimant knew or ought reasonably to have known of the claim, which in a delayed-diagnosis case is many years after the consultation [12].

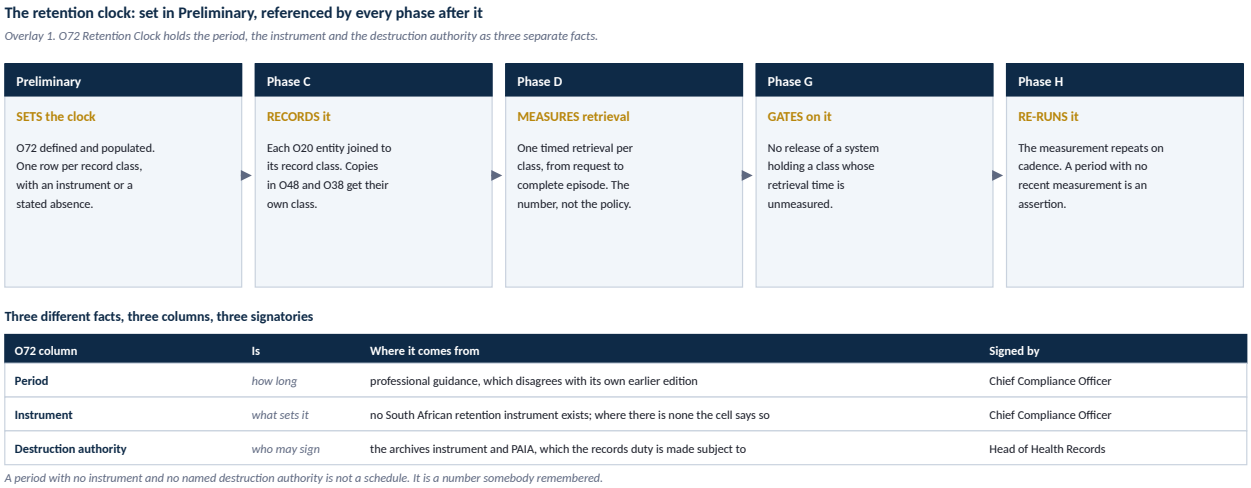


Figure 2.3 Overlay 1, which starts here. The Preliminary Phase sets the clock on O72; every later phase references the same rows rather than re-deciding them. The three facts in the lower band are the reason the sheet has eight columns rather than two.

The measured consequence is what buys the work. The Auditor-General's audit of 27 facilities across nine provinces, reported to Parliament in 2026, found 15 reporting missing or lost records, 17 not tracking records after issue and 25 with inadequate storage — and in **three of nine paid medico-legal claims reviewed, departments conceded because they could not locate the medical record**, against a provincial claims balance of R50,2bn at 31 March 2026 and R1,57bn paid to claimants in 2025/26 [13]. The recommendation uses the operative word: secure **retrievable** records.

One more reason the sheet has eight columns rather than four. A period attaches to a record class and not to every copy of it, and the supplied file is full of copies with no period at all: `O48 Environment`'s four environments, `O38 Data Set`'s six sets, `A12 Clinical Data Warehouse`, `A11 Integration Engine`'s message store, and whatever backup regime nobody has written down. **The retention period of a copy is a function of the copy's purpose, not of the source record's class** — volume 1, chapter 13 argues it — and `System of record` plus `Hold rule` make the distinction recordable.

> **`O72 Retention Clock — new; the supplied repository does not have it`**
>
> **Prefix** `RTN-` · **Domain** Governance · **Place** after `O71 Disclosure Event`, at the end of the object sheets and before `EXT Industry Extension` · **Rows for a first pass** 10 to 16 · **Signatory** the Chief Compliance Officer, countersigned by the Head of Health Records
>
> Columns, in sheet order: `Ref` · `Record class` · `Period` · `Clock starts at` · `Instrument` · `System of record` · `Hold rule` · `Destruction authority`
>
> One row per record class, not per entity and not per system. `Clock starts at` is an event — last contact, dormancy, a 21st birthday, a date of treatment, a date of death — because a period with no trigger cannot be computed. `Instrument` names what sets the period and its edition, and where there is none it reads `none — local decision` plus the date and whoever decided. `Hold rule` is what suspends the clock: litigation, a complaint, an inspection, an access request. `Destruction authority` names who signs, which for a provincial record is not the same post as for a private one.

2.7 — Define O72 Retention Clock and decide what a record class is

Sheet `O72`, new · **Input** step 1.4's six empty tabs · **Output** rows 1, 2 and 4 populated, one row on `00 Index`, and a written definition of a record class · **Signed by** the Chief Compliance Officer · **Time** half a day

1. Write row 1 as `Retention Clock`, row 2 as the organisation, `Governance domain` and `0 records`, row 3 blank, row 4 as the eight column names above.
2. Add `O72 Retention Clock ~ 0 records · Governance` to `00 Index`.
3. Decide, in writing, what a record class is here. The professional guidance's own classes are a defensible starting set and are not exhaustive: they say nothing about an access log, a message store or a de-identified research extract.
4. Record two rules. Every row names an `Instrument` or says there is none — a blank is how a schedule built on a secondary source gets signed. And `Destruction authority` names a post, not a body, because a committee cannot sign a disposal authority.

2.8 — Set the clock: one row per record class

Sheet `O72` · **Input** the current edition of the professional guidance, obtained from the council rather than quoted from a secondary account; the organisation's own policy if it has one; `O48` and `O38` for the copies · **Output** 10 to 16 rows, each with a period, an instrument and a destruction authority · **Signed by** the Chief Compliance Officer, countersigned by the Head of Health Records · **Time** a week

1. Obtain the current edition of the guidance first. Two are in circulation and which is in force could not be established for volume 1 [10][11]. Record the edition and date you built against in the `Instrument` column of every row that relies on it; a schedule citing no edition is unauditable.
2. Write the five classes the guidance covers — general adult, minors including obstetric, mentally incapacitated, occupational health, long-latency exposure — each with its period, trigger and instrument.

3. Write the classes it does not cover, which are the ones that will be argued about: the access log, the integration message store, the warehouse extract, the de-identified research dataset, the backup set, and the billing record, which the guidance requires to be kept separate from the care record [10] and which therefore has its own class and period.
4. Set each of those periods from the copy's purpose. A message store holding clinical messages for replay has a purpose measured in days and is routinely kept for years, because nobody assigned it one.
5. Fill `Destruction authority` differently per organisation type. For Rietvlei District Health Service it is the Deputy Director-General authorisation plus a disposal authority issued against an approved schedule under the archives instrument; for Driehoek Health Group it is the organisation's own schedule with a signed disposal authority and a register of destroyed records.
6. Set `Hold rule` on every row — a schedule with no hold rule will delete the record that was about to be subpoenaed — and automate no deletion. Item 11 does that, after retrieval has been measured in Phase D and gated in Phase G. In a client file, ask for the schedule and the authority cited per period: an empty but visible `Instrument` column is a better deliverable than a confident wrong one.

Form 2-E — O72 Retention Clock, part one: identify

Feeds `O72`, which the supplied repository does not have · **Rows for a first pass** 10 to 16, one per record class · **Signed by** the Chief Compliance Officer · **Ratified by** `GB-01`

All five columns go straight into Excel; form 2-F carries the other four, keyed on `Ref`.

Ref	Record class	Period	Clock starts at	Instrument
RTN-02	Minors, including obstetric — example row	to 21st birthday	date of birth	professional guidance, cl. 9.3.1

Form 2-F — O72 Retention Clock, part two: assess

Feeds `O72`, which the supplied repository does not have · **Rows for a first pass** the same 10 to 16 rows as form 2-E · **Signed by** the Head of Health Records · **Ratified by** `GB-01`

`Ref` and `Record class` repeat from form 2-E as the key; the last three go straight into Excel.

Ref	Record class	System of record	Hold rule	Destruction authority
RTN-02	Minors, including obstetric	A01 plus paper — example row	any open claim or complaint	Deputy Director-General

FIVE QUESTIONS THAT SET A RECORD CLASS'S PERIOD

Ask the Head of Health Records and the Chief Compliance Officer together. Where the two disagree, that is the finding.

Which edition of the professional guidance is in force here, and who holds a copy?

What event starts the clock for this class, and which system records that event?

Who has ever authorised a destruction here, and what did they sign?

Where is the register of destroyed records, and what does it hold per entry?

What suspends the clock, and who is told when it is suspended?

Classification inherits downward: appoint the owner of the check

Overlay 2 runs the length of this volume and item 19 states it as a principle. The rule: an attribute may be classified more tightly than the entity it belongs to, and never more loosely.

The supplied file breaks it in 26 of 60 rows on `O37`: `ATR-001`, the patient identifier, is `Public` beneath an entity classified `Restricted`, every identifying and dating attribute of Diagnosis is `Public` beneath an entity classified `Internal`, and nineteen of the 60 are `Public` outright — all against the file's own `PRN-14` at `O05` row 18, "Classification Travels with Data ~ Classification is a property of the data, not its location".

Item 5 fixes that. Preliminary does one thing, and it takes ten minutes once the bodies are staffed: name the person who runs the check and the gate it runs at. It cannot wait, because nothing in the file would catch the breach — `STD-03` is `Current` with conformance `Not assessed`, `EP-04 Data platform` records "Classification not applied at read", and `O64`'s four security policies are all `Breached`. A rule nobody checks is a preference.

2.9 — Name the owner of the classification check and the gate it runs at

Sheet `O05`, `O32`, `O65`, and the gate list on `O58` · **Input** step 2.1's membership · **Output** one named person, one gate, one cadence · **Signed by** the Chief Data Officer · **Time** an hour

1. Name the person who runs the check: not the security architect and not the data steward, but whoever `GB-03` says owns classification — a decision `GB-03` has not taken.
2. Name the gate. `GT-1 G1 Design` is the right one, because an attribute's classification is set in design and inherited from then on.
3. Set the cadence: every design at `GT-1`, plus a full sweep of `O37` once a quarter until the 26 rows are zero. Record that `STD-03`'s conformance is `Not assessed` and that assessing it is now somebody's named job, with a date.

What Preliminary signs, and what it refuses to decide

Every item in the pack has one signature and one date.

Artefact	Where it lives	Signed by
Governance body membership, with first meeting dates	`O57` plus form 2-A	Chief Enterprise Architect
Two new roles and what they sign	`O10` , `O11` , `R21`	Chief Enterprise Architect
Principles re-owned and re-statused, with the healthcare gap recorded	`O05`	Chief Enterprise Architect
Sourcing route per application and the shared services not owned	the sourcing register, form 2-B	Chief Enterprise Architect
`O69 Code System` , defined and populated	`O69` , forms 2-C and 2-D	the named data owner per entity
`O72 Retention Clock` , defined and set	`O72` , forms 2-E and 2-F	Chief Compliance Officer with the Head of Health Records
The classification check, with an owner and a gate	`O05` , `O32` , `O58`	Chief Data Officer

2.10 — Sign the Preliminary pack, and record the refusals on the same page

Sheet `O60` and `O31` · **Input** the seven artefacts above · **Output** one decision row per artefact on `O60` , and a one-page refusal list · **Signed by** the Chief Enterprise Architect · **Time** half a day

1. Add one row to `O60 Governance Decision` per artefact, naming the body and a status that is not `Outstanding` . The sheet holds four `Outstanding` rows and no mechanism for closing one. Record on `O31 Deliverable` that this phase produced artefacts its twenty-row list does not include: `O31` has no Preliminary row and its `Phase` values run A to G plus `RM` .
2. Write the refusals on the same page as the decisions, in the form volume 1, chapter 22 gives: this decision has a named prerequisite, here it is, here is the date we expect it, and here is what we will do meanwhile.
3. Refuse four here. The terminology binding, until a national release centre operates [7][8]. The retention periods, until the current edition of the guidance is in hand [10][11]. The target application estate, until identity, terminology and integration are decided. And any automated deletion, until retrieval has been measured.
4. Circulate the pack with the date on it, and version it from the first week.

Two readings this item cannot resolve, carried here because every later item inherits them. `O46` 's `T01` to `T07` resolve to nothing, and the file supports two explanations — a mis-keyed pointer to `O26` , or a dropped sheet. The cells favour the first; neither can be proved. And `O66 Architecture Increment` dates its six increments "Month 10" to "Month 56" while `O56 Project` carries absolute dates, with no programme baseline anywhere in the 105 sheets. Report `INC-1` 's internal ordering problem — the increment landing at month 10 is delivered by the latest-starting of the six projects — and do not invent a programme start date to make the arithmetic work. An invented baseline is finding 1 in a new column.

THE CHAPTER IN TEN LINES

- South Africa specifies almost nothing an architect needs, so the Preliminary Phase is the thickest part of the engagement rather than the thinnest.
- Four sheets of the supplied file record that retention is unmanaged, and no sheet holds a column in which a period could be written.
- O57 names five governance bodies and O58 four gates, and O57, O58, O60 and O61 appear in no relationship sheet at all.
- One gate of four has ever been used: all three rows of O59 test a building block at G2 and all three read Non-compliant with no evidence produced.
- Two posts have to be created before this volume can be worked, because its steps name them as signatories and the file has no row for either.
- O05 twenty-four principles are the generic set, all owned by a committee that is neither an actor nor a role, and all of them orphans.
- Item 19 owns the principles starter set; Preliminary fixes ownership and status and records the healthcare gap.
- The four-way sourcing question is a Preliminary decision, because a state service you consume sets a recovery objective Phase D must design to.
- O69 Code System exists because code validity is effective-dated and the file records one component name five times and no edition anywhere.
- O72 Retention Clock exists because the period, the instrument and the destruction authority are three separate facts and the file holds none of them.
- Every period in circulation traces to professional guidance that disagrees with its own earlier edition, so six years is wrong three times over.
- The destruction authority for a public record sits with the archives instrument the record-keeping duty is made subject to, not with the facility.
- Classification inherits downward and never loosens; Preliminary names the person who runs the check and the gate it runs at.

SOURCES FOR THIS CHAPTER

- **1.** Hospital Association of South Africa Ltd v Minister of Health and Another, ZAGPPHC 69 of 2010, judgment 28 July 2010 (the reference price list regulations declared invalid and set aside; no national tariff since) https://www.saflii.org/za/cases/ZAGPPHC/2010/69.html
- **2.** National Health Act 61 of 2003, consolidated text (s13 record-keeping duty, expressly subject to the National Archives and Record Service of South Africa Act 43 of 1996 and the Promotion of Access to Information Act 2 of 2000; s17 access control and the s17(2) offences) https://www.saflii.org/za/legis/consol_act/nha2003147.pdf
- **3.** User guide to the ICD-10 Master Industry Table, January 2014, National Department of Health (South African start and end dates per code; added, modified or discontinued status; revision history; biennial update on 1 July with three months for operational change and six for system change) https://www.health.gov.za/wp-content/uploads/2021/02/cd-10mit_userguide_january2014.pdf
- **4.** PHISC CCSA Coding Standards and Guidelines, version 11, October 2024, CCSA Technical Workgroup (local copyright held by the South African Medical Association and the underlying procedural terminology copyright by the American Medical Association; the standards apply to coding for data and not for billing) https://www.phisc.net/system/files/PHISC%20CCSA%20Coding%20Standards%20and%20Guidelines%20Version%2011%20(October%202024).pdf?download=PHISC+Standards+%28Public%29
- **5.** PHISC, About us: remit and subcommittees (the body does not recommend the local procedural adaptation as a national procedural coding standard; the Diagnosis Related Group subcommittee declared dormant) https://www.phisc.net/about-us
- **6.** LOINC, getting started (Regenstrief Institute; royalty-free, released twice a year) https://loinc.org/start
- **7.** SNOMED International announcement, South Africa adopts SNOMED CT, 14 May 2026 (no National Release Centre named and no effective membership date given) https://forums.snomed.org/t/major-milestone-for-africa-south-africa-adopts-snomed-ct-to-drive-digital-health-transformation/1332
- **8.** SNOMED International, Get SNOMED CT: licensing in member and non-member territories (a member licence covers use within the member country) https://www.snomed.org/get-snomed
- **9.** Health Normative Standards Framework 2021, Notice 2667, Government Gazette 47337, 21 October 2022, published under National Health Act s74(1) (names neither SNOMED CT nor LOINC; ICD-10 in the acronym list only; conformance testing at a department-approved laboratory with an attestation) https://www.health.gov.za/wp-content/uploads/2022/10/HNSF_Gazette_21_October_2022.pdf
- **10.** HPCSA Booklet 9, Guidelines on Keeping Patient Records, September 2016 (retention at least six years from dormancy; minors to the 21st birthday; mentally incapacitated for the patient lifetime; occupational health 20 years; long-latency 25 years; cl. 9.5 Deputy Director-General authorisation; cl. 9.8 statutory precedence; para 13.7 billing records separate from care records) https://www.hpcsa-blogs.co.za/wp-content/uploads/2022/08/Booklet-9-Guidelines-on-Patient-Records.pdf
- **11.** HPCSA Booklet 9, Keeping of Patient Records, revision marked September 2022 (general records retained ideally indefinitely, particularly if electronic; which edition is in force could not be established) https://www.hpcsa-blogs.co.za/wp-content/uploads/2022/11/Booklet-9-Keeping-of-Patient-Records_Review-Draft_-FINAL_Sept-2022.pdf
- **12.** Retention of medical records in South Africa, South African Orthopaedic Journal (Medical Protection Society reprint; the Prescription Act 68 of 1969 periods against the retention guidance, and the running of the three-year period from knowledge of the claim) - headline source, secondary https://scielo.org.za/pdf/saoj/v12n1/14.pdf
- **13.** Auditor-General findings on provincial health, reported via MedicalBrief, 14 October 2026 (27 facilities across nine provinces: 15 reporting missing or lost records, 17 not tracking records after issue, 25 with inadequate storage; three of nine paid medico-legal claims conceded for want of the record; claims balance R50,2bn at 31 March 2026 and R1,57bn paid in 2025/26; the recommendation to secure retrievable records) - headline source, secondary https://www.medicalbrief.co.za/health-sector-crippled-by-recurring-systemic-weaknesses-ag-warns/

ITEM 3

Architecture Vision

Phase A is the only phase in which the absence of a clinical function is cheap to fix, and the business case that funds the fix is medico-legal before it is clinical.

What Phase A owes, and who signs it

Three things. A stakeholder map that names people who exist. A scope statement that says what is in and out, with a date. And a vision with a business case somebody with a budget can approve or refuse.

The repository says all three are done. `O31 Deliverable` rows 5, 6 and 7 are `DLV-01 Architecture Vision`, `DLV-02 Statement of Architecture Work` and `DLV-03 Stakeholder Map`, all phase `A`, all `Approved` — and **no relationship anywhere in the file touches `O31`**, one of the twenty object sheets appearing in no relationship sheet at all. Its four columns are `Ref`, the deliverable name, `Phase` and `Status`. There is nowhere to put the approver and nowhere to put the date.

Say that out loud, because it sets the standard for the rest of this item: **a status column is not a signature.** Chapter 21 gives the five-day read of this repository and this item does not repeat it. What follows is the Phase A work, in order, on the ten sheets Phase A touches.

The count that opens the engagement

3.1 — Count the organisation twice before you draw it once

Sheet `O09 Organization Unit`, `O10 Actor`, `O11 Role` · **Input** the three sheets as supplied · **Output** a one-page count with your own figures beside the file's · **Signed by** nobody; this is yours · **Time** one hour

1. Sum `O09` column 5, `Headcount`, across all twelve rows. The answer is 1,607.
2. Read `O09` column 2 down all twelve rows and mark every unit that could employ a clinician. The answer is none: Executive, Finance, Risk & Compliance, Human Resources, Technology, Data & Analytics, Operations, Commercial, Customer Service, Supply & Logistics, Internal Audit, Legal.
3. Sum `O12` column 4, `Scale`, across its eight rows: 1,769 people, of whom 1,329 sit at locations typed `Care` or `Diagnostics` in column 3.
4. Count `O10`. Eighteen actors — seven executives, five architects, a delivery head, two operations heads, a data steward, a security architect, an internal auditor. No clinician, no pharmacist, no clinical safety officer, no medical records manager.
5. Compare `O11` to `O10` row by row. All eighteen names match one for one and `O11` column 4, `Performed by`, reads `Person` on all eighteen. `O11` adds nothing.
6. Look each `O09` column 4 `Head` value up in `O10` column 2. Six of twelve do not appear: Chief People Officer, Chief Commercial Officer, Head of Service, Head of Supply, Head of Internal Audit, General Counsel.
7. Then the other direction. `ACT-07 Chief Compliance Officer` heads no unit, and owns all seven regulatory drivers in `O01` column 6.

Write the seven numbers down. They are the whole of your standing for the next four phases.

The organisation the repository describes, counted two ways

O09 Organization Unit · 12 rows

Column 4 is Head, column 5 Headcount

ORG-01	Executive	138	non-clinical
ORG-02	Finance	187	non-clinical
ORG-03	Risk & Compliance	165	non-clinical
ORG-04	Human Resources	67	non-clinical
ORG-05	Technology	249	non-clinical
ORG-06	Data & Analytics	180	non-clinical
ORG-07	Operations	56	non-clinical
ORG-08	Commercial	24	non-clinical
ORG-09	Customer Service	225	non-clinical
ORG-10	Supply & Logistics	52	non-clinical
ORG-11	Internal Audit	45	non-clinical
ORG-12	Legal	219	non-clinical
	Total headcount	**1,607**	**clinical: 0**

O12 Location · 8 rows

Column 3 is Type, column 4 Scale

LOC-01	Corporate Office	291 people	Corporate
LOC-02	Acute Hospital	276 people	Care
LOC-03	Community Hospital	287 people	Care
LOC-04	Outpatient Clinic Network	223 people	Care
LOC-05	Diagnostic Laboratory	271 people	Diagnostics
LOC-06	Imaging Centre	272 people	Diagnostics
LOC-07	Primary Data Centre	53 people	Technology
LOC-08	Secondary Data Centre	96 people	Technology
	Total by location	**1,769 people**	
	In care or diagnostic settings	**1,329 people**	

And the stakeholder map does not join to itself

Six of the twelve heads in O09 column 4 are not actors in O10: Chief People Officer, Chief Commercial Officer, Head of Service, Head of Supply, Head of Internal Audit, General Counsel.

And ACT-07 Chief Compliance Officer, who owns all seven regulatory drivers in O01, heads no unit at all.

Figure 3.1 Finding 14, counted. **O09 holds 1,607 people in twelve units and not one unit is clinical**, while O12 puts 1,329 of its 1,769 people at locations typed Care or Diagnostics. The two halves of the organisation model describe different organisations, and the stakeholder map does not join to itself: six of O09's twelve heads are not actors in O10, and the Chief Compliance Officer who owns every regulatory driver in O01 heads no unit.

Step 3.1 produces one sentence for a chief executive, needing no model: *your repository describes an organisation of 1,607 people, none of them clinical, operating 1,329 clinical and diagnostic staff at six sites.* Checkable in ten minutes by anyone who doubts it, which is the property a Phase A finding needs.

The absence, and the five defects downstream of it

Finding 14 is not a filing error. It causes most of the others, and Phase A is the only phase in which it is cheap.

Trace it. `R22` pairs ten organisation units with ten capabilities in row order, so Human Resources owns Medicines Management. That is usually read as sloppiness. It is not: **there is no clinical unit for `R22` to point at**, and a positional join is what you get when the correct target does not exist in the list you are joining to. The same mechanism makes Human Resources the owner of the imaging system and Internal Audit the owner of theatre scheduling in `O23` column 10. It produces the eight `Head of ...` titles in `O20` that appear nowhere else in the file, because those posts have no unit to sit in, which leaves ten of fourteen data entities owned by a title the repository does not define. It produces `O62`'s eight risks, **not one of them about a patient**. And it produces `O63` row 8, `CO-04 Clinical safety standards`, evidence `In breach`, with nobody in the 105 sheets who could sign a safety case.

So the absence is the root and the rest are branches. Fix the root in Phase A and most of them become ordinary work. Leave it and Phase B's capability owners, Phase C's data owners, Phase D's device custodians and Phase G's gate signatories are all drawn from twelve units that can supply none of them.

3.2 — Add the clinical units, and get the headcount for each

Sheet `O09 Organization Unit` · **Input** the client's own establishment structure, the nursing service manager's roster, the clinical directorate's terms of reference · **Output** six to ten new `ORG-` rows with a named head and a headcount · **Signed by** the chief executive, or in the public sector the hospital chief executive with the clinical manager countersigning · **Time** two days

1. Continue the existing numbering from `ORG-13`. Do not renumber `ORG-01` to `ORG-12`; later phases cite them.
2. Fill all five columns the sheet has: `Ref`, `Organisation unit`, `Type`, `Head`, `Headcount`. `Type` on the supplied rows takes `Group`, `Corporate`, `Line` and `Assurance`. Add `Clinical` and say in the Phase A document that you added a value, because a new value in an unvalidated column is how a taxonomy rots.
3. The head is a **post title that exists in the client today**, not an aspiration. If the post is vacant, write the title and record the vacancy in the risk list under step 3.9.
4. Reconcile the new headcounts against `O12` column 4 before you stop. Until `O09` and `O12` agree within a stated tolerance, the organisation model is two organisations.

Form 3-A — The clinical organisation units the repository does not have

Feeds `O09 Organization Unit` · **Rows a first pass needs** six to ten · **Signed by** the chief executive

Ref	Organisation unit	Type	Head	Headcount
ORG-13 (example)	Medical Directorate	Clinical	Clinical Manager	214

Chapter 5 gives the cast list. The units that must exist before a clinical capability has an owner are a medical directorate, nursing, pharmacy, allied health, clinical governance and safety, diagnostics and the records function — seven, of which the last is the one everybody forgets and the one the business case below rests on.

3.3 — Name the actors and the roles, and stop mirroring

Sheet `O10 Actor`, `O11 Role` · **Input** Form 3-A, the client's credentialling committee membership, the registered Information Officer · **Output** new `ACT-` rows and new `ROL-` rows that are not a copy of each other · **Signed by** the chief executive and the Information Officer · **Time** two days

1. Add actors for the clinical posts: clinical director, nursing service manager, chief pharmacist, clinical safety officer, medical records manager, clinical coding lead. Six rows, four columns each: `Ref`, `Actor`, `Type`, `Status`.
2. Set `Status` honestly. The supplied eighteen all read `Filled`. If a post is vacant, write `Vacant` and let `ZZ Checks` row 17 be wrong in your favour rather than wrong in the file's.

3. Add the three owner titles the file uses and does not define — `Group Executive`, `Head of Procurement` and Chief Commercial Officer. Either they are actors or the rows naming them are unowned.
4. Then stop short of `O11`. Do not add a role for every actor. Add one only where the role is held by more than one actor, or by a different actor on a different shift, or by a committee — and `O11` column 4 reads `Person` on all eighteen supplied rows, which a clinical governance committee is not. Chapter 5 gives the reason: authority here is role plus credential plus privilege, and a one-to-one actor-to-role map carries none of the three.

ASK THE SPONSOR — THE CLINICAL HALF OF THE ORGANISATION

Which clinical functions exist as management units today, and who heads each one by name?

Who is accountable for the existence and retrievability of a clinical record at each site, by name and by post?

Who is the registered Information Officer, and is there a clinical safety officer for information systems?

Which of the eight Head of titles in O20 are real posts, and which were invented by the template?

The motivation layer was written for a retailer, and half of it is salvageable

`O01 Driver` has thirteen rows and splits cleanly in two. `DRV-01` to `DRV-07` are typed `Regulatory`, origin `External`, pressure `High`, owner Chief Compliance Officer, and they are healthcare: health data protection, clinical records retention, medical device regulation, clinical safety standards, professional registration and scope, funder and coding rules, incident reporting duties. Those seven are the best content in this file's motivation layer. They map one-to-one onto `O06` rows 5 to 11 and onto `O63`'s seven control objectives, and they are the spine of the business case below.

`DRV-08` to `DRV-13` are typed `Business`, origin `Internal`, pressure `Medium`, owner `Group Executive`, and they are a generic template: cost pressure, customer expectation, legacy estate, data fragmentation, talent scarcity, sustainability reporting. `DRV-09` reads "Customers expect digital-first service at the pace of other sectors" — but in the public sector patients may attend any establishment by statute, and in the private sector they are steered by scheme design [4]. The split runs down the chain and gets worse at each level, because the retail half carries the measures.

Sheet	The seven that work	The rows to retire or re-point
`O01`	`DRV-01` to `DRV-07`, regulatory, owned	`DRV-09` customer expectation; `DRV-13` sustainability reporting
`O02`	`GOL-01` protect the licence to operate	`GOL-03` grow revenue from existing customers, owned by a title `O10` does not define
`O03`	`OBJ-01`, `OBJ-02` evidence without manual reconciliation	`OBJ-05` single view of customer; `OBJ-06` cross-sell rate
`O04`	`MSR-01` obligations with evidence, baseline 33%	`MSR-05` single customer view coverage; `MSR-06` cross-sell ratio, baseline 48
`O06`	`REQ-001` to `REQ-007`, one per regulatory driver	`REQ-022` to `REQ-027`, six capabilities told to be measurable

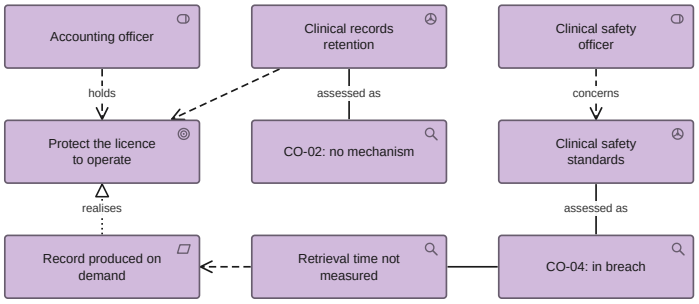


Figure 3.2 The motivation chain Phase A has to be able to draw, built on the half of O01 that was written for this industry. Two of the nine elements do not exist in the repository: the accounting officer is nobody's actor, and **O10 has no clinical safety officer**, which is why CO-04's evidence value is in breach with nobody able to sign a safety case. The two assessments are the file's own evidence values from O63.

`O04` is the sheet to look at hardest, because it is the one the sponsor will ask about. Six columns: `Ref`, `Measure`, `Unit`, `Domain`, `Baseline`, `Cadence`. **There is no target column, no owner column and no definition column.** `MSR-13 Recovery objectives tested` has unit `count` and baseline 43 for a measure with twelve possible subjects; `MSR-09 Change lead time` has baseline 54 days with no statement of which two events those days run between; and `MSR-15` and `MSR-16` appear in no relationship sheet, so two of sixteen measure no objective at all.

That is chapter 4's Measure Definition arriving as a Phase A problem rather than a Phase C one, and it is why item 5 creates `O70 Measure Definition`. Phase A does not build the sheet. It refuses to carry a baseline whose definition is unknown, and records against every measure it keeps the two events the interval runs between.

3.4 — Retire the retail chain and re-point what is left

Sheet `O01 Driver`, `O02 Goal`, `O03 Objective`, `O04 Measure` · **Input** the table above, the client's own board pack · **Output** a marked-up copy of each sheet with retirements and replacements, and a list of orphaned downstream rows · **Signed by** the chief executive for `O02`, the chief financial officer for `O04` · **Time** three days

1. Mark `DRV-09`, `GOL-03`, `OBJ-05`, `OBJ-06`, `MSR-05` and `MSR-06` for retirement. Do not delete them. A retired row with a date is evidence of a decision; a deleted row is a repository that lies about its own history.
2. Follow each chain. `GOL-03` carries `OBJ-05` and `OBJ-06`, which carry `MSR-05` and `MSR-06` through `R10`, and `R11` joins goals to objectives. Retiring a goal without its objectives leaves `R11` rows pointing at a retired parent.
3. Replace `GOL-03` with a goal a provider has. The candidate comes from the client, but the shape is: care delivered to the standard the regulator inspects against, with the evidence to show it. `O02` has five columns — `Ref`, `Goal`, `Description`, `Owner`, `Status` — and the owner must be an actor that exists after step 3.3.
4. For every measure you keep, write the two events the interval runs between and the population it is computed over into the vision document, not into `O04`, which has nowhere to put them. Carry it to item 5.
5. Leave `DRV-01` to `DRV-07` alone. They are right, they are owned, and they are the only part of this layer written for the industry.

`O06` stops part-way down every list it derives from

Thirty-two requirements, seven columns, every one resolving cleanly to a driver. And a pattern worth more than the rest of the sheet:

- `REQ-001` to `REQ-007`: one per regulatory driver. **Seven of seven. Complete.**
- `REQ-008` to `REQ-015`: *mastered once* for `E01` to `E08`. **Eight of fourteen entities** — Clinical Protocol, Practitioner, Care Setting, Claim, Consent and Medical Device get none.
- `REQ-016` to `REQ-021`: *integration managed* for `A01` to `A06`. **Six of twelve applications**, and nothing for `A11 Integration Engine`, so the integration platform is under no requirement to manage its own interfaces.
- `REQ-022` to `REQ-027`: *measurable* for `C01` to `C06`. **Six of ten capabilities.** Then five technology adoptions, all `Could`, all `Proposed`.

The generator stopped at the end of a screen in four of the five blocks. And `Priority` is a function of `Category` and nothing else: every `Regulatory` and `Data` row is `Must` and `Approved`, every `Integration` and `Business` row `Should`, every `Technology` row `Could`. So `REQ-030 Adopt Master patient index`, which chapter 11 shows is the precondition for every other data requirement on the sheet, is a `Could` while `REQ-024 Diagnostics measurable` is a `Should`. Identity resolution ranks below a dashboard because of the word in column 4.

3.5 — Complete the four truncated requirement blocks and re-prioritise

Sheet `O06 Requirement` · **Input** `O20`'s fourteen entities, `O23`'s twelve applications, `O13`'s ten level-1 capabilities · **Output** the twenty-two missing rows, and a priority column reset against consequence · **Signed by** the chief enterprise architect, countersigned by the Chief Compliance Officer for the regulatory block · **Time** three days

1. Fill the gaps mechanically first: six mastering requirements for `E09` to `E14`, six integration requirements for `A07` to `A12`, four measurability requirements for `C07` to `C10`. Twenty-two rows, same seven columns, continuing from `REQ-033`.

2. Then read the six new mastering rows and stop. Consent and Medical Device cannot be *mastered once* until there is somewhere to master them, which is items 5 and 7. Write the requirement and mark it blocked on the new sheet by name: `O67 Consent`, `O68 Medical Device`.
3. Re-set `Priority` by consequence: if this is not met, what is the worst documented outcome. `REQ-030` and `REQ-002` answer at the top; `REQ-024` does not.
4. Add the requirement the sheet has no row for and the file states the obligation for: that the complete record of a named episode of care can be produced, with provenance, inside a stated time. `CO-02`'s evidence value is `No mechanism`, so the obligation is recorded and the requirement is not.
5. Do not touch `R09`. It pairs all thirty-two requirements to building blocks by row position and zero of thirty-two pair by subject. Adding rows to a positional join makes it longer, not better. Item 12 owns `R09`.

Scope, and the four lines that have to be dated

Scope here is decided along four lines, three of them invisible on an organogram. Which **establishments**, because the licence, the records duty and the inspection all attach to the establishment rather than the group [4]. Which **care settings**, because chapter 4 makes care setting a kind of care rather than a place, so "the acute hospitals" is not a scope line. Which **record classes**, because the retention clock and the destruction authority differ by class. And which **counterparties**, because the laboratory, the switch, the funders and the chronic-medicine distribution network are organisations the client neither owns nor directs.

`O12 Location` gives eight locations and no establishment identifiers, licence references or bed counts. Driehoek Health Group's nine acute hospitals and two day clinics are separately licensed, each with its own bed count and expiry; Rietvlei District Health Service's district hospital, three community health centres, 22 fixed clinics and mobile service hold no private licence at all, because that regime is for private facilities.

One trap in the time axis. `O03` column 4, `Horizon`, holds `12 months` through `3 years` and **there is no anchor date anywhere on the sheet** — the third relative timebase in this file, alongside `O66`'s increments dated `Month N` and `O30` and `O56`'s absolute dates from 2027-01. Two of the three have no origin. Report that the file holds incompatible timebases and do not invent a start date to reconcile them; fix your own instead, with a dated baseline on the scope statement and every horizon in the vision stated as a date.

3.6 — Write the scope statement as four lists and a date

Sheet `O12 Location`, `O09 Organization Unit` as extended by step 3.2 · **Input** the client's licence register, the establishment list, the counterparty list · **Output** four explicit in-scope lists, four explicit out-of-scope lists, and a baseline date · **Signed by** the sponsor and the accounting officer or board · **Time** two days

1. List the establishments in scope by name and licence or certificate reference, then the establishments out of scope, by name. An unenumerated exclusion is not an exclusion.
2. List the care settings in scope. Chapter 4 has the set, and it is a classification of service rather than of place.
3. List the record classes in scope and name the retention instrument each is governed by. The Preliminary phase sets this clock; Phase A states which classes the vision covers.
4. List the counterparties whose failure stops clinical work and which the client does not own — at least the laboratory, the claims switch, the funders and the national patient index.
5. Write the baseline date on the front page. Every horizon then becomes a date, and the three incompatible timebases become one.

Form 3-B — Scope decision record

Feeds no sheet: there is none for it, because `O31` has no approver and no date column. This form is the record of the decision; keep it with the vision and carry it into Phase G. **Rows a first pass needs** one per scope line · **Signed by** the sponsor

Scope line	In scope	Out of scope	Decided by	Date
Establishments (example)	Rietvlei District Hospital; three community health centres	22 fixed clinics; mobile service	Head of Department	2027-02-15

The business case is medico-legal before it is clinical

This is the part of Phase A the industry makes different, and it goes first in the document rather than last.

The instinct from every other sector is to argue the clinical benefit: better records improve care. It is true, unmeasurable inside a planning cycle, and it loses to the clinical improvements with a clinician's name on them. The argument that works is not about care at all. **It is that the organisation is paying cash, now, for records it cannot produce, and the person who signs the business case is already personally accountable for that cash.**

Volume 1 chapter 13 carries the arithmetic; this item turns it into an instrument. The figures come with their dates, because an undated figure in this sector is an opinion.

The Auditor-General audited 27 facilities across nine provinces and found 15 reporting missing or lost records, 17 not properly tracking records after they were issued, 25 with inadequate storage and 18 with too few records-management staff — 2021/22 records-audit findings. And **in three of nine paid medico-legal claims reviewed, the department conceded because it could not locate the medical record** [1]. Not because the care was indefensible. Because the record was not there.

Provincial health departments' recorded medico-legal claims balance was R50,2bn at 31 March 2026 on the Auditor-General's figures and R50,6bn for 2025/26 on National Treasury's — about R400 million apart on substantially the same quantity, and chapter 2 teaches why both are quoted rather than averaged [1][2]. What leaves the fiscus is smaller and more usable: R1,57bn paid to claimants in 2025/26 and R4,6bn over three years across the nine provinces, with Treasury putting the long-run average near R1,5bn a year [1][2]. Gauteng budgeted R179m in 2024 and paid about R700m, and the shortfall came out of medical supplies, maintenance and operating costs [7]. Against which the direct National Health Insurance grant for 2026/27 carries a line for "Digital Health System" of **R6,5 million** [3] — and chapter 13 gives two published readings of the allocation that line sits inside, which disagree. Carry both. The argument is unaffected either way.

The Phase A business case, as arithmetic rather than argument

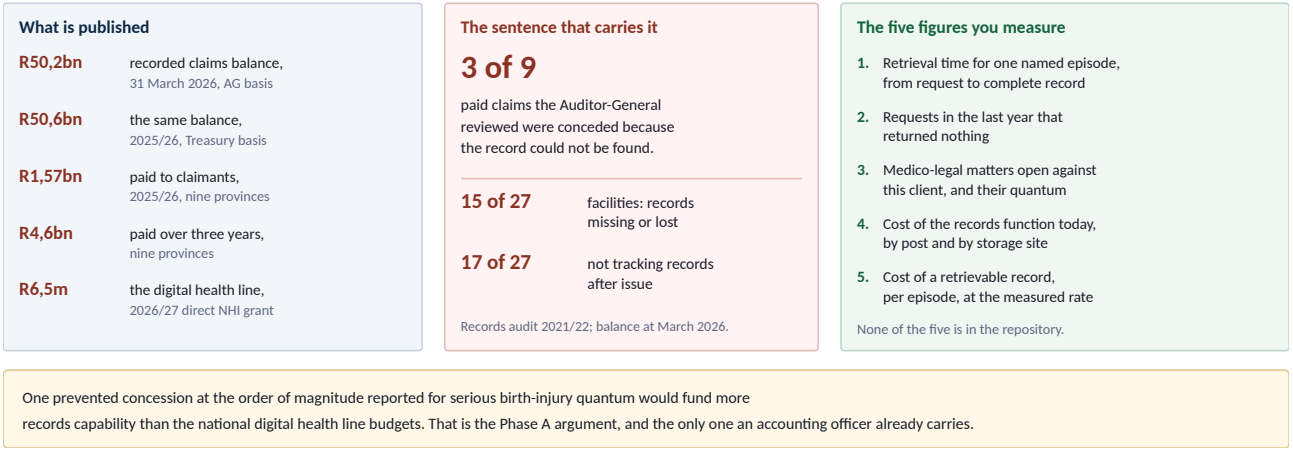


Figure 3.3 The Phase A business case as an instrument rather than an argument. The left column is published and dated; the middle is the sentence that carries it; the right is the five figures the reader measures locally, **none of which the repository has anywhere to hold**. Volume 1 chapter 13 carries the arithmetic and the two readings of the 2026/27 allocation that disagree.

The operative word in the Auditor-General's own recommendation is **retrievable** [1]. That is not storage. It is the ability to produce the complete record of a specific episode of care, for a named patient, from up to twenty-one years ago, under litigation, with provenance — a different capability from an electronic record, bought from a different budget, owned by a post the repository does not contain.

> **THE PHASE A MOVE, IN ONE SENTENCE**
>
> Do not ask the accounting officer to fund an improvement. Show them a cash outflow they already carry, a measured retrieval time that explains it, and the price of changing that one number — then show them the national programme's own digital health line against it.

Two cautions a client's finance function will find if you do not. **The exposure period is indeterminate, so a retention period is not a defence**; chapter 13 has the prescription rules and the unverified items in them. **And the period itself is a professional guideline rather than a statute, disagreeing with itself between editions**: at least six years from dormancy for a general adult record in the September 2016 edition [5], retained ideally indefinitely and particularly if electronic in the revision marked September 2022 [6], with no way to establish which is in force. Obtain the current booklet before quoting a period and record which edition your figures were built against. And note where the decision sits: the record-keeping duty under the National Health Act is expressly subject to the National Archives and Record Service of South Africa Act 43 of 1996 and to the access-to-information regime, so a public body's question is not what the period is but what it may do when the period expires, and who signs [4].

3.7 — Measure the retrieval time before you write the business case

Sheet no sheet; this is the measurement the repository has nowhere to hold · **Input** five closed episodes of care chosen by the client, at least one over five years old and one obstetric · **Output** five measured retrieval times with their failure modes · **Signed by** the records manager added in step 3.3, countersigned by the clinical manager · **Time** one week elapsed, half a day of effort

1. Pick the five episodes with the client, not for them, and get the selection minuted. A sample the client chose cannot be argued with later.

2. For each, request the **complete** record of that episode: the notes, the documents that arrived from elsewhere, the results, the observation charts, the consent forms, the operation note, the nursing notes and the medicines chart. A retrieval that returns the discharge summary and nothing else has not answered the request, and the result must say so.
3. Record the clock from request to complete delivery, in hours, and what was missing at the end of it.
4. Record the failure mode: not found, found incomplete, found but not tracked back after a previous issue, held off site, held in a practitioner's rooms, electronic but not reconstructable as it stood.
5. Count how many of the five could be rendered **as the record stood on a stated date**. That is the form a court asks for, and the number most clients have never measured.
6. Carry the result into Phase D as the baseline. Overlay 1 sets the clock in Preliminary, Phase C records the class, Phase D measures retrieval, Phase G gates on it and Phase H re-runs it; this is the first measurement in that chain and the only one Phase A needs.

Form 3-C — Medico-legal exposure and retrieval worksheet

Feeds no sheet: there is none for it, because `004` has no target or definition column and `062` no financial column. It is the Phase A instrument the accounting officer signs. **Rows a first pass needs** five measured episodes plus four exposure lines · **Signed by** the chief financial officer and the accounting officer

Line	What is measured	Source and date	Figure	Who confirmed
Retrieval, episode 1 (example)	Hours from request to complete record	Records office log, 2027-02-09	61 hours; chart missing	Records Manager

ASK THE ACCOUNTING OFFICER OR THE BOARD — THE FOUR QUESTIONS THAT DECIDE THE BUSINESS CASE

What did this organisation pay out on medico-legal matters in each of the last three years, and what is the recorded balance today?

In how many of those matters was the record produced in full, and in how many was it not?

What does the records function cost today, by post and by storage site, and who holds that budget?

Who may authorise the destruction of a clinical record here, and where is the register of records already destroyed?

The fourth question is the one that turns the conversation. A missing record and a lawfully destroyed record are indistinguishable without a register of destroyed records and a signed disposal authority, and in three of nine paid claims that distinction cost the state the case [1]. The register is cheap. The alternative has a measured price.

`054` , and the number you must not inherit

`O54 Architecture Alternative` holds five rows and five columns — `Ref` , `Alternative` , `Description` , `Cost` , `Recommendation` . Do nothing beyond compliance at $24m and business value first at $52m and big bang replacement at $88m are `Rejected` ; `ALT-B` foundations first, $46m, is `Recommended` ; `ALT-E` compliance led, $31m, is `Partly adopted` . Three things are wrong with that sheet and all three are Phase A's to fix.

The recommended alternative is costed at $46m and the plan that implements it totals $61.3m. `O30` 's twelve work packages, `O56` 's twelve projects and `R14` all agree on $61.3m, and each project's cost, start and end equal its work package's in all twelve rows — the second cleanest arithmetic in the file. So the recommendation was made on a figure 33% below the plan, and above three of the five alternatives including two rejected as cheaper. No sheet reconciles the $15.3m.

Two recommendations are live at once. `ALT-B` is `Recommended` , `ALT-E` `Partly adopted` with nothing recording which parts or what they cost, and `O60` row 5, `DEC-01 Adopt the recommended architecture alternative` , is `Outstanding` . Recommended, partly adopted and undecided at once.

The currency and the date are both absent. `O54` , `O30` and `O56` carry costs as dollar strings with no currency column, no base date and no rate anywhere in the 105 sheets. For a South African provider that is not a formatting preference: a 2026 rand business case and an undated dollar plan are not comparable, and the gazetted medicine price adjustment, the public fee schedule and the injury-compensation tariff each move on their own dates. Phase A either converts with a stated rate and date, or restates in rand. It does not carry both.

And `O54` is one of the twenty sheets appearing in no relationship sheet at all: five alternatives, one recommended, none joined to a work package, a project, a gap, a requirement or a decision.

3.8 — Re-cost the alternatives in one currency on one date, and close the decision

Sheet `O54 Architecture Alternative` , read against `O30 Work Package` , `O56 Project` and `O60 Governance Decision` · **Input** the $61.3m roll-up, the client's own cost base, the business case from step 3.7 · **Output** a re-costed `O54` with a currency and a base date, and `DEC-01` closed with a named body and a date · **Signed by** the investment committee, or the accounting officer where there is no committee · **Time** one week

1. State the currency and the base date before you touch a figure, then restate all five alternatives on that basis.
2. Reconcile $46m to $61.3m, or re-cost `ALT-B` . Do not carry both; one is wrong and Phase E will build a roadmap on whichever you leave.
3. Resolve `ALT-E` . Either it is adopted, in which case say which parts and what they cost, or it is not. `Partly adopted` with no enumeration is not a decision.
4. Add the alternative `O54` does not contain: records retrievability first. On step 3.7's arithmetic it is the cheapest option on the sheet and the only one addressing a measured cash outflow. If it loses, it loses on the record, with a reason.
5. Close `DEC-01` with a named body and a date. An `Outstanding` decision in `O60` beside an `Approved` vision in `O31` is the contradiction Phase G will find.

`062` has eight risks and not one of them is a patient

The last Phase A sheet, and the quickest. `O62 Delivery Risk` has five columns — `Ref` , `Risk` , `Likelihood` , `Impact` , `Owner` — and eight rows, all programme-generic. Three are `Certain` , one is `Possible` with impact `Critical` , and `RSK-06` 's owner is Head of Procurement, which is not an actor in `O10` .

Missing is the whole clinical and records risk set, for which the file states every obligation: `CO-04` clinical safety standards `In breach` , `CO-02` records retention `No mechanism` , `CO-05` professional registration and scope `No mechanism` , `CO-01` health data protection `In breach` . Four recorded obligations with no risk against them.

3.9 — Add the risks the file states the obligation for

Sheet `O62 Delivery Risk` · **Input** `O63` 's seven control objectives and their evidence values, the retrieval measurement from step 3.7, the vacancies from step 3.3 · **Output** six to eight new `RSK-` rows with owners who are now actors · **Signed by** the chief risk officer, countersigned by the clinical manager for the clinical rows · **Time** half a day

1. Write one risk per control objective whose evidence value is `No mechanism` or `In breach` . Four rows, minimum.
2. Add the one the measurement gives you: the complete record of a named episode cannot be produced within the stated time. Set `Likelihood` from step 3.7's result, not from judgement.
3. Add the vacancy risks from step 3.3. A clinical safety officer post that does not exist is a delivery risk with a named owner, which is worth more than a register entry about capacity.
4. Check every `Owner` against `O10` as extended by step 3.3, and take the `Impact` scale as the sheet spells it. A risk owned by a title the repository does not define is an unowned risk.

What goes to the sponsor, and in what order

One document, and the order matters more than the content.

First, the arithmetic. The seven numbers from step 3.1 and the five measurements from step 3.7. Nothing argumentative, nothing needing a model, every figure with its source and its date. This buys the standing for everything after it.

Second, the absence, as a question rather than a finding. Which clinical functions should appear as organisation units, and who in each owns data. Asked in week one it is a governance question the client answers; asserted in month four it is a criticism of their repository.

Third, the business case, medico-legal first. The cash outflow, the measured retrieval time, the price of changing it, against the national digital health line [3]. Then the clinical argument, which is now supported rather than carrying the weight.

Fourth, the scope statement with a baseline date, four lists in and four lists out. **Fifth, the decisions you need and who signs each:** the clinical units, the retired retail chain, the re-costed alternatives in one currency on one date, and `DEC-01` closed.

What does not go in: a target architecture, a capability heatmap, a roadmap, or any diagram built on the eight positional joins. Phase A has no business asserting a capability owner from `R22` or a service-to-application pairing from `R04`; item 4 re-derives both from evidence.

One closing discipline, carried from chapter 2: where the file gives two readings, give both. The record count is 820 and the file says 843. Control objectives without evidence are two literally, six on the regulatory report's own stated test, and seven of seven on the test an auditor would apply. Components past vendor support are one today and two from January 2027. A Phase A document that picks one reading and buries the other will be contradicted by the first person who reads the sheet.

THE CHAPTER IN TEN LINES

- Phase A owes three things: a stakeholder map of people who exist, a dated scope statement, and a vision with a business case somebody can refuse.
- O31 marks the Architecture Vision, the Statement of Architecture Work and the Stakeholder Map all Approved, has no approver or date column, and is touched by no relationship in the file.
- Count before you draw. O09 holds 1,607 people in twelve units and no clinical unit; O12 puts 1,329 of 1,769 people in care and diagnostic locations.
- Six of O09's twelve unit heads are not actors in O10, and the Chief Compliance Officer who owns all seven regulatory drivers heads no unit.
- The absence of a clinical function is the root of R22's positional ownership, O23's misplaced application owners, O20's eight undefined owner titles and O62's eight non-clinical risks.
- DRV-01 to DRV-07 are healthcare and worth keeping. The business half of the motivation layer is a retail template, and the retail half carries the measures.
- O04 has a baseline and a cadence and no target, owner or definition column, and two of its sixteen measures measure no objective at all.
- O06 stops part-way down four of the five lists it derives from, and Priority is a function of Category, which makes identity resolution a Could.
- The business case is medico-legal before it is clinical: three of nine paid claims the Auditor-General reviewed were conceded because the record could not be found.
- Measure the retrieval time on five real episodes before writing the case, and count how many could be rendered as the record stood on a stated date.
- O54 recommends a 46 million alternative that the plan implements at 61,3 million, carries two live recommendations, and states its costs in a currency with no base date.
- O62 holds eight programme risks and none about a patient, while four control objectives on O63 have no mechanism or are in breach.
- Where the file gives two readings, give both. A Phase A document that picks one and buries the other will be contradicted by the first person who reads the sheet.

SOURCES FOR THIS CHAPTER

- **1.** Auditor-General findings on the health sector, via MedicalBrief, 14 October 2026 (records audit of 27 facilities; provincial liability, payouts, and claims conceded for want of a record) https://www.medicalbrief.co.za/health-sector-crippled-by-recurring-systemic-weaknesses-ag-warns/
- **2.** National Treasury 2026 Budget Review, Chapter 6 (provincial medico-legal contingent liabilities and average annual settlements) https://archive.treasury.gov.za/documents/National%20Budget/2026/review/Chapter%206.pdf
- **3.** BusinessTech report on the October 2026 Medium Term Budget Policy Statement adjustment to the direct National Health Insurance grant, 30 October 2026 https://businesstech.co.za/news/trending/882160/nhi-hits-a-roadblock-in-south-africa/
- **4.** National Health Act 61 of 2003, consolidated text (sections 13, 17 and 44, and the record-keeping duty's subjection to the National Archives Act and the access-to-information regime) https://www.saflii.org/za/legis/consol_act/nha2003147.pdf
- **5.** HPCSA Booklet 9, Guidelines on the Keeping of Patient Records, September 2016 edition (retention periods; billing records kept separate from patient care records) https://www.hpcsa-blogs.co.za/wp-content/uploads/2022/08/Booklet-9-Guidelines-on-Patient-Records.pdf
- **6.** HPCSA Booklet 9, Keeping of Patient Records, review draft marked September 2022 (general records retained ideally indefinitely, particularly if electronic) https://www.hpcsa-blogs.co.za/wp-content/uploads/2022/11/Booklet-9-Keeping-of-Patient-Records_Review-Draft_-FINAL_Sept-2022.pdf
- **7.** Auditor-General of South Africa, briefing to the NCOP Social Services Committee, 5 November 2025 (Gauteng budgeted against paid; facilities not tracking records after issue) https://pmg.org.za/committee-meeting/41983/

ITEM 4

Business Architecture

Twenty-four stages are declared, twenty exist, and the right answer is thirty-six.

What Phase B owes, and the three sheets that say it is done

`031 Deliverable` rows 8, 9 and 10 are `DLV-04 Business Architecture`, `DLV-05 Capability Map` and `DLV-06 Value Stream Definition`, phase `B`, all `Approved`. Beneath those three approvals sit 100 capability rows, five value streams, twenty processes, ten organisational functions, ten services, ten business building blocks, five products and ten vendor agreements.

Phase B's job is not to build a business architecture. It is to work out which of those 170 rows carries information, and to replace the rest. The answer is the ten level-1 capability names with their descriptions, and the five value stream names with their scope lines. Fifteen rows out of 170; everything else was generated from them or inherited from a template. That is the normal condition of a repository populated to satisfy a template, and the reason Phase B here is a naming exercise before it is a modelling one.

The capability model is arithmetically perfect and semantically empty

Start with what works, because it is real and you should keep it.

`013 Business Capability` holds 100 rows in nine columns: `Ref`, `Capability`, `Level`, `Parent`, `Description`, `Criticality`, `Current`, `Target`, `Disposition`. Ten at level 1, thirty at level 2, sixty at level 3. `R01` holds the ninety parentages and **every one agrees with `013` column 4.** `RPT03 Capability Heatmap` recomputes `Target` minus `Current` on the ten level-1 rows and is correct in all ten. This is the cleanest arithmetic in the repository and the only decomposition in it that adds up.

Now read the names.

The thirty level-2 rows are ten words — Administration, Optimisation, Onboarding, Reporting, Assurance, Support, Planning, Execution, Monitoring, Design — each used exactly three times, appended to a stem taken from the parent. The sixty level-3 rows append " — Define" or " — Operate", thirty of each, and `Description` follows the same machine. Not one of the ninety names a thing a provider does: no triage, no theatre scheduling, no specimen handling, no medicines reconciliation, no coding, no identity resolution, no records retrieval.

O13 Business Capability: 100 rows, and what is actually in them

What adds up

Level 1	10 rows	written for healthcare
Level 2	30 rows	ten suffixes, 3 uses each
Level 3	60 rows	Define and Operate only

R01 holds 90 parentages and every one agrees with O13 column 4. The decomposition is the cleanest arithmetic in the file. It is also generated.

The thirty level-2 names are ten words, three times each

Administration ×3	Optimisation ×3	Onboarding ×3
Reporting ×3	Assurance ×3	Support ×3
Planning ×3	Execution ×3	Monitoring ×3
Design ×3		

None names a thing a provider does. Level 3 appends " — Define" or " — Operate" to each.

Three checks you can run on the sheet in ten minutes

36 of 90	child rows are rated more critically than their parent. Criticality does not inherit in this file, any more than classification does: 18 at level 2 under level 1, and 18 at level 3 under level 2.
100 of 100	rows have a maturity gap of exactly 1 or exactly 2, and Disposition is Sustain on every gap of 1 and Build on every gap of 2, with no exception. Disposition carries no judgement at all.
13 of 100	rows are rated Critical and not one of them is a level-1 capability. Diagnostics is Low; "Diagnostics Monitoring — Define" is Critical.

Figure 4.1 O13's hundred rows, read two ways. The decomposition is the cleanest arithmetic in the repository and **every one of the ninety child names was generated**: ten suffix words used three times each at level 2, and Define or Operate at level 3. The three checks below are runnable on the sheet in ten minutes, and the second is the one to take to a workshop.

Three checks follow, each runnable on the sheet in ten minutes, and the second is the one to take to a workshop.

Criticality does not inherit. Thirty-six of the ninety child rows are rated more critically than their parent: eighteen level-2 rows above their level-1 parent, eighteen level-3 above their level-2. `C01.01 Patient Access Administration` is `Critical` beneath a `High` parent; `C03.03.01 Diagnostics Monitoring — Define` is `Critical` beneath a `Low` grandparent. This is overlay 2 in the business layer — a parent capability cannot be less critical than a thing it is made of — and the file breaks the rule in 26 of 60 data attributes and 36 of 90 capability children. The same defect twice in two layers, with the same remedy both times: a rule, applied at entry, with a check that fails.

`Disposition` is a restatement of the maturity gap. Every one of the 100 rows has `Target` minus `Current` equal to exactly 1 or exactly 2. Every gap of 1 reads `Sustain`, 58 times; every gap of 2 reads `Build`, 42 times; there is not one exception. So the column carries no judgement: nothing is at target, nothing is more than two levels short, and nothing is being stopped. A real assessment has all three.

Thirteen rows are `Critical` and not one of them is a level-1 capability. `C02 Clinical Care Delivery` is `High`. `C03 Diagnostics` is `Low`. `C08 Workforce & Rostering` is `Low`. The `Critical` ratings sit on generated sub-rows: `Population Health Monitoring — Define`, `Estates Planning — Define`, `Medicines Management Design — Operate`.

4.1 — Cut levels 2 and 3 and re-cut level 2 from the value streams

Sheet `O13 Business Capability` , `R01` · **Input** `O13` 's ten level-1 rows and descriptions, the five value streams from step 4.7 · **Output** a level-2 decomposition of eight to fourteen children per level-1 capability, named after things the organisation does · **Signed by** the business owner of each level-1 capability, who is named in step 4.4 · **Time** two workshops of half a day each

1. Keep all ten level-1 rows, their `Ref` values and their descriptions. They are the only part of this sheet written for the industry.
2. Archive the ninety child rows with a date. Do not delete them: a later reader needs to see that they were generated, not that they never existed.
3. Re-cut level 2 from the stage list step 4.7 produces, not from a template. A level-2 capability is something a person is responsible for being able to do, and it should be nameable without the parent's name in it.
4. Stop at level 2 unless a level-3 row answers a question somebody is asking, and renumber beneath the existing level-1 refs so every citation to `C01` through `C10` still resolves.

4.2 — Re-score criticality with an inheritance rule that fails

Sheet `O13 Business Capability` · **Input** the re-cut level 2 from step 4.1, the resilience position in `O49` , the safety obligations in `O63` · **Output** a criticality value per row and a check that rejects a child above its parent · **Signed by** the chief operating officer, countersigned by the clinical manager · **Time** one day

1. Score level 1 first, and clinically rather than financially. A capability whose failure stops clinical work is `Critical` whatever its cost line says.
2. Then score children, with the rule: a child may be scored no higher than its parent. If a child needs to be higher, fix the parent.
3. Run the check after every edit — join `O13` to itself on column 4 and compare column 6, and the answer must be zero rows — and record the rule in the same words as the classification rule in overlay 2, because it is the same rule.

4.3 — Re-baseline maturity so that the gap can be 0, 3 or 4

Sheet `O13 Business Capability` · **Input** the re-cut level 2, the client's own evidence for each score · **Output** `Current` and `Target` per row with the evidence behind each `Current` , and a `Disposition` independent of the gap · **Signed by** the capability owner for `Current` , the sponsor for `Target` · **Time** one workshop per level-1 capability

1. Score `Current` against evidence that exists and write the evidence beside the score. A maturity score with no evidence is an opinion with a decimal point.
2. Allow a gap of 0, because the capability already at target is the one you do not fund; and allow a gap above 2, because the repository's maximum of 2 is why the plan reads as twelve similar projects.
3. Set `Disposition` from a decision rather than arithmetic, with values that can differ from the gap: `Sustain` , `Build` , `Buy` , `Partner` , `Retire` .
4. Keep `RPT03` 's computation. It is correct, and after this step it computes over figures that mean something.

Ten names, four sheets, nothing added

`O13` 's ten level-1 capabilities are Patient Access & Scheduling, Clinical Care Delivery, Diagnostics, Medicines Management, Care Coordination, Clinical Governance & Safety, Revenue Cycle & Claims, Workforce & Rostering, Estates & Medical Devices, and Population Health & Analytics.

Those same ten names appear, in the same order, on three more sheets.

`O17 Business Function` has ten rows and four columns — `Ref` , `Business function` , `Capability` , `Description` — with column 2 holding the capability's name unchanged and column 4 reading "Organisational function performing this capability." on **all ten rows**, identically. It appears in no relationship sheet.

`O15` has ten rows, one per level-1 capability, joined one-to-one by `R03` , its column 4 reading "Externally consumable service realising patient access & scheduling" and so on down the ten. `R03` is worth a second look. It pairs `C01` to `C10` with `BSV-01` to `BSV-10` in row order, exactly as the eight positional joins do — and it is **right**, because the ten services were generated from the ten capabilities in the first place. A relationship that is correct by construction tells you nothing about the estate, and it is the row a reader samples to reassure themselves that the file's relationships are sound.

`O35 Business Building Block` has ten rows and five columns — `Ref` , the building block name, `Capability` , `What it must do` , `Acceptance criteria` — where column 2 is "<capability> capability uplift", column 4 "<capability> operating at target maturity." and column 5 "<capability> measured and owned."

An acceptance criterion that cannot fail is not an acceptance criterion. "Diagnostics measured and owned" cannot be tested, has no threshold, names no measure and names no owner — and `O59 Compliance Assessment` holds three rows against thirty-seven building blocks, all three verdicts non-compliant with evidence recorded as "no evidence produced". So the file has unfalsifiable criteria and no assessment of them, which is consistent rather than contradictory.

4.4 — Name the owner of each level-1 capability, from the units Phase A added

Sheet `O13 Business Capability`, `O17 Business Function`, `R22` · **Input** the clinical organisation units from item 3's step 3.2, the heads named in its step 3.3 · **Output** one named post per level-1 capability, and `O17` either populated with something or retired · **Signed by** the chief executive · **Time** one workshop of half a day

1. For each of the ten level-1 capabilities ask which post is accountable for the organisation being able to do it. One post, not a committee, and a post that exists.
2. Where none exists, record it as a gap rather than assigning the nearest available unit. That assignment is exactly what produced `R22`.
3. Where the answer is genuinely more than one post — Estates & Medical Devices has at least three, per chapter 5 — record the candidates and the decision required, not an answer.
4. Then decide what `O17` is for. Either a function is a durable grouping distinct from the capability, in which case say what it does, or it is a copy of `O13` and should be retired.

4.5 — Write acceptance criteria that can fail

Sheet `O35 Business Building Block` · **Input** the re-baselined maturity from step 4.3, the measures kept from item 3's step 3.4 · **Output** two to four testable criteria per building block, each with a measure, a threshold and a date · **Signed by** the capability owner from step 4.4, countersigned by whoever will test it at the Phase G gate · **Time** one day

1. For each of the ten rows replace "<capability> measured and owned" with criteria of the form: this measure, computed this way, at or above this threshold, on this date, evidenced by this artefact.
2. Name the measure from `O04` as corrected in Phase A, or state that it does not exist yet and which new sheet will carry its definition — `O70 Measure Definition`, from item 5 — and name the owner as a post from step 4.4 rather than `Data Steward`.
3. Write the criterion so that a reasonable person could record a fail. If nobody can imagine the fail, rewrite it. Where the capability is one the health standards regulator inspects — health records and responses to adverse events both sit inside its clinical governance domain [1] — write the criterion to the evidence an inspection would ask for.
4. Hand the set to Phase G. Item 10 gates on what was reconciled, by whom, against what, and an untestable criterion cannot be gated on.

`O15` contradicts `O13` on criticality for six of ten

`O15` has six columns: `Ref`, the service name, `Capability`, `Description`, `Exposure`, `Criticality`. Read its criticality against the criticality of the capability each row names in column 3:

Capability	`O13` criticality	`O15` criticality	Exposure recorded
C01 Patient Access & Scheduling	High	Medium	Internal
C02 Clinical Care Delivery	High	High	External
C05 Care Coordination	Medium	Critical	External
C07 Revenue Cycle & Claims	Medium	High	External
C08 Workforce & Rostering	Low	Critical	Internal
C10 Population Health & Analytics	High	Medium	Internal

Six of the ten pairs disagree. The two rows rated `Critical` on `O15` are Care Coordination and Workforce & Rostering, whose capabilities are `Medium` and `Low`, and nothing in the file is `Critical` on both sheets. One sheet is wrong on every disagreeing row and nothing records which.

`Exposure` is the other problem. Column 4 describes every row as "Externally consumable" and column 5 then types four `External` and six `Internal`. For Workforce & Rostering the description and the value contradict each other on the same row; for Diagnostics the `External` value is right for a reason the sheet does not record — the laboratory is a different organisation, which chapters 1 and 9 both make the defining fact of this estate.

4.6 — Reconcile criticality once, in one place, and define exposure

Sheet `O13 Business Capability`, `O15`, read against `O49 Resilience Tier` · **Input** step 4.2's criticality scores, `O49`'s tiers and recovery objectives · **Output** one criticality value per capability, consumed by `O15` rather than restated, and an exposure value with a stated test · **Signed by** the chief operating officer · **Time** half a day

1. Decide which sheet owns criticality. The capability does, because the service is a way of delivering it, so `O15` carries none of its own or carries a copy with a note saying so.
2. Resolve the six disagreements with the capability owner in the room. Each takes two minutes and the record of the decision is the deliverable.
3. Define `Exposure` with a test: who outside this organisation consumes this directly. On that test Workforce & Rostering is `Internal` and Diagnostics is `External`.

4. Then check against `O49`, where the clinical record is `Medium` criticality at a 24-hour recovery objective and never tested, and the tier label carries no information at all. Item 7 owns that sheet; Phase B's duty is to hand over a defensible criticality before item 7 builds on it.

Finding 13: twenty-four declared, twenty present, thirty-six real

This is the spine of the item.

`O14 Value Stream` has five rows and five columns: `Ref`, `Value stream`, `Scope`, `Stages`, `Capabilities`. The five names are right and the five scope lines are reasonable. Column 4, `Stages`, holds the strings `4 stages`, `5 stages`, `6 stages`, `4 stages`, `5 stages`. Twenty-four declared.

`O16 Business Process` holds twenty rows: four per value stream, named `Referral to Treatment — stage 1` through `— stage 4`, `Order to Result — stage 1` through `— stage 4`, and so on, **regardless of what `O14` says**. Three of the five streams therefore contradict their own decomposition inside the same workbook: `VS02` declares five and has four, `VS03` declares six and has four, `VS05` declares five and has four.

The shortfall is four. The real shortfall is sixteen.

Finding 13, drawn: three counts of the same five journeys

Value stream	O14 declares	O16 holds	Chapter 9 names	The shortfall you inherit
VS01 Referral to Treatment	4	4	7	none declared; 3 stages unmodelled
VS02 Order to Result	5	4	7	short 1 declared; 3 unmodelled
VS03 Prescribe to Administer	6	4	6	short 2 declared; 2 unmodelled
VS04 Encounter to Payment	4	4	8	none declared; 4 stages unmodelled
VS05 Incident to Improvement	5	4	8	short 1 declared; 4 unmodelled
Total	**24**	**20**	**36**	and not one of the 20 has a name

O14 column 4 holds the strings "4 stages", "5 stages", "6 stages". It is a count, not a list, so nothing can be wrong in it and nothing can be checked against it. O16 names every row "— stage 1" to "— stage 4" whatever O14 says, and carries an Automation value for each: an assessment of automation attached to a step with no name.

Figure 4.2 Finding 13 with the third column the file does not have. O14 declares twenty-four stages, O16 holds twenty, and volume 1 chapter 9 names thirty-six. **VS01 and VS04 are the dangerous rows**: they declare four and hold four, so every check the file runs on them passes, and they are short by three and four stages respectively.

Volume 1 chapter 9 names the stages of all five streams, and it names thirty-six. Those names are the deliverable. Reuse them exactly; do not re-derive them, because the chapter did the work of establishing which stage touches no system and where each handoff fails, and a re-derivation loses that.

Value stream	The stages, as chapter 9 names them	Declared	Present
VS01 Referral to Treatment	Recognise and decide; Compose; Transmit; Accept, redirect or decline; Book; Attend and treat; Back-refer and close	4	4
VS02 Order to Result	Request; Collect and label; Transport; Accession and check; Analyse and authorise; Report and route; Acknowledge and act	5	4
VS03 Prescribe to Administer	Prescribe; Verify; Dispense or supply; Administer, or record not given; Monitor; Reconcile	6	4
VS04 Encounter to Payment	Establish liability; Authorise; Deliver and document; Code; Assemble and price; Submit; Adjudicate; Reconcile and resubmit	4	4
VS05 Incident to Improvement	Detect and report; Take statements; Classify; Escalate; Investigate; Decide and act; Verify closure; Aggregate	5	4

Two things to notice before you touch the sheet.

The column is a count, not a list, so nothing in it can be wrong and nothing can be checked against it. `4 stages` is unfalsifiable in the way `O35`'s acceptance criteria are, and for the same reason: a summary figure maintained by one hand against a decomposition maintained by another. Chapter 21 calls this the defect most likely to appear in your own work.

`VS01` and `VS04` are the dangerous rows, not the three that disagree. They declare four and hold four, so every check the file runs on them passes — and chapter 9 names seven and eight. A stream whose declared count matches its decomposition and is wrong by three stages is harder to find than one that contradicts itself, and the three missing stages in `VS01` are Transmit, Book and Back-refer and close, of which Transmit is the stage with no system behind it at all and Back-refer and close is where the referral loop closes or does not.

4.7 — Replace the stage count with the stage names

Sheet `O14 Value Stream`, `O16 Business Process` · **Input** volume 1 chapter 9's stage lists for all five streams · **Output** `O14` column 4 holding a list rather than a count, and `O16` holding thirty-six rows · **Signed by** the owner of each value stream, named in step 4.4 · **Time** one day

1. Rewrite `O14` column 4 as a comma-separated list of stage names in sequence. The header stays `Stages`; the content stops being a number. A count that cannot be checked is worse than no count.
2. Extend `O16` from twenty rows to thirty-six, continuing from `BPR-21`, and rename `BPR-01` to `BPR-20` to the stage each actually is, in sequence, so existing citations survive.

3. Name each row with the stage name, not the stream name plus a number. `Accept, redirect or decline` can be argued with; `stage 3` cannot, and the argument is the architecture.
4. Where a stage has no system behind it, write that in the description. `Transmit` in `VS01` is carried by the patient; `Transport` and `Acknowledge and act` in `VS02` touch nothing in most deployments. Those three rows are the most valuable on the sheet.
5. Do not add the three streams chapter 9 names and the repository lacks — emergency arrival and triage, admission to discharge, transfer between facilities — until the five that exist are named. Record them as a scope decision instead.

Form 4-A — Value stream stage register

Feeds `O14 Value Stream` column 4 and `O16 Business Process` · **Rows a first pass needs** thirty-six, one per stage · **Signed by** the value stream owner

Ref	Business process	Value stream	Description	Automation
BPR-06 (example)	Collect and label	VS02	Identity verified at the patient; specimen type and collection time recorded	Manual

4.8 — Fix the capability column against `R02`, and account for C08 to C10

Sheet `O14 Value Stream` column 5, `R02` · **Input** the re-cut capabilities from step 4.1, the stage names from step 4.7 · **Output** one agreed capability set per value stream, held in one place · **Signed by** the value stream owner and the capability owner jointly · **Time** half a day

1. Compare `O14` column 5 against `R02` row by row. `VS01` to `VS03` agree. `VS04` reads `C01, C02, C03` on `O14` and `C04, C05, C06` in `R02`; `VS05` reads `C02, C03, C04` against `C05, C06, C07`.
2. Both readings of `VS04` are wrong on their merits: "episode through coding, claim, adjudication and settlement" uses `C07 Revenue Cycle & Claims` on neither sheet, and `C07` appears in a value stream only under `R02`'s `VS05`.
3. Account for `C08`, `C09` and `C10`, which appear in no value stream on either sheet. Either they support streams the file does not model or they are enabling capabilities no stream consumes directly. Say which, in writing.
4. Then delete one of the two places. A capability set held twice will diverge again, and the relationship sheet is the one a tool can traverse.

`O16`: twenty processes, no name, no performer, no system

`O16` has five columns: `Ref`, `Business process`, `Value stream`, `Description`, `Automation`. Column 4 reads "Stage 1 of referral to treatment." down all twenty rows in the obvious pattern. Column 5 takes `Automated`, `Manual` and `Partly automated`, and it is filled on every row.

The file carries an automation assessment for twenty steps that have no name. `BPR-07` and `BPR-08` both read `Manual`; on chapter 9's naming those are Transport and Accession and check, of which one touches no system and one is the laboratory's own system in another organisation. The assessment may be right by luck. It cannot be right by derivation, because nothing records what the step is.

And `O16` is one of the twenty sheets appearing in no relationship sheet at all. No relationship joins a process to a role, an actor, an application, a data entity or a requirement, so the repository has a process layer that nothing performs, nothing supports and nothing needs.

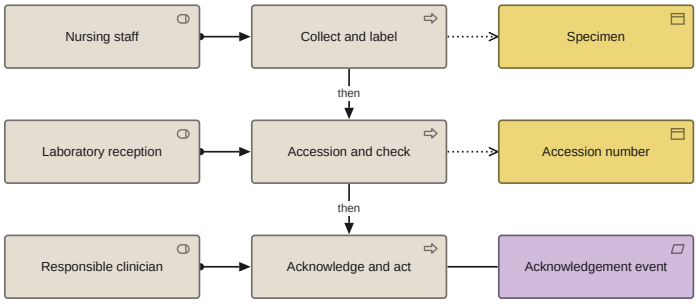


Figure 4.3 What step 4.9 produces, drawn for three of VS02's seven stages as chapter 9 names them. Each stage carries a performer by role, the object the step touches and what it must record. **The third row is the one that matters**: Acknowledge and act touches no system in most deployments, so its requirement is an event that does not yet exist — and the repository has no relationship sheet joining a process to a role at all.

A named stage has to carry four things, from chapter 9: who performs it by role rather than job title, which system it touches, where the handoff fails, and what it requires of the data. The repository has a column for none of them, so Phase B produces them as a form and carries them into Phase C and Phase D rather than forcing them into five columns.

4.9 — Give every stage a performer, a system and a failure mode

Sheet `O16 Business Process` as extended by step 4.7 · **Input** chapter 9's four questions per stage, the roles added in item 3's step 3.3, `O23`'s twelve applications · **Output** Form 4-B completed for all thirty-six stages · **Signed by** the value stream owner, countersigned by the operational manager who runs the step · **Time** one workshop per value stream, half a day each

1. Name the performer by role. Chapter 5 is the authority: a clinical step's performer is defined by registration and scope of practice rather than job title, so record the role and the credential it requires.
2. Name the system from `O23` by `Ref`. Where the answer is paper, write paper; where it is nothing, write nothing, and count how many stages that is. In a low-maturity public estate it will be five or six of thirty-six.
3. Name the handoff that fails. Almost every documented failure here is a handoff rather than a step, and each stream has a known one: a referral sent and never acknowledged, a result reported and never acted on, a dose not given with no reason recorded, a funder claim short-paid rather than rejected, an action agreed and never implemented.
4. Record what the stage requires of the data as a plain sentence and carry it to Phase C as a candidate entity rather than resolving it here.
5. Re-derive `Automation` last. On an unnamed step it is noise; on a named step with a named system it is a finding.

Form 4-B — Stage definition, part A: identify

Feeds `O16 Business Process`. Columns 1 to 5 are typed straight into the sheet; part B's four answers do not fit `O16` and are carried in this form to Phase C. **Rows a first pass needs** thirty-six, one per stage · **Signed by** the value stream owner

Ref	Business process	Value stream	Description	Automation
BPR-24 (example)	Acknowledge and act	VS02	Named recipient records that the result was seen and what was done	Manual

Form 4-C — Stage definition, part B: assess

Feeds no sheet. The repository has no relationship sheet joining a process to a role or to an application, and no column for a handoff, so this form is the Phase B record and the input to Phase C and Phase D. **Rows a first pass needs** thirty-six, one per stage · **Signed by** the operational manager who runs the step

Ref	Performer, by role	System touched	Where the handoff fails	What the data must carry
BPR-24 (example)	Responsible clinician, or whoever covers	None in most deployments	No acknowledgement event exists	An actor and a timestamp

ASK THE VALUE STREAM OWNER — THE FOUR QUESTIONS CHAPTER 9 ASKS OF EVERY STAGE

Which stages of this stream touch no system at all, and what carries the information instead?

At which handoff does this stream most often fail here, and how do you find out when it has?

Who is responsible for the step after the handoff, and does that responsibility follow a roster?

What would have to be recorded for you to be able to measure the interval between two stages?

Re-deriving a positional join, and the two that are Phase B's

Eight relationship sheets in this file were built by pairing two lists in row order: `R04`, `R07`, `R09`, `R13`, `R20`, `R22`, `R23` and `R25`. Every reference resolves, every name matches its reference, and almost nothing means anything. Chapter 21 establishes the finding and item 1 opens on it; item 10 gates on it. Two of the eight belong to Phase B, and the instruction is **re-derive them from evidence, not repair them in place.**

The distinction matters. Repairing in place means changing the targets until they look right, which produces a sheet nobody can check: a reader cannot tell which rows were inherited and which decided, and the next person to doubt it redoes the whole exercise. Re-deriving means building the sheet again from evidence, recording the evidence per row, and keeping the original beside it.

4.10 — The re-derivation method, and the convention for recording it

Sheet any of the eight positional joins · **Input** the original sheet, and at least two independent sources of evidence per row · **Output** a new tab carrying the re-derived rows with their evidence, and the original untouched · **Signed by** the owner of each end of the relationship · **Time** half a day per ten rows

1. Copy the sheet to a new tab named with the sheet's reference plus `RD` — `R22 RD`, `R04 RD`. The original stays as supplied: it is the evidence of what was inherited, and Phase H will want it.

2. Keep the original row-4 headers exactly, in order, and add three columns to the right: the evidence relied on, the person who confirmed it, the date.
3. Derive each row from evidence, never from the other sheet, with two independent sources where you can get them — for ownership, who holds the budget and who signs; for support, what the application's own description says and what the interface list shows it moving.
4. Where the evidence does not resolve, leave the target blank and record why. A blank row with a reason is worth more than a plausible row with none, and it is the row that gets the decision made.
5. Count the rows that changed and put the count in the Phase B document: ten of ten on `R22`, eight of ten on `R04`.

R04, ten rows: the positional join, and what the evidence gives instead

Row	Service in column 1	Reached by row order	What O23 column 4 says it is	Evidence gives
5	Patient Access & Scheduling	A01 EHR / EPR	the clinical record of truth	A02 PAS
6	Clinical Care Delivery	A02 PAS	registration and scheduling	A01 EHR / EPR
7	Diagnostics	A03 LIS	laboratory orders and results	**A03 + A04**
8	Medicines Management	A04 RIS / PACS	imaging workflow and store	A05 ePrescribing
9	Care Coordination	A05 ePrescribing	prescribing and administration	A01, and no owner
10	Clinical Governance & Safety	A06 Theatre Management	theatre scheduling	A09 Incident & Risk
11	Revenue Cycle & Claims	A07 Revenue Cycle / Claims	coding, billing, adjudication	**A07 — holds**
12	Workforce & Rostering	A08 Rostering	staff scheduling to demand	**A08 — holds**
13	Estates & Medical Devices	A09 Incident & Risk	safety incidents and action	no application
14	Population Health & Analytics	A10 Patient Portal	appointments and results	A12 Warehouse

Two rows survive, one is partly right

Rows 11 and 12 are correct by coincidence: two lists happened to align at the seventh and eighth item.

A11 and A12 are never reached

Ten rows against twelve applications, so the integration engine and the warehouse serve nothing.

Figure 4.4 R04 re-derived from evidence rather than repaired in place. Six of the ten pairings contradict the application's own description in O23 column 4; **two are correct by coincidence**, where two lists happened to align at the seventh and eighth item. Ten rows against twelve applications is also why RPT13 reports the integration engine and the warehouse as the file's only two orphans.

`R22`, ten rows, and why it cannot be completed before item 3's step 3.2

`R22` has four columns — `Org unit`, `Unit name`, `Capability`, `Capability name` — and pairs `ORG-01` through `ORG-10` with `C01` through `C10` in order. So Finance owns Clinical Care Delivery, Risk & Compliance owns Diagnostics, Human Resources owns Medicines Management, Commercial owns Workforce & Rostering and Supply & Logistics owns Population Health & Analytics, while `ORG-11 Internal Audit` and `ORG-12 Legal` own nothing.

Not one of the ten is right, and the reason is item 3's: there is no clinical unit on `O09` for eight of these capabilities to belong to. So `R22 RD` cannot be completed until `O09` holds the rows item 3's step 3.2 adds. If Phase A has not done it, Phase B stops here and says so rather than assigning Clinical Care Delivery to Finance with better handwriting.

4.11 — Re-derive `R22` from budget, signature and system ownership

Sheet `R22`, into a new `R22 RD` tab · **Input** `O09` as extended by item 3's step 3.2, the capability owners from step 4.4, the client's delegation of authority · **Output** ten to fourteen re-derived rows with evidence, and a blank row with a reason wherever the owner is contested · **Signed by** the chief executive · **Time** one day

1. For each level-1 capability ask three questions: which unit's budget pays for it, who signs when it fails, and which unit owns the supporting system in `O23`. Where all three answer the same unit, the row is derived.
2. Where they disagree, record all three answers and leave the target blank. Chapter 5 names the entities with no single plausible owner, and Estates & Medical Devices is one: estates, nursing operations, clinical engineering and finance all have a claim.
3. Where the owning unit does not exist on `O09`, write the name the client uses and raise it against step 3.2 rather than substituting a unit that does exist.
4. Give Internal Audit and Legal no capability, deliberately, and record that as a decision. An assurance function owning an operational capability is the defect in `O23` row 10, where Internal Audit owns theatre scheduling.

`R04`, ten rows, and the three that survive by coincidence

`R04` is the sheet whose tab name ends in a trailing space, so a lookup written against the visible name will not find it. Its four columns pair the ten entries of `O15` with `A01` through `A10` in row order, which means `A11 Integration Engine` and `A12 Clinical Data Warehouse` are never reached — and that is precisely the two orphans `RPT13` reports, correctly, as the only two orphans in the file.

Read it against `O23` column 4, which carries each application's own description, and six of the ten pairings are plainly wrong. Clinical Care Delivery reaches `A02 PAS`, "registration, scheduling and patient administration", while `A01 EHR / EPR` is "the clinical record of truth". Medicines Management reaches `A04 RIS / PACS`, imaging. Clinical Governance & Safety reaches `A06 Theatre Management` while `A09 Incident & Risk` is "safety incidents,

investigation and action". Population Health & Analytics reaches `A10 Patient Portal` while `A12 Clinical Data Warehouse` is "governed analytics across the care record".

Three land correctly — Diagnostics to `A03 LIS`, Revenue Cycle & Claims to `A07`, Workforce & Rostering to `A08` — and the teaching point is that **they are correct by coincidence**, two lists happening to align at the third, seventh and eighth item. A reader who samples three rows and finds them right concludes the sheet is sound. Chapter 21's sampling rule exists for that reason: twenty rows, three questions, a pass rate.

4.12 — Re-derive `R04` from the application's own description and the interface list

Sheet `R04`, into a new `R04 RD` tab · **Input** `O23` columns 2 and 4, `R06`'s twenty-three interfaces, `O19`'s ten vendor agreements, the service definitions from step 4.6 · **Output** re-derived rows with evidence, one or more applications per service, and a blank where no application supports the service · **Signed by** the capability owner from step 4.4 and the application owner from `O23` column 10 · **Time** one day

1. Quote the sheet as `R04` and never by its full tab name, which carries a trailing space. Say so once in the Phase B document, because the next person to write a lookup needs to know.
2. For each service, read `O23` column 4 for all twelve applications and pick on description, not row order. Then check the choice against `R06`: an application supporting a service moves the data that service's stages need.
3. Allow more than one application per service, and allow none. The supplied sheet forces exactly one, which is why Diagnostics reaches `A03 LIS` and not also `A04 RIS / PACS`.
4. Record that `A11` and `A12` support no service directly, which is the right answer: an integration engine and a warehouse serve other applications. `RPT13` called that an orphan; it is a platform, and the correction belongs in the Phase B document.
5. Count the changes. Eight of ten rows change, and that count is the sentence that gets the rest of the work funded.

`O18` and `O19`: five products that are settings, and ten agreements that are one per application

`O18 Product` has five rows in four columns — `Ref`, `Product`, `Type`, `Description`: Inpatient Care, Outpatient Care, Emergency Care, Diagnostic Services, Community Care, typed `Care` four times and `Diagnostics` once.

Those are care settings, which chapter 4 defines as a kind of care rather than a place, and they are not products in any sense a funder or a patient would recognise. In the private sector what the organisation may sell is also capped by an instrument the file has no column for: a provincial facility licence is site-specific and bed-count-specific, and in KwaZulu-Natal it runs for twelve months from issue and no further than 31 December of that year [2]. A private provider's products are what it is paid for — an admission under an authorisation, a theatre list, a day case, a diagnostic study — and a public provider's are the service packages the hospital category determines, which chapter 3 shows cannot be derived from a bed count. `R25` then pairs the five rows with applications in row order, so Emergency Care is supported by the billing system and Diagnostic Services by the patient portal, while the laboratory and imaging systems support no product at all.

`O19` holds ten rows in five columns: `Ref`, the agreement's name, `Counterparty`, `Type` and `Expiry`. The header on column 2 is the word ArchiMate 4 no longer carries as an element, which is why this volume names the sheet and not the header. The ten rows are one per application for `A01` through `A10`, named "<application> vendor agreement", with `Counterparty` holding the application's own name rather than a supplier's. `Type` takes `Managed service`, `Perpetual` and `Subscription`. `Expiry` runs from 2028-04 to 2030-09.

Three observations. The counterparty is not a counterparty: you cannot write to `PAS` and ask it to renew. `A11` and `A12` have no agreement, matching `R04`'s reach exactly, so two applications sit in the estate with no commercial instrument recorded. And `O19` appears in no relationship sheet, so no agreement is joined to the application it governs or the service that depends on it — while the earliest expiry, April 2028, falls after the earliest work package start of January 2027 and inside the programme.

4.13 — Separate product from care setting, and give `O19` a real counterparty

Sheet `O18 Product`, `O19`, `R25` · **Input** the client's price list or service package, the supplier register, the five care settings from item 3's step 3.6 · **Output** `O18` holding things the organisation is paid for, `O19` holding a named legal counterparty and a renewal date per row · **Signed by** the chief commercial officer for `O18`, the head of procurement for `O19`; neither is an actor in `O10`, so item 3's step 3.3 must have added them · **Time** two days

1. Move the five care settings into the scope statement, where chapter 4's definition puts them, and repopulate `O18` with what the organisation is paid for: the authorisation-bounded unit privately, the service package the hospital category sets publicly.
2. Rewrite `O19` column 3 with the legal entity's name. An agreement whose counterparty is the name of the software it licenses cannot be renewed, escalated or terminated by anybody.
3. Add rows for `A11` and `A12`. An application in production with no recorded commercial instrument is a finding for Phase D and a risk for item 3's step 3.9.
4. Then re-derive `R25` by the method in step 4.10. It is the third positional join in the business layer and the same two-sources rule applies.

ASK THE COMMERCIAL AND PROCUREMENT OWNERS

What is this organisation actually paid for, and at what unit is the price set?

Which agreements expire inside the programme window, and who is accountable for each renewal?

For the integration engine and the warehouse, what commercial instrument is in force and where is it held?

What goes to the business owner

Phase B's deliverable to the business is four pages and a form pack.

The fifteen rows that carry information, and the 155 that do not. Said plainly, with the three checks from the capability section beside it, because they are checkable in ten minutes and buy the standing for the rest.

The thirty-six stage names, reused from chapter 9 and not re-derived, with the sixteen the repository does not have marked. Then Form 4-A, Form 4-B and Form 4-C, as paper, into the workshops.

The two re-derived joins, each as a new tab beside the original, each with its evidence column filled and its change count stated: ten of ten on `R22`, eight of ten on `R04`. And the one blocked dependency: `R22 RD` cannot be completed until `O09` holds the clinical units, which is Phase A's step 3.2 and not Phase B's to do.

The decisions you need and who signs each. One owner per level-1 capability, or a recorded contest where there is no single plausible owner. One place where criticality lives. An exposure test. A product definition separate from care setting. And one named owner per value stream, because a stream with twenty unnamed stages and no owner is what the next five phases get built on.

One discipline to carry forward. Every correction here is checkable against a cell, and every one is a caricature of a pattern a reader will meet within a month: a summary count maintained separately from the decomposition beneath it, an acceptance criterion written so it cannot fail, two sheets holding the same judgement and diverging, and a relationship built by joining two lists because the right target was in neither. Name the pattern when you report the defect. A client who hears "your spreadsheet is wrong" defends the spreadsheet; a client who hears "this is what happens when a count lives in one place and the detail in another" fixes the process.

THE CHAPTER IN TEN LINES

- Of the 170 business-layer rows, fifteen carry information: the ten level-1 capability names with their descriptions, and the five value stream names with their scope lines.
- O13's decomposition adds up perfectly and every one of the ninety child names was generated from ten suffix words and the pair Define and Operate.
- Thirty-six of ninety capability children are rated more critically than their parent. Criticality inherits downward exactly as classification does, and the file breaks both rules.
- Disposition is a restatement of the maturity gap: Sustain on every gap of 1 and Build on every gap of 2, with no exception in a hundred rows.
- The same ten names appear on O13, O15, O17 and O35, and O35's acceptance criteria cannot fail, which is why three assessments against thirty-seven building blocks produced no evidence.
- O15 and O13 disagree on criticality for six of ten pairs, and O15 describes staff rostering as externally consumable.
- O14 declares twenty-four stages, O16 holds twenty, and volume 1 chapter 9 names thirty-six. Reuse those names exactly rather than re-deriving them.
- VS01 and VS04 declare four stages and hold four, so every check passes, and they are short by three and four stages. They are harder to find than the three that contradict themselves.
- O14's capability column contradicts R02 for VS04 and VS05, and C08, C09 and C10 appear in no value stream on either sheet.
- O16 carries an automation value for twenty steps that have no name, performer or system, and appears in no relationship sheet at all.
- Re-derive a positional join into a new tab named with the sheet reference plus RD, keep the original as supplied, and record the evidence and the confirmer per row.
- R22 changes on ten of ten rows and cannot be completed until O09 holds the clinical units Phase A adds. R04 changes on eight of ten.
- O18's five products are care settings, and O19's ten agreements name the software as the counterparty while two applications have no agreement at all.

SOURCES FOR THIS CHAPTER

- **1.** Office of Health Standards Compliance, 2023/24 Annual Performance Report presentation, 16 October 2024 (inspection domains, including health records and responses to adverse events) https://static.pmg.org.za/241016Updated_OHSCs_Presentation_-_16_Oct_2024.pdf
- **2.** KwaZulu-Natal Department of Health, licensing of private health care facilities (site-specific and bed-count-specific licences, twelve months from issue and not beyond 31 December) https://www.kznhealth.gov.za/licensing-of-private-health-care-facilities/

ITEM 5

Data Architecture

Fifteen steps, three new sheets, and one rule: classification may tighten downward and may never loosen.

Phase C is one phase and two items. This is the data half, and in a provider it is the half that carries the legal risk. It decides who may hold a record, how tightly it is graded, which system wins when two disagree, and whether a figure the board publishes can be reproduced next year. **Ravensmere Health Network** answers all four wrongly, consistently, and in a way that passes every integrity check the workbook performs on itself.

Fourteen steps. Three create a sheet the repository does not have: `O67 Consent`, `O70 Measure Definition` and `O71 Disclosure Event`.

> **READ THESE FOUR BEFORE YOU OPEN A SHEET**
>
> Volume 1 does the teaching; this item does the filling. Do not look any of it up twice.
>
> - **Chapter 10.** The conceptual model, the sixteen omitted entities, the four fields and a date that replace an owner where ownership is contested, and the two entities that exist only as a row.
> - **Chapter 11.** Identity. Steps 5.6 and 5.7 assume all of it.
> - **Chapter 12.** Coding. Item 2 turned that register into `O69 Code System`; this item binds entities to it.
> - **Chapters 13 and 18.** The clinical record as three instruments, and the regime. Chapter 13 introduced Disclosure event and Measure Definition; steps 5.10 and 5.11 make them sheets.

What this item fills

Eleven object sheets, two relationship sheets, one report, and three sheets you create.

Sheet	Rows	What this item does to it
`O20 Data Entity`	14	Regrades 7, re-seats 14 master systems, names an owner
`O21`, `O22`	12, 12	`O21` gains the two it stops short of; `O22` is confirmed
`O36`, `R19`	11, 11	Rebuilt from the clinical model, and made to agree
`O37 Data Attribute`	60	Regrades 26, adds identifier and code-binding rows
`O38`, `O39`, `O40`	6, 6, 12	Re-derived from evidence; four data sets replaced
`O42 Metadata Element`	14	Corrects 3 that contradict other sheets; records the clock
`O43`, `O44`	10, 10	Made to agree with `O20`, extended, re-cut
`R05`, `RPT05`	28, 14	`R05` re-derived; `RPT05` prints `O20` and nothing else

Four Phase C deliverables already read `Approved` on `O31 Deliverable`: `DLV-07 Data Architecture`, `DLV-08 Data Model`, `DLV-09 Application Architecture` and `DLV-10 Interface Catalogue` — approved before any of this work was done. Set all four to `Draft` in step 5.1.

The governing rule, and the one principle the file already holds

An attribute may be classified more tightly than the entity it belongs to. It may never be classified more openly. An attribute is the data and the entity is a container for it, so a looser attribute grade is an instruction to handle the same bytes two ways depending on which sheet somebody read.

The repository states the rule itself. `PRN-14 Classification Travels with Data` sits at `O05` row 18, status `Approved`: "Classification is a property of the data, not its location." Twenty-six of the sixty rows on `O37 Data Attribute` break it.

Figure 5.1 Every attribute O37 holds, with the grade it carries, against the grade its entity carries on O20. Twenty-six cells are looser than their parent, against principle PRN-14 at O05 row 18. Four entities have no attributes at all, two of which carry the most legal risk in the file.

Two things in that figure are worth saying aloud before any step.

The patient identifier is `Public`. `ATR-001`, entity `Patient`, `Key` = `Yes`, beneath an entity graded `Restricted`. So is `ATR-006 Classification`: the field saying how tightly to handle the row is the most openly handled field on it.

And the two entities that pass the inheritance test pass it for the wrong reason. `Encounter` and `Procedure` have no attribute looser than the entity because the entity is graded `Public`, the floor. A report counting breaches by entity shows those two as clean, and they are the two worst rows on the sheet.

Carry all the readings when you report the count, because the count depends on the test:

Test	Rows	Note
Attribute looser than its entity	26 of 60	The inheritance breach. `PRN-14`
Attribute graded `Public` outright	19 of 60	Includes every key attribute of `Diagnosis`
Entity graded `Public` or `Internal`	6 of 14	All six patient-identifying or clinical
Attributes present at all	60 of a possible 84	Four entities have none

The last row gets missed. `O37` holds six attributes each for `E01` to `E10` and nothing for `E11 Care Setting`, `E12 Claim`, `E13 Consent` or `E14 Medical Device` — and two of those four carry the most legal risk in the file. Nothing here would catch any of it: `STD-03 Data classification standard` on `O32` is `Current` with conformance `Not assessed`, `EP-04 Data platform` on `O65` records "Classification not applied at read", and `POL-01` on `O64` is already `Breached`. The twenty-six rows are not a data-entry slip; they are what a classification scheme looks like when nothing reads it.

5.1 — Set the scope, and put the four Phase C deliverables back to draft

Sheet `O31 Deliverable`, `O20 Data Entity` · **Input** Phase B as you left it, and the fourteen rows of `O20` as delivered · **Output** a one-page scope note and four deliverables at `Draft` · **Signed by** the Chief Data Officer (`ACT-05`), noted by `GB-03 Data Governance Council` · **Time** half a day

1. Set `DLV-07`, `DLV-08`, `DLV-09` and `DLV-10` to `Draft`, with the date and your name in the scope note. A deliverable that arrives `Approved` before the work is done is the first thing a client's auditor asks about, and the only defensible answer is that you changed it.
2. Mark every row of `O20` in scope, out of scope or deferred, by value stream rather than by sheet: `VS01 Referral to Treatment` alone puts `E01`, `E02`, `E07` and `E08` in scope.
3. Set the attribute boundary in writing, now. `O37` holds sixty rows and will hold six hundred if nobody draws a line. The boundary this item uses: identifying attributes, dating attributes, the classification attribute, every attribute carrying a code, and nothing else.
4. Record which entities you are **not** going to give a single owner, and why. Volume 1 chapter 10 names six for which a single owner teaches a falsehood; here `E10 Practitioner`, `E11 Care Setting` and `E08 Care Plan` are three of them.
5. Note that `O21` and `O22` hold twelve rows each against `O20`'s fourteen, and that the two missing from `O21` are `E13 Consent` and `E14 Medical Device` — step 5.9 and item 7, not this step.

Done when four deliverables read `Draft`, every row of `O20` is marked, the attribute boundary is a written list, and three entities carry the words "ownership contested" with a date.

Goes wrong when scope is set sheet by sheet. Three weeks then go into `O37` while `Owner` on `O20` is still a job title nobody holds.

5.2 — Read the three sheets that each name a system of record, and settle every entity

Sheet `O20 Data Entity`, `O43 Master Data Rule`, `R05 Application manages Data En` · **Input** the three columns side by side, and `O23 Application Component` column 4 for what each application actually does · **Output** one system of record per entity, decided in a room · **Signed by** `GB-03 Data Governance Council`, with the entity owner · **Time** two half-day workshops

> **`O20 Data Entity`**
>
> `Ref`, `Data entity`, `Class`, `Definition`, `Classification`, `Owner`, `Master system`, `Lineage`. `Master system` holds an application name, not a `Ref`, on all fourteen rows, so a lookup against `O23` has to match on name. `Lineage` takes `Full`, `Partial` and `Absent`.

1. Read `O20.Master system` against `O23`'s description of each application. `E01 Patient` is mastered in `ePrescribing & Medicines Admin` while `A01 EHR / EPR` is "The clinical record of truth". `E03 Clinical Observation` is mastered in `Revenue Cycle / Claims` and `E12 Claim` in `EHR / EPR` — exactly swapped.
2. Note the three mastered in an analytics platform: `E02 Encounter`, `E08 Care Plan` and `E10 Practitioner` all name `Clinical Data Warehouse`, the downstream consumer of a record, declared master of three `Master`-class entities.
3. Read `R05` rows 5 to 18, the `Master` rows. They reproduce all fourteen assignments exactly. **Two sheets agreeing is not corroboration here; it is one error copied.** That is why nobody checked.
4. Now read `O43.Golden source` and find a third answer. It disagrees with `O20` on eight of the ten entities it covers, agreeing only on `E04 Diagnosis` and `E06 Medication Order` — both of which `O20` masters in `PAS`, whose disposition on `O23` is `Retire`.
5. Settle each entity in the room and record the decision, the date and the person. Where the defensible master has disposition `Retire`, `Replace` or `Consolidate`, say so on the row: nine of the fourteen are in that position.
6. Correct `O20`, then `R05`, then `O43`, in that order, and make `O43.Rule type` fit the rule you wrote. `MDR-01 Patient` is typed `Golden source`; a patient master needs `Match rule`, and volume 1 chapter 11 is why.
7. Extend `O43` to fourteen rows. It stops at `MDR-10 Practitioner`, so `Care Setting`, `Claim`, `Consent` and `Medical Device` have no mastering rule.

Done when all three sheets give the same answer for all fourteen entities, every change has a minuted decision behind it, and every retiring master system carries a destination.

Goes wrong when the team repairs the sheet easiest to edit and leaves the other two. `RPT05` reads `O20` only, so the report prints clean over two sheets that still disagree.

Three sheets name a system of record. Two agree and are wrong; the third disagrees.

R05 rows 5 to 18 reproduce O20.Master system on all fourteen rows. O43.Golden source differs on eight of the ten entities it covers.

DATA ENTITY	O20.Master system = R05	O43.Golden source	AGREE?	O23 SAYS THAT SYSTEM IS
E01 Patient	ePrescribing	Rostering	differ	prescribing
E02 Encounter	Clinical Data Warehouse	Incident & Risk	differ	analytics
E03 Clinical Observation	Revenue Cycle / Claims	RIS / PACS	differ	billing
E04 Diagnosis	PAS	PAS	same	registration
E05 Procedure	ePrescribing	LIS	differ	prescribing
E06 Medication Order	PAS	PAS	same	registration
E07 Diagnostic Order / Result	Rostering	Theatre Management	differ	rostering
E08 Care Plan	Clinical Data Warehouse	PAS	differ	analytics
E09 Clinical Protocol	Patient Portal	PAS	differ	portal, retiring
E10 Practitioner	Clinical Data Warehouse	EHR / EPR	differ	analytics
E11 Care Setting	Patient Portal	no rule	absent	portal, retiring
E12 Claim	EHR / EPR	no rule	absent	clinical record
E13 Consent	Integration Engine	no rule	absent	message router
E14 Medical Device	Theatre Management	no rule	absent	theatre scheduling

The two rows where all three sheets agree name PAS, whose disposition on O23 is Retire. O43 stops at MDR-10, so four entities have no mastering rule.

Figure 5.2 The three sheets that each name a system of record. O20 and R05 are identical by construction, so their agreement is one error copied rather than corroboration; O43 supplies a third answer and differs on eight of the ten entities it covers. The two rows where all three agree name an application that is being retired.

5.3 — Regrade the seven entities whose grade contradicts their own definition

Sheet `O20 Data Entity` · **Input** `O20` columns 4 and 5, read together, one row at a time · **Output** a grade per entity with the reason written beside it · **Signed by** the Chief Compliance Officer (`ACT-07`) and `GB-03` · **Time** one half-day workshop

1. Read column 4 and column 5 together, aloud, for all fourteen rows. Six entities that are patient-identifying or clinical by their own definition are graded `Public` or `Internal`: `E02 Encounter`, `E05 Procedure`, `E12 Claim` and `E14 Medical Device` are `Public`; `E04 Diagnosis` and `E07 Diagnostic Order / Result` are `Internal`.

2. Read the seventh in the other direction. `E09 Clinical Protocol` — "An evidence-based pathway the organisation follows" — is `Restricted`, the tightest value on the sheet, for a document the organisation publishes and which identifies nobody.
3. Fix each grade against the regime rather than against a feel. Health information about a patient is confidential under section 14 of the National Health Act, with third-party disclosure only on written consent, a court order or a law, or where non-disclosure poses a serious threat to public health [1]. The prohibition on processing health information is lifted for a provider by section 32 of the privacy Act only for the data subject's proper treatment and care or for administering the practice, and only where the responsible party is bound by confidentiality [2].
4. Grade `E12 Claim` carefully. A funder claim carries diagnosis and procedure codes for a named person, so the content is clinical even though the object is financial. Booklet 9 requires billing records to be kept separate from patient care records [4]; separate custody is not a looser grade.
5. Re-grade `E09` down, and note where it sits: `A10 Patient Portal`, disposition `Retire`, in service since 2010, 303 users — the most of any application in the estate.
6. Write the instrument beside every grade you changed. A grade without an instrument is an opinion, and the next architect will change it back.

Done when no row's grade contradicts its own definition and every changed grade names an instrument.

Goes wrong when the team grades `Medical Device` as an asset record. A device identifier bound to a patient implant is patient-identifying. Item 7 builds `O68 Medical Device`; the grade is decided here.

Form 5-A — Regrade, re-seat and re-own the fourteen entities

Real `O20` columns in sheet order, less `Class` and `Definition`, which are already right and are typed straight into Excel. One row per entity; a first pass needs all fourteen. Signed by the entity `Owner` and `GB-03`.

Ref	Data entity	Classification	Owner	Master system	Lineage
E12	Claim — EXAMPLE, delete this row	Confidential		Revenue Cycle / Claims	Absent

5.4 — Apply the inheritance rule to all sixty rows of `037`

Sheet `O37 Data Attribute` · **Input** `O37` column 6 against the grade settled in 5.3, entity by entity · **Output** sixty rows where no attribute is looser than its entity, and a list of the ones you made tighter on purpose · **Signed by** the Chief Compliance Officer (`ACT-07`), noted by the Security Architect (`ACT-17`) · **Time** one day

> **`037 Data Attribute`**
>
> `Ref`, `Entity`, `Attribute`, `Type`, `Key`, `Classification`. Sixty rows: ten entities, six attributes each — `Identifier`, `Name`, `Status`, `Effective date`, `Owner`, `Classification`. `Entity` holds the entity name, not its `Ref`. Twenty rows carry `Key` = `Yes`.

1. Sort by `Entity` and put the settled entity grade in a working column beside it. Do not edit until it is sorted, because the breaches cluster: `Clinical Protocol` breaks on all six of its rows, `Patient` on five, `Practitioner` on five, `Diagnosis` on four, `Clinical Observation` on three.
2. Raise every attribute to at least its entity's grade — twenty-six rows: `ATR-001`, `003`, `004`, `005`, `006`, `014`, `016`, `018`, `019`, `020`, `022`, `024`, `036`, `038`, `043`, `049`, `050`, `051`, `052`, `053`, `054`, `055`, `056`, `057`, `058` and `059`.
3. Do `ATR-001` first and in front of the room. The patient identifier, `Key` = `Yes`, graded `Public` beneath a `Restricted` entity. Nothing else on the sheet explains the problem as fast.
4. Then the `Diagnosis` block, `ATR-019` to `ATR-024`: every identifying and dating attribute of a coded condition attributed to a named patient is `Public`, and only `Status` and `Owner` are not.
5. Then `Practitioner`. `ATR-056`, `ATR-057` and `ATR-058` are `Public` beneath a `Confidential` entity. Registration status is in fact published by the councils, which makes this the one place on the sheet where `Public` is arguable — so write the argument down rather than leaving the cell to carry it.

6. Record the tightenings separately. An attribute tighter than its entity is allowed and often right. That list is the evidence the rule was applied rather than a global find-and-replace.
7. Add the rows 5.1's boundary requires and `O37` lacks: the identifier rows from 5.6, and a code-system attribute for every coded entity. `E04 Diagnosis` and `E05 Procedure` are both defined as coded and neither has a code, a code system or a code version attribute. Bind each to a row of `O69 Code System` from item 2.

Done when no row is looser than its entity, the twenty-six corrections are listed with dates, the deliberate tightenings are listed separately, and every coded entity is bound to `O69`.

Goes wrong when somebody sets the whole column to `Confidential`. That satisfies the rule, destroys the information, and will be undone by the first team needing a `Public` field.

Form 5-B — The attribute inheritance sweep

All six real `O37` columns in sheet order. Work one entity at a time, writing the entity's settled grade at the head of each block. A first pass needs twenty-six rows, one per breach, plus the identifier and code-system rows from 5.4 and 5.6. Signed by the Chief Compliance Officer.

Ref	Entity	Attribute	Type	Key	Classification
ATR-001	Patient — EXAMPLE, delete this row	Identifier	String	Yes	Restricted

5.5 — Name a person against every entity, or write down that there is none

Sheet `O20 Data Entity` column 6 · **Input** `O20.Owner` against `O09 Organization Unit`, `O10 Actor` and `O11 Role` · **Output** fourteen owners, each a named person or a recorded absence · **Signed by** `GB-03 Data Governance Council`, whose mandate on `O57` is to designate data owners and resolve disputes · **Time** one day, plus however long an answer takes

1. Look each of the nine distinct titles in `O20.Owner` up in `O10 Actor` and `O11 Role`. Eight resolve to nothing — every `Head of …` title on the sheet. Only `Data Steward` resolves, to `ACT-16` and `ROL-16`. Ten of the fourteen entities are owned by a title the repository does not define.
2. Note why, because it is structural and not fixable here. None of `O09`'s twelve organisation units is clinical — no medical directorate, no nursing, no pharmacy, no clinical governance unit, no records function — so the eight titles have no unit to sit in. Item 4 raises that; this step records the consequence for `O20`.
3. Replace each title with a named person, or record the absence explicitly. `PRN-13 Data Ownership is Individual` at `O05` row 17 says an owner is a person, not a committee, and `OBJ-10` on `O03` — "Every core entity has a named individual owner" — is `Not started`. Four rows read `Data Steward`, which is a role.
4. Where ownership is genuinely contested, do not pick. Write volume 1 chapter 10's four fields and a date on one page per entity.
5. Check the two misplacements before you leave. `E06 Medication Order` is owned by `Head of Estates` and `E13 Consent` by `Head of Revenue Cycle` — facilities owns the prescription record and billing owns consent.
6. Add the records function to the scope page as a stakeholder; volume 1 chapter 10 gives the published evidence that it is widely failing its statutory obligations.

Done when `O20.Owner` holds a person on every row it can and a dated absence on every row it cannot, and `ZZ Checks` row 17 is annotated: "vacant architecture roles: 0" is true of `R21`'s eighteen and silent on these eight titles.

5.6 — Put the patient's identifiers on `O37` as rows, not as one column

Sheet `O37 Data Attribute` · **Input** `ATR-001`, and the registration practice at each care setting · **Output** one `O37` row per identifier type the organisation holds, each graded · **Signed by** the Domain Architect, Data (`ACT-10`), with the records function · **Time** one day

1. Look at what the file has. `E01 Patient` has one attribute called `Identifier`, type `String`, while the extension sheet's own definition of the type is "An identified individual receiving care, **with their identifiers**", plural. There is no structure for a second identifier, one issued by another provider, a match rule or a duplicate.
2. Add one row per identifier type, not one column. Volume 1 chapter 11 gives the four layers; what belongs here is a row for each, with its own `Type`, `Key` and `Classification`, and the assigning authority and validity period recorded beside the sheet.
3. Include a provisional type for the unidentified patient. A trauma case is registered, treated, imaged and billed before anyone knows who they are, and a scheme with no provisional type forces the clerk to invent one.
4. Grade every identifier row at the entity grade or tighter, including the folder number. A local identifier is identifying within the organisation that issued it, which is the organisation holding the record.
5. Record the check digit as a control. It makes correctness partly testable, so implement it as a warning at capture and count the warnings — a free measure, defined in step 5.11.
6. Write down the measured reason, because a client will ask. In a Tshwane cohort four per cent of infants matched exactly on four demographic details, and the national linking algorithm found 4,7% of the true matches at 87,7% accuracy [6]. That ends the "we will match on demographics" conversation in one slide, with a South African source and a date.
7. Do not create a new numbered sheet. The six are fixed at `067` to `072`: identifier types are `037` rows, and the link table is a physical structure belonging to a component in `022`.

Done when `037` holds one row per identifier type, each graded at `Restricted` or tighter, a provisional type exists, and the check-digit warning count has an owner.

5.7 — Specify merge and unmerge as operations, and the audit record a merge needs

Sheet `043 Master Data Rule`, `044 Data Building Block` · **Input** the settled patient master from 5.2 and the identifier rows from 5.6 · **Output** a merge specification, an unmerge specification and an audit record definition, all referenced from `DBB-01 Patient mastering` · **Signed by** a named clinical risk owner, with `GB-03` · **Time** two days

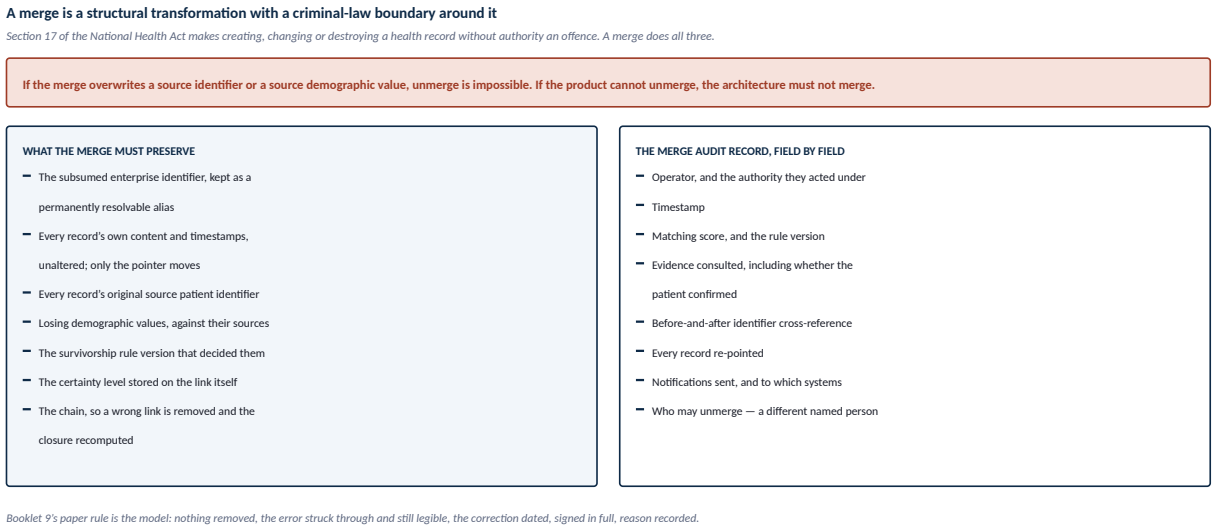


Figure 5.3 What a merge must preserve for an unmerge to remain possible, and the fields of the audit record that separate a lawful merge from the offence in section 17 of the National Health Act.

1. Change `MDR-01` on `043` from `Golden source` to `Match rule` and write the rule. The only `Match rule` rows the sheet holds are `MDR-04 Diagnosis` and `MDR-08 Care Plan`, neither of which needs one.
2. Write the thresholds into the rule and have them approved by a named clinical risk owner, not by a configuration consultant. A false merge puts two people's clinical data in one record; a false duplicate splits one person's across two. Volume 1 chapter 11 carries the asymmetry and why the first is the worse error.
3. Specify the link, not the merge: a link table with the certainty level stored on the link, chains revised by transitive closure, a blacklist of placeholder identifier values, historical links reused. That is what the Western Cape provincial data centre runs — mainly deterministic with an assigned certainty level, and a probabilistic scorer built and deliberately not in production [8].
4. Specify the merge. A surviving enterprise identifier; the subsumed one kept as a permanently resolvable alias; every record re-pointed without altering its content or timestamps; conflicts resolved by the survivorship rule with losing values retained against their sources; the merge recorded as an event and propagated.
5. Specify the unmerge as a precondition, not a later phase. It is strictly harder, and it requires that every record still carries its original source patient identifier — so **if the merge overwrites source identifiers, unmerge is impossible**. Prove it before go-live, on real linked data.
6. Define the audit record against the offence. Section 17 of the National Health Act makes it an offence to create, change or destroy a health record without authority, to fail to do so when properly required, and to connect a user's identifying details to their condition, treatment or

history without authority — a fine, up to a year's imprisonment, or both [1]. **A merge does all three, lawfully.** The only thing separating a lawful merge from an offence is the authority and the record of it.

7. So the audit record carries: the operator and the authority they acted under; the timestamp; the matching score and the rule version that produced it; the evidence consulted, including whether the patient confirmed; the before-and-after state of the identifier cross-reference; every record re-pointed; and the notifications sent. Booklet 9's paper rule is the model — nothing removed, an error struck through leaving the original legible, the correction dated, signed in full and its reason recorded [4].
8. Make the merge role and the unmerge role different people, and replace `DBB-01`'s acceptance criterion — "No patient record without a named owner" tests nothing about identity — with the unmerge proof.

Done when `MDR-01` is a match rule with approved thresholds, the audit record is defined field by field against section 17, unmerge has been demonstrated on real data, and two different named people hold the two roles.

Goes wrong when the product cannot unmerge and the project merges anyway. If the product cannot unmerge, the architecture must not merge.

5.8 — Rebuild `O36`, and make `R19` agree by correction rather than by copy

Sheet `O36 Data Entity Relationship`, `R19 Data Entity relates to Data` · **Input** both sheets side by side, and volume 1 chapter 10's spine · **Output** a relationship set a clinician would recognise, present identically on both sheets · **Signed by** the Domain Architect, Data (`ACT-10`) · **Time** one half-day workshop

> **`O36 Data Entity Relationship` and `R19 Data Entity relates to Data`**
>
> `O36` headers: `Ref`, `From entity`, `Relationship`, `To entity`, `Cardinality`. `R19` headers: `Ref`, `From`, `Relationship`, `To`, `Cardinality`. The column names differ; the eleven rows `REL-01` to `REL-11` are byte-for-byte identical. Correct both, or the next reader finds the old model on whichever one they open.

1. Read the eleven rows as a chain: `Patient` to `Encounter` to `Clinical Observation` to `Diagnosis`, and on down `O20`'s row order to `Care Setting` to `Claim`. Each entity appears as `From` exactly once. It is a daisy chain over the entity list, not a model.
2. Note which rows survive. The first three are defensible. Everything from `REL-04` on is the chain: `Procedure is derived from Medication Order`, `Clinical Protocol is composed of Practitioner`, `Practitioner is derived from Care Setting`, `Care Setting has Claim` at `1:1`.
3. Put `Patient` back at the hub: in a clinical model most relationships radiate from it, and here it has exactly one. Fix the cardinalities wrong in direction as well as in subject — `REL-04 Diagnosis has Procedure` is `N:1`, many diagnoses to one procedure.
4. Add the relationships `E13 Consent` and `E14 Medical Device` need, and the one `E02 Encounter` needs: `Encounter` has none to `E11 Care Setting`, although its own extension definition is "A contact between patient and service, with setting and outcome", and `Care Setting` is joined to `Claim` instead.
5. Correct `O36`, then type the same rows into `R19` under its own header names. Do not paste `O36`'s headers across.

Done when both sheets hold the same relationship set, `Patient` is the hub, and no entity appears as `From` exactly once by construction.

The three sheets this item creates

Six sheets are added across this volume and the numbering is fixed. Item 2 built `O69 Code System` and `O72 Retention Clock`; item 7 builds `O68 Medical Device`. Three are built here, and each is necessary because of something you can point at in the file.

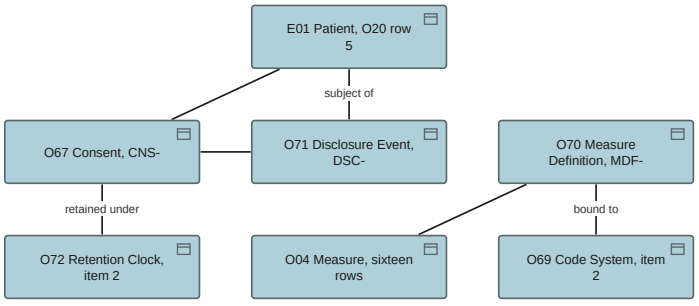


Figure 5.4 Where the three sheets this item creates attach, with the two item 2 created. A consent is retained under a retention class; a disclosure is a fact about a patient's record; a measure definition reads a code system edition and governs a published figure.

`O67 Consent`, because consent to treatment and consent to disclosure are different objects with different lifecycles and the file holds one attributeless row for both. `E13 Consent` appears on exactly three sheets: `O20` row 17, `R05` rows 17 and 31, and `RPT05`. No logical data component, no attribute, no mastering rule, no building block, no migration, no mapping, no relationship. Its definition is "Recorded permission for treatment and data sharing" — two objects in one row — and its lineage reads `Full`, the strongest value on the sheet, for an entity in no mapping and no data set. The only governance hook is `MTD-07 Consent basis` on `O42`, coverage `Partial`, which is metadata about consent rather than an instance of one.

O71 Disclosure Event, because the openness and participation rights in the privacy regime, and a withdrawal of consent, are not actionable without a record of what was disclosed to whom. Nothing in the file records a disclosure. SEC-04 Attributable access on O41 is Absent, SEC-05 Interface logging is Partial, and POL-03 — "Material activity is logged and observable" — is Breached. A log line is not the entity: a disclosure is a fact about the record, with a recipient, a purpose, an authority and a date, and it must outlive any log retention window.

O70 Measure Definition, because O04 Measure holds a unit and a baseline and nowhere to say what was counted. MSR-11 Lineage coverage of regulatory reports carries baseline 23 against a sheet where lineage is Full on six of fourteen entities (43%) and Full or Partial on nine (64%). Neither is 23. MSR-13 has unit count and baseline 43 for a measure with twelve possible subjects, and MSR-10 is 23% of fourteen entities, which is not a whole number of entities. Three irreproducible figures, and nowhere to write the rule down.

5.9 — Create O67 Consent

Sheet O67 Consent, new · **Input** the registration and admission paperwork in use at each care setting, and the research approvals on file · **Output** one row per consent instance, with E13 Consent on O20 re-stated as the parent of three types · **Signed by** the Chief Compliance Officer (ACT-07) and the clinical governance lead · **Time** three days

> **O67 Consent — create this sheet**
>
> Prefix CNS-. Columns, in this order: Ref, Consent type, Subject, Scope, Granted, Expires, Withdrawn, Evidence, System of record. Row 1 title, row 2 organisation and count, row 4 headers, data from row 5, as every other object sheet.

1. Set Consent type to one of three values and no others: Treatment, Disclosure, Research. They are different legal acts with different withdrawal effects, and volume 1 chapter 4 defines all three. Three values on one sheet with a disciplined Scope column is the smallest structure that works in a repository of this shape.
2. Make Scope structured, not prose. For a Disclosure row it is a set of triples — information classes, by recipients, by purposes — over a period. Free text makes enforcement impossible; triples make a proposed disclosure checkable against the consent in force on the date of the disclosure.
3. Fill Granted and Expires as dates and treat a missing Expires as a decision, not a blank. O61 on this file shows what an unfilled Expires column does: four of its five rows are blank. Make Withdrawn a date and an actor, not a flag — the organisation's defence to a complaint is the record that it stopped on a date.
4. Fill Evidence with what you actually hold. Section 14 of the National Health Act requires consent to third-party disclosure to be **in writing** [1], so a verbal consent recorded in a note does not satisfy it, and Evidence must name the document, its location and its retrievability.
5. Record the capacity assessment in Evidence where the subject is a child. Under section 129 of the Children's Act a child over 12 with sufficient maturity and the mental capacity to understand the benefits, risks and implications may consent to their own treatment [9] — so capacity is a function of age and of an assessed maturity, and the assessment is itself a clinical act that must be recorded.
6. Set System of record from step 5.2's decision, not from the file. O20 masters E13 Consent in A11 Integration Engine, which O23 describes as "HL7 and FHIR message routing" — a message router holding the consent master.
7. Write the prohibition on the sheet's cover note: **consent is not the access-control mechanism.** The lawful basis for clinical processing at a provider is the statutory authorisation under section 32 [2]; consent governs disclosure outside the care relationship, and research. A system gating clinical access on a consent flag will block emergency care; one with no consent object cannot govern disclosure or research. Both failures are common and opposite.
8. Re-point O20 row 17, add O21 row LDC-13, and reference O67 from the O44 consent building block, which does not exist today.

Done when O67 holds every consent instance for the in-scope care settings, every Disclosure row's Scope is a set of triples, and no row has a blank Expires without a written reason.

Form 5-C — `067 Consent`, identify

`067 Consent` does not exist in the repository. This form creates it. Five of the nine fixed `067` columns, in sheet order; Form 5-D carries the lifecycle four. One row per consent type per care setting — at Rietvlei District Health Service, about twenty rows. Signed by the Chief Compliance Officer.

Ref	Consent type	Subject	Scope	System of record
CNS-001	Disclosure — EXAMPLE, delete this row	Patient, E01	Discharge summary / referring practitioner / continuity of care / 12 months	

Form 5-D — `067 Consent`, lifecycle

This form feeds `067 Consent`, which the supplied repository does not have. Photocopied on its own it still needs that warning.

The remaining four `067` columns, in sheet order, keyed on Form 5-C's `Ref`. `Withdrawn` is a date and an actor, never a tick. Signed by the Chief Compliance Officer; `Evidence` countersigned by the records function.

Ref	Granted	Expires	Withdrawn	Evidence
CNS-001	2026-03-14 — EXAMPLE, delete this row	2027-03-13		Signed form, registry file, retrieval measured at 4 days

5.10 — Create `071 Disclosure Event`

Sheet `071 Disclosure Event`, new · **Input** the subject access requests, funder queries, court orders, statutory notifications and research extracts of the last twelve months · **Output** one row per disclosure, and a standing capture point for the next one · **Signed by** the records function, with the Chief Compliance Officer · **Time** three days to establish, then continuous

> **`071 Disclosure Event` — create this sheet**
>
> Prefix `DSC-`. Columns, in this order: `Ref`, `Record subject`, `Disclosed to`, `Purpose`, `Authority`, `Date`, `Consent reference`, `Logged in`.

1. Start with the twelve months you can reconstruct and stop there. A disclosure log cannot be built backwards over years, which is why this sheet has to exist before the next disclosure rather than after the next complaint.
2. Make `Disclosed to` a named recipient, not a category. "The funder" is not a recipient; a named administrator on a date is. Make `Authority` one of the four the regime allows and name the provision: written consent under section 14, a court order, a law that compels it, or a serious threat to public health [1]. Internal care-necessary sharing by a duty-bound health worker is not a disclosure for this sheet — it is processing, and section 15 permits it [1].
3. Point `Consent reference` at a `CNS-` row from `067`. A `Disclosure` row on `067` with no `DSC-` rows under it is a permission nobody used; a `DSC-` row with no `CNS-` reference and no other `Authority` is the finding.

4. Fill `Logged in` with the system holding the evidence, and test the retrieval. The 2026 health-information regulations require records to be kept secure and confidential and disposed of so that they cannot afterwards be accessed or disclosed [3] — a destruction requirement with an evidentiary standard, and the same standard applies to proving a disclosure happened.
5. Use the sheet for the three things it is for: telling a patient who received their data, acting selectively on a withdrawal, and proving disclosure stopped on a date. Withdrawal stops future disclosure and cannot unsend what was sent, so the only honest answer to a patient is the list.
6. Add a research block. Where research does not reveal the patient's identity the three authorisations under section 16 are not required [1], so a de-identified extract is a materially different legal object from a pseudonymised one. Record which you released, under which ethics reference, with an expiry.

Done when twelve months of disclosures are on the sheet, every row names a recipient and a provision, and every withdrawal on `O67` has been tested against the sheet for disclosures made after it.

Form 5-E — `O71 Disclosure Event`, the disclosure

`O71 Disclosure Event` does not exist in the repository. This form creates it. Five of the eight fixed `O71` columns, in sheet order. One row per disclosure; a first pass covers twelve months. Signed by the records function. Form 5-F carries the authority.

Ref	Record subject	Disclosed to	Purpose	Date
DSC-001	Patient, E01 — EXAMPLE, delete this row	Named scheme administrator	Funder claim substantiation	2026-04-02

Form 5-F — `O71 Disclosure Event`, the authority

This form feeds `O71 Disclosure Event`, which the supplied repository does not have. Photocopied on its own it still needs that warning.

The remaining three `O71` columns plus `Ref`, in sheet order. A row with no `Authority` is a finding, not a blank. Signed by the Chief Compliance Officer.

Ref	Authority	Consent reference	Logged in
DSC-001	Written consent, NHA s14 — EXAMPLE, delete this row	CNS-001	Revenue Cycle / Claims

5.11 — Create `O70 Measure Definition`

Sheet `O70 Measure Definition`, new · **Input** `O04 Measure`'s sixteen rows, and every figure the organisation published in the last year · **Output** one row per published figure, and a rule that no figure leaves the building without one · **Signed by** the measure `Owner`, with `GB-03` · **Time** two days for the sixteen, then one hour per new measure

> **`O70 Measure Definition` — create this sheet**
>
> Prefix `MDF-`. Columns, in this order: `Ref`, `Measure`, `Numerator`, `Denominator`, `Exclusions`, `Boundary rule`, `Period`, `Definition version`, `Owner`.

1. Do the sixteen rows of `O04` first, because three of them cannot be reproduced from anything on the file, and that is the argument for the sheet.
2. Write `Numerator` and `Denominator` as counts of rows on named sheets, with the filter. "Entities with a named owner" is a count of `O20` rows whose `Owner` resolves to an `O10 Actor`, over fourteen — 4 of 14, or 29%, not the 23 on `MSR-10`. Make `Exclusions` explicit and never empty: an empty exclusion list is an assertion that there are none, and it must be made deliberately.
3. Fill `Boundary rule` with a named reference to one of volume 1 chapter 4's constructs. This is the column the sheet exists for. The published evidence is that the four theatre-utilisation calculation methods have no simple mathematical relationship, with raw utilisation of 80% against touch-time utilisation of 60% on the same list over fifteen weeks in one specialty [7]. There is no conversion between them, and the same holds for occupancy, length of stay, readmission and mortality.
4. Version every row, and make the version effective-dated. A measure changes when a code system edition changes, so bind each `MDF-` row to the `O69 Code System` rows its numerator reads. A case-mix figure depends on a grouper version and a readmission rate on a planned-readmission rule set; neither is reproducible without the version.
5. Settle the three figures the file cannot explain, recording both readings where both are defensible rather than averaging. Then make the rule a gate condition: no figure goes into a board pack without an `MDF-` reference beside it. That is a one-line change to the Phase G pack, and the only control on this sheet that will survive you.

Done when all sixteen `O04` rows have an `MDF-` row, every `MDF-` row names a boundary rule and a version, and the three irreproducible baselines have a written reading or a written correction.

Form 5-G — `O70 Measure Definition`, the rule

`O70 Measure Definition` does not exist in the repository. This form creates it. `Ref` plus the four rule columns, in sheet order; Form 5-H carries the identity four. Sixteen rows, one per `O04` measure. Signed by the measure `Owner`.

Ref	Numerator	Denominator	Exclusions	Boundary rule
MDF-10	O20 rows whose Owner resolves to an O10 Actor — EXAMPLE, delete this row	14 O20 rows	None	Not applicable, entity count

Form 5-H — `O70 Measure Definition`, the identity

This form feeds `O70 Measure Definition`, which the supplied repository does not have. Photocopied on its own it still needs that warning.

`Ref` plus the four identity columns, in sheet order, keyed on Form 5-G's `Ref`. `Owner` is a person. `Version` increments whenever any column on 5-5A changes, and the old version is kept. Signed by `GB-03`.

Ref	Measure	Period	Version	Owner
MDF-10	Entities with a named owner — EXAMPLE, delete this row	Quarterly	1.0, from 2026-11-01	

5.12 — Record the clock, and stop `O42` contradicting the sheets beneath it

Sheet `O42 Metadata Element` · **Input** `O42`'s fourteen rows against `O20`, `O37` and the `O72 Retention Clock` rows from item 2 · **Output** a metadata catalogue whose coverage values are true, and a retention class against every entity · **Signed by** the Chief Data Officer (`ACT-05`) · **Time** one day

1. Correct the three rows that contradict other sheets. `MTD-02 Business owner` is `Absent` while `O20` names an owner on all fourteen rows; `MTD-10 Primary key` is `Recorded, not held` while `O37` flags twenty keys; `MTD-08 Lineage` is `Full` while `O20` records five `Absent` and three `Partial`.
2. Fill `MTD-06 Retention rule`, coverage `Absent`, by pointing every `O20` row at an `RTN-` row on `O72 Retention Clock`. Phase C records the clock: Preliminary set it, Phase D measures the retrieval, Phase G gates on it.
3. Carry the instrument problem rather than hiding it. There is no South African retention instrument: every period in circulation traces to Booklet 9, whose 2016 edition says at least six years from the date the record became dormant [4] and whose September 2022 review draft says ideally indefinitely, particularly where records are electronic [5]. Section 13 of the National Health Act makes record-keeping subject to the National Archives Act 43 of 1996 and to the access-to-information Act [1] — which is where the destruction authority sits.
4. Fill `MTD-07 Consent basis`, coverage `Partial`, from `O67`. It was the only consent hook the file had, and it now has an instance sheet behind it. Set `MTD-05 Sensitivity` coverage from the state of `O37` after step 5.4, not before.

Done when no `O42` coverage value contradicts the sheet it describes, and every `O20` row names a retention class, an instrument and a destruction authority.

5.13 — Re-derive `O39` and `O40`, and replace the four data sets that are not healthcare

Sheet `O39 Data Migration`, `O40 Source To Target Mapping`, `O38 Data Set` · **Input** the settled master systems from 5.2 and item 6's interface register · **Output** movements somebody has planned, and data sets that exist · **Signed by** the Head of Delivery (`ACT-13`) and the Chief Data Officer (`ACT-05`) · **Time** one day

1. Check the arithmetic before the content. On `O39`, `From system` is the application two positions after the entity's own index and `To system` four after, for all six rows. On `O40`, `Source field` and `Target field` step one application forward per row and wrap at `MAP-11` and `MAP-12`. Both are joins by row position.
2. Read the consequences aloud. `MIG-01` migrates `Patient` from `PAS` to `RIS / PACS` — the patient master into the imaging system. `MAP-12` maps `Claim` into `PAS`, the application `O23` is retiring.
3. Clear both sheets and re-derive from the decisions in 5.2 and from item 6's register. Where no migration is planned, leave the sheet short: six fabricated rows are worse than two real ones. Give `O40` real field names — `PAS field` is not a field name, and twelve rows of it cannot be used for anything.
4. Read `O38`'s six rows. `DST-02 Customer analytics set`, `DST-04 Finance consolidation set` and `DST-05 Risk measurement set` — "Exposure and position data", a trading-book phrase. No clinical set, no patient set, no safety set.
5. Replace them with what the organisation actually submits or extracts: the statutory aggregate return, the funder submission, the research extract with its ethics reference and expiry, and the safety and incident set. Bind every set to the `MDF-` rows that read it and the `CNS-` rows that permit it.

5.14 — Re-cut `O44`, and raise the gaps Phase C actually found

Sheet `O44 Data Building Block`, `O29 Gap`, `RPT05 Data Ownership` · **Input** everything above · **Output** ten to fourteen building blocks with testable acceptance criteria, and a gap list that matches what you found · **Signed by** `GB-01 Architecture Board` at `GT-1 G1 Design` · **Time** one day

1. Read `O44`'s ten acceptance criteria. All ten read "No <entity> record without a named owner", which tests the `Owner` column and nothing the block does. Replace each with a criterion a tester can fail: for `DBB-01 Patient mastering`, the demonstrated unmerge from step 5.7.
2. Extend `O44` to cover `E11` to `E14`. It stops at `DBB-10 Practitioner`, so consent, the device register, care setting and the funder claim have no building block.
3. Raise the gaps. `O29 Gap` holds two records, both business-domain capability uplifts, on a file reporting eight data entities without full lineage, twenty-six attribute classification breaches, ten entities owned by an undefined title, four entities with no attributes, three sheets disagreeing on the system of record, and four metadata elements recorded `Absent`. Phase C alone should take `O29` into double figures.
4. Check what `RPT05 Data Ownership` prints: `O20` less `Definition`, with `Class` moved, and all fourteen rows whatever your scope was — so 5.1's scope page is the only place the boundary exists. Its footnote, "5 of 14 entities have no lineage from report back to source", is true on the test it states and silent on the three `Partial` rows. Give both readings: five `Absent`, eight without `Full`. Then take `DLV-07` and `DLV-08` back through `GT-1`.

PHASE C, THE DATA HALF — THE FIVE QUESTIONS A REVIEWER WILL ASK

Which entity's system of record was hardest to settle, and what did GB-03 decide, on what date?

Which instrument did you cite when you re-graded the funder claim, and which did you refuse to cite?

How many of the twenty-six attribute breaches closed by tightening, and how many by re-grading the entity?

Which O67 rows have no expiry, and what is the written reason for each?

Which three published figures still have no MDF- row, and who owns each?

What leaves the room

Two deliverables and a list. `DLV-07 Data Architecture` carries the fourteen settled entities with a grade, an owner, a master system and a retention class each, plus the three new sheets. `DLV-08 Data Model` carries the rebuilt `O36`, the identifier rows, and the merge and unmerge specification with its audit record. The list is the gap list, and a client reads it first.

Say one thing out loud at sign-off. Every cell corrected here resolved perfectly before it was corrected: every `Ref` matched its name, every lookup returned a row, and `RPT05` printed cleanly over all of it. A repository can pass every integrity check it owns and still tell a patient's lawyer that the diagnosis attributed to them was classified `Public` by the organisation's own data architecture. Integrity is not meaning, and Phase C is where the difference gets decided.

THE CHAPTER IN TEN LINES

- Phase C has two items. This is the data half, and it is the half that carries the legal risk.
- Four Phase C deliverables arrive Approved before any work is done. Set them to Draft first.
- Classification inherits downward. An attribute may be tighter than its entity, never looser.
- O37 breaks that on 26 of 60 rows, including the patient identifier, Public under a Restricted entity.
- Encounter and Procedure show no breach only because both entities are graded Public, the floor.
- Three sheets name a system of record. O20 and R05 are the same error copied; O43 gives a third answer.
- Not one of the fourteen entities is mastered in a system that could own it. Settle each in a room.
- Ten of fourteen entities are owned by a job title the repository does not define anywhere.
- Identifiers are rows on O37, not one column. Do not invent a seventh new sheet for them.
- A merge creates, changes and connects records. Section 17 makes doing that without authority an offence.
- If the merge overwrites a source identifier, unmerge is impossible. Prove unmerge before go-live.
- O67 Consent, O70 Measure Definition and O71 Disclosure Event are created here, with fixed columns.
- Integrity is not meaning. Every cell this item corrects resolved perfectly before it was corrected.

SOURCES FOR THIS CHAPTER

- **1.** National Health Act 61 of 2003, consolidated text; sections 13 to 17 https://www.saflii.org/za/legis/consol_act/nha2003147.pdf
- **2.** Protection of Personal Information Act, section 32, authorisation concerning a data subject's health or sex life https://popia.co.za/section-32-authorisation-concerning-data-subjects-health-or-sex-life/
- **3.** Moonstone, New POPIA regulations on health information now in force, reporting Government Gazette No. 54268 of 6 March 2026 https://www.moonstone.co.za/new-popia-regulations-on-health-information-now-in-force/
- **4.** HPCSA Booklet 9, Guidelines on the Keeping of Patient Records, September 2016 edition https://www.hpcsa-blogs.co.za/wp-content/uploads/2022/08/Booklet-9-Guidelines-on-Patient-Records.pdf
- **5.** HPCSA Booklet 9, Keeping of Patient Records, review draft marked Revised September 2022 https://www.hpcsa-blogs.co.za/wp-content/uploads/2022/11/Booklet-9-Keeping-of-Patient-Records_Review-Draft_-FINAL_Sept-2022.pdf
- **6.** Haeri Mazanderani, Technau, Hsiao and others, on infant identification in the prevention-of-mother-to-child-transmission programme, South African Medical Journal 2018;108(9):729-733 https://scielo.org.za/scielo.php?script=sci_arttext&pid=S0256-95742018001000017
- **7.** Zhang and Pandit, British Journal of Anaesthesia 2023;131(1):130-134, theatre utilisation calculation methods https://www.sciencedirect.com/science/article/pii/S0007091223001794
- **8.** Coetzee and others, data resource profile of the Provincial Health Data Centre of the Western Cape, International Journal of Population Data Science https://ijpds.org/article/view/1143
- **9.** Children's Act 38 of 2005, section 129, consent to medical treatment and surgical operation https://www.acts.co.za/childrens/129_consent_to_medical_treatment_and_surgical_operation.php

ITEM 6

Application Architecture

One relationship sheet with real content that contradicts itself, and four that resolve perfectly and describe nothing.

The application half of Phase C is where a client expects to argue about vendors and discovers that the argument it can actually afford is about seams. **Ravensmere Health Network** has twelve applications, twelve exposed services, twenty-three interfaces and six relationship sheets joining them to everything else. One of those sheets has real content and contradicts itself; four were built by pairing two lists in row order and cannot be repaired at all, only re-derived; and the sixth is half keyed by reference and half by name.

Twelve steps. None of them creates a sheet. All of them replace a cell that already resolves.

> **READ THESE THREE BEFORE YOU OPEN A SHEET**
>
> - **Volume 1, chapter 14.** The conceptual application model: the three questions per function, the seam as a first-class object with an owner and a version, and what survives a change of vendor. Step 6.4 builds exactly the inventory that chapter says an organisation should be able to produce by its sixth week.
> - **Volume 1, chapter 15.** The real estate, system by system, the four tiers, and where each departmental system meets the record. Use it to sanity-check whether an application in `O23` could plausibly do what a relationship sheet says it does.
> - **Volume 1, chapter 16.** What governs each seam, layer by layer, and the last column of its seam map — the one where not a single entry is a standard. Steps 6.4 and 6.6 depend on that reading.

What this item fills

Three object sheets, five relationship sheets and two reports.

Sheet	Rows	What this item does to it
`O23 Application Component`	12	Corrects 5 dispositions and 12 owners
`O24 Information System Service`	12	Re-styled, re-described, and honestly marked
`O45 Application Building Block`	10	Re-cut; acceptance criteria made testable
`R06 Application interfaces with`	23	The register. Rebuilt test by test, not row by row
`R04` , `R07` , `R20` , `R25`	10, 12, 12, 5	Cleared and re-derived from evidence
`R05 Application manages Data En`	28	Its half-keyed first column fixed
`RPT02` , `RPT04`	12, 23	Read-only. Both print what the object sheets hold

`DLV-09 Application Architecture` and `DLV-10 Interface Catalogue` on `O31 Deliverable` both read `Approved` . Item 5 set them to `Draft` in step 5.1; if you are working this item first, do it here.

The one sheet with real content, and the one thing to do with it

`R06 Application interfaces with` is the only relationship sheet in this repository that carries information rather than arithmetic. Twenty-three rows, each with a source, a target, a transport type, a frequency, and two governance flags. Every `Ref` resolves. Every name matches its `Ref` . `RPT04 Interface Catalogue` prints the same twenty-three rows with the names resolved, and its footnote — "13 of 23 interfaces are unmanaged and 10 carry no logging" — is arithmetically correct.

And the rows contradict themselves, because `Type` , `Managed` and `Logged` are three independent columns describing one fact. So the Phase C application step is not "check the interfaces resolve". It is: **declare the test, apply it to all twenty-three rows, and publish the count with the test beside it.**

R06 Application interfaces with Application: the four tests, row by row

A direct store read marked Managed · B Managed and unlogged · C Unmanaged and logged · D transport type against its frequency

REF	SOURCE	TARGET	TYPE	FREQ	MANAGED	LOGGED	A	B	C	D	FAILS
INT-001	EHR / EPR	PAS	File	Hourly	Managed	No		■			1
INT-002	EHR / EPR	Revenue Cycle	Event	Nightly	Managed	Yes					—
INT-003	LIS	EHR / EPR	API	Nightly	Unmanaged	Yes			■		1
INT-004	LIS	PAS	Event	Hourly	Unmanaged	No					—
INT-005	LIS	Data Warehouse	Event	Daily	Unmanaged	Yes			■		1
INT-006	RIS / PACS	EHR / EPR	File	Nightly	Unmanaged	No					—
INT-007	RIS / PACS	Revenue Cycle	File	Daily	Managed	No		■			1
INT-008	ePrescribing	Rostering	File	Hourly	Unmanaged	Yes			■		1
INT-009	ePrescribing	Patient Portal	Batch	Daily	Managed	Yes					—
INT-010	Revenue Cycle	EHR / EPR	Direct DB read	Daily	Managed	No	■	■			2
INT-011	Revenue Cycle	PAS	API	Daily	Unmanaged	Yes			■		1
INT-012	Rostering	EHR / EPR	File	Nightly	Managed	No		■			1
INT-013	Rostering	Patient Portal	API	Daily	Managed	Yes					—
INT-014	Rostering	Integration Engine	Direct DB read	Hourly	Unmanaged	No					—
INT-015	Incident & Risk	PAS	Batch	Real time	Managed	Yes				■	1
INT-016	Incident & Risk	RIS / PACS	Batch	Daily	Unmanaged	No					—
INT-017	Incident & Risk	Revenue Cycle	API	Hourly	Unmanaged	No					—
INT-018	Patient Portal	PAS	Event	Real time	Unmanaged	Yes			■		1
INT-019	Patient Portal	LIS	API	Daily	Unmanaged	Yes			■		1
INT-020	Integration Engine	LIS	File	Real time	Managed	No		■		■	2
INT-021	Integration Engine	Theatre Mgmt	Batch	Hourly	Unmanaged	Yes			■		1
INT-022	Data Warehouse	LIS	Direct DB read	Hourly	Managed	Yes	■				1
INT-023	Data Warehouse	Integration Engine	File	Daily	Unmanaged	Yes			■		1

15 of 23 rows fail at least one test. INT-010 and INT-020 fail two each. Eight rows survive all four: INT-002, 004, 006, 009, 013, 014, 016, 017.

Add a fifth test — Event carrying a scheduled frequency — and INT-002 and INT-004 join the list: seventeen fail, six survive. Declare the test.

Figure 6.1 All twenty-three rows of R06 against four declared tests. Fifteen rows fail at least one; INT-010 and INT-020 fail two each; eight survive all four. A fifth test on event interfaces carrying a scheduled frequency takes the count to seventeen. The number is a property of the test, which is why the test goes at the head of the register.

Four tests, each one defensible, each one a different number.

Test	What it catches	Rows
`Type` = `Direct DB read` and `Managed` = `Managed`	A store read cannot be routed integration	2 of 3
`Managed` = `Managed` and `Logged` = `No`	Managed and unobservable	5
`Managed` = `Unmanaged` and `Logged` = `Yes`	Point-to-point links producing correlated logs	8
`Type` incompatible with `Frequency`	`Batch` at real time, `File` at real time	2

The union of those four is **fifteen rows**, because `INT-010` fails two of them and `INT-020` fails two. Eight rows survive all four: `INT-002`, `004`, `006`, `009`, `013`, `014`, `016` and `017`. Add a fifth test — `Type` = `Event` carrying a scheduled `Frequency`, which an event-driven interface by definition does not — and `INT-002` and `INT-004` join the list, taking it to seventeen and leaving six survivors.

> **REPORT THE TEST, NOT THE NUMBER**
>
> Fifteen, seventeen, or sixteen if you run a different fifth test. All three are defensible and the spread is the finding. A reviewer who is given a number without the test cannot check it, and the next architect will produce a different one from the same sheet. `ZZ Checks` rows 10, 11 and 12 each give one number against a threshold of zero and name no test at all. That is the pattern to break here, and item 10 gates on it.

One figure in `R06` is reportable and nothing in the file reports it: **five rows are both `Unmanaged` and `Logged` = `No`** — `INT-004`, `006`, `014`, `016` and `017`. Those are the interfaces nobody governs and nobody can reconstruct. `MSR-08 Managed interfaces` on `004 Measure` carries baseline 25%, and ten of twenty-three is 43%. No row in the file reconciles the two, which is what `070 Measure Definition` from item 5 exists for.

6.1 — Freeze the portfolio, and separate what `023` records from what it asserts

Sheet `023 Application Component`, `RPT02 Application Portfolio` · **Input** `023`'s twelve rows, and one named person per application who can confirm them · **Output** a portfolio page where every column is either confirmed or marked unconfirmed · **Signed by** the Domain Architect, Application (`ACT-11`) · **Time** one day

> **`023 Application Component`**
>
> `Ref`, `Application`, `Type`, `Description`, `Disposition`, `Criticality`, `Health`, `In service`, `Users`, `Owner`. Ten columns. `RPT02 Application Portfolio` is the same sheet less `Description`, so correcting `RPT02` does nothing; correct `023`.

1. Mark each column confirmed or unconfirmed. `In service` and `Users` are countable and should be confirmed from the systems themselves. `Disposition`, `Criticality` and `Health` are judgements and currently have no author and no date.

2. Record what the estate says about itself before you change anything: `A10 Patient Portal` carries 303 users, the most of any application, and disposition `Retire`; `A12 Clinical Data Warehouse` carries 58, the fewest, and `Consolidate`; `A01 EHR / EPR` has been in service since 2013 with 145 users and is `Medium` criticality.
3. Write the vintage line. Four of twelve applications entered service in 2013 or earlier, and `A10` in 2010. Volume 1 chapter 15 carries the provincial and private estates' real vintages; use them to say whether this is unusual.
4. Note that `O23.Type` holds only `Package`, `Custom` and `Platform` — nine, one and two. There is nowhere to record a shared national service, a bureau arrangement or a departmental system bought outside the technology budget, and volume 1 chapter 15's four tiers need all three.
5. Do not correct `RPT02`. It is a report, and every value on it comes from `O23`.

Done when every `O23` cell is confirmed or marked unconfirmed with a date, and the three judgement columns name the person who made the judgement.

6.2 — Fix the five dispositions that contradict criticality, health or mastering

Sheet `O23 Application Component` · **Input** `O23` columns 5, 6 and 7 read across, against `O20.Master system` as item 5 settled it · **Output** a disposition per application with a reason and a date · **Signed by** `GB-02 Investment Committee`, because a disposition is a funding statement · **Time** one workshop, half a day

1. Read the three columns across for all twelve rows. `A09 Incident & Risk` is `Retain` at `Critical` criticality in `Poor` health — and it is the system holding "Safety incidents, investigation and action", against `DRV-07 Incident reporting duties` whose control objective `CO-07` is `In breach`. `A04 RIS / PACS` is `Retain` in `Poor` health, in service since 2013.
2. Settle the criticality question that drives everything else. `A01 EHR / EPR` — "the clinical record of truth" — is `Medium`, while `A07 Revenue Cycle / Claims` and `A09` are both `Critical`. The clinical record is recorded as less critical than the billing system. Volume 1 chapter 1 is why that ranking is wrong in a provider, and item 7 carries what it does to recovery objectives.
3. Check every disposition against what the application masters, now that item 5 has settled it. Nine of fourteen data entities are mastered in an application whose disposition is `Retire`, `Replace` or `Consolidate`. `A02 PAS` is `Retire` and masters `E04 Diagnosis` and `E06 Medication Order`; `A10 Patient Portal` is `Retire` and masters `E09 Clinical Protocol` and `E11 Care Setting`.
4. For each of those nine, write the destination on the row. A disposition of `Retire` with no recorded destination for the data it masters is the single most expensive omission in this sheet, because it surfaces in Phase F as a migration nobody scoped.
5. Note the one that cannot be read at all: `MAP-12` on `O40` maps `E12 Claim` **into** `PAS`, the application being retired. Item 5 clears `O40`; this step records why.

Done when no disposition contradicts its own criticality and health without a written reason, and every retiring application that masters an entity names a destination.

6.3 — Give each application an owner who could actually run it

Sheet `O23 Application Component` column 10 · **Input** `O23.Owner` against `O09 Organization Unit`, `O10 Actor` and `O11 Role` · **Output** twelve named owners, or twelve recorded absences · **Signed by** `GB-04 Executive Committee`, as the escalation point when bodies cannot agree · **Time** one day

1. Read the twelve owners. Every value is an `O09` organisation unit, cycled through the unit list rather than assigned: `A04 RIS / PACS` is owned by `Human Resources`; `A05 ePrescribing & Medicines Admin` by `Customer Service`; `A06 Theatre Management` by `Internal Audit`, which `O09` types as `Assurance` — the auditor owns what it audits; `A09 Incident & Risk` by `Data & Analytics` rather than `Risk & Compliance`.
2. Note that not one clinical application is owned by a clinical function, and that this is not fixable inside this sheet: `O09` holds no clinical unit to own one. Item 4 raises it; this step records the consequence.
3. Replace the unit with a named person. `PRN-13 Data Ownership is Individual` at `O05` row 17 — "An owner is a person, not a committee" — is a data principle by category and an estate principle in practice. All twelve current values breach it.
4. Separate the two owners every application actually has, and record both: the business owner who decides what it does, and the service owner who keeps it running. `O23` has one column, so put the second in the portfolio page and say so.

Done when `O23.Owner` names a person on every row, and the applications whose owner is contested carry a dated absence rather than a unit name.

Form 6-A — The portfolio re-grade

Six of the ten real `O23 Application Component` columns, in sheet order. `Type`, `Description`, `In service` and `Users` are confirmed from the systems and typed straight into Excel. Twelve rows, one per application. Signed by `GB-02 Investment Committee`; `Owner` countersigned by `GB-04`.

Ref	Application	Disposition	Criticality	Health	Owner
A09	Incident & Risk — EXAMPLE, delete this row	Invest	Critical	Poor	

6.4 — Build the interface register from `R06`, test by test

Sheet `R06 Application interfaces with`, `RPT04 Interface Catalogue` · **Input** all twenty-three rows, plus one named engineer per interface · **Output** a register with a declared test, a count against that test, and a named owner and contract version per row · **Signed by** the Domain Architect, Application (`ACT-11`), with the Head of Service Management (`ACT-15`) · **Time** three days

> **`R06 Application interfaces with`**
>
> `Ref`, `Source`, `Source name`, `Target`, `Target name`, `Type`, `Frequency`, `Managed`, `Logged`. `Type` takes `File`, `API`, `Event`, `Batch` and `Direct DB read`. `Frequency` takes `Real time`, `Hourly`, `Daily` and `Nightly`. `Managed` takes `Managed` and `Unmanaged`; `Logged` takes `Yes` and `No`. The tab name is cut at thirty-one characters, so match on the truncated string.

1. Write the test down before you count anything. Use the four above, or four of your own, and put them at the head of the register. A count without a declared test is the defect this step exists to remove.
2. Apply them. Fifteen rows fail at least one; `INT-010` and `INT-020` fail two each. Eight survive. Mark each failing row with which test it failed, not with a generic flag.
3. Do the three `Direct DB read` rows in front of the room. `INT-010 Revenue Cycle / Claims` reads `EHR / EPR` daily and is marked `Managed` with `Logged = No`; `INT-022 Clinical Data Warehouse` reads `LIS` hourly and is `Managed` with `Logged = Yes`; only `INT-014 Rostering` to `Integration Engine` is honestly `Unmanaged`. `RPT01` row 12 glosses unmanaged as "Not routed through managed integration", which is the definition that makes the first two incoherent.
4. Add the columns the register needs and `R06` does not have, in the register rather than the sheet: the owner of the interface, the contract document and its version, the date the contract was last confirmed, and what breaks when the interface stops. Volume 1 chapter 16's seam map ends in a column where not one entry is a standard, and that column is the contract.
5. Add the closed loops as register entries, not as assumptions. The three that must be designed at a seam are results acknowledgement, referral acknowledgement and transfer acceptance. The published control set for the first is specific: track every order through collected, completed, reported and acknowledged; identify the ordering clinician and whoever is responsible for follow-up; monitor time to acknowledgement; flag abnormal results distinctly and critical values differently again; escalate automatically when nobody responds [2].
6. Record where the loop is structurally unavailable rather than merely absent. Where laboratory results reach clinicians through a web results viewer rather than an interface into a clinical system [4], acknowledgement tracking and escalation cannot be implemented at all, and the risk has to be carried somewhere else and named.
7. Publish the five rows that are both `Unmanaged` and `Logged = No` as a named finding: `INT-004`, `006`, `014`, `016` and `017`. No sheet in this repository reports that figure, and it is the one a security reviewer asks for first. Section 17 of the National Health Act requires controls against unauthorised access to a health record and makes the specified failures offences [3], so an interface that carries patient data and records nothing is a control gap with a statutory edge rather than a housekeeping item.
8. Do not edit `RPT04`. It is `R06` with names resolved; its footnote is correct and will stay correct.

Done when the register has a declared test at its head, every row names an owner and a contract version, the five unmanaged-and-unlogged rows are a finding with a date, and `MSR-08`'s baseline has either been reconciled or been given an `MDF-` row on `O70 Measure Definition`.

Goes wrong when the team repairs the `Managed` column so the count improves. Nothing changed in the estate, and the next reviewer counts the same rows and gets a different answer.

Form 6-B — The interface register, one row per interface

Six of the nine real `R06 Application interfaces with` columns, in sheet order. `Source name`, `Target name` and `Frequency` resolve from the refs and are typed straight into Excel. Twenty-three rows; write the test you declared at the head of the sheet before the first row. Signed by the Domain Architect, Application.

Ref	Source	Target	Type	Managed	Logged
INT-010	A07 — EXAMPLE, delete this row	A01	Direct DB read	Unmanaged	No

6.5 — Route what must be routed, and count what bypasses the engine

Sheet `R06 Application interfaces with`, `O23 Application Component` · **Input** the register from 6.4 · **Output** a routing decision per interface, and a named owner for the nineteen that bypass the integration engine · **Signed by** `GB-01 Architecture Board` at `GT-1 G1 Design` · **Time** two days

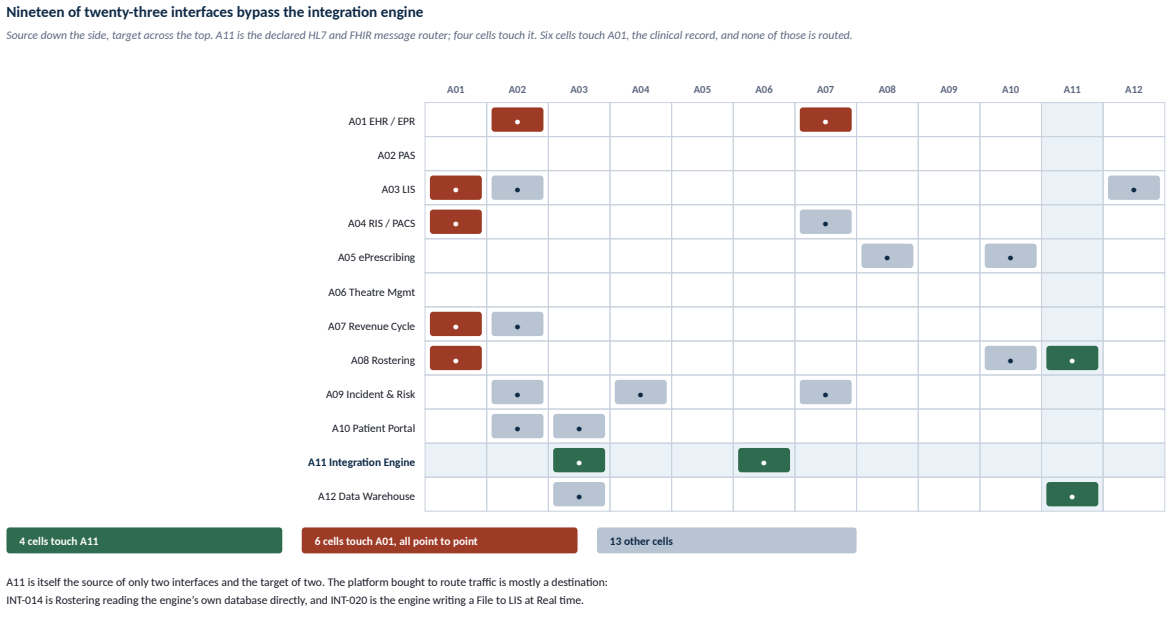


Figure 6.2 The twenty-three interfaces as a source-by-target matrix. Four cells touch A11 Integration Engine, the declared message router; six touch A01, the clinical record, and not one of those is routed. Nineteen interfaces bypass the engine the estate bought to route them.

1. Count what touches `A11 Integration Engine`, which `O23` describes as "HL7 and FHIR message routing" with disposition `Consolidate`. Four interfaces of twenty-three: `INT-014`, `INT-020`, `INT-021` and `INT-023`. Nineteen bypass it.
2. Say which nineteen, and start with the six that matter most: every interface into or out of `A01 EHR / EPR` — `INT-001`, `002`, `003`, `006`, `010` and `012` — goes point to point. The clinical record of truth has no managed route in or out.
3. Note which direction the four that do touch the engine run. `INT-014` is `Rostering` reading the engine's own database directly; `INT-020` is the engine writing a `File` to `LIS` at `Real time`; `INT-022` is the warehouse reading `LIS` directly, not through the engine. The platform bought to route traffic is mostly a destination.
4. Make the routing decision per interface and record it on the register: route through `A11`, keep point to point with a named owner and a stated reason, or retire. Do not write "route everything": `REQ-016` to `REQ-021` on `O06` already require managed, published and logged interfaces for six applications and are all `Should`, not `Must`.

5. Cite the framework that governs the seam rather than reproducing it. The national interoperability framework is a gazetted instrument [1], and volume 1 chapter 16 says which layer of it governs which seam. The contract is still the local specification.
6. Raise the gap. `OBJ-08 Every new interface managed and published` on `O03` is `In progress` and there is no gap record against it anywhere in `O29`.

Done when every one of the twenty-three rows carries a routing decision with an owner, and the nineteen bypassing interfaces are a single numbered finding rather than nineteen separate ones.

6.6 — Stop `O24` describing a file drop as a published service

Sheet `O24 Information System Service` · **Input** `O24`'s twelve rows against `R06`'s types · **Output** twelve services with a real description, a real style and an honest publication state · **Signed by** the Domain Architect, Application (`ACT-11`) · **Time** one day

> **`O24 Information System Service`**
>
> `Ref`, `Information system service`, `Application`, `Description`, `Style`, `Published`. All twelve `Description` values read "Service exposed by <application>." `Style` takes `File`, `API`, `Event` and `Batch`. `Published` takes `Published` and `Unpublished`.

1. Read the sheet's shape first. Twelve services for twelve applications, one each, named `<application> Service` — which is a sheet generated from the application list, not a service catalogue. A real estate has applications exposing several services and applications exposing none.
2. Read the four marked `Published`. `ISS-01` to `ISS-04` are all `Style` = `File` — file drops described as published services, against `OBJ-08`. Seven of the twelve, `ISS-05` to `ISS-11`, are `Unpublished`.
3. Replace every `Description`. "Service exposed by PAS" tells a consumer nothing: a description has to say what crosses, in which direction, and what the consumer may rely on.
4. Set `Style` from `R06`'s actual types for that application, and where they disagree say which is true. `A01 EHR / EPR` has `ISS-01` styled `File` while its six interfaces in `R06` are `File`, `Event`, `API` and `Direct DB read`.
5. Delete the rows for applications that expose no service, and add rows where one application exposes several. Twelve-for-twelve is the shape of a generated sheet.

Done when no `Description` is template text, `Style` agrees with `R06`, and `Published` means a consumer could find and call the service without asking an engineer.

Form 6-C — The service catalogue

All six real `O24 Information System Service` columns, in sheet order. Expect more than twelve rows when you are finished, and fewer applications. `Published` means a consumer can find and call it without asking an engineer. Signed by the Domain Architect, Application.

Ref	Information system service	Application	Description	Style	Published
ISS-01	EHR / EPR Service — EXAMPLE, delete this row	A01		File	Unpublished

The four sheets that cannot be repaired

`R04`, `R07`, `R20` and `R25` were built by pairing two lists in row position. They are not wrong in the way a typographical error is wrong; they are wrong in the way a correct calculation over the wrong inputs is wrong. Every `Ref` resolves, every name matches its `Ref`, and the arithmetic is visible once you look for it.

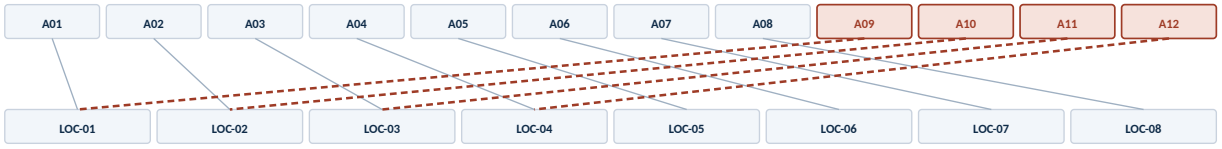
Four sheets joined by row position, and every reference in them resolves

Each was built by laying one list against another in index order. Integrity is perfect; the estate described does not exist.

SHEET	LEFT LIST	RIGHT LIST	THE JOIN	RIGHT BY CHANCE
R04	O15's ten service rows	the twelve applications	BSV-0n to A0n, straight down	3 of 10 land correctly
R07	the twelve applications	O27's twelve components	A0n to PTC-0n, straight down	2 of 12
R20	the twelve applications	O12's eight locations	A0n to LOC-0n, wrapping at eight	not testable; see below
R25	O18's five products	the twelve applications	striding A01, A04, A07, A10, A01	1 of 5

R20 drawn out, because the wrap is the part nobody notices

Twelve applications laid against eight locations in index order. The last four restart at the top of the location list.



A09 to A12 restart at LOC-01. The Patient Portal is sited at the Acute Hospital because it is the tenth row, not because anyone looked.

LOC-01 Corporate Office · LOC-02 Acute Hospital · LOC-03 Community Hospital · LOC-04 Outpatient Clinic Network · LOC-05 Diagnostic Laboratory
LOC-06 Imaging Centre · LOC-07 Primary Data Centre · LOC-08 Secondary Data Centre. R07 plus O27 site the same applications differently for eleven of twelve.

Figure 6.3 The four relationship sheets built by laying one list against another in index order, with the rule each used and how many rows land correctly by chance. R20 is drawn out because its wrap at eight locations is the part a reader does not notice.

So do not repair them. Clear them and re-derive each from evidence somebody can produce — a deployment record, a site survey, a product definition, a capability owner's own statement. A repaired positional join is still a positional join with some of the rows moved, and it carries the same credibility risk: the first reader who checks one row will check all of them.

Three further things that figure makes visible, and all three are the point of the four steps that follow. `R20` wraps: twelve applications against eight locations, so `A09` to `A12` restart at `LOC-01`. `R25` strides: five products against `A01`, `A04`, `A07`, `A10` and `A01` again, which is the same four-application cycle as `R05`'s consumer rows. And `R07` and `R20` describe the same estate and contradict each other for eleven of twelve applications, because `O27 Physical Technology Compone` sites each component and `R20` sites each application, and only `A12` agrees.

6.7 — Re-derive `R04` from what each capability owner can demonstrate

Sheet `R04` · **Input** `O15`'s ten service rows, `O13 Business Capability`, and the application each capability owner actually uses · **Output** ten or more rows joining a service to the application that realises it · **Signed by** the capability owner, with `GB-01` · **Time** one day

> **`R04`, and the three other positional joins**
>
> `R04` has four columns: the `O15` ref and its name, then the application ref and its name. Its first column header is the element name this volume does not print, so match on position rather than on the header. `R20 Application located at Loca` holds `Application`, `Application name`, `Location`, `Location name`. `R25 Application supports Produc` holds `Product`, `Product name`, `Application`, `Application name`. None of the four has a `Ref` column.

1. Note the tab name before you write a formula against it. `R04`'s tab name ends in a **trailing space**, because the full name was cut at thirty-one characters after a word boundary. A lookup written without the trailing space will not find the sheet.
2. Read the ten rows. `BSV-0n` is joined to `A0n`, straight down. Six are plainly wrong: `BSV-02 Clinical Care Delivery Service` reaches `A02 PAS`, which `O23` describes as registration and scheduling, while `A01 EHR / EPR` is the clinical record; `BSV-04 Medicines Management Service` reaches `A04 RIS / PACS`, an imaging system; `BSV-06 Clinical Governance & Safety Service` reaches `A06 Theatre Management` while `A09 Incident & Risk` holds safety incidents; `BSV-10 Population Health & Analytics Service` reaches `A10 Patient Portal` while `A12 Clinical Data Warehouse` is the analytics platform.
3. Note the three that happen to land correctly — `BSV-03` to `A03 LIS`, `BSV-07` to `A07`, `BSV-08` to `A08` — and the one that is arguable, `BSV-01 Patient Access & Scheduling Service` to `A01` where `A02 PAS` fits better. Three correct out of ten is what a ten-by-ten positional join produces by chance, and saying so stops a client reading the three as evidence of care.
4. Clear the sheet and re-derive. Ask each capability owner which application they open to do the work, and record the answer with their name. Expect more than ten rows, because a service is usually realised by two or three applications.
5. Record the capabilities that reach no application. All ten level-1 capabilities are realised by a service, and none of the ninety level-2 and level-3 capabilities is realised by anything at all. Ninety of a hundred capability records sit with nothing beneath them, which is a Phase B finding this step hands back.

Done when every row names the capability owner who confirmed it, the count is no longer exactly ten, and the three rows that happened to be right are marked as confirmed rather than left as inherited.

6.8 — Re-derive `R07` from deployment evidence, not from the component list

Sheet `R07 Application deployed on Tec` · **Input** the deployment records, and `O27 Physical Technology Compone` · **Output** one row per application-to-component deployment, from a record somebody can produce · **Signed by** the Domain Architect, Technology (`ACT-12`) · **Time** one day

> **`R07 Application deployed on Tec`**
> `Application` , `Application name` , `Platform component` , `Component name` , `Environment` . No `Ref` column — this sheet is keyed by its first two columns, as the other relationship sheets are. `Environment` reads `Production` on all twelve rows.

1. Read the twelve rows as `A0n` against `PTC-0n` . Ten are wrong. `A02 PAS` is deployed on `PTC-02 PACS image store` ; `A04 RIS / PACS` on `PTC-04 Integration engine servers` ; `A07 Revenue Cycle / Claims` on `PTC-07 Medical device gateways` ; `A10 Patient Portal` on `PTC-10 Backup and archive` ; and the component actually named "Integration engine servers" hosts the imaging system while `A11 Integration Engine` sits on `PTC-11 Cloud landing zone` . Only `A01` to `PTC-01` and `A03` to `PTC-03` survive.
2. Note that `Environment` reads `Production` on all twelve rows, so nothing in the file records a non-production deployment — on a repository where `O48 Environment` holds four environments and `SEC-07 Non-production masking` is `Absent` . Item 7 owns that; record it here as a cross-reference.
3. Clear the sheet and re-derive from deployment evidence: a configuration record, a build pipeline, a change ticket, or a named engineer's written statement. One row per application per component per environment, so expect three or four times twelve rows.
4. Check the support dates as you go. `PTC-04 Integration engine servers` has a support end of 2026-05 and position `Past support` ; `PTC-06 Clinical workstation estate` ends 2026-12, which is two months after the reference date and is therefore **not** past support today. `ZZ Checks` row 13 says two components are past support; the defensible figure today is one, and item 7 carries it.
5. Where no deployment record exists, write "no record" on the row and leave it. A re-derived sheet with eight evidenced rows and four marked "no record" is a better artefact than twelve rows nobody can stand behind, and the four are a gap with an owner.

Done when every row names its evidence, `Environment` carries more than one value, and the contradiction with `R20` has been resolved or recorded by step 6.9.

Form 6-D — Re-deriving a positional join

All five real `R07 Application deployed on Tec` columns, in sheet order. **This sheet has no `Ref` column.** Use the same method for `R04` , `R20` and `R25` , whose columns differ: name the evidence for every row, and leave the sheet short rather than filling it. Signed by the Domain Architect, Technology.

Application	Application name	Platform component	Component name	Environment
A04	RIS / PACS — EXAMPLE, delete this row	PTC-02	PACS image store	Production

6.9 — Re-derive `R20` , and record the contradiction you cannot resolve

Sheet `R20 Application located at Loca` · **Input** `O12 Location` 's eight rows, `R07` as re-derived in 6.8, and `O27` column 7 · **Output** application locations from a site record, and a written statement of which of the two sheets is authoritative · **Signed by** the Domain Architect, Technology (`ACT-12`) · **Time** half a day

1. Read the twelve rows as `A0n` against `LOC-0n` , wrapping at eight so `A09` to `A12` restart at `LOC-01` . `A03 LIS` is at `LOC-03 Community Hospital` while `LOC-05` is the `Diagnostic Laboratory` ; `A04 RIS / PACS` is at `LOC-04 Outpatient Clinic Network` while `LOC-06` is the `Imaging Centre` .
2. Then read `R20` against `R07` plus `O27` column 7, which sites each physical component. They disagree for eleven of twelve applications and agree only for `A12` . `A01 EHR / EPR` is at `Corporate Office` on `R20` and runs on `PTC-01` , which `O27` sites at `Outpatient Clinic Network` .

3. Decide what the sheet is for before you fill it, because the two readings are different facts. An application's location can mean where its users are, or where its compute runs. `O12.Type` separates `Corporate`, `Care`, `Diagnostics` and `Technology`, which suggests the second; `O12.Scale` is "N people", which suggests the first.
4. Pick one reading, write it at the head of the sheet, and re-derive against it. Then record the other as a gap rather than resolving it silently: a repository that answers "where does this application run" and "where are its users" from one column will be wrong for one of the two questions every time it is asked.
5. Carry the unresolved part explicitly. `R07` plus `O27` and `R20` are two statements about one estate and they agree on one application of twelve. You can re-derive both from evidence and they may still disagree, because the two questions have different answers; what you cannot do is leave the file holding both without saying which one a reader should use.

Done when the sheet has a stated reading at its head, every row names a site record, and the eleven disagreements with `R07` plus `O27` are either closed or recorded as one gap.

6.10 — Re-derive `R25`, and name the eight applications supporting no product

Sheet `R25 Application supports Produc` · **Input** `O18 Product`'s five rows and the clinical service lines behind them · **Output** one row per product per application that supports it · **Signed by** the product owner, with `GB-01` · **Time** half a day

1. Read the five rows. They stride `A01`, `A04`, `A07`, `A10`, `A01` — the same four-application cycle as `R05`'s consumer rows. `PRD-03 Emergency Care` is supported by `A07 Revenue Cycle / Claims`, the billing system. `PRD-04 Diagnostic Services` is supported by `A10 Patient Portal`, while `A03 LIS` and `A04 RIS / PACS` support no product at all.
2. List the eight applications supporting no product: `A02`, `A03`, `A05`, `A06`, `A08`, `A09`, `A11` and `A12`. Two of those — `A11` and `A12` — are the two orphans `RPT13 Orphan Analysis` did find, and it found only those two.
3. Clear and re-derive. One row per product per supporting application; five products across twelve applications should produce fifteen to twenty-five rows, not five.
4. Record the sheets that have no `Ref` column as you go. `R04`, `R07`, `R20` and `R25` are all keyed by their first two columns, so a row cannot be referenced from anywhere else in the repository — which is part of why nobody ever cited one of them and nobody ever checked them.
5. Settle what a product is here before you fill the sheet, because `O18`'s five rows are clinical service lines rather than things anybody buys. Volume 1 chapter 6 carries how the money actually moves in both sectors, and a product that cannot be priced or contracted will not survive the first commercial review.

Done when every row names the product owner who confirmed it, the eight unsupported applications are a numbered finding, and `O18`'s own definition of a product is written down.

6.11 — Fix `R05`'s half-keyed first column

Sheet `R05 Application manages Data En` · **Input** rows 5 to 32 · **Output** a sheet where the first column holds a `Ref` on all twenty-eight rows · **Signed by** the Domain Architect, Data (`ACT-10`) · **Time** one hour

> **`R05 Application manages Data En`**
> `Application`, `Application name`, `Data entity`, `Entity name`, `Role`. `Role` takes `Master` and `Consumer`, fourteen rows each. No `Ref` column, so the first two columns are the key — which is what makes the defect below fatal rather than untidy.

1. Look at the first column. Rows 19 to 32, the `Consumer` rows, hold `A01`, `A04`, `A07` and `A10` — refs, correctly. Rows 5 to 18, the `Master` rows, hold the application's **name** in the column headed `Application`: "ePrescribing & Medicines Admin", "Clinical Data Warehouse", "PAS".
2. So a lookup on that column returns fourteen rows of twenty-eight. This is the only sheet in the repository where a key column is half refs and half names, and it means `R05` cannot be joined to `O23` without special-casing the file.
3. Replace the fourteen names with their refs. Nothing else on those rows changes.
4. Then notice what the `Consumer` half is. Fourteen rows cycling `A01`, `A04`, `A07`, `A10` over `E01` to `E14` in order — a positional join inside a sheet that is otherwise real. `A04 RIS / PACS` is recorded as the consumer of `E14 Medical Device` and of `E10 Practitioner`; `A10 Patient Portal` as the consumer of `E12 Claim`. Re-derive the consumer half from the interface register in 6.4, which is the only evidence of what actually reads what.

Done when all twenty-eight first-column values are refs, and the fourteen `Consumer` rows come from the interface register rather than from a four-application cycle.

6.12 — Re-cut `O45`, and raise the application gaps

Sheet `O45 Application Building Block`, `O29 Gap` · **Input** everything above · **Output** building blocks with testable acceptance criteria, and a gap list that matches the register · **Signed by** `GB-01 Architecture Board` at `GT-1 G1 Design` · **Time** one day

1. Read `O45`'s ten acceptance criteria. All ten read "No unmanaged interface to or from <application>", which is a single test applied ten times and is already failed by thirteen of twenty-three interfaces on the day the sheet was written.

2. Keep the test and make it specific per block. For `ABB-01 EHR / EPR target`, the criterion is the six named interfaces in step 6.5 carrying a routing decision and a contract version. A criterion a tester cannot fail is not a criterion.
3. Extend `O45` to twelve rows. It stops at `ABB-10 Patient Portal`, so `A11 Integration Engine` and `A12 Clinical Data Warehouse` have no building block — the same two applications `RPT13` found as orphans and the same two that support no product.
4. Raise the gaps: the nineteen bypassing interfaces as one, the five unmanaged-and-unlogged rows as one, the four positional joins as one each, the twelve application owners as one, and the five contradicting dispositions as one. `O29 Gap` holds two records, both business-domain capability uplifts. Phase C's application half alone adds nine.
5. Take `DLV-09` and `DLV-10` back through `GT-1`. `DLV-10 Interface Catalogue` is the register from 6.4, not `RPT04`.

PHASE C, THE APPLICATION HALF — THE FIVE QUESTIONS A REVIEWER WILL ASK

Which test did you declare on the interface register, and what count did it produce?

Which of the twenty-three interfaces survived every test, and who owns each one?

Which nineteen bypass the integration engine, and which of those are you leaving point to point?

For R20, did you choose users or compute, and where is that written down?

Which four sheets did you clear rather than repair, and what evidence replaced each?

What leaves the room

`DLV-09 Application Architecture` carries the twelve applications with a confirmed disposition, a named owner and a settled criticality, plus the service catalogue and the twelve building blocks. `DLV-10 Interface Catalogue` carries the register: twenty-three rows, a declared test, an owner and a contract version against each, and the routing decision. Neither is `RPT02` or `RPT04`; those two are reports and they will print whatever the object sheets hold.

The gap list is the third artefact and the one a client acts on. Nine entries from this item, against an `O29 Gap` sheet that arrived holding two. And one sentence to put at the top of it: four of the six relationship sheets in this item resolve perfectly, have always resolved perfectly, and describe an estate that does not exist. Item 10's Phase G question is not whether a relationship resolves. It is what was reconciled, by whom, and against what.

THE CHAPTER IN TEN LINES

- R06 is the only relationship sheet here with real content, and its three governance columns contradict each other.
- Fifteen of its twenty-three rows fail at least one of four declared tests. Seventeen fail on five.
- Report the test, not the number. ZZ Checks gives three interface counts and names no test at all.
- Five rows are both Unmanaged and unlogged. No sheet in the repository reports that figure.
- Nineteen of twenty-three interfaces bypass A11, including every interface into or out of the EHR.
- MSR-08 carries baseline 25% where ten of twenty-three is 43%. Give it an MDF- row on O70.
- R04, R07, R20 and R25 are positional joins. Clear and re-derive them; do not repair them.
- None of those four has a Ref column, so no row in them can be cited from anywhere else.
- R04's tab name ends in a trailing space. A lookup written without it will not find the sheet.
- R07 plus O27 and R20 disagree about where eleven of twelve applications are. Pick a reading and say so.
- R05's first column holds names on the Master rows and refs on the Consumer rows: fourteen of twenty-eight.
- The clinical record of truth is Medium criticality while the billing system is Critical.
- DLV-10 Interface Catalogue is the register you build, not RPT04, which only prints what R06 holds.

SOURCES FOR THIS CHAPTER

- **1.** Health Normative Standards Framework for Digital Health Interoperability in South Africa 2021, Notice 2667, Government Gazette 47337, 21 October 2022 https://www.health.gov.za/wp-content/uploads/2022/10/HNSF_Gazette_21_October_2022.pdf
- **2.** SAFER Guide 8, Test Results Reporting and Follow-Up, United States Office of the National Coordinator for Health Information Technology https://healthit.gov/wp-content/uploads/2025/06/SAFER-Guide-8.-Test-Results-Reporting-Final.pdf
- **3.** National Health Act 61 of 2003, consolidated text; section 17 on controls against unauthorised access https://www.saflii.org/za/legis/consol_act/nha2003147.pdf
- **4.** Zharima, Griffiths and Goudge, on the barriers and facilitators to implementing electronic health records in a middle-income country, Frontiers in Digital Health 5, 4 August 2023 https://www.frontiersin.org/journals/digital-health/articles/10.3389/fdgth.2023.1207602/full

ITEM 7

Technology Architecture, devices and the downtime path

Thirty-nine technology records and not one device. The phase where the read-only record stops being a principle and gets a stopwatch against it.

Phase D in this repository is four sheets and thirty-nine records. `O25 Technology Service` holds twelve, `O26` seven, `O27 Physical Technology Component` twelve and `O28 Platform Service` eight. Not one of the thirty-nine is a clinical device. The estate that keeps patients alive — the pumps, the ventilators, the monitors, the analysers, the anaesthetic machines — is represented by a single row called "Medical device gateways", typed `Compute`, and by a component name, "Medical device integration", repeated verbatim on four sheets. The word "calibration" appears twice in the whole file. "Risk class" appears once.

That is the first of the two jobs in this item. The second is the downtime path: a read-only record that volume 1 chapter 19 argued for as a principle becomes, here, a Phase D deliverable with a number against it — how long it actually takes to produce one patient's record when the primary is gone.

Phase D is also where the repository's resilience arithmetic is at its worst and most teachable. `O49 Resilience Tier` gives the clinical record of truth a 24-hour recovery objective and has never tested it. `O48 Environment` records production as holding no personal data. Neither is a typing error. Both are what happens when a technology architecture is assembled from a template by somebody with no clinical question in front of them.

What Phase D has to hand over

Five artefacts, and the repository has places for four of them.

Artefact	Where it lands	Signed by
Corrected technology currency position	`O27`, `RPT10`	Domain Architect, Technology
Corrected deployment record	`R07`	Head of Service Management
A device register	`O68 Medical Device`, which you create	Head of Estates, countersigned by a clinical engineer
Clinically derived recovery objectives	`O49`	Chief Operating Officer
The read-only record, with a measured retrieval time	`O26`, `O49`, and the measurement sheet	Chief Operating Officer

`DLV-11 Technology Architecture` and `DLV-12 Technology Standards` on `O31 Deliverable` are both already `Approved` for phase `D`. They were approved over a technology domain with no devices in it, a resilience tiering whose label carries no information and a deployment record built by pairing two lists in row order. Re-opening an approved deliverable is the conversation this item exists to make winnable, and the argument that wins it is arithmetic rather than opinion.

7.1 — Read the technology domain before you change a cell of it

Sheet `O25`, `O26`, `O27`, `O28`, `RPT10` · **Input** the workbook · **Output** a marked-up copy with every disagreement circled · **Signed by** Domain Architect, Technology · **Time** half a day

1. Open `O25 Technology Service`. Columns are `Ref`, `Technology service`, `Description`, `Category`. Read column `Category` against column `Technology service` row by row. Seven of twelve disagree: `TSV-01 Compute` is categorised `Data`, `TSV-02 Storage` is `Security`, `TSV-09 Data Platform` is `Security` and `TSV-11 Security Operations` is `Platform`. The data platform is categorised as security and security operations as platform. Circle all seven.
2. Open `O26`. Its tab name is cut at thirty-one characters and the truncation has reached the sheet's own row 1, which reads "Logical Technology Componen". Columns are `Ref`, `Logical technology component`, `Rationale`, `Status`. Seven rows, `LTC-01` to `LTC-07`.
3. Compare `O26`'s `Status` against `O47 Reference Model Entry`'s `Status` for the same seven names. Four of seven disagree: terminology services is `Current` on `O26` and `Candidate` on `O47`; clinical decision support is `Emerging` and `Adopted`; zero-downtime clinical systems is `Target` and `Candidate`; de-identification and secure enclave is `Target` and `Adopted`. Nothing anywhere in the file records any of the seven as running: no row of `O24 Information System Service`, `O25` or `O28` names them, and `TBB-01` to `TBB-07` on `O55 Building Block` all carry the acceptance criterion "in production for at least one service".
4. Open `O27`. Columns are `Ref`, `Physical technology component`, `Vendor`, `Class`, `Support end`, `Position`, `Location`. Note that `Vendor` holds the literal string "Vendor" in eight of twelve rows. The three that name anything name a class of product rather than a supplier — "Oracle-class", "VMware-class", "Hyperscaler" — and one names a company. A vendor column with no vendors in it cannot support a support-expiry conversation, which is the only reason the column exists.
5. Open `O28 Platform Service`. Columns are `Ref`, `Platform service`, `Description`. Eight rows, no status, no owner, no position. Nothing in the file relates to any of them.
6. Write the counts on the front of the marked-up copy: 39 technology records, 0 devices, 0 of the seven logical components evidenced in production, 7 of 12 categories wrong, 4 of 7 statuses in conflict.

Why `O25`, `O26` and `O28` cannot be fixed by editing them

All three are outside the model entirely. No relationship sheet in this repository touches a technology service, a logical technology component or a platform service, by `Ref` or by name — they sit alongside `O05 Principle`, `O07 Constraint`, `O16 Business Process`, `O19`, `O31 Deliverable`, `O33 Architecture Contract`, `O38 Data Set`, `O42 Metadata Element`, `O52 Implementation Factor`, `O54 Architecture Alternative` and the whole governance layer in the set of twenty object sheets that appear in no relationship at all: 206 records hanging off nothing. Correcting `O25`'s `Category` column is worth half an hour. It does not make `O25` part of the architecture. That takes a relationship, and item 12 gates it.

7.2 — Correct the currency position, and date it

Sheet `O27`, `RPT10` · **Input** 7.1's marked-up copy · **Output** a dated currency statement · **Signed by** Domain Architect, Technology · **Time** two hours

1. `RPT10 Technology Currency` is a verbatim copy of `O27` with the columns renamed `Component` instead of `Physical technology component`. Its row 18 footer reads "2 components are running past vendor support today".
2. Test the footer against the sheet. `O27 Position` reads `Past support` on two rows: `PTC-04` Integration engine servers, `Support end` 2026-05, and `PTC-06` Clinical workstation estate, `Support end` 2026-12. On a reference date of 25 October 2026, 2026-12 has not arrived.
3. **The correct figure is one, not two**, and it is one because of the date you read it on. Write the date into the statement. `PTC-06` becomes the second on 1 January 2027 and your statement is wrong from that morning unless it says when it was taken.
4. Carry the same correction into `ZZ Checks` row 13 and `RPT01 Dashboard` row 15, which both say 2.
5. Record the two support-end dates against the applications that run on the components rather than against the components. That is 7.3's job, and it turns a currency number into a clinical one.

BEFORE YOU LEAVE 7.2

What date was the currency statement taken on, and where is that date written?

Which component becomes unsupported next, and on what date?

Who owns the renewal decision for it, by name, and when were they told?

7.3 — Reconcile the deployment record against itself

Sheet `R07`, `O27`, `O23` · **Input** `R07`'s twelve rows · **Output** a corrected `R07` and a list of contradictions you cannot resolve from the file · **Signed by** Head of Service Management · **Time** one day

1. `R07` columns are `Application`, `Application name`, `Platform component`, `Component name`, `Environment`. Twelve rows. Read them in order. `A01` joins `PTC-01`, `A02` joins `PTC-02`, and so on to `A12` and `PTC-12`. That is not a deployment record. It is two lists zipped together.
2. Ten of twelve are wrong on their face. The imaging system `A04` is deployed on `PTC-04` Integration engine servers while the Integration Engine `A11` is deployed on `PTC-11` Cloud landing zone. The billing system `A07` is deployed on `PTC-07` Medical device gateways. The patient portal `A10` is deployed on `PTC-10` Backup and archive. The patient administration system `A02` is deployed on `PTC-02` PACS image store. Only `A01` on `PTC-01` EHR application tier and `A03` on `PTC-03` Laboratory instruments network survive.
3. Note the `Environment` column: every one of the twelve rows reads `Production`. No application in this file is deployed to anything else, which means the four rows of `O48 Environment` are deployed to by nothing at all.
4. Now cross `R07` against `O27`'s own `Location` column and against `R20`, which sites applications directly. They agree for one application of twelve, `A12`. `A01` runs on `PTC-01`, which `O27` sites at Outpatient Clinic Network, while `R20` sites `A01` at Corporate Office. **Two sheets site the same clinical record in two different places and neither is marked provisional.**
5. Fill Form 7-A from the service management records, not from the repository. Where the organisation has no record either, leave the row blank and write "no source" in `Component name`. A blank with a reason is a finding; a guess is a defect you have added.
6. The correction you cannot make from the file is the location. Take it to the data centre and facility owners and get it signed. Volume 1 chapter 22 has the worked hosting design and the decision order.

Form 7-A — Deployment record, feeding `R07`

Twelve rows for a first pass, one per application on `O23`. Row 1 is the worked example — Driehoek Health Group. Signed by the Head of Service Management. Columns are `R07`'s own, spelled as the sheet spells them; `R07` is a relationship sheet and has no `Ref` column.

Application	Application name	Platform component	Component name	Environment
A01	EHR / EPR	PTC-01	EHR application tier	Production

Typed straight into Excel: `Application name` and `Component name`, because both are lookups from `O23` and `O27` and should never be keyed by hand. Key the three that carry meaning.

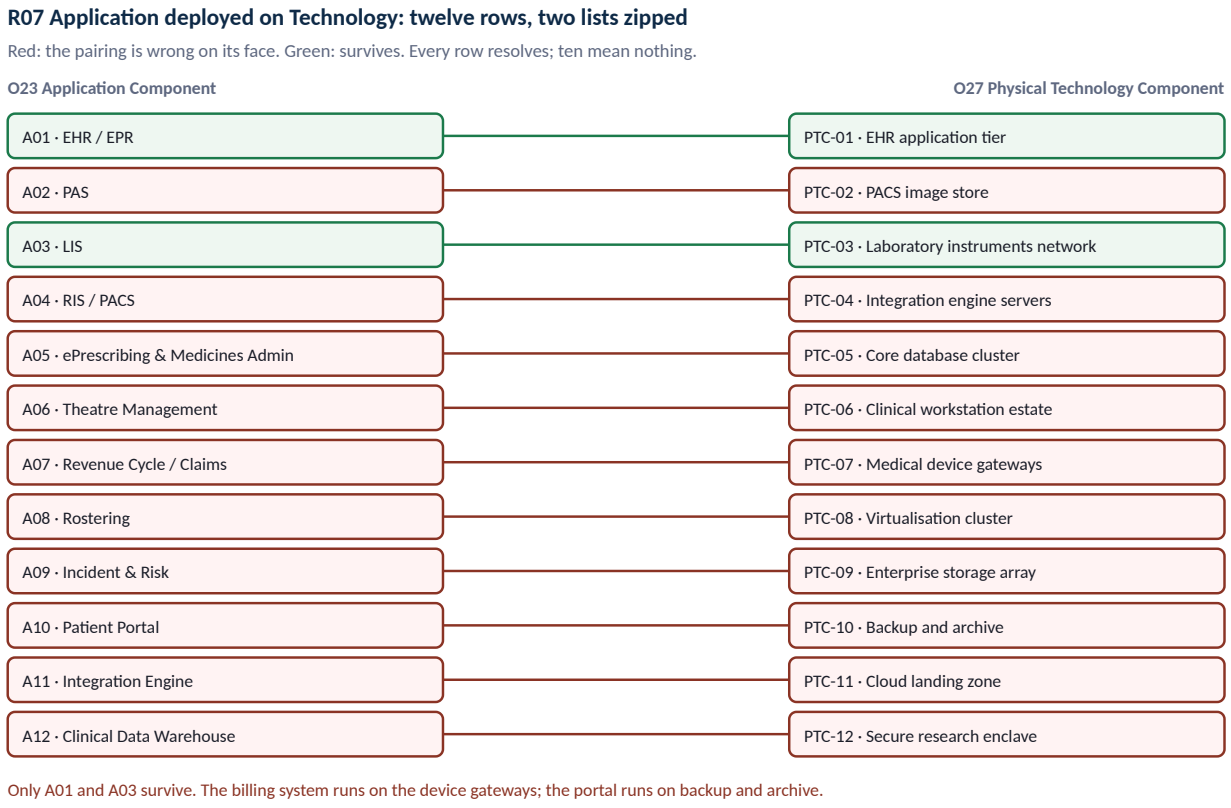


Figure 7.1 Every row of `R07` resolves and ten of the twelve mean nothing. This is the repository's dominant defect seen in the technology domain: integrity is perfect and the content is a coincidence. The pattern it caricatures is a real one — a deployment record generated from two exports and never walked through with service management.

The device the file has no place for

`EXT Industry Extension` declares six industry types. `X05 Medical Device` reads: "A clinical device with calibration, maintenance and risk class." That sentence is the specification, written by the person who built this repository, of a thing the repository then failed to build.

Follow the entity. `E14 Medical Device` on `O20 Data Entity` exists as one row: `Class Master`, `Definition` "A clinical device with its maintenance and calibration.", `Classification Public`, `Owner` "Head of Diagnostics", `Master system` Theatre Management, `Lineage Partial`. It appears on exactly three sheets — `O20`, `R05` and `RPT05 Data Ownership`, the last of which is a copy of `O20` with the `Definition` column dropped. It has no

logical data component on `O21`, no attributes on `O37`, no master data rule on `O43`, no data building block on `O44`, no migration on `O39`, no mapping on `O40` and no entity relationship on `O36`. Every other `Master` entity except `E11 Care Setting` has at least four.

So the file says the type holds calibration and risk class, and then: **"calibration" appears twice in the whole workbook** — in `E14`'s definition and in `X05`'s — **and "risk class" appears once**, in `X05`'s. Neither phrase appears in a column header, a cell value or a row anywhere else. There is no serial number, no manufacturer, no model, no software version, no calibration date, no maintenance schedule and no risk class in 820 records.

The nearest thing to a device record in the repository is infrastructure. `PTC-07` on `O27` is "Medical device gateways", `Vendor` "Vendor", `Class` `Compute`, `Support end` 2029-04, `Position` `Supported`, `Location` Secondary Data Centre. It is the thing bedside devices talk *through*, and `O27`'s `Class` values — `Application`, `Storage`, `Network`, `Database`, `End user`, `Compute` — have no value a device could take. A ventilator entered on `O27` becomes a compute component with a vendor support date. That is the whole argument for a separate sheet, and it is an argument from the file rather than from practice: **`O27`'s columns record vendor support, and a device's governing dates are calibration and maintenance, which are not vendor support and do not move with it.**

Three more facts in the file make the sheet unavoidable.

`DRV-03` on `O01 Driver` is "Medical device regulation", `Regulatory`, "Software as a medical device and post-market surveillance", owned by the Chief Compliance Officer, impact `High`. `REQ-003` on `O06 Requirement` is "Evidence Medical device regulation", priority `Must`, status `Approved`. `CO-03` on `O63 Control Objective` is the obligation's control, `Evidence` `Partial` — and it is the *only* one of the seven obligations that is not `In breach` or `No mechanism`, which means the repository rates its device compliance as its best regulatory position while holding no device data at all. The evidence `CO-03` is partial on is, through `R16`, `SEC-03` Privileged access management. A privileged access control is not evidence of device regulation.

And `C09 Estates & Medical Devices` on `RPT03 Capability Heatmap` is `Current` 1, `Target` 3, `Gap` 2, `Disposition` `Build` — **the lowest current maturity of all ten level-1 capabilities**. The repository knows the capability is the weakest thing it has and has given it nothing to work with.

> **WHAT THE REGULATOR DOES AND DOES NOT GIVE YOU**
>
> South Africa regulates devices through an establishment licensing regime under section 22C of the Medicines and Related Substances Act rather than a product registration regime, and the registration call-up plan has not been published [4]. Three consequences for this sheet. **There is no national device registry to reference**, so the device register you build is the only one there is. **Risk class is the manufacturer's classification with the regulator's determination** across four classes, A to D, by principle rather than by a software-specific rule [2] — so the `Risk class` cell holds a value with a rationale behind it, and the rationale belongs in the file the cell points at. And **the software state of a regulated device belongs to its manufacturer**, with the organisation owning the risk of running it on its network, which is why `Software version` and `Network segment` are columns on the same row [3][5]. Volume 1 chapter 18 carries the regime; do not restate it in your deliverable, cite it.

7.4 — Create `O68 Medical Device` and populate it

Sheet `O68 Medical Device`, which the repository does not have · **Input** the biomedical engineering asset list, the clinical engineering maintenance system and the network inventory · **Output** one row per device class in scope · **Signed by** Head of Estates, countersigned by a clinical engineer · **Time** three weeks for a first pass, and it will not be complete

1. Add the sheet at `O68`, after `O66 Architecture Increment` and `EXT`. Keep the file's own conventions: row 1 the sheet title, row 2 organisation, domain and record count, row 4 the headers, data from row 5, first column `Ref`. Prefix every reference `DEV-`.
2. The columns, in this order and spelled this way: `Ref`, `Device`, `Model`, `Risk class`, `Software version`, `Calibration due`, `Maintenance due`, `Network segment`, `Owner`.
3. Scope the first pass by consequence, not by completeness. Start with every device that writes into the clinical record or takes a reading a clinician acts on without re-checking it. At Driehoek that was 11 device classes across nine acute sites; at Rietvlei it was four, and the four were the whole conversation.
4. Fill Form 7-B first for every device class — identity and accountability. Fill form 7C second — the operating state. They are two passes over the same list because the two sets of facts live in different systems and are signed by different people.
5. `Risk class` takes the manufacturer's class with the regulator's determination [2]. Where you cannot establish it, write `Not established` and raise it as a procurement question, because an unclassified device is a device nobody has assessed.
6. `Software version` is the version the manufacturer has released and validated, not the version you would like. Record it as the manufacturer states it. Where a device is at a version the manufacturer no longer supports, that is a currency finding and it belongs on `O27` as well, against `PTC-07`.
7. `Network segment` takes its value from 7.5 and from nowhere else. A free-text segment name that is not on the segment list is a blank.
8. `Owner` is a person. `PRN-13` on `O05 Principle` — "Data Ownership is Individual / An owner is a person, not a committee" — is `Approved`, and `O20` breaks it fourteen times by naming eight "Head of" titles that exist nowhere else in the file. Do not add a fifteenth. Name the post and the human in it.

Form 7-B — Device register, identify. Feeding `068 Medical Device`

This form feeds a sheet the repository does not have. Create `068` before you type it in. Ten to twelve rows for a first pass. Row 1 is the worked example — Driehoek Health Group, theatre complex. Signed by the Head of Estates.

Ref	Device	Model	Risk class	Owner
DEV-01	Volumetric infusion pump, general ward	as the plate reads	C, manufacturer classified	named clinical engineer

Typed straight into Excel: nothing on this form. Every cell is read off a device or out of a maintenance system by a person standing next to it, which is why the form is on paper.

Form 7-C — Device register, operating state. Feeding `068 Medical Device`

This form feeds a sheet the repository does not have. The same rows as Form 7-B, joined on `Ref`. Ten to twelve rows for a first pass. Row 1 is the worked example. Signed by the clinical engineer, with the network segment countersigned by the Security Architect.

Ref	Software version	Calibration due	Maintenance due	Network segment
DEV-01	as the manufacturer states it	2027-03	2027-01	unpatchable clinical devices

Typed straight into Excel: `Calibration due` and `Maintenance due`, where a maintenance management system already holds them and can export. Everything else is keyed from the form.

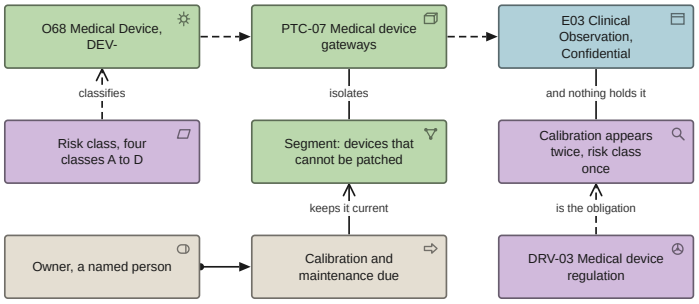


Figure 7.2 What `O68 Medical Device` joins to, and why none of it can live on `O27`. The gateway is infrastructure, the observation is the clinical consequence, and the obligation is stated by the file with nothing behind it. `O27` 's dates are vendor support; a device's governing dates are calibration and maintenance, which do not move with vendor support.

7.5 — Name the network segments, then write them into `O68`

Sheet `O50 Network Link`, `O68`, `O32 Standard` · **Input** the network inventory and the device list from 7.4 · **Output** a named segment list and a dependency note per segment · **Signed by** Security Architect, countersigned by Domain Architect, Technology · **Time** one week

1. `O50` columns are `Ref`, `From`, `To`, `Bandwidth`, `Diverse path`. Ten rows, and it is a link list, not a segment list. There is no segment anywhere in this repository.
2. Read `O50` for what it does say. Two location pairs are each carried twice with different figures: `NWL-01` Acute Hospital to Community Hospital at 10 Gbps and `NWL-09`, the same pair, at 40 Gbps; and `NWL-02` Community Hospital to Outpatient Clinic Network with `Diverse path` `Yes` against `NWL-10`, the same pair, `No`. The file gives no basis for choosing between either pair. Carry both readings and mark the sheet as unreconciled.
3. Then read the two that matter clinically. `NWL-05` Imaging Centre to Primary Data Centre and `NWL-06` Primary Data Centre to Secondary Data Centre are both 100 Mbps with `Diverse path` `No` — **the slowest and least resilient links in the estate are the one carrying imaging to the data centre and the one carrying the data centre to its own secondary site**, against `PRN-04` Business Continuity, "Operations continue despite the loss of any single component", status `Approved`.
4. Write the segment list. Eight is a defensible starting architecture and volume 1 chapter 19 has the reasoning; the three that are most often missing are in the table below. Make the argument on clinical safety grounds, not on information security grounds, because the standard that assigns the duty assigns it to the health care delivery organisation across safety, effectiveness and security together [5].
5. Before you segment anything, do dependency discovery. Devices depend on network services in undocumented ways — time, name resolution, address assignment, multicast behaviour, and in older devices hard-coded addresses. Segmenting first and discovering afterwards is how a clinical network goes down on a Tuesday.
6. `STD-11 Network segmentation standard` on `O32` is `Current` with `Conformance Breach`. You are not introducing segmentation; you are closing a recorded breach of a current standard, and that is a stronger funding argument than a new initiative.

Segment	Why it is separate	What it must not reach
Devices that cannot be patched	Agents cannot be installed on a regulated device; monitoring replaces them	General internet; any administrative segment
Devices that can be patched	A managed cadence governed by the manufacturer's releases	Guest and patient wireless
Operational technology	Building management, pneumatic tube, nurse call, access control, theatre integration — frequently flat, frequently reachable	Clinical servers; the internet
Backup and recovery infrastructure	It must survive the compromise of what it protects	Every segment it backs up

THE SEGMENTATION QUESTIONS A CLINICIAN HAS TO ANSWER, NOT AN ENGINEER

Which clinical workflows cross a segment boundary, and at what point in the workflow?

What is the emergency override, who can invoke it, and where is its use recorded?

Which devices lost function the last time a network change was made, and was that written down anywhere?

Two findings on the file, and what you do about each

`048 Environment` says production holds no personal data

7.6 — Repair `048`, and say which reading you took

Sheet `048`, `041 Security Control`, `064 Security Policy`, `065 Enforcement Point` · **Input** the provisioning and refresh records · **Output** a corrected `048` and a dated masking position · **Signed by** Chief Data Officer, countersigned by Security Architect · **Time** half a day to read, three months to fix

1. `048` columns are `Ref`, `Environment`, `Description`, `Personal data present`. Four rows. `ENV-01 Production` reads `No`. `ENV-02 Pre-production` reads `Yes`. `ENV-03 Test` reads `No`. `ENV-04 Development` reads `Yes`.
2. A provider's live patient-record estate is recorded as holding no personal data, while two of the three non-production environments are recorded as holding it. Note the fourth inconsistency the headline finding does not name: `ENV-03 Test` is `No` while `ENV-02` and `ENV-04` are `Yes`, on an estate where the refresh process is the same for all three.
3. Read the rest of the file before you decide what the column means. `SEC-07` on `041` is "Non-production masking / Personal data masked outside production", `Position Absent`. `EP-05` on `065` is "Non-production provisioning / Every environment refresh", `Position` "Most refreshes are unmasked". `POL-01` on `064` is "Personal information is protected wherever it is", `Scope` "Applies across production and non-production", `Position Breached`.
4. **Two readings, and you must write down which one you took.** Either the column means "unmasked copies present", in which case `ENV-01`'s `No` is the only honest cell on the sheet and the three non-production rows are the finding; or the column means what its header says, in which case `ENV-01` is false. The header text — `Personal data present` — supports the second. Take the second, record the first as the alternative, and do not quietly pick one in a steering pack.
5. Correct `ENV-01` to `Yes` and date the correction. Then correct `ENV-03` only when somebody has looked at a test refresh, not because the other two say `Yes`.
6. `STD-13 Environment standard` on `032` is `Current` with `Conformance Breach`. Attach the corrected `048` to that breach rather than raising a new finding.

Form 7-D — Environment position, feeding `048 Environment`

Four rows go into `048`, which has four; the rules below them are for the second pass, when an environment is added or a shared one is split out and has to be assessed on its own. Row 1 is the worked example — Rietvlei District Health Service. Signed by the Chief Data Officer. The fifth column is not on `048` and is the reason you can defend the fourth; keep it on the form and in the covering note.

Ref	Environment	Description	Personal data present	Date last refreshed, and masked or not
ENV-01	Production	Live service	Yes	n/a — production is not refreshed from anywhere

Typed straight into Excel: `Ref`, `Environment` and `Description`, which already exist and should not be retyped. Key only `Personal data present`, and keep the fifth column out of the sheet.

FINISH 7.6 WITH THESE

On what date was each non-production environment last refreshed from production?

Was that refresh masked, and who can produce the evidence that it was?

Which reading of `Personal data present` did you take, and where is that decision recorded?

`049` gives the clinical record of truth twenty-four hours

7.7 — Re-derive the recovery objectives clinically

Sheet `049 Resilience Tier`, `023 Application Component`, `003 Objective`, `004 Measure` · **Input** the clinical function decomposition in step 6 below · **Output** a recovery objective per clinical function, and `049` rewritten against it · **Signed by** Chief Operating Officer, on clinical advice · **Time** two workshops and a fortnight

1. `049` columns are `Ref`, `Application`, `Tier`, `Recovery objective`, `Tested`. Twelve rows, one per application. Read the `Tier` column against the `Recovery objective` column. `RES-01 EHR / EPR` is `Tier 1` at `24 hours`. `RES-02 PAS` is `Tier 3` at `15 min`. `RES-09 Incident & Risk` is `Tier 3` at `15 min`. The loosest objective in the file sits in the tightest tier.
2. Now read the whole column rather than the four rows that make the point, because the honest statement is sharper than "inverted". `Tier 1` holds `24 hours` twice, `4 hours` twice, `1 hour` once and `15 min` once. `Tier 2` holds `1 hour`, `15 min` and `4 hours`. `Tier 3` holds `15 min` twice and `1 hour` once. **The tightest objective in the file appears in all three tiers, and the loosest appears only in Tier 1.** The tier label carries no information at all, and in the one place it carries any it points the wrong way. Say that, not "inverted" — somebody will check.
3. Cross `049` against `023`'s `Criticality` column. `A07 Revenue Cycle / Claims` is `Critical` and sits at `Tier 2`. `A09 Incident & Risk` is `Critical` and sits at `Tier 3`. `A01 EHR / EPR`, described on `023` as "The clinical record of truth", is `Medium` criticality, `Health At risk`, `In service` since 2013 — and `Tier 1`, `24 hours`, `Never tested`.
4. Count the testing. `Tested` reads `Tested` on three rows — `RES-04 RIS / PACS`, `RES-05 ePrescribing & Medicines Admin`, `RES-12 Clinical Data Warehouse`. **Nine of twelve have never been tested, including both `Critical` applications and the clinical record.** `OBJ-13` on `003` is "Recovery objectives tested on cadence", `Horizon` 2 years, status `Not started`. `MSR-13` on `004` is "Recovery objectives tested", `Unit count`, `Baseline` 43, `Cadence Monthly` — a baseline of 43 for a measure with twelve possible subjects.
5. Then throw the sheet away and derive the objectives again, by clinical function rather than by application. **A recovery objective derived from financial loss per hour is the wrong answer here, because the loss function is neither monetary nor linear.** Volume 1 chapter 19 has the six-step method; the steps that change `049` are these three.

6. Decompose by clinical function. Not "the record", but identity resolution; allergy and current medication retrieval; order placement; result delivery; result acknowledgement and escalation; imaging retrieval; medication administration recording; the theatre list; blood product cross-match validity. `049` has no row for a function, and that is the sheet's real defect — twelve applications where the organisation needed nine functions.
7. For each function ask the clinical question — how long before a patient is harmed — and then ask what the manual substitute is and how long it holds. The South African evidence on the second question is that manual collation of laboratory results worked and produced a backlog that outlived the incident [7]. A manual process that holds for four hours does not hold for four weeks.
8. Express the recovery point objective in clinical terms too. "We may lose one hour of data" means a clinician may not see the last hour of observations, orders and administrations. For some functions the acceptable recovery point is zero, and that is a design constraint rather than an aspiration.
9. Rewrite `049` with the derived objectives, then put a tier label on it afterwards so that the tier is a consequence of the objective rather than a decoration on top of it. State the degraded-mode design for each function and get it clinically approved; the approval is the artefact that makes the design defensible.

Form 7-E — Recovery objectives, feeding `049 Resilience Tier`

Twelve rows for a first pass against the current sheet, then one row per clinical function once the decomposition in step 6 is agreed. Row 1 is the worked example — Driehoek Health Group. Signed by the Chief Operating Officer on clinical advice. The sixth column is not on `049`; it is the derivation, and without it the fourth column is an assertion.

Ref	Application	Tier	Recovery objective	Tested	How long before a patient is harmed
RES-01	EHR / EPR	Tier 1	15 min read, 4 hours write	2027-02-11, partial	minutes for allergy and medication retrieval

Typed straight into Excel: `Ref` and `Application`, which already exist. `Tier` is keyed last, after the objective, and the sixth column goes into the covering note rather than the sheet.

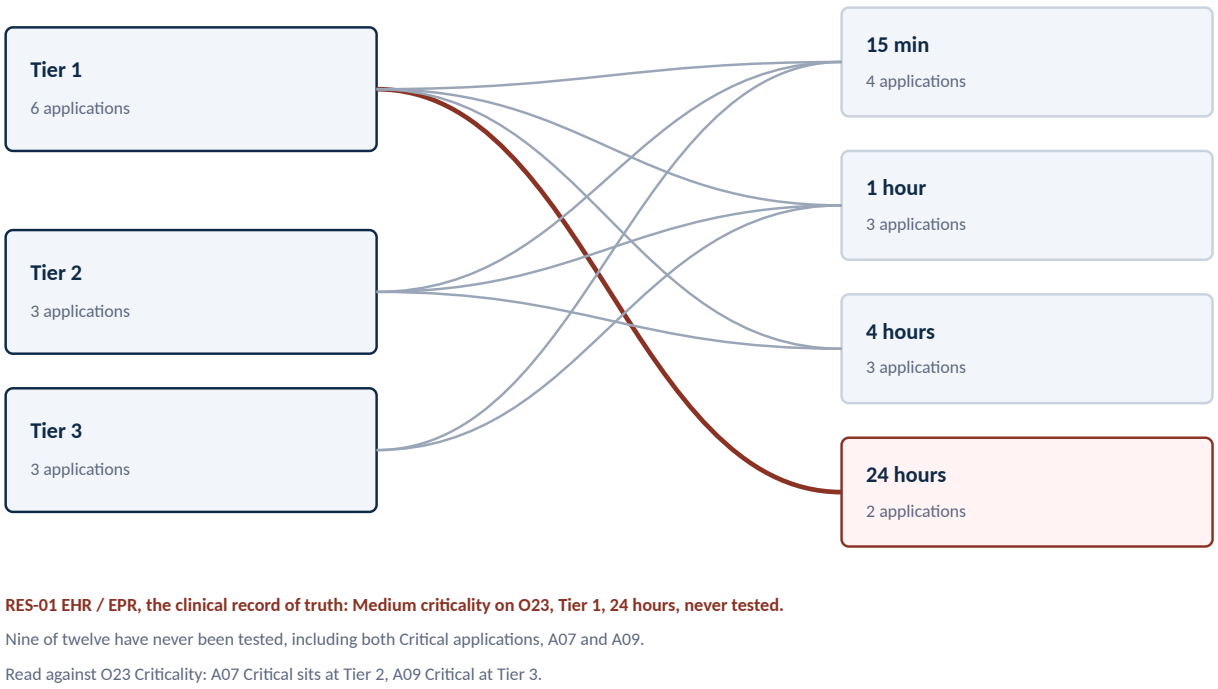


Figure 7.3 `049` read in full rather than four rows at a time. The honest statement is not that the tiering is inverted but that the tier label carries no information: the 15-minute objective appears in all three tiers, and only the 24-hour one is confined to Tier 1 — where the clinical record sits, untested since it went into service in 2013.

The read-only record becomes a deliverable with a number

Volume 1 chapter 19 argued the read-only record as the highest-value single resilience investment in a clinical estate, and the one most often omitted because it looks like duplication. The specification is published and specific: **a read-only backup record, updated at least hourly, tested weekly, and able to print** [1].

In Phase D it stops being a principle. It becomes a row on `O26` with a status, a row on `049` with a tested date, and **a measurement** — the number of minutes it takes to put one patient's record in a clinician's hand with the primary system gone. That number is the Phase D measurement of overlay 1, and the overlay only closes when there is a number rather than an intention. Preliminary set the retention clock; Phase C recorded it; Phase D measures retrieval; Phase G gates on the measurement; Phase H re-runs it. Everything before this item is a commitment. This is where it gets a stopwatch against it.

The reason it must be measured rather than asserted is in the audit record. The Auditor-General found 15 of 27 facilities with missing or lost records, 17 not tracking records after issue, and **three of nine paid medico-legal claims conceded because the record could not be found** [8]. A record that cannot be produced is, for every practical purpose, a record that does not exist — and the organisation pays for it in a courtroom rather than in an outage.

7.8 — Measure the retrieval, then publish the number

Sheet `026` , `049` , and a measurement record the repository has nowhere to hold · **Input** the read-only record as built · **Output** a dated retrieval time with a method and a worst case · **Signed by** Chief Operating Officer · **Time** one afternoon per site, repeated

1. `LTC-06` on `026` is "Zero-downtime clinical systems / Resilience appropriate to life-critical use", `Status` `Target` . `047` records the same component as `Candidate` . Nothing records it as running. `TBB-06` on `055` carries the acceptance criterion "Zero-downtime clinical systems in production for at least one service" and `R13` assigns it to `WP-12 Supplier and partner integration` , which is the round-robin at work and not a plan.
2. Build to the five properties, each of which is a yes or no and not a judgement. **Hourly currency** — achievable with log shipping or snapshot replication. **Weekly test** — a restore and a read, not a replication check. **Printable** — because the next step after reading is carrying the information to a bedside. **Independently reachable** — separate authentication, separate network path, separate hosting; a read-only copy inside the same failure domain as the primary is decoration. And **audited like the primary**, because unauthorised access to the copy is as criminal under the National Health Act as unauthorised access to the original [6], and a downtime copy with relaxed access control is a standing privacy risk rather than a resilience asset.
3. Then measure. Ten trials, on the live read-only record, with the primary unreachable from the workstation used. Pick the patients at random from the admitted list. Start the clock when a clinician is asked for a named patient's current medication and allergy list; stop it when they have it in their hand on paper. Record every trial, not the median only, because the worst case is the one that harms somebody.
4. Record four numbers: the median, the 95th percentile, the worst trial, and the number of trials that failed to produce a record at all. The fourth number is the one that gets attention.

5. Write the currency lag separately. "Hourly" is a design intent; the measurement is the age of the oldest record in the copy at the moment of the trial.
6. Then do it unannounced, with clinicians who have not used it, because a drill run by the team that built the thing measures the team and not the estate.
7. Put the result on `O49` as `Tested` with a date, and on `O26` as `Status Current` only once the number exists. Until then `LTC-06` stays `Target` and you say so out loud.

THE RETRIEVAL MEASUREMENT, PHASE D

Site, and the date the measurement was taken

Median time from request to a printed record, over ten trials

95th percentile, and the worst single trial

Trials that produced no record at all, out of ten

Age of the oldest record in the read-only copy at the moment of the trial

Was the drill announced, and had the clinician used the copy before

Name and signature of the clinician who held the stopwatch

THE NUMBER IS THE DELIVERABLE, NOT THE ARCHITECTURE

An architecture document that says "a read-only record is maintained" has said nothing. One that says "median 9 minutes, 95th percentile 31 minutes, worst trial 54 minutes, one of ten trials produced no record, oldest record in the copy 71 minutes, unannounced, 19 November 2026" has said everything, and the reader can act on it the same afternoon. Those figures are the shape of a first measurement at a single site and are offered as a shape, not a benchmark — the only number that matters is the one taken at the site in front of you, on a date written next to it. Re-run it after every change to the primary, and Phase H re-runs it on a cadence.

7.9 — Close the security position Phase D is accountable for

Sheet `O41`, `RPT11`, `O64`, `O65` · **Input** 7.6's corrected `O48` and 7.5's segment list · **Output** a dated control position with three named owners · **Signed by** Security Architect · **Time** one day

1. `O41` columns are `Ref`, `Security control`, `What it does`, `Position`, `Enforced at`. Eight rows. `RPT11 Security Position` is a verbatim copy with `Control` in place of `Security control`, and its row 14 footer — "3 controls are absent. A control that is absent is not a control." — is **correct**, which makes it one of the few footers in this workbook that is.
2. The three are `SEC-02 Design review`, `Enforced at Design gate`; `SEC-04 Attributable access`, `Enforced at Application`; and `SEC-07 Non-production masking`, `Enforced at Provisioning`. Each has an enforcement point named for a control that does not exist.
3. Read `O65` against them. `EP-01 Design review gate` is "Newly established". `EP-02 Identity provider` "Enforces authentication, not attribution". `EP-03 API gateway` "Product not yet selected". `EP-04 Data platform` "Classification not applied at read". `EP-05 Non-production provisioning` "Most refreshes are unmasked". **All five enforcement points record a defect in the column that says what they enforce**, and all four policies on `O64` read `Breached`.
4. Two of the three absent controls are Phase D's to close and one is not. `SEC-07` closes with the provisioning change from 7.6. `SEC-04` closes at the application, not the platform, and belongs to item 6's phase — but the device register from 7.4 is where a device's access becomes attributable at all, because an unidentified device on a clinical segment is an unattributable access path.
5. Carry overlay 2 through this step. Classification inherits downward: an attribute may be tighter than its entity and may never be looser. `EP-04`'s "Classification not applied at read" is the enforcement failure that makes the 26 `O37` attributes classified more openly than their parent entity *effective* rather than merely wrong on a sheet. And `E14 Medical Device` is classified `Public` on `O20` while the observations devices produce land in `E03 Clinical Observation` at `Confidential`. **A device record at `Public` feeding an entity at `Confidential` is the overlay broken in the technology layer**, and the check is one line: no row of `O68` may be classified more openly than the entity its readings land in.
6. Date the control position. A control position with no date is a control position somebody will quote in eighteen months.

7.10 — Re-open and re-sign the Phase D deliverables

Sheet `O31 Deliverable` , `O58 Governance Gate` , `O61 Governance Change Request` · **Input** everything from 7.1 to 7.9 · **Output** two re-signed deliverables and one change request · **Signed by** Architecture Board at `G1` · **Time** one board meeting

1. `DLV-11 Technology Architecture` and `DLV-12 Technology Standards` on `O31` are both `Phase D` , `Status Approved` . Neither can be left approved on the evidence in 7.1 to 7.9.
2. `GT-1` on `O58` is "G1 Design / Before build / Design meets principles, standards and reference architecture". `O32` records `STD-11 Network segmentation standard` and `STD-13 Environment standard` as `Current` with `Conformance Breach` . A design cannot pass `G1` against a standard it breaches, so either the breach closes or the gate records a dispensation with an expiry — which is what `GB-01 Architecture Board` on `O57` is mandated to do.
3. Raise the change request on `O61` . Its columns are `Ref` , `Change request` , `Against` , `Type` , `Status` , `Expires` . Follow the file's own pattern: `CR-G03` is raised `Against Repository` with `Type Correction` , so a correction to the technology domain takes the same shape. Set an expiry; `CR-G02` and `CR-G03` have an empty `Expires` cell and `CR-G03` has sat at `Raised` .
4. Attach five numbers to the board paper and nothing else: components past support with the date counted; devices on the register and devices with `Risk class Not established` ; applications with a clinically derived recovery objective, out of twelve; applications tested, out of twelve; and the retrieval median with its worst trial. Each one dated, each one moving.
5. Do not take the repository to the board. Take the five numbers; keep the repository for architects.

WHAT PHASE D CHANGED, IN ONE LINE EACH

`RPT10` 's footer is 1 and carries a date. `R07` is a deployment record rather than two zipped lists. `O68 Medical Device` exists, with nine columns and a named human in the `Owner` cell of every row. Segments are named and `O68` points at them. `O48` 's production row is `Yes` and the reading taken is written down. `O49` 's objectives come from clinical tolerance and carry a tested date. And the read-only record has a median, a 95th percentile, a worst trial and a count of failures, taken on a stated date at a stated site. Items 8 to 12 inherit all seven and none of them re-derives any of it.

THE CHAPTER IN TEN LINES

- Phase D here is four sheets and thirty-nine records, and not one of them is a clinical device.
- RPT10's footer says two components are past support; on 25 October 2026 the answer is one, and the statement needs the date it was taken on.
- R07 pairs application n with platform component n, so the billing system runs on the medical device gateways and the portal runs on backup and archive.
- The extension sheet says a medical device holds calibration, maintenance and risk class; calibration appears twice in the whole workbook and risk class once.
- O68 Medical Device carries Ref, Device, Model, Risk class, Software version, Calibration due, Maintenance due, Network segment and Owner, prefix DEV-.
- There is no national device register to point at, so the one you build is the only one there is.
- Segment on clinical safety grounds, do dependency discovery before you segment, and write the segment name into O68 from the list and nowhere else.
- O48 records production as holding no personal data while two of three non-production environments hold it; say which reading of the column you took.
- O49's tier label predicts nothing about the recovery objective beside it, and nine of twelve applications have never been tested, including both Critical ones.
- Recovery objectives are derived from clinical tolerance by function, not from financial loss by system, and the degraded mode is clinically approved.
- The read-only record becomes a Phase D deliverable with a median, a 95th percentile, a worst trial and a count of failures, taken unannounced on a stated date.
- Classification inherits downward and never loosens: no row of O68 may be classified more openly than the entity its readings land in.
- Re-open DLV-11 and DLV-12, take five dated numbers to the board, and leave the repository with the architects.

SOURCES FOR THIS CHAPTER

- **1.** SAFER Guide: Contingency Planning, August 2024, Office of the National Coordinator for Health Information Technology. United States guidance; the read-only record specification, the drill cadence and the response-time thresholds https://www.healthit.gov/sites/default/files/playbook/pdf/2-contingency-planning-final.pdf
- **2.** SAHPRA, SAHPGL-MD-04 version 5, Guideline for the Classification of Medical Devices and IVDs, effective 27 February 2025. The four risk classes and the classification responsibility; described, not reproduced https://www.sahpra.org.za/wp-content/uploads/2025/03/SAHPGL-MD-04_v5-Guideline-for-Classification-of-MD-and-IVDs.pdf
- **3.** SAHPRA, Communication to industry MD08-2025/2026, Regulatory Requirements of Artificial Intelligence and Machine Learning Enabled Medical Devices, 26 September 2025. The locked algorithm version and the change-control position https://www.sahpra.org.za/wp-content/uploads/2025/09/MD08-20252026_-SAHPRA-Communication-to-Industry-AI-Medical-devices_Acknowledgements.pdf
- **4.** SAHPRA, Medical devices and in-vitro diagnostics. The establishment licensing regime and the unpublished registration call-up plan https://www.sahpra.org.za/medical-devices/
- **5.** IEC 80001-1:2021, Application of risk management for IT networks incorporating medical devices, part 1. Catalogue entry; the standard's content is described, not reproduced https://www.iso.org/standard/72026.html
- **6.** National Health Act 61 of 2003, consolidated text; section 17(2) on unauthorised access to a health record https://www.saflii.org/za/legis/consol_act/nha2003147.pdf
- **7.** Cassim and Chapanduka, on the National Health Laboratory Service ransomware incident and its operational consequences, South African Medical Journal 2024;114(12):e2549 https://www.scielo.org.za/pdf/samj/v114n12/06.pdf
- **8.** Auditor-General findings on the health sector, reported by MedicalBrief, 14 October 2026; missing and untracked records, and medico-legal claims conceded for want of a record https://www.medicalbrief.co.za/health-sector-crippled-by-recurring-systemic-weaknesses-ag-warns/

ITEM 8

Opportunities and Solutions

Two gap records against twenty-six the reports already compute, and a programme that costs a third more than the plan it implements.

`029 Gap` holds two records. The same workbook reports ten capabilities with a maturity gap, six standards in breach, three absent controls, six obligations with no evidence and a component running past vendor support. Both gaps on `029` say the same thing in the same words — "capability uplift not in place" — and both are `Business` domain, both `Open`, both pointed at a capability building block. Nothing on the sheet is about data, applications, technology, security or regulation, and the file has findings in all five.

That is the Phase E problem in this estate, and it has a specific shape worth naming: **the gap register was written from the first two rows of a building-block list rather than from the reports.** So the sequence in this item inverts the usual one. You do not read `029` and cost it. You derive the gap set from the reports, write it to `029`, and only then cost it — because a gap register assembled before the reports were read will always be the size of whoever's patience ran out.

The repository knows. `CR-G03` on `061 Governance Change Request` reads "Raise unfunded building blocks as gaps", `Against Repository`, `Type Correction`, `Status Raised`, with an empty `Expires` cell. The organisation raised a change request against its own repository to do exactly what this item does, and left it at `Raised`.

Five kinds of gap, and only one of them is on the sheet

Before the derivation, a distinction the sheet's `Domain` column half-makes and then drops. The reports in this workbook produce five different kinds of gap, and they are not interchangeable, because each is closed by a different thing and signed by a different person.

Kind	What the report gives you	What closes it
Capability	A current and a target maturity, and the arithmetic between them	A business change, usually people and process before systems
Conformance	A standard with a breach against it	A design change, or the standard's withdrawal and reissue
Control	A control recorded `Absent` or `Partial` with an enforcement point named	A mechanism, built once and then operated
Obligation	An obligation with no evidence behind it	Evidence, produced on a cadence, which is a running cost and not a project
Currency	A component past vendor support, with a date	Money and an outage window

`029`'s two rows are both the first kind. The other four kinds are absent from the sheet and present in the reports, which is why reading the reports first changes the answer rather than merely enlarging it. And the fourth kind is the one architects most often mis-plan, because an obligation gap closes into a permanent operating cost rather than into a delivered work package, and `030` has no column for a running cost.

What Phase E has to hand over

Artefact	Where it lands	Signed by
The derived gap set	`029 Gap`	Chief Enterprise Architect
Gap to work package mapping	`R12`	Head of Delivery
Work package to building block mapping	`R13`	Head of Delivery
A reconciliation of the plan to the chosen alternative	`054`, and a covering note	Investment Committee
Implementation factors that are about this industry	`052 Implementation Factor`	Chief Operating Officer

`DLV-13 Consolidated Gaps` on `031 Deliverable` is `Phase E`, `Status Approved`. `DLV-14 Architecture Roadmap` and `DLV-15 Work Packages` are both `Draft`. The approved one is the one that is wrong.

8.1 — Derive the gap set from the reports, not from `029`

Sheet `RPT03`, `RPT12`, `RPT11`, `RPT09`, `RPT10`, and `029` last · **Input** the five reports · **Output** a candidate gap list with a source reference on every row · **Signed by** Chief Enterprise Architect · **Time** one day

1. Open `RPT03 Capability Heatmap`. Columns are `Ref`, `Capability`, `Criticality`, `Current`, `Target`, `Gap`, `Disposition`. Ten rows, one per level-1 capability, and the `Gap` column is `Target` minus `Current` and correct in all ten. **Every one of the ten has a gap of 1 or 2.** Four are `Build`: `C03 Diagnostics`, `C06 Clinical Governance & Safety`, `C07 Revenue Cycle & Claims`, `C09 Estates & Medical Devices` and `C10 Population Health & Analytics` — five, and the arithmetic is worth checking yourself rather than taking from me. `C09` is `Current` 1, the lowest in the file.
2. Open `RPT12 Standards Conformance`, which is `032 Standard` copied whole. Six rows read `Conformance Breach`: `STD-05`, `STD-06`, `STD-11`, `STD-13`, `STD-15`, `STD-16`. Two of the six are `Status Deprecated` as well — `STD-15 Records management standard` and `STD-16`

`Supplier security standard`. **A breach of a withdrawn standard is not a gap anybody can close**, so those two become a different kind of gap: the standard needs reissuing before the breach means anything. Carry both as gaps and say which is which.

3. Open `RPT11 Security Position`. Three rows read `Position Absent`: `SEC-02 Design review`, `SEC-04 Attributable access`, `SEC-07 Non-production masking`. The footer is correct.
4. Open `RPT09 Regulatory Coverage`, which is `O63 Control Objective` copied whole. Columns are `Ref`, `Obligation`, `Source`, `Control objective`, `Evidence`. **Give all three readings and do not pick one.** Two rows are literally without evidence; six of seven are without evidence on the report's own stated test, every one except `CO-03 Medical device regulation` at `Partial`; and on the plain question "is any obligation evidenced", the answer is none of the seven. Three numbers, one sheet, and a steering pack that quotes one of them without the other two is misleading whichever one it picks.
5. Open `RPT10 Technology Currency`. One component is past vendor support on a reference date of 25 October 2026 — `PTC-04`, `Support end` 2026-05 — and a second, `PTC-06` at 2026-12, joins it in January. The report's own footer says two. Item 7 has the correction and the date rule.
6. Add up what the reports alone give you: 10 capability gaps, 6 standards in breach, 3 absent controls, 6 obligations without evidence on the report's stated test, 1 component past support. **Twenty-six candidate gaps against `O29`'s two**, every one of them computed from a sheet in the same workbook, and not one of them requiring an opinion.
7. Write the derivation down as a table before you write a single `O29` row, because the table is what makes the number defensible and the row is what makes it actionable.

Report	Column read	Test	Count
`RPT03` Capability Heatmap	Gap	Greater than zero	10 of 10
`RPT12` Standards Conformance	Conformance	Equals Breach	6 of 16
`RPT11` Security Position	Position	Equals Absent	3 of 8
`RPT09` Regulatory Coverage	Evidence	Not evidenced, on the report's stated test	6 of 7
`RPT10` Technology Currency	Support end	Earlier than the date of reading	1 of 12

1. Only now open `O29 Gap`. Columns are `Ref`, `Gap`, `Domain`, `Building block`, `Severity`, `Status`, `Due`. `GAP-01` is `Critical`, `Due` 2029-11, pointed at `BBB-01`. `GAP-02` is `Medium`, `Due` 2029-08, pointed at `BBB-02`. `RPT07 Gap Analysis` is `O29` plus `R12`'s work package and is correct as a copy. Neither gap is overdue on the reference date, which is the one thing `RPT01`'s "Gaps overdue: 0" gets right.

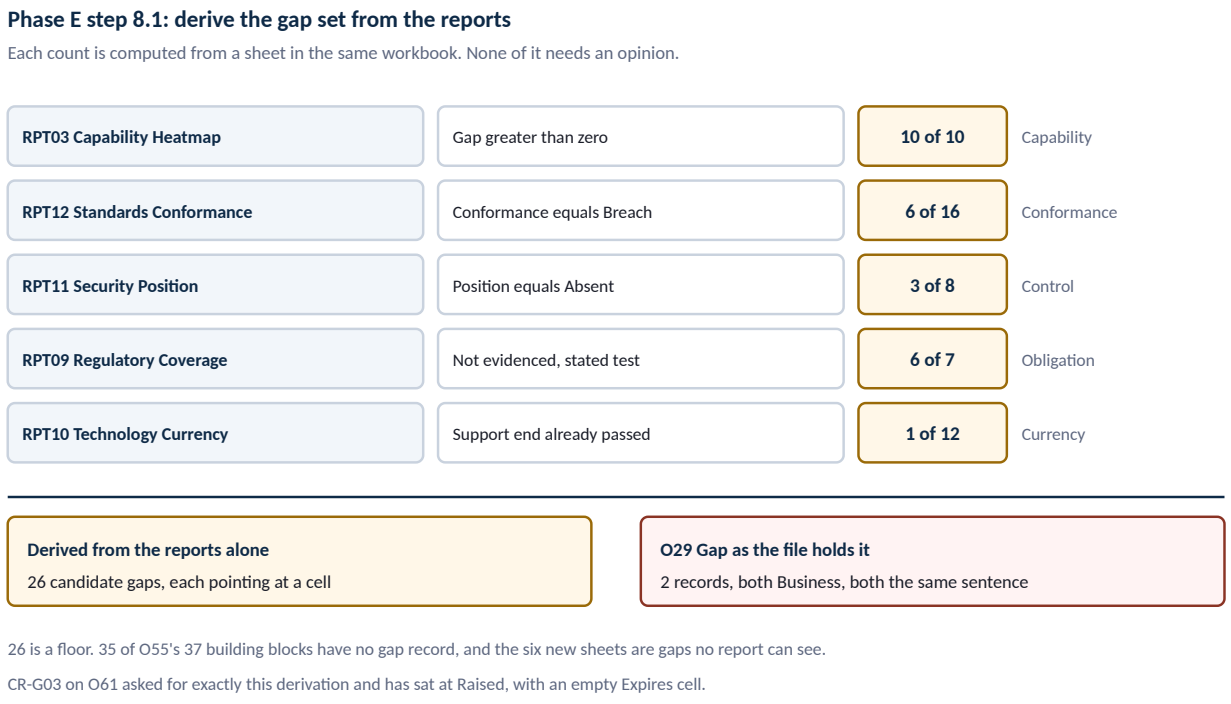


Figure 8.1 The Phase E step in one page. Five reports, five tests, five counts, and a candidate set an order of magnitude larger than the sheet it replaces — derived rather than argued. The pattern this caricatures is the gap register written in the first week and never revisited once the reports existed.

8.2 — Add the gaps no report in this file can see

Sheet `O55 Building Block`, `O20`, `O48`, `O49`, `RPT13` · **Input** 8.1's candidate list · **Output** the gap set, with the unreportable gaps flagged · **Signed by** Chief Enterprise Architect · **Time** half a day

1. Twenty-six is a floor, not a census. The reports in this workbook compute almost nothing: nine of the thirteen are a single object sheet with one column dropped or one footer added, and only `RPT03` and `RPT13` compute anything at all. A gap that no sheet holds a column for cannot appear in a report.
2. `O55` holds 37 building blocks. `O29` points at two of them. **Thirty-five building blocks have no gap record**, and `R12` closes both of the two it has, so on the sheet's own terms gap coverage is 100% and ten of the twelve work packages close nothing.
3. Note what `CR-G03` cannot actually test. It asks for "unfunded building blocks" to be raised as gaps — but **no sheet in this repository records funding against a building block.** `O30 Work Package` costs twelve work packages; `R13` maps packages to blocks; nothing costs a block. So the change request asks for a derivation the file cannot support, and the first thing Phase E owes it is a cost per block or a statement that there is none. Write that into the change request before you close it.
4. Add the gaps the new sheets create. Items 2, 5 and 7 tell the reader to create `O67 Consent`, `O68 Medical Device`, `O69 Code System`, `O70 Measure Definition`, `O71 Disclosure Event` and `O72 Retention Clock`. Each is a gap in this phase, because each is a thing the architecture needs and the repository has nowhere to hold. The argument is already made in those items; Phase E's job is to get them onto `O29` with a severity and a due date so that somebody funds them.
5. Add the gaps items 5, 6 and 7 established that no report reaches. Not one of the 14 data entities on `O20` is mastered in a system that could own it. Twenty-six of 60 attributes on `O37` are classified more openly than their parent entity, against the file's own `PRN-14`. `O48` records production as holding no personal data. `O49`'s tier labels carry no information about the recovery objectives next to them, and nine of twelve applications have never been tested. `RPT13 Orphan Analysis` reports two orphans; twenty object sheets, 206 records, appear in no relationship at all.
6. Add one Phase E gap of its own, and it is the one in step 8.6: the plan costs 33% more than the alternative it implements, and no sheet reconciles the difference.
7. Flag every row with its source. A gap with no source reference is a gap somebody will argue with, and the whole value of this derivation is that each row points at a cell.

Form 8-A — The derived gap set, feeding `O29 Gap`

Twenty-six rows minimum for a first pass, one per candidate from 8.1, before 8.2's additions. Row 1 is the worked example — Ravensmere, read straight off `RPT03`. Signed by the Chief Enterprise Architect. The sheet has seven columns and this form carries six; `Building block` is the seventh and is typed in Excel from `O55` after the row exists.

Ref	Gap	Domain	Severity	Status	Due
GAP-03	Estates & Medical Devices at maturity 1 against target 3	Business	Critical	Open	

Typed straight into Excel: `Building block`, which is a lookup from `O55`, and the source reference, which `O29` has no column for and which goes in the covering note keyed by `Ref`. Key the six above by hand, because each one is a judgement about severity that a lookup cannot make.

BEFORE `029` IS REWRITTEN

How many gaps did the reports alone give you, and how many did you add from items 5 to 7?

Which rows have no source reference, and why are they on the list?

Who agreed the severity scale, and is it the same scale `GAP-01` and `GAP-02` were set on?

8.3 — Re-point `R12` so that the mapping is not a coincidence

Sheet `R12` , `O29` , `O30` · **Input** the rewritten `O29` · **Output** a gap-to-package mapping with no unmapped gap and no package that closes nothing · **Signed by** Head of Delivery · **Time** half a day

1. `R12 Gap closed by Work Package` columns are `Gap` , `Gap name` , `Severity` , `Work package` , `Package name` . Two rows. `GAP-01` to `WP-01 Governance foundations` ; `GAP-02` to `WP-02 Data ownership and mastering` .
2. Read that pairing. `GAP-01` is the Patient Access & Scheduling capability uplift, closed by a governance foundations package. `GAP-02` is the Clinical Care Delivery capability uplift, closed by a data ownership package. Both are the row-order pairing that runs through this repository: gap *n* joined to package *n*.
3. Re-point both, then map every new gap. A gap with no package is unfunded and should say so in `Status` . A package closing no gap is a package with no stated purpose, and ten of twelve are in that position today.
4. Expect the mapping to be many-to-many and resist the sheet's shape, which is one row per pair. One package will close several gaps and one gap will need several packages — the segmentation gap from item 7 needs the technology currency package and the access and security package, and neither closes it alone. Add rows; do not collapse the truth into the shape.

8.4 — Rebuild `R13` so that a package delivers the blocks its own name describes

Sheet `R13` , `O30` , `O55` · **Input** `R13` 's 37 rows · **Output** a corrected mapping · **Signed by** Head of Delivery · **Time** one day

1. `R13 Work Package delivers Building Block` columns are `Work package` , `Package name` , `Building block` , `Domain` , `Block name` . Thirty-seven rows, which is exactly `O55` 's count, with `WP-01` to `WP-12` cycled through the list in order and wrapping three times.
2. The consequence is that no work package delivers the blocks its name describes. **`WP-04 Technology currency` delivers no technology building block at all** — it gets `BBB-04` Medicines Management capability uplift, `DBB-06` Medication Order mastering and `ABB-08` Rostering target. `WP-02 Data ownership and mastering` delivers one data block of the ten, `DBB-04 Diagnosis mastering` . `WP-03 Integration platform` delivers no integration block. **`WP-11 Customer experience` , a `Business` package at $2.7m, delivers `DBB-01 Patient mastering` , `ABB-03 LIS target` and `TBB-05 Medical device integration` .**
3. That last row is the one to show a steering committee, because it is checkable in ten seconds and it makes the general point without a lecture: medical device integration is being delivered by the customer-experience project.
4. Rebuild it by reading `O55` 's `Domain` column against `O30` 's `Domains` column and pairing on domain first, then on subject. The arithmetic is below, and it will not balance. Where it does not, the finding is a missing work package rather than a spare building block — and three `Business` packages carrying ten capability uplifts is the honest reading of why `C09 Estates & Medical Devices` has sat at maturity 1.

Domain	Building blocks on O55	Work packages on O30	Blocks per package
Business	10	3: `WP-01` , `WP-08` , `WP-11`	3 to 4
Data	10	3: `WP-02` , `WP-06` , `WP-10`	3 to 4
Application	10	3: `WP-03` , `WP-07` , `WP-12`	3 to 4
Technology	7	3: `WP-04` , `WP-05` , `WP-09`	2 to 3

1. Check the `Domains` column itself as you go. The header is plural and every cell is singular, so a package spanning two domains has nowhere to say so — `WP-05 Access and security` and `WP-10 Compliance mechanisms` arguably both do, and both are keyed to one domain.
2. Then check the seven technology blocks specifically, because they are the ones item 7 depends on. `TBB-06 Zero-downtime clinical systems` carries the acceptance criterion "in production for at least one service" and is currently delivered by `WP-12 Supplier and partner integration` . The read-only record and its measured retrieval time from item 7 is what that block means, and it belongs with the resilience package.

Thirty-seven building blocks and four sentences

8.5 — Make the acceptance criteria testable

Sheet `O55 Building Block` , `O35` , `O44` , `O45` , `O46` , `O58` · **Input** `O55` 's 37 rows · **Output** an acceptance criterion per block that a gate can fail · **Signed by** Chief Enterprise Architect, countersigned by the Head of Delivery · **Time** three days

1. `O55` columns are `Ref` , `Domain` , `Building block` , `What it must do` , `Acceptance criteria` . Thirty-seven rows in four blocks: `BBB-01` to `BBB-10 Business` , `DBB-01` to `DBB-10 Data` , `ABB-01` to `ABB-10 Application` , `TBB-01` to `TBB-07 Technology` .
2. Read the `Acceptance criteria` column down the sheet. **All thirty-seven are one of four sentences with a name substituted into it.** A capability block reads "<name> measured and owned." A data block reads "No <name> record without a named owner." An application block reads "No unmanaged interface to or from <name>." A technology block reads "<name> in production for at least one service." The same four frames appear on `O35` , `O44` , `O45` and `O46` for the same blocks.
3. Three of the four frames are not testable as written, and the fourth is testable and trivially passed. "Measured and owned" names no measure and no owner. "No record without a named owner" is a statement about every row of a table and has no sampling rule, no date and nothing that says who checks. "No unmanaged interface" is testable against `R06` , which records 13 of 23 interfaces as unmanaged — so the criterion is checkable and currently failed on ten of the twelve applications. And "in production for at least one service" passes the moment anything at all is switched on, which is why `TBB-01` to `TBB-07` can be marked complete without the capability existing anywhere a clinician can reach. Count the one that is checkable: `R06` records 13 of its 23 interfaces as `Unmanaged` , and those thirteen touch **every one of the twelve applications** as source or target, so the application frame is currently failed by all ten blocks that carry it.
4. `GT-2` on `O58` is "G2 Build complete / Before release / Every acceptance criterion evidenced". A gate that tests thirty-seven instances of four sentences tests nothing, and that is the Phase E defect that does the most damage downstream, because it is the defect that lets Phase G sign.
5. Rewrite each criterion to the same four-part shape, which is the convention the rest of this volume uses: **a named artefact, a named signatory, a date or a cadence, and a number that can fail.** "Medical device integration in production for at least one service" becomes " `O68` holds a row for every device class writing into the record at the three largest sites, each with a `Software version` and a named `Owner` , counted and signed by the Head of Estates on a stated date, with zero rows at `Risk class` `Not established` ."
6. Do the seven technology blocks first, then the ten data blocks, and leave the ten capability blocks last — they need a measure definition before they can have a criterion, and item 5 is where the measure definition is created.
7. Where a block's criterion cannot be made to fail, the block is not a building block. Say so and delete it rather than carrying a row that will be signed off in a meeting nobody remembers.

The plan costs more than the plan it implements

8.6 — Reconcile `O54` to `O30` , or stop the programme

Sheet `O54 Architecture Alternative` , `O30 Work Package` , `O56 Project` , `R14` · **Input** the four cost columns · **Output** a reconciliation, or a decision to re-baseline · **Signed by** Investment Committee · **Time** one week, and it is the most valuable week in Phase E

1. `O54` columns are `Ref` , `Alternative` , `Description` , `Cost` , `Recommendation` . Five rows. `ALT-A Do nothing beyond compliance` $24m `Rejected` . `ALT-B Foundations first` $46m `Recommended` . `ALT-C Business value first` $52m `Rejected` . `ALT-D Big bang replacement` $88m `Rejected` . `ALT-E Compliance led` $31m `Partly adopted` .
2. Add up `O30` 's `Cost` column. Twelve work packages total $61.3m. `O56` 's `Cost` column totals the same $61.3m, and `PRJ-n` 's cost, start and end equal `WP-n` 's in all twelve rows. `R14` 's cost column totals $61.3m. **Three sheets agree with each other and all three disagree with the recommendation.**
3. The recommended alternative was costed at $46m. The work packages that implement it cost $61.3m. That is **$15.3m, or 33%, above the figure the recommendation was made on** — and above `ALT-C` at $52m and `ALT-E` at $31m, both of which were set aside, one of them as too expensive. No sheet in the workbook reconciles the two numbers. There is no contingency row, no inflation row, no scope note and no variance column anywhere.
4. Note the state of the decision as well as the number. `DEC-01` on `O60 Governance Decision` is "Adopt the recommended architecture alternative", `Status Outstanding` — while `O54` already marks `ALT-B Recommended` and `ALT-E Partly adopted` . **Something is partly adopted before the decision to adopt anything has been taken**, and the body that owns the decision is `GB-02 Investment Committee` on `O57 Governance Body` , which holds "G0 funding and G3 benefit confirmation" and meets quarterly.
5. Fix the currency before you take it anywhere. Every cost in this repository is in US dollars, with no exchange rate, no rate date and no base year on any sheet — for a South African provider whose obligations, staff and suppliers are in rand. **A programme figure with no currency date is not a figure.** Restate the whole column in rand at a stated rate on a stated date, keep the dollar column beside it if the board is used to it, and say which is authoritative.
6. Then do the arithmetic that settles it, which is a reconciliation and not an argument. Four lines: what was in `ALT-B` 's $46m; what is in the twelve work packages; what is in the second and not the first; and what is in the first and not the second. One of those four lines is empty today and it is the second-to-last, because nobody has written the scope of `ALT-B` down anywhere.
7. Three outcomes are legitimate and a fourth is not. Re-cost `ALT-B` at $61.3m and re-take `DEC-01` on the real number. Or de-scope the work packages to $46m and say which gaps go unclosed. Or re-open the alternatives, because at $61.3m the comparison that produced `ALT-B` no longer holds — `ALT-C` was rejected at $52m. **What is not legitimate is proceeding with a recommendation made on a number the plan does not meet**, and the provincial funding position makes that the conversation of the year rather than a technicality [2].

Form 8-B — Cost reconciliation, feeding the covering note to `054`

Four lines at the top level, and the sheet has no column for any of them — **this form feeds no sheet the repository has**, and the reconciliation lives in the covering note to `054` with a row reference. The rules below the four are for the sub-lines each one breaks into, because "what is in the packages and not in the alternative" is never one line. Row 1 is the worked example — Ravensmere, read off `054` row 6. Signed by the Investment Committee.

Line	Amount, with currency and rate date	Source
The recommended alternative as costed	$46m, no rate, no date	`054` row 6, `ALT-B`

Typed straight into Excel: nothing. The reconciliation is four lines and a signature, and it is the one Phase E artefact that should never be in a spreadsheet, because the second it is a formula somebody will change an input.

Figure 8.2 The recommendation and the plan, on one axis. `ALT-B` was chosen at 46 and the work packages that implement it total 61,3 — above `ALT-C`, which was rejected as too expensive. Three outcomes are legitimate and proceeding is not one of them.

BEFORE COSTING ANYTHING, CHECK THE DELIVERY STATES

Seven of the twelve work packages on `O30` read `Delivered` in the `Delivery` column, and the earliest `Start` on the sheet is 2027-01 with the earliest `End` 2029-02. **Nothing has started.** `WP-05 Access and security` reads `Not started` for `Approval` and `Delivered` for `Delivery` on the same row. Three rows read `Past end date` and `ZZ Checks` reports "work packages past end date: 3", which is **0** on any date before 2029-02. Costing a plan whose status column is fiction produces a cost with the same standing, so set every `Delivery` cell to its real value before you add up a single column.

8.7 — Make `O52` about this industry

Sheet `O52 Implementation Factor` , `O62 Delivery Risk` , `O60 Governance Decision` · **Input** `O52` 's eight rows · **Output** a factor list a hospital chief operating officer recognises · **Signed by** Chief Operating Officer · **Time** one workshop

1. `O52` columns are `Ref` , `Implementation factor` , `Category` , `Deduction` . Eight rows, `Category` taking `Dependency` , `Constraint` , `Resource` , `Assumption` and `Risk` .
2. Read `IF-03` : "Peak trading freeze", `Constraint` , "No change in the peak window". **A hospital has no peak trading window.** It is template residue from a retail estate, and it is the single clearest evidence in the workbook that the implementation factors were not written for this organisation. The factor a provider has in that slot is different in kind: winter respiratory pressure, a public holiday weekend, the pay-week clinic surge, a strike, a mass casualty incident — each of which is a *clinical load* constraint rather than a revenue one, and none of which can be scheduled around a release window the way a retail freeze can.
3. Read `IF-08` : "Site access windows", `Constraint` , "Physical work limited to shutdowns". This one is right and under-stated. A ward does not shut down. Device work, cabling and segmentation at a clinical site happen around care, not instead of it, and the implementation factor is the escort, the infection control clearance and the clinical sign-off rather than the window.
4. Cross `IF-01` and `IF-02` against the decision sheet. `IF-01 Vendor selection not made` sits with `DEC-02` "Select the integration product", `Status Outstanding` , `RSK-01` "Integration product not selected" at `Certain` / `High` , and `EP-03 API gateway` "Product not yet selected" — while `WP-03 Integration platform` reads `Delivered` and `TA-2 Integrable` on `O53 Transition Architecture` describes "Managed integration carrying production traffic". **The integration platform is delivered and carrying production traffic before the product has been chosen.** `IF-02 Data owners not designated` has the same shape against `DEC-04` and `O20` , which names an owner for all fourteen entities.
5. Add the factors this industry actually imposes and the file has nowhere for. The shared services no provider owns and no provider can mitigate alone — volume 1 chapter 19 names four for South Africa and requires every architecture to declare them with the clinical function each one stops. The clinical training load, because a release that needs clinicians trained competes with a roster. And the paper fallback supply, which is a logistics dependency rather than an information technology one and is the thing that fails in week three of an extended outage [1][3].

Form 8-C — Implementation factors, feeding `O52 Implementation Factor`

Eight rows to replace the eight that are there, plus the additions from step 5. Row 1 is the worked example — Driehoek Health Group, replacing `IF-03` . Signed by the Chief Operating Officer. All four columns are `O52` 's own.

Ref	Implementation factor	Category	Deduction
IF-03	Winter respiratory pressure, June to August	Constraint	No clinical-system release in the window; device work deferred

Typed straight into Excel: `Ref` only. The other three are the workshop's output and should be keyed from the signed form so that there is a paper original of what was agreed.

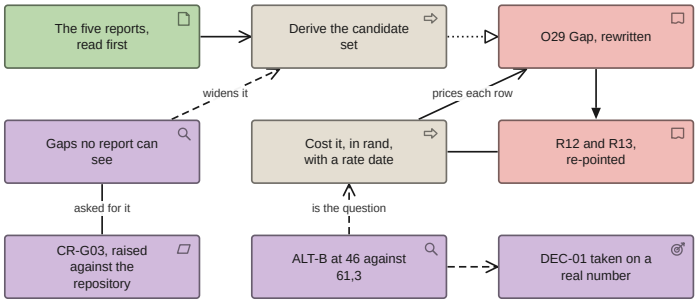


Figure 8.3 The order of work in Phase E, which is the reverse of the order the repository implies. Reports first, then the gaps no report can see, then `O29`, then the cost — and the cost question is the alternative, not the packages.

8.8 — Sequence the gap set into the transition architectures

Sheet `O53 Transition Architecture`, `O30`, `O66 Architecture Increment` · **Input** the rewritten `O29` and the corrected `R13` · **Output** a sequence with exit criteria · **Signed by** Chief Enterprise Architect · **Time** two days

1. `O53` columns are `Ref`, `Transition architecture`, `Description`, `Work packages`, `Exit criteria`. Three rows: `TA-1 Governed and supportable` with `WP-01, WP-02, WP-04, WP-05`; `TA-2 Integrable` with `WP-03`; `TA-3 Business visible` with `WP-07, WP-11`. **Five of the twelve work packages are in no transition architecture at all** — `WP-06`, `WP-08`, `WP-09`, `WP-10`, `WP-12` — including `WP-09 Resilience uplift`, which is where item 7's work lives.
2. The `Exit criteria` column holds a count, not a criterion: "5 exit criteria", "6 exit criteria", "7 exit criteria". Eighteen criteria are referred to and none is written down anywhere in the workbook. Write them. A transition architecture whose exit criteria are a number cannot be exited.
3. **Carry both time axes and do not reconcile them.** `O66 Architecture Increment` dates its increments "Month N" with no baseline date anywhere in the file; `O56 Project` uses absolute dates from 2027-01 to 2031-08. The workbook holds two incompatible time axes and nothing in it establishes a programme start, so do not invent one. Report the internal ordering problem instead: `INC-1` lands at month 10 and is delivered by the latest-starting of the six projects `R23` pairs to increments.
4. Sequence on dependency rather than on value. `IF-01` blocks integration work until the vendor decision is taken and `IF-02` blocks mastering until data owners are designated, and both decisions are `Outstanding` on `O60`. **Two outstanding decisions gate most of the programme**, which means the first thing on the roadmap is two decisions and not a work package.
5. Put the retention clock and the read-only record early, because they are cheap and they are the two the organisation is most exposed on. `CO-02 Clinical records retention` carries `Evidence` `No mechanism`, `STD-04 Data retention standard` is `Emerging` / `Not assessed`, and `MTD-06 Retention rule` has `Coverage` `Absent` — four sheets recording that retention is unmanaged. The retention period, its instrument and its destruction authority are three different facts and the file holds none of them, the periods in circulation disagree with each other [4], and the read-only record's specification is published and specific [1]. Neither needs a vendor decision, and together they move the two measures that a board understands.

8.9 — Close `CR-G03` and re-sign `DLV-13`

Sheet `O61`, `O31`, `O58 Governance Gate` · **Input** everything from 8.1 to 8.8 · **Output** a closed change request and a re-signed deliverable · **Signed by** Architecture Board · **Time** one board meeting

1. `CR-G03` asked for unfunded building blocks to be raised as gaps. It can now be closed, with the note from 8.2 point 3 attached: the derivation it asked for was not available from the file because nothing costs a building block, and what was done instead was to derive the gap set from the reports. Set `Status` and set `Expires`, because three of the five rows on `O61` have an empty `Expires` cell and `CR-G01` is the only one with a date.
2. `DLV-13 Consolidated Gaps` is `Approved` over a two-row gap sheet. Re-open it, re-sign it against the derived set, and date the signature.
3. `DLV-14 Architecture Roadmap` and `DLV-15 Work Packages` are `Draft` and should stay `Draft` until `DEC-01` is taken on a real cost and `DEC-02` is taken on a vendor. A roadmap signed over an unreconciled $15.3m is a roadmap that will be re-cut.
4. `GT-0` on `O58` tests that "Scope, acceptance criteria and contract exist" before work starts. Four of the twelve work packages have no architecture contract — `WP-09`, `WP-10`, `WP-11`, `WP-12` — so a third of the programme could not pass its own first gate today. That is a Phase E finding even though the gate belongs to Phase G, and item 10 is where it is enforced.
5. Take four numbers to the board and leave the rest in the repository: gaps on `O29` before and after; the reconciliation line that is still empty; work packages whose `Delivery` status was corrected; and the rand figure with its rate date.

THE FOUR QUESTIONS PHASE E HAS TO ANSWER OUT LOUD

How many gaps are on `O29` now, and how many of them have a source reference to a cell?

What is the programme cost in rand, at what rate, on what date?

Which of the three legitimate outcomes in 8.6 was taken, and by which body, on what date?

Which work packages are in no transition architecture, and who agreed that they are not?

WHAT PHASE E CHANGED

`O29` is derived from the reports and every row points at a cell. `R12` maps gaps to packages that could close them. `R13` gives each package the blocks its name describes, and medical device integration is no longer delivered by the customer-experience project. The $15.3m is on one page with four lines and a signature block, in rand, with a rate date. `O52` is about winter pressure and ward access rather than a peak trading freeze. And `CR-G03` is closed by the thing it asked for, with the reason the derivation it wanted was not available written into the closure. Item 9 takes the roadmap from here and item 10 gates it.

THE CHAPTER IN TEN LINES

- O29 Gap holds two records; the reports in the same workbook compute twenty-six, and twenty-six is a floor.
- The reports produce five different kinds of gap and O29's two rows are both the first kind.
- An obligation gap closes into a permanent operating cost, and O30 has no column for a running cost.
- Give all three readings of control objectives without evidence: two literally, six on the report's stated test, none of seven evidenced.
- CR-G03 asked for unfunded building blocks to be raised as gaps, but no sheet in the file costs a building block, so the derivation it wanted is not available.
- All thirty-seven acceptance criteria on O55 are one of four sentences with a name substituted in, and three of the four cannot fail.
- R13 cycles twelve work packages through thirty-seven building blocks in order, so medical device integration is delivered by the customer-experience project.
- O30, O56 and R14 agree on 61,3m; ALT-B, marked Recommended, is 46m, and no sheet reconciles the 33 per cent difference.
- Every cost in the file is in US dollars with no rate and no rate date; restate the column in rand before the board sees it.
- Seven work packages read Delivered before any has reached its start date, so correct the Delivery column before costing anything.
- O52's peak trading freeze is template residue from a retail estate; a provider's constraint is winter pressure, strike action and ward access.
- Five of the twelve work packages sit in no transition architecture, and O53's exit criteria are a count rather than a criterion.
- Carry both time axes: O66 dates increments Month N with no baseline and O56 uses absolute dates, so report INC-1's ordering problem and invent no programme start.

SOURCES FOR THIS CHAPTER

- **1.** SAFER Guide: Contingency Planning, August 2024, Office of the National Coordinator for Health Information Technology. United States guidance; the read-only record specification and the paper-stock replenishment figures https://www.healthit.gov/sites/default/files/playbook/pdf/2-contingency-planning-final.pdf
- **2.** National Treasury Medium Term Budget Policy Statement on provincial health finances, reported by MedicalBrief, 28 October 2026 (headline report) https://www.medicalbrief.co.za/treasury-flags-perilous-state-of-provincial-health-finances/
- **3.** Cassim and Chapanduka, on the National Health Laboratory Service ransomware incident and the backlog manual collation produced, South African Medical Journal 2024;114(12):e2549 https://www.scielo.org.za/pdf/samj/v114n12/06.pdf
- **4.** HPCSA Booklet 9, Guidelines on the Keeping of Patient Records, September 2016 edition; the retention period in circulation, which the September 2022 review draft contradicts https://www.hpcsa-blogs.co.za/wp-content/uploads/2022/08/Booklet-9-Guidelines-on-Patient-Records.pdf

ITEM 9

Migration Planning

Phase E chose the work. Phase F dates it, states it and sequences it — and this file holds two timebases, one of which has no origin.

What Phase F has to produce, and what the file gives you to do it with

Phase E chose the alternative and drew the roadmap. Phase F puts dates on the work, places each piece of it inside a state the estate can be said to have reached, names who delivers it, and decides what has to be proved before each cutover. That is the whole of it. If you can hand a programme director a sequenced, costed, owned plan whose states nest and whose exit criteria are written down, Phase F is done.

Two things change the shape of that work in a provider, and neither is in the method.

The first is that **the clock travels with the data.** A migration that moves, copies or replaces a class of clinical record moves the retention obligation and the retrieval obligation with it, and creates a copy that has neither until somebody writes them down. Volume 1 chapter 13 sets out why: the period in circulation traces to a professional guideline that gives at least six years from dormancy in its 2016 edition and ideally indefinite, particularly if electronic, in its September 2022 review draft [1][2], and the record-keeping duty under section 13 of the National Health Act is expressly subject to the National Archives Act and the access-to-information regime, which is where the authority to destroy actually sits [3]. None of that is a Phase F decision. All of it is a Phase F **date**.

The second is that **some of the dates are not yours.** A public provider's year ends on 31 March and its accounting officer answers to the Auditor-General on that date. A private group's funder agreements renew annually and the tariff conversation runs to its own calendar. Volume 1 chapter 8 has the cadences. Nothing in this repository records a single external date: `O52 Implementation Factor` holds `IF-04 Regulatory deadline / Fixed external date` and names no date at all.

Here is the inventory of what Phase F touches in this repository, with the state each sheet is actually in.

Sheet	Rows	Should hold	Holds
`O30 Work Package`	12	Scoped, costed, dated, sequenced work	Twelve rows where seven read `Delivered` before any has reached its start date
`O31 Deliverable`	20	The artefacts, each owned and dated	Four columns: `Ref`, `Deliverable`, `Phase`, `Status`. No owner, no date
`O53 Transition Architecture`	3	States the estate passes through, with the criteria that let you leave	Three states covering seven of the twelve packages, and a count where the criteria belong
`O56 Project`	12	Delivery vehicles carrying work packages	The same twelve names, costs and dates again, plus a `Lead` column where six leads do not match the work
`O66 Architecture Increment`	6	Increments inside a state, on the plan's calendar	Six increments dated `Month 10` to `Month 56`, against no baseline anywhere in the file
`R14`	12	Project to work package	A one-to-one join that is correct, and tells you nothing `O56` did not
`R15`	7	Work package inside a state	Seven rows for twelve packages
`R23`	6	Increment to the project that delivers it	Six rows pairing `INC-1` to `INC-6` with the odd-numbered projects in order
`RPT08 Roadmap`	12	Work against state, increment and external constraint	`O30` with the `Domains` column dropped, and a footnote that is arithmetically wrong

Three of those nine are the work of this item. The other six you check and move on.

The arithmetic you redo before you accept a single date

Do this first and do it with a pen. It takes twenty minutes and it changes who is in the room for the rest of Phase F.

`O30`'s last four columns are `Start`, `End`, `Approval` and `Delivery`. The reference date on this file is 25 October 2026. **The earliest `Start` on the sheet is 2027-01** (`WP-06 Analytics and reporting`) and **the earliest `End` is 2029-02** (`WP-11 Customer experience`). Not one of the twelve work packages has reached its start date, let alone its end date.

Against that, the `Delivery` column reads `Delivered` on seven rows — `WP-01`, `WP-03`, `WP-04`, `WP-05`, `WP-07`, `WP-09` and `WP-10` — and `Past end date` on three: `WP-02`, `WP-06` and `WP-08`. Read `WP-05 Access and security` straight across: `Approval` is `Not started` and `Delivery` is `Delivered`, two columns apart on the same row, for work beginning in March 2027. `WP-01`, `WP-07` and `WP-09` read `In flight` for approval and `Delivered` for delivery — delivered while the approval to start is still running.

So the check that `ZZ Checks` row 18 reports as "work packages past end date: 3" is **0**, and the same wrong figure is printed in `RPT01`'s dashboard and again as `RPT08`'s own footnote line. Three sheets carry one error consistently, which is what makes it a good teaching instrument rather than a typo: a consistent wrong number survives review because every place you check agrees with every other place.

THE ERROR IS NOT CONFINED TO THE DELIVERY LAYER

`O33 Architecture Contract` holds eight rows, one per work package `WP-01` to `WP-08`. Its `Compliance status` column reads `Non-compliant` on exactly three — `ACO-02`, `ACO-06` and `ACO-08` — and those are the contracts for `WP-02`, `WP-06` and `WP-08`, the three packages wrongly marked `Past end date`. The other five read `Not yet assessable`, and those are the five packages marked `Delivered`. The compliance status is not an assessment. It is the delivery column recoded. Item 10 takes that apart; what you need here is that an arithmetic defect in `O30` has already propagated into the governance layer and is being read there as a compliance position.

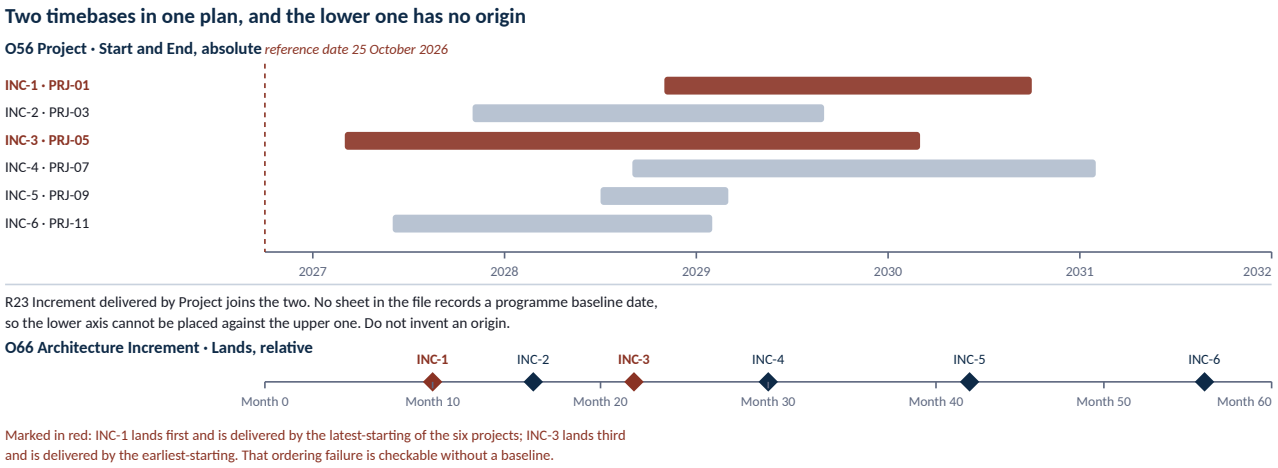


Figure 9.1 The two timebases the file holds. Above, the six projects R23 names, on the calendar O56 gives them. Below, the six increments O66 dates in relative months against no recorded origin. The ordering check that needs no baseline fails on two of the six.

Two timebases, and one of them has no origin

`O66 Architecture Increment`'s third column is `Lands`, and its six values are `Month 10`, `Month 16`, `Month 22`, `Month 30`, `Month 42` and `Month 56`. `O56 Project` dates its twelve projects absolutely, from 2027-01 to 2031-08. `R23` joins the two sheets. **There is no programme baseline date anywhere in this file** — not on `00 Index`, not on `O30`, not on `O53`, not on `O56` — so `Month 10` cannot be resolved to a calendar month and the two sets of timings cannot be reconciled at all.

Carry both readings and pick neither. Either the increments were planned relative to a start date that was never written down, or they were planned against a different plan and carried across. The cells support neither over the other, and the honest entry in your findings log is that the file holds two timebases, one of which has no origin.

What you *can* check without a baseline is internal ordering, and it fails. `R23` maps `INC-1` to `INC-6` onto `PRJ-01`, `PRJ-03`, `PRJ-05`, `PRJ-07`, `PRJ-09` and `PRJ-11`, whose start dates are 2028-11, 2027-11, 2027-03, 2028-09, 2028-07 and 2027-06. The sequence is not monotonic. **`INC-1`, which lands first at month 10, is delivered by the latest-starting of the six projects** — `PRJ-01 Governance foundations`, November 2028 — and `INC-3`, which lands at month 22, is delivered by the earliest-starting, `PRJ-05 Access and security`, March 2027. That is checkable arithmetic on the sheet's own figures and it needs no origin to state.

DO NOT BACK-SOLVE A BASELINE

The temptation is to take the earliest start on `O56`, 2027-01, call it month zero and move on. Try it and watch what happens. `INC-1` then lands in November 2027, delivered by a project that starts in November 2028 — a year after the increment it delivers was due. `INC-6 Core renewal` lands in September 2031, one month after the latest end date on the whole sheet. **An invented origin makes the plan arithmetically consistent and semantically worse**, because it converts a visible absence into an invisible error. Record the absence, raise the decision, and leave `Lands` relative until somebody with the authority to set a baseline sets one.

9.1 — Redo the delivery arithmetic and correct the three sheets that carry it

Sheet `O30 Work Package` · **Input** `RPT08`, `ZZ Checks`, a written reference date · **Output** a corrected `Delivery` value on all twelve rows and a one-page errata · **Signed by** `ROL-13 Head of Delivery` · **Time** half a day

1. Write the reference date at the top of the page and in the errata. Every statement below is true only against a date, and the file's own checks do not carry one.
2. Sort `O30` by `Start` ascending. Record the earliest `Start`, the earliest `End` and the latest `End`: 2027-01, 2029-02 and 2031-08.
3. Count `Delivery` by value. Seven `Delivered`, three `Past end date`, two `Not started`. Count how many rows have a `Start` at or before the reference date: none.
4. Set `Delivery` to `Not started` on all twelve rows and keep the original values in the errata, not in a comment. A comment is not evidence.
5. Read `Approval` against `Delivery` row by row and list every pair that cannot both be true. `WP-05` is the clearest: `Not started` and `Delivered`. `WP-01`, `WP-07` and `WP-09` are `In flight` and `Delivered`.

6. Correct `ZZ Checks` row 18 from 3 to 0, `RPT01`'s equivalent dashboard row, and `RPT08`'s footnote. While you are in `ZZ Checks`, correct the record count from 843 to 820 and the platform components past support from 2 to 1 — `PTC-04` only, because `PTC-06`'s support ends 2026-12, two months after the reference date. Item 1 explains all three; do not re-derive them here.
7. Hand the errata to whoever signs the roadmap before they sign it.

Done when. No `Delivery` value on `O30` describes a state the dates cannot support, and the three sheets that printed the wrong count print the same corrected one.

Goes wrong when. You correct `O30` and leave the reports. Nine of this repository's thirteen reports are a copy of one object sheet with a column dropped or a footnote added, so a correction that stops at the object sheet leaves the wrong number in the only place an executive reads.

Form 9-A — Work package, re-dated and re-stated

One row per work package. Feeds `O30 Work Package`. The `Domains` and `Cost` columns are typed straight into Excel from the existing sheet and are not on this form. A first pass needs twelve rows, one per existing package. Signed by `GB-02 Investment Committee`.

Ref	Work package	Start	End	Approval	Delivery
WP-05	EXAMPLE — delete this row. Access and security	2027-03	2030-03	Not started	Not started

9.2 — Record that there is no programme baseline, and raise the decision

Sheet `O66 Architecture Increment` · **Input** `O56`, `R23`, `O60 Governance Decision` · **Output** a marked `Lands` column and a dated decision request · **Signed by** `GB-02 Investment Committee` · **Time** two hours

1. Search the file for a baseline date. `O30` and `O56` carry absolute months; `O53` carries no dates at all; `O66` carries relative months; `00 Index` carries none. Record the search, not just the result, because the next person will ask.
2. Leave `Lands` as it is and append what it is relative to. `Month 10 (relative; no baseline recorded)` is an honest cell. `2027-11` is not.
3. Run the ordering check in the prose above and write the two inversions down: `INC-1` at month 10 delivered by the latest-starting project, `INC-3` at month 22 by the earliest-starting.
4. Add a row to `O60 Governance Decision`: `Set the programme baseline date`, body `Investment Committee`, status `Outstanding`. There is no such row today, and `O60` has no date column at all — step 9.7's note on added columns applies here too.
5. Tell the committee what the decision costs them if they defer it: every increment date, every transition exit date and every gate date in Phase G hangs off it, so Phase F cannot be signed without it.

Done when. `O66`'s `Lands` cells say what they are relative to and `O60` carries an outstanding decision naming the body that will set the origin.

Goes wrong when. Somebody in a workshop says "call it January 2027" and nobody writes down that it was called.

9.3 — Write the eighteen exit criteria the file only counted

Sheet `O53 Transition Architecture` · **Input** `R15`, `R13` as re-derived in 9.4, `O55 Building Block` · **Output** a written, numbered criteria set per state · **Signed by** `GB-01 Architecture Board` · **Time** one day

1. Read `O53`'s `Exit criteria` column. It holds `5 exit criteria`, `6 exit criteria` and `7 exit criteria`. Search the file for the criteria themselves. Eighteen are counted and not one is written down anywhere.
2. Do step 9.4 first. The route from a state to its criteria runs `R15` to the work packages, `R13` to the building blocks, then `O55`'s `Acceptance criteria` column — and `R13` as it stands is a positional join, so criteria derived through it belong to the wrong state.
3. Throw out every criterion that cannot be run by one person in one afternoon. Ten of `O55`'s thirty-seven blocks carry the sentence frame `<capability> measured and owned.`, which names no measurement and no owner. The application blocks are testable as written: `No`

`unmanaged interface to or from EHR / EPR` is a count against `R06`.

4. Write five for `TA-1 Governed and supportable` that are provably true or false on the day. These five are testable against the file in front of you, and all five currently fail: every `Owner` value in `O20 Data Entity` resolves to a row in `O11 Role` (four of fourteen do — the other ten are `Head of ...` titles that exist nowhere in the file); no row in `O27` has a `Support end` before the date of the test; no row in `O37 Data Attribute` carries a `Classification` looser than its entity's in `O20` (twenty-six of sixty do); `O72 Retention Clock` holds a row for every record class the state touches, each with a measured retrieval time; and `O63 Control Objective` names an instrument against all seven obligations.
5. Write three for `TA-2 Integrable` and six for `TA-3 Business visible` the same way, from the blocks those states actually deliver.
6. Replace the count in the cell with the numbered list, and make the count equal the list length.

Done when. Each `Exit criteria` cell holds a numbered list, and the number of items equals the number the cell claims.

Goes wrong when. You write aspirations. "The estate is governable" is not an exit criterion. "Fourteen of fourteen `O20` rows name an owner that resolves to `O11`" is one, and it fails today.

Form 9-B — Transition architecture and its exit criteria

One row per state. Feeds `O53 Transition Architecture`; the `Exit criteria` cell takes the numbered list, not a count. A first pass needs five rows — the three states the file holds plus the two step 9.6 adds. Signed by `GB-01 Architecture Board`.

Ref	Transition architecture	Description	Work packages	Exit criteria
TA-2	EXAMPLE — delete this row. Integrable	Managed integration carrying production traffic	WP-03	1. Zero unmanaged interfaces on R06. 2. …

9.4 — Re-derive `R13` so a work package delivers the blocks its own name describes

Sheet `R13` · **Input** `O30`, `O55 Building Block` · **Output** thirty-seven re-pointed rows · **Signed by** `ROL-08 Chief Enterprise Architect` · **Time** half a day

1. Read `R13` as a sequence rather than as a model. It holds thirty-seven rows and it is `WP-01` to `WP-12` cycled through `O55`'s thirty-seven blocks in sheet order, three times round and a remainder. Every reference resolves. Nothing means anything.
2. Check four rows and you have the pattern. `WP-02 Data ownership and mastering` delivers one of the ten data blocks and it is `DBB-04 Diagnosis mastering`. `WP-04 Technology currency` delivers no technology block at all. `WP-03 Integration platform` delivers no integration block. `WP-11 Customer experience` delivers `DBB-01 Patient mastering`, `ABB-03 LIS target` and `TBB-05 Medical device integration`.
3. Re-point on `O55`'s `Domain` column first, because that is a mechanical pass and it is most of the work. The ten `DBB` rows belong to `WP-02`. The seven `TBB` rows distribute across `WP-03`, `WP-09` and `WP-10`. The ten `ABB` rows distribute across `WP-03`, `WP-07` and `WP-12`.
4. Then re-point on name, which is the judgement pass. `TBB-01 HL7 FHIR interoperability` and `TBB-02 Terminology services` belong to `WP-03 Integration platform`; `TBB-06 Zero-downtime clinical systems` to `WP-09 Resilience uplift`; `TBB-07 De-identification and secure enclave` to `WP-10 Compliance mechanisms`.
5. Count. Every one of the thirty-seven blocks must appear exactly once and every work package must deliver at least one. If a package delivers nothing after the pass, either it is not in the plan or a block is missing — both are findings, neither is a rounding error.
6. Note which blocks nobody wants. `TBB-03 Master patient index`, `TBB-04 Clinical decision support` and `TBB-05 Medical device integration` satisfy no requirement in `R09` either. A block that no package wants and no requirement needs is a candidate for deletion, not for a work package.

Done when. Thirty-seven rows, each block once, no empty work package, and the block's `Domain` consistent with the package's `Domains`.

Goes wrong when. You preserve the row count and change nothing else. Referential integrity was already perfect, which is exactly why nobody caught this.

9.5 — Re-derive `R23` so an increment is delivered by the project that delivers it

Sheet `R23` · **Input** `O66`, `O56`, `R14` · **Output** re-pointed rows and a list of projects delivering no increment · **Signed by** `ROL-13 Head of Delivery` · **Time** two hours

1. Read `R23`'s six rows. `INC-1` to `INC-6` are paired with `PRJ-01`, `PRJ-03`, `PRJ-05`, `PRJ-07`, `PRJ-09`, `PRJ-11` — the odd-numbered projects in order. The join is positional and all six names disagree with the project said to deliver them.
2. Read them side by side and the absurdities surface on their own. `INC-5 Analytics and evidence` is delivered by `PRJ-09 Resilience uplift`, while `PRJ-06 Analytics and reporting` delivers nothing. `INC-6 Core renewal` is delivered by `PRJ-11 Customer experience`, while `PRJ-07 Core platform renewal` delivers `INC-4`.
3. Re-point on name. `INC-1` from `PRJ-01`; `INC-2` from `PRJ-03`; `INC-3 Compliant by mechanism` from `PRJ-05` and `PRJ-10 Compliance mechanisms`; `INC-4` from `PRJ-11`; `INC-5` from `PRJ-06`; `INC-6` from `PRJ-07`. The sheet becomes seven rows, not six, because one increment takes two projects — and that is the point: a one-to-one join was never going to be right.
4. List the projects that still deliver no increment after the pass: `PRJ-02 Data ownership and mastering`, `PRJ-04 Technology currency`, `PRJ-08 Process automation` and `PRJ-12 Supplier and partner integration`. Either they sit inside `INC-1` and you say so, or the increment set is short and you add to it.
5. Leave the `Lands` column exactly as step 9.2 left it. Re-pointing the join does not supply a baseline.

Done when. No row in `R23` pairs an increment with a project whose name describes other work, and every project either delivers an increment or is recorded as inside one.

Goes wrong when. You repair `R23` and leave `R13`. They are the same defect at two levels and the exit criteria depend on both.

Form 9-C — Increment, re-pointed to the project that delivers it

One row per increment-to-project pair; an increment may take more than one row. Feeds `R23`. `Lands` is copied from `O66` unchanged, including its relative form. A first pass needs seven rows. Signed by `ROL-13 Head of Delivery` and `GB-01 Architecture Board`.

Increment	Increment name	Lands	Project	Project name
INC-5	EXAMPLE — delete this row. Analytics and evidence	Month 42 (relative)	PRJ-06	Analytics and reporting

Where the money sits, against the state it is meant to reach

The five that sit in no transition architecture

WP-06	Analytics and reporting	**$9.2m**
WP-08	Process automation	**$7.4m**
WP-09	Resilience uplift	**$7.4m**
WP-10	Compliance mechanisms	**$9.4m**
WP-12	Supplier and partner integration	**$1.5m**

WP-10 Compliance mechanisms is the most expensive package on the sheet and the one the retention clock and the consent object depend on.

57% of the programme by cost reaches no declared state of the estate.

R15 holds seven rows. Twelve work packages exist. The gap is not a modelling nicety: an unplaced package has no exit criteria and so cannot be refused at a gate.

Figure 9.2 Cost against state. Seven work packages sit in a transition architecture and five do not, and the five are the larger half of the plan. The dashed line is the cost of the alternative the plan is supposed to implement.

9.6 — Place the five work packages that sit in no state at all

Sheet `O53` and `R15` · **Input** `O30`, `R15` · **Output** two new states and twelve of twelve packages placed · **Signed by** `GB-01 Architecture Board` · **Time** half a day

1. Read R15. It holds seven rows covering WP-01, WP-02, WP-03, WP-04, WP-05, WP-07 and WP-11. Five packages appear in no state: WP-06, WP-08, WP-09, WP-10 and WP-12.
2. Add their Cost values. $9.2m, $7.4m, $7.4m, $9.4m and $1.5m come to **$34.9m of the $61.3m plan — 57% of the programme by cost reaching no declared state of the estate**, and the largest single package on the sheet, WP-10 Compliance mechanisms, is among them.
3. Check whether the three states you have nest. On R15's own membership and O30's dates, TA-1 runs 2027-03 to 2031-07, TA-2 runs 2027-11 to 2029-09 and TA-3 runs 2027-06 to 2031-02. The third state begins before the second and the first finishes after both. A transition architecture is a state the estate passes through; on these dates it reaches state three before it has finished reaching state one.
4. Decide which you are fixing — the membership or the dates — and fix one. Doing both in one pass produces a plan nobody can review.
5. Add TA-4 Compliant by mechanism and TA-5 Analytics and evidence, naming them to match INC-3 and INC-5 so the states and the increments share a vocabulary. Then place all twelve: TA-1 takes WP-01, WP-02, WP-04, WP-05 and WP-09; TA-2 keeps WP-03; TA-3 takes WP-07, WP-08, WP-11 and WP-12; TA-4 takes WP-10; TA-5 takes WP-06.
6. Write R15 out to twelve rows and make O53's Work packages cell agree with it exactly. The two agree today and that is worth preserving.
7. Rename as you go. TA-3 Business visible is glossed First capability the customer notices, and so is INC-4. A hospital has patients. The phrase is residue from the generic base this file was built on, and WP-11, PRJ-11 and ALT-C carry the same word. Fix it here or it goes into the board pack.

Done when. R15 holds twelve rows, one per work package, and every state's earliest start is at or after its predecessor's.

Goes wrong when. You place the five and leave the dates crossed, which gives you a complete plan that still runs backwards.

9.7 — Give every deliverable an owner and a date

Sheet O31 Deliverable · **Input** O30, O56, O11 Role · **Output** an owner and a due date on twenty rows · **Signed by** ROL-08 Chief Enterprise Architect · **Time** two hours

1. Read O31's columns: Ref, Deliverable, Phase, Status. There is no owner and no date. **Add Owner and Due, and say in the change note that they are additions the file did not have.** Columns you add silently are the next reader's defect.
2. Read Status. Thirteen rows are Approved and seven are Draft, and the seven are DLV-14 through DLV-20.
3. Date the two that are this phase's own output first: DLV-16 Migration Plan and DLV-17 Project Charters, both Draft.
4. Then raise the problem above them. DLV-14 Architecture Roadmap and DLV-15 Work Packages are Phase E deliverables and still Draft. Phase F cannot be signed off a draft roadmap, so either they are approved this week or the Phase F dates are provisional and labelled as such.
5. Take every owner from O11 Role by reference, not from a job title you invent in the workshop. Ten of O20's fourteen owners are Head of ... titles that appear nowhere else in the file; do not repeat the pattern on a fresh sheet.
6. Note for item 14 that O31 appears in no relationship sheet and in no report. Twenty deliverables are invisible to the whole reporting layer, which is why nobody noticed that seven are drafts.

Done when. Twenty rows, each with a Due date and an Owner that resolves to a row in O11.

Goes wrong when. The owner is a committee. A committee cannot be chased.

Form 9-D — Deliverable, owned and dated

One row per deliverable. Feeds O31 Deliverable. **Owner and Due are columns this repository does not have; adding them is part of the step.** A first pass needs twenty rows — or seven, if you start with the drafts. Signed by ROL-08 Chief Enterprise Architect.

Ref	Deliverable	Phase	Status	Owner	Due
DLV-16	EXAMPLE — delete this row. Migration Plan	F	Draft	ROL-08	

9.8 — Reconcile the plan to the alternative it is supposed to implement

Sheet `O30` and `O54 Architecture Alternative` · **Input** `R14`, `O56`, `O60` · **Output** a reconciliation or a re-baseline request · **Signed by** `GB-02 Investment Committee` · **Time** half a day

1. Add `O30`'s `Cost` column. It comes to **$61.3m** across twelve packages. Add `O56`'s `Cost`: the same, row for row. Add `R14`'s `Cost`: the same again. The three sheets agree, which is worth saying out loud before the next sentence.
2. Read `O54`. `ALT-B Foundations first` is `$46m` and marked `Recommended`. `ALT-E Compliance led` is `$31m` and marked `Partly adopted`. `ALT-A` at `$24m`, `ALT-C` at `$52m` and `ALT-D` at `$88m` are `Rejected`.
3. State the gap plainly: **the plan is $15.3m, or 33%, above the figure the recommendation was made on**, and above two of the three options that were rejected for being the wrong answer. No sheet in the file reconciles the two numbers.
4. Read `O60` while you are there. `DEC-01 Adopt the recommended architecture alternative` is `Outstanding`, while `O54` already marks one alternative `Recommended` and a second `Partly adopted`. Something has been partly adopted before anything was adopted.
5. The file is denominated in dollars throughout, which is not a South African provider's currency. Say so in the first paragraph of anything you circulate, and either convert at a stated rate on a stated date or keep the dollars and label them. Do not let the two mix in one table.
6. Produce one of two artefacts and not neither: a reconciliation that itemises which packages fall outside `ALT-B`'s scope, or a re-baseline request to `GB-02` at $61.3m. A plan approved against an unreconciled 33% gap is approved against nothing.

Done when. `O60` carries a dated decision on the alternative, and the plan total either equals the alternative's cost or the difference is itemised package by package.

Goes wrong when. The reconciliation is done in a spreadsheet nobody keeps. Put it in `O52 Implementation Factor` as a dependency beside `IF-01 Vendor selection not made`.

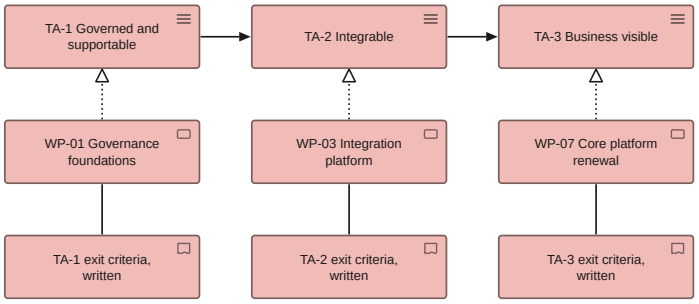


Figure 9.3 What a transition state looks like once step 9.3 has run: a state the estate reaches, the work that realises it, and a written criteria set that says when it has been reached. The file holds the top two rows and a count where the bottom row belongs.

9.9 — Rebuild `RPT08` so it reports a sequence rather than a list

Sheet `RPT08 Roadmap` · **Input** `O30`, `R15`, `R23`, `O52` · **Output** a roadmap carrying state, increment and external constraint · **Signed by** `ROL-08 Chief Enterprise Architect` · **Time** two hours

1. Compare `RPT08` to `O30`. It is the same twelve rows with the `Domains` column dropped, plus a footnote. It carries no transition, no increment, no dependency and no external date. A roadmap with no sequence in it is a list of work sorted by nothing.
2. Add three columns: `Transition` from `R15`, `Increment` from `R23` as re-derived in 9.5, and `Depends on` from `O52 Implementation Factor`.
3. Correct the footnote to 0.
4. Read `O52` properly while you are populating `Depends on`, because two of its eight rows contradict the plan. `IF-01 Vendor selection not made / Blocks integration work until resolved` sits against `WP-03 Integration platform`, which the file marks `Delivered`. `IF-02 Data owners not designated / Blocks mastering work` sits against `WP-02`, which the file marks `Past end date`. Both blocked packages are recorded as finished.
5. Fix the two rows of `O52` that are not about this industry. `IF-03 Peak trading freeze / No change in the peak window` is a retailer's constraint; a provider has winter pressure, theatre lists and funder year-end, and those are the windows to name. `IF-04 Regulatory deadline / Fixed external date` names no date — give it one or delete it.
6. Put the reference date in the report header. A roadmap without one cannot be read six months later.

Done when. Every row of `RPT08` names its state, its increment and its blocking factor, and the footnote is arithmetically true on the stated date.

Goes wrong when. You add the columns and leave them empty on the five packages step 9.6 placed.

9.10 — Date the retrieval measurement before the cutover that depends on it

Sheet `O30` and `O66` · **Input** `O58 Governance Gate`, `O72 Retention Clock`, volume 1 chapters 13 and 19 · **Output** three dated milestones per record-moving package · **Signed by** `GB-01 Architecture Board` · **Time** half a day

1. List every work package that moves, copies or replaces a class of clinical record. On this plan that is `WP-02`, `WP-06`, `WP-07` and `WP-10` at minimum, and `WP-03` carries the message store that most people forget is a copy.
2. For each one, put three dated milestones in the plan, in this order: the `O72 Retention Clock` row exists for every record class the package touches; the retrieval time for that class has been **measured**, not estimated; and the `G2` verdict for the package's building blocks is recorded in `O59`. Item 2 defines `O72` and item 10 runs the gate. Phase F's job is only the dates.
3. Make the measurement a milestone with an owner, not a line in a standard. Volume 1 chapter 19 specifies the read-only record — hourly currency, weekly testing, printable output and an independent network and authentication path [5] — and the first measured retrieval against it is a date somebody can be held to.
4. Put the reason in the plan narrative in one sentence, because it is the sentence that gets the milestone funded. The Auditor-General's 2026 health-sector findings record fifteen of twenty-seven facilities with missing or lost records, seventeen not tracking records after issue, and **three of nine paid medico-legal claims conceded because the record could not be found** [4]. That is the measured price of an unmeasured retrieval time.
5. Date the destruction decision too. The retention period, the instrument and the destruction authority are three different facts, and the authority is the one that stops a cutover: section 13 of the National Health Act makes the record-keeping duty subject to the National Archives Act and the access-to-information regime [3], and the guideline the periods come from gives at least six years from dormancy in one edition and ideally indefinite in the next [1][2]. Phase F cannot settle that. It can put `Set the destruction authority for each record class` into `O60` with a date and a body.
6. Sequence the classification work ahead of the platform work. `O65 Enforcement Point` records `EP-04 Data platform / Classification not applied at read`, and `WP-06 Analytics and reporting` is the package that lands the platform. Land the platform first and the twenty-six attributes that are classified more openly than their entity are inherited at reporting scale. Classification inherits downward and never loosens; in Phase F that rule is a sequencing constraint, not a principle.

Done when. No cutover in the plan is dated earlier than the retrieval measurement for the record classes it moves.

Goes wrong when. The measurement is scheduled after the cutover, which is the usual order, because the only system that can be measured before a cutover is the one being retired.

What the two running examples do with this

WORKED ANSWERS — THE SAME STEP, TWO ORGANISATIONS

Driehoek Health Group — nine acute hospitals, two day clinics, about 1,750 registered beds. Driehoek's step 9.8 reconciliation found three packages outside the recommended alternative's scope and worth $13.1m of the $15.3m gap. The Investment Committee re-baselined rather than cut, and the minute records both figures. Its step 9.10 list of record-moving packages ran to six, because its group-level data warehouse is a copy nobody had classed as one.

Rietvlei District Health Service — one district hospital, three community health centres, 22 fixed clinics, a mobile service. Rietvlei's plan hit the destruction authority first, not the retention period: as a public body it cannot destroy records on its own authority at all, so its step 9.10 output was a single dated decision request naming the approval it has to obtain, and its first transition state carries one exit criterion the private group does not need — that the disposal authority exists in writing.

BEFORE YOU LEAVE PHASE F

Six questions. Each answer is a document reference or a date, or the phase is not finished.

On what date was the delivery state of every work package last re-stated, and against what reference date?

What is the programme baseline date, and who set it?

For each transition state, where are its exit criteria written down?

Which work packages are in no transition state, and what do they cost?

What reconciles the plan total to the cost of the alternative recorded as recommended?

For every package that moves a record class, what is the date of its retrieval measurement?

THE CHAPTER IN TEN LINES

- Not one of the twelve work packages has reached its start date. The earliest start on O30 is 2027-01 and the earliest end 2029-02.
- Seven rows nonetheless read Delivered, and WP-05 reads Not started for approval and Delivered for delivery on the same row.
- ZZ Checks reports three work packages past their end date. The correct figure on the reference date is zero, and RPT01 and RPT08 carry the same wrong number.
- O33's compliance status is the delivery column recoded: the three Non-compliant contracts are the three wrongly dated packages, the five Not yet assessable ones the five marked Delivered.
- O66 dates its increments in relative months and O56 in calendar months, and no sheet records a baseline. Carry both readings and do not invent an origin.
- The ordering check that needs no baseline fails: INC-1 lands first and is delivered by the latest-starting project, INC-3 third by the earliest-starting.
- Eighteen exit criteria are counted on O53 and not one is written down anywhere in the file.
- R13 and R23 are positional joins. Re-derive both before deriving any exit criterion through them.
- Five work packages sit in no transition state and they are 34.9 of the 61.3 million dollars, including the most expensive package on the sheet.
- The three states do not nest: the third begins before the second and the first finishes after both.
- The plan costs 33% more than the alternative recorded as recommended, and the decision to adopt any alternative is still outstanding.
- O31 has no owner column and no date column, seven of its twenty deliverables are drafts, and no report reads it.
- For every package that moves a record class, the retrieval measurement is a dated milestone before the cutover, not a principle after it.

SOURCES FOR THIS CHAPTER

- **1.** HPCSA Booklet 9, Guidelines on the Keeping of Patient Records, September 2016 edition (retention periods; destruction of provincial hospital records only with Deputy Director-General authorisation) https://www.hpcsa-blogs.co.za/wp-content/uploads/2022/08/Booklet-9-Guidelines-on-Patient-Records.pdf
- **2.** HPCSA Booklet 9, Keeping of Patient Records, review draft marked September 2022 (general records retained ideally indefinitely, particularly if electronic) https://www.hpcsa-blogs.co.za/wp-content/uploads/2022/11/Booklet-9-Keeping-of-Patient-Records_Review-Draft_-FINAL_Sept-2022.pdf
- **3.** National Health Act 61 of 2003, consolidated text, section 13 (the record-keeping duty, expressly subject to the National Archives and Record Service of South Africa Act 43 of 1996 and the Promotion of Access to Information Act) https://www.saflii.org/za/legis/consol_act/nha2003147.pdf
- **4.** Auditor-General findings on the health sector, via MedicalBrief, 14 October 2026 (records audit of 27 facilities; three of nine paid medico-legal claims conceded because the record could not be found) https://www.medicalbrief.co.za/health-sector-crippled-by-recurring-systemic-weaknesses-ag-warns/
- **5.** SAFER Guide: Contingency Planning, August 2024, Office of the National Coordinator for Health Information Technology (read-only backup record, updated at least hourly, tested weekly, able to print). United States guidance https://www.healthit.gov/sites/default/files/playbook/pdf/2-contingency-planning-final.pdf

ITEM 10

Implementation Governance

Eight relationship sheets resolve perfectly and mean nothing, so the gate question is not whether it resolves. It is what was reconciled, by whom, against what.

The question this phase asks, and the question it must stop asking

Phase G has one job. It decides whether what was built is what was designed, and writes the answer somewhere a third party can find it. The bodies, the gates, the contracts and the dispensations all exist to make that answer durable.

In most repositories the phase goes wrong in the same way, and this one shows it at full size. **Eight relationship sheets in this file resolve perfectly and mean nothing.** `R04`, `R07`, `R09`, `R13`, `R20`, `R22`, `R23` and `R25` were each built by pairing two lists in row order: Human Resources owns Medicines Management, the billing system supports Emergency Care, and all thirty-two rows of `R09` reach a building block that has nothing to do with the requirement. Every reference resolves, by reference and by name, so every integrity check a tool can run comes back clean.

So the gate question cannot be *does it resolve*. It has to be:

> THE PHASE G QUESTION
>
> **What was reconciled, by whom, against what?**
>
> Three answers, not one. The *what* is the artefact or the count. The *by whom* is a named person holding a role that exists. The *against what* is the independent thing the claim was checked against — and if there is no independent thing, the gate records that, which is itself a verdict.

That is not a style preference. It is the only form of the question this repository cannot answer falsely. `059 Compliance Assessment` holds three rows, all at gate `G2`, all `Non-compliant`, all with the `Evidence` value **"No evidence produced"**. Those three rows are the most honest cells in the file; read them as the specification for the other hundred and forty-five.

Volume 1 chapter 18 owns the regulators and their powers. Do not restate it in a gate pack. What you own here is the gate: the question, the evidence standard behind it, and the record that survives the person who ran it.

What this phase has to work with

Sheet	Rows	Should hold	Holds
`033 Architecture Contract`	8	A contract per work package, assessable	Eight rows whose compliance status is the delivery column of `030` recoded
`057 Governance Body`	5	Bodies with a mandate and a cadence	Five sound rows, and no clinical or records mandate among them
`058 Governance Gate`	4	The gates, and what each tests	Four well-formed gates, three of which have never been run
`059 Compliance Assessment`	3	A verdict per building block per gate	Three rows, all at `G2`, all `Non-compliant`, all with no evidence
`060 Governance Decision`	7	Decisions, dated, with a body	Four `Outstanding` and three `Approved`, and no date column at all
`061 Governance Change Request`	5	Dispensations and corrections, with expiries	Five rows, of which four have no `Expires` value
`063 Control Objective`	7	The obligation, its instrument, the control	Seven rows whose control objective restates the obligation
`064 Security Policy`	4	Policy and its position	Four policies, all `Breached`
`065 Enforcement Point`	5	Where a policy is actually applied	Five points, each recording its own defect in the `Position` column
`R16`	7	Objective to the control that meets it	One control per objective, joined in row order, with the objective's position copied into the control's
`R17`	4	Policy to enforcement point	Four rows, one per policy, in row order; `EP-05` enforces nothing
`R24`	3	The test behind each assessment	`059`'s three rows again, with one column dropped
`RPT09 Regulatory Coverage`	7	Obligation against evidence	`063` reprinted, with a footnote defining a test the other sheets then label differently
`RPT11 Security Position`	8	Control, position, enforcement	`041` reprinted, footnote correct
`RPT12 Standards Conformance`	16	Standards against what conforms	`032` reprinted, with one application's verdict as the standard's position

Fifteen sheets, four of which are a copy of another sheet. The phase's real content is on five of them.

The compliance position that nobody assessed

Start here. It takes ten minutes and it settles whether this repository's governance layer has ever been run at all.

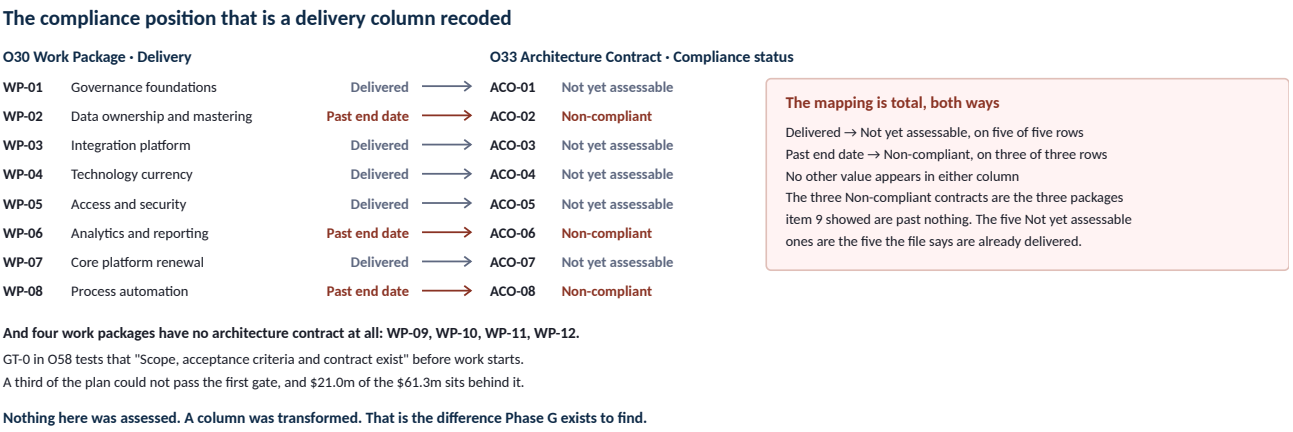


Figure 10.1 O33's compliance status against O30's delivery state. The mapping is total in both directions, which means no assessment took place: a column was transformed, and it carried an arithmetic error from three sheets away into the governance layer.

`O33 Architecture Contract` holds eight rows, `ACO-01` to `ACO-08`, pointing at `WP-01` to `WP-08`. Its `Compliance status` column takes exactly two values: five rows read `Not yet assessable` and three read `Non-compliant`. Line those three up against `O30`'s `Delivery` column and they are `WP-02`, `WP-06` and `WP-08` — the three packages item 9 showed are past no end date at all. Line the five up and they are the five the file marks `Delivered`.

The mapping is total in both directions. `Delivered` becomes `Not yet assessable`; `Past end date` becomes `Non-compliant`. **No assessment took place. A column was transformed**, and the transformation inherited an arithmetic error from a sheet three layers away. Both halves of it are also backwards on their own terms: work recorded as delivered is recorded as not yet assessable, and work whose end date is between two and five years in the future is recorded as non-compliant.

Then count what is missing. Four work packages have no architecture contract at all — `WP-09` to `WP-12`, **$21.0m of the $61.3m plan**. `GT-0 G0 Charter` tests that "Scope, acceptance criteria and contract exist" before work starts. A third of the programme by count could not pass the first gate, and nothing in the file notices.

The evidence standard, and the arithmetic of the gate

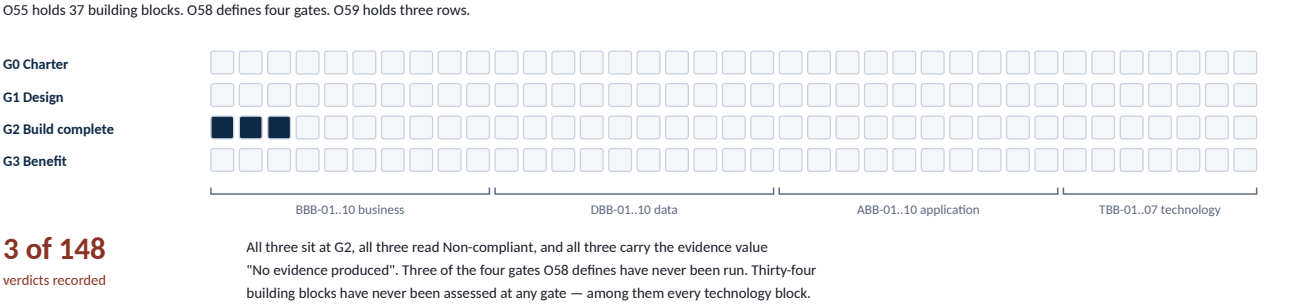


Figure 10.2 Thirty-seven building blocks against four gates. Each square is a verdict the gate regime requires; the three filled ones are the verdicts that exist, all at one gate and all recording that no evidence was produced.

`O55` holds thirty-seven building blocks and `O58` defines four gates, so a gate regime that means anything produces a hundred and forty-eight verdicts. `O59` holds three, all at `G2` — so `G0`, `G1` and `G3` have never been run, and **thirty-four of the thirty-seven blocks have never been assessed at any gate**, including all seven technology blocks.

Three cells of a hundred and forty-eight is not an incomplete assessment. It is a sheet that was started and abandoned, and the useful consequence is that you can set the standard for all hundred and at once, before anybody has a vested interest in a verdict.

The standard has three parts and no more, because a standard nobody can hold in their head is not applied.

One. The artefact. A verdict cites a document, a count, a screenshot or a dated test result. "The team confirmed" is not an artefact. `O59`'s `Evidence` column is where it goes, and "No evidence produced" is a legitimate value — the only illegitimate value is a verdict with an empty evidence cell.

Two. The person. A verdict names a person holding a role that exists in `O11 Role`. `O05 Principle`'s twenty-four principles are all owned by "Architecture Board", which is neither an actor nor a role in this file — it is `GB-01` on `O57`. A body cannot sign; a person signs for it.

Three. The independent thing. The claim is checked against something the claimant did not produce. For an interface count that is `R06`; for an ownership claim, `O11`; for a retrieval time, a stopwatch. Where there is nothing independent, the verdict says so, and that sentence is the most valuable one the gate produces.

10.1 — Write the evidence standard before you assess anything

Sheet `O59 Compliance Assessment` · **Input** `O55`, `O58`, `O11 Role` · **Output** a one-page evidence standard, minuted · **Signed by** `GB-01 Architecture Board` · **Time** half a day

1. Write the three parts above on one page, in the organisation's own words, with an example of each taken from this repository rather than from a method.
2. Decide and record what `Verdict` may contain. Four values are enough: `Compliant`, `Non-compliant`, `Compliant with dispensation`, `Not assessed`. `Not yet assessable` is the value `O33` uses and it should not survive, because it describes the assessor rather than the thing.
3. Decide what happens to a verdict with no evidence. The only defensible rule is that it is recorded as `Non-compliant` with the evidence value stating what was missing — which is what `O59`'s three existing rows already do, correctly.
4. Name the assessor and the decider separately. `O57` gives you both: `GB-05 Architecture Function` "publishes criteria, assesses compliance, maintains the repository", and `GB-01 Architecture Board` "holds `G1` and `G2`". The same person must not do both.
5. Minute it. An evidence standard that is not minuted is overruled in the first difficult meeting.

Done when. The standard is one page, it is minuted, and it names the four verdict values and the assessor-decider split.

Goes wrong when. The standard is written after the first contested verdict, when it will be written to fit it.

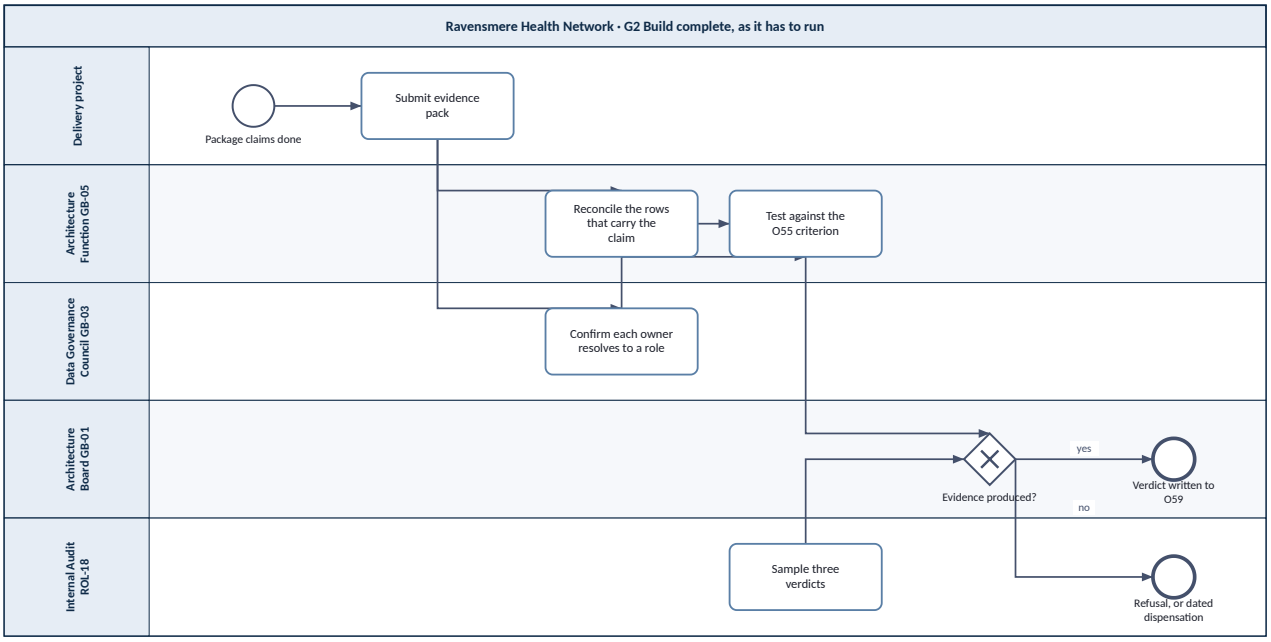


Figure 10.3 G2 with the three answers built in. The assessor and the decider are different bodies, the reconciliation happens before the verdict rather than after it, and a refusal is a legitimate exit — as is a dispensation, provided it carries a date.

10.2 — Set the reference space for `O59`, derive its rows, and make `R24` carry the test

Sheet `O59` and `R24` · **Input** `O55`, `O58` · **Output** the reference convention and the full derived row set · **Signed by** `GB-05 Architecture Function` · **Time** one day

1. The existing references are `CA-01` to `CA-03`. Continue at `CA-04` and do not renumber: three existing rows are cited in `R24` and nothing is gained by breaking them.
2. Derive the row set mechanically. For each of `O55`'s thirty-seven blocks, and for each gate in `O58` that applies to it, one row. Where a block is not subject to a gate, write the row with `Verdict` = `Not assessed` and say in `Evidence` why not. An absent row is indistinguishable from an oversight; a present row with a reason is a position.
3. Fill `Building block` and `Block name` from `O55` by reference, and `Acceptance criterion` from `O55`'s `Acceptance criteria` column verbatim. If the criterion is untestable as written, raise it against `O55` rather than improving it here: an assessment against a quietly edited criterion is not an assessment.
4. Flag the ten business blocks now. `BBB-01` to `BBB-10` carry the criterion `<capability> measured and owned.`, which names no measure and no owner — and the only three blocks this file has ever assessed are three of those ten.
5. Then separate `R24` from `O59`. `R24` today is `O59`'s three rows with `Evidence` dropped, which makes it a duplicate rather than a join. Give it the one thing `O59` has no column for: what the criterion was tested **against**. Its five columns are `Assessment`, `Building block`, `Block name`, `Criterion`, `Verdict`; add `Tested against` and `Assessed by`, and record both as additions the repository did not have.
6. Count at the end and write the count on the sheet: verdicts recorded, verdicts outstanding, blocks never assessed.

Done when. `O59` holds a row for every block-and-gate pair, and every row's `Evidence` cell is non-empty.

Goes wrong when. You derive the rows and leave the criteria untouched, which gives you a hundred and forty-eight verdicts against ten sentences that cannot be tested.

Form 10-A — Compliance assessment A, identify

One row per building block per gate. Feeds `O59 Compliance Assessment`, columns one to four. The remaining three columns are on Form 10-B. A first pass needs one row per block at the gate you are running — thirty-seven for a full gate. Signed by `GB-05 Architecture Function`.

Ref	Building block	Block name	Acceptance criterion
CA-04	EXAMPLE — delete this row. ABB-01	EHR / EPR target	No unmanaged interface to or from EHR / EPR.

Form 10-B — Compliance assessment B, assess

The same rows as Form 10-A, carried forward by `Ref`. Feeds `O59 Compliance Assessment`, columns five to seven. **`Tested against` and `Assessed by` are columns this repository does not have**; step 10.2 adds them to `R24`. `Block name` is typed straight into Excel from `O55`. A first pass needs the same count as Form 10-A. Signed by `GB-01 Architecture Board`.

Ref	Gate	Verdict	Evidence	Tested against	Assessed by
CA-04	EXAMPLE — delete this row. G1	Non-compliant	19 of 23 interfaces bypass the engine	R06, counted 14 Nov	ROL-11

10.3 — Reconcile the rows that carry the claim, and record the reconciliation

Sheet `R24` · **Input** `R04`, `R07`, `R09`, `R13`, `R20`, `R22`, `R23`, `R25` · **Output** a reconciliation record per gate · **Signed by** `GB-05 Architecture Function` · **Time** one day per gate

1. For the gate you are running, list every relationship row the claim depends on. A `G1` verdict on `ABB-07 Revenue Cycle / Claims target` depends on `R07` for what it runs on, `R20` for where it is, `R25` for what it supports and `R18` for which standards apply.
2. Check each row by subject, not by reference. The test is one sentence read aloud: does the pairing describe something true of this organisation? `R25` makes `PRD-03 Emergency Care` supported by `A07 Revenue Cycle / Claims` — the billing system supporting the emergency department — while the laboratory system supports no product at all.
3. Record three counts per sheet, not one: rows checked, rows that survive the reading, rows re-pointed. A reconciliation with one number in it is a row count, and a row count was never the thing in doubt.
4. Write the result into `R24`'s `Tested against` column for every affected verdict, naming the sheet and the date.
5. Where a claim depends on a sheet nobody has checked, the verdict is `Non-compliant` and the evidence says which sheet. Keep the reconciliation record outside the workbook as well as in it: it is the document a regulator, an auditor or a successor asks for, and `O59` is not a document.

Done when. Every verdict names the relationship sheets it depends on, the date they were checked and the counts.

Goes wrong when. Somebody runs the integrity check instead. Referential integrity in this file is perfect and always was. A tool cannot tell you that Human Resources does not own Medicines Management.

10.4 — Stop `O33` recoding `O30`, and write the four contracts that do not exist

Sheet `O33 Architecture Contract` · **Input** `O30`, `O55`, `R13` as re-derived in item 9 · **Output** twelve assessable contracts · **Signed by** `GB-01 Architecture Board` · **Time** one day

1. Prove the recoding to yourself first, in the two columns, eight rows. Then stop using the `Compliance status` column until it has been rebuilt, and say in the change note that its previous values were derived rather than assessed.
2. Re-populate it from `O59` and nowhere else. A contract's status is the aggregate of the verdicts against the blocks the contract covers, which is why step 10.2 comes first.
3. Add the four missing contracts: `ACO-09` for `WP-09 Resilience uplift`, `ACO-10` for `WP-10 Compliance mechanisms`, `ACO-11` for `WP-11` and `ACO-12` for `WP-12`. Twelve work packages, twelve contracts.
4. Make each contract carry the acceptance criteria of the blocks its package delivers, by reference to `O55`. `O33`'s four columns do not hold them, so the criteria live in `O59` and the contract points at the rows. Say which.
5. Check `GT-0` against the result. It tests that scope, acceptance criteria and a contract exist before work starts. On the file as handed over, four packages fail on the contract and ten fail on the criteria.
6. Note `O33`'s other problem for item 14: it appears in no relationship sheet and no report. Eight contracts governing $40.3m are invisible to the whole reporting layer.

Done when. Twelve contracts exist, and every `Compliance status` value traces to named `O59` rows.

Goes wrong when. You rebuild the column and keep the old values alongside "for comparison". There is nothing to compare; one of them was never a measurement.

10.5 — Replace the seven tautologies in `O63` with objectives that name an instrument

Sheet `O63 Control Objective` · **Input** `O41`, `R16`, volume 1 chapters 18 and 19 · **Output** seven rewritten control objectives, each naming its instrument · **Signed by** `ROL-07 Chief Compliance Officer` · **Time** one day

1. Read `O63`'s fourth column. All seven values are the obligation name dropped into a frame: `Control that evidences Health data protection.` A control objective that restates the obligation states no control, and it cannot fail, which is why all seven can sit at `In breach`, `No mechanism` or `Partial` without anything being actionable.
2. Read the third column, `Source`, and notice that it names a subject rather than an instrument. "Special-category data, consent and lawful basis" is a topic. An instrument is a named law, rule, guideline or contract with a provision reference.
3. Name the instrument for each of the seven. These are the instruments volume 1 chapters 18 and 19 establish, and they are not interchangeable:

- `CO-01 Health data protection` — the general prohibition on processing health information, the authorisation that lifts it for care, and the Act's own confidentiality rule requiring **written** consent before disclosure to a third party [1][2][3].
- `CO-02 Clinical records retention` — the record-keeping duty, subject to the archives and access regime, over a professional guideline giving at least six years from dormancy in one edition and ideally indefinite in the next [1][4][5]. Record which edition you built against.
- `CO-03 Medical device regulation` — the device classification guideline, and the 2025 communication on devices whose algorithm updates, which locks the version at licensing and requires an approved change plan with a drift trigger [6][7].
- `CO-04 Clinical safety standards` — and here say plainly that there is no South African instrument. The nearest usable things are the published risk-management standard for IT networks carrying medical devices, described in volume 1 and not to be reproduced [8], and the establishment inspection regime [9]. An obligation whose instrument does not exist is a different finding from one whose control is absent.
- `CO-05 Professional registration and scope` — registration with the relevant council and the ethical rule setting one named individual's scope of practice, with the council registers as the authority of record [10][11]. The architectural point is that registration status is a live attribute with an expiry.
- `CO-06 Funder and coding rules` — the funder agreements themselves, which live on `O19`. No national instrument specifies the code sets, so this obligation's instrument is a contract, and that is a finding rather than a gap. Volume 1 chapter 12 has the detail.
- `CO-07 Incident reporting duties` — the establishment standards regime and its inspection and complaints powers [9].

1. Rewrite the `Control objective` cell as a control. `CO-02` becomes something a person can fail: "Every record class has a retention period, a named instrument, a destruction authority and a measured retrieval time, recorded in `O72 Retention Clock`."
2. Add a column `Instrument` and a column `Last assessed`. `O63` has neither, and without the second the `Evidence` column is undated, which is the reason seven obligations can sit in breach for a year.
3. Give `CO-02` and `CO-03` a reference into `O72 Retention Clock` and `O68 Medical Device`. Both sheets are created elsewhere in this volume — items 2 and 7 — and both exist because this obligation has nowhere to land.

Done when. Each of the seven rows names an instrument with a provision reference, and each `Control objective` cell describes something that can be failed.

Goes wrong when. The instrument column holds "POPIA" or "the NHA". A statute is not a provision.

Form 10-C — Control objective, with the instrument behind it

One row per obligation. Feeds `O63 Control Objective`. **`Instrument` and `Last assessed` are columns this repository does not have; adding them is part of step 10.5.** The `Source` column is typed straight into Excel from the existing sheet. A first pass needs seven rows, one per existing obligation. Signed by `ROL-07 Chief Compliance Officer`.

Ref	Obligation	Control objective	Instrument	Evidence	Last assessed
CO-02	EXAMPLE — delete this row. Clinical records retention	Every record class has a period, an authority and a measured retrieval time in O72	NHA s13, read with the archives regime	No mechanism	

10.6 — Break the one-control-per-objective join in `R16`

Sheet `R16` · **Input** `O63`, `O41`, `RPT11` · **Output** a many-to-many join and two corrected `Position` columns · **Signed by** `ROL-17 Security Architect` · **Time** half a day

1. Read `R16`'s seven rows. `CO-01` to `CO-07` are paired with `SEC-01` to `SEC-07` in order. One control per objective, and the pairings are a positional join: identity lifecycle is given as the control for health data protection, periodic access review as the control for accurate coding and claim substantiation, and non-production masking as the control for the statutory reporting of serious safety events.
2. Read the `Position` column. All seven values are `O63`'s `Evidence` value copied across, which means the column reports the objective's position and not the control's. On four rows it therefore contradicts `O41`: `SEC-03 Privileged access management` and `SEC-06 Periodic access review` are `In place` on `O41` and arrive through `R16` as `Partial` and `In breach`.
3. Re-point on substance, allowing many-to-many. `CO-01` needs `SEC-01`, `SEC-04 Attributable access`, `SEC-07 Non-production masking` and `SEC-08 Data-at-rest encryption`. `CO-02` needs no control on `O41` at all, which is why its evidence reads `No mechanism` — record the absence rather than inventing a pairing.
4. Fix `SEC-08`. It appears in no `R16` row and meets no control objective, although it is the only control standing between a lost device and a reportable breach.
5. Set `Position` from `O41` and only from `O41`, and leave the objective's position where it belongs, on `O63`.
6. Do the same reading on `R17`. Four rows pair `POL-01` to `POL-04` with `EP-01` to `EP-04` in order, and `EP-05 Non-production provisioning` enforces nothing — although `POL-01`'s own scope text says it "applies across production and non-production", which makes `EP-05` that policy's enforcement point by the policy's own words.

Done when. No `Position` value on `R16` or `R17` disagrees with the source sheet for the object it names, and every control and every enforcement point appears at least once.

Goes wrong when. You keep one control per objective because the sheet is tidier that way. Seven obligations with one control each is how a control framework comes to have no depth.

10.7 — Give the three loosely labelled checks all their readings

Sheet `ZZ Checks` · **Input** `O63`, `RPT09`, `O20`, `O42`, `O11`, `R21` · **Output** a corrected checks sheet carrying every defensible reading · **Signed by** `ROL-08 Chief Enterprise Architect` · **Time** half a day

1. Three of the sheet's sixteen checks are labelled in a way that admits more than one true answer. Print every reading. A single figure against a loose label is how a repository comes to argue with itself in public.
2. **Control objectives without evidence.** `ZZ Checks` and `RPT01` both say 6 and label it "without evidence". `RPT09`'s footnote says "6 obligations have no mechanism or a mechanism in breach", which is a different and stated test. Literally without evidence — `No mechanism` — is **2** (`CO-02`, `CO-05`). On `RPT09`'s stated test it is **6**. Not fully evidenced is **7 of 7**: not one control objective on the sheet is evidenced. Print 2, 6 and 7, each with its test.
3. **Data entities without lineage.** `O20`'s `Lineage` column is `Absent` on **5** and `Partial` on a further 3, so **8 of 14 lack full lineage** — and the patient master is among them. Meanwhile `O42`'s `MTD-08 Lineage` records coverage `Full`, contradicting both figures on the same file. Print 5 and 8 and the contradiction.

4. **Vacant architecture roles.** The claim of 0 is true on its stated scope: `R21` records all eighteen roles as filled and `O10` agrees. It is silent on the eight data-ownership titles in `O20`'s `Owner` column — `Head of Patient Access`, `Head of Estates` and six others — which exist nowhere in the file, leaving **ten of fourteen data entities owned by a title that does not exist**. Print 0 with the scope, then the ten.
5. Correct the three checks that are plainly wrong while you are there: records 843 to **820**, work packages past end date 3 to **0**, platform components past support 2 to **1**. Items 1 and 9 carry the derivations.
6. Add the one check the sheet does not have and the gate needs: verdicts recorded against verdicts required. On the file as handed over it reads 3 against 148.

Done when. No check on the sheet has one number against a label that supports two.

Goes wrong when. You pick the reading that makes the client look best, or the one that makes them look worst. Both are advocacy.

10.8 — Date every governance decision, and add the body no gate has a holder for

Sheet `O60 Governance Decision` and `O57 Governance Body` · **Input** `O54`, `O52`, `O62` · **Output** dated decisions and one new body · **Signed by** `GB-04 Executive Committee` · **Time** half a day

1. `O60` has four columns: `Ref`, `Decision`, `Body`, `Status`. There is no date, no decision-by date and no record of what the decision depends on. **Add `Required by` and `Decided on`** and record that they are additions.
2. Read the four `Outstanding` rows against the rest of the file and date each one from its dependency, not from a committee calendar. `DEC-02 Select the integration product` is blocked by nothing and blocks `WP-03`, which the file marks `Delivered`; `IF-01` on `O52` and `RSK-01` on `O62` both record the same thing as a dependency and a certain risk. `DEC-04 Designate data owners` is the decision ten of fourteen data entities are waiting for.
3. `DEC-01 Adopt the recommended architecture alternative` is still `Outstanding` while `O54` already marks `ALT-B Recommended` and `ALT-E Partly adopted`. Date it or withdraw the adoption.
4. Now read `O57`'s five bodies for what they do not cover. `GB-01` holds `G1` and `G2`, `GB-02` holds `G0` funding and `G3` benefit, `GB-03` designates data owners, `GB-04` escalates and `GB-05` assesses. **No body in this file holds clinical safety, and no body holds the records duty.** `CO-04 Clinical safety standards` is `In breach` and nobody owns the gate that would catch it; `O62`'s eight delivery risks contain none that is clinical.
5. Add `GB-06 Clinical Safety and Records Group`, with a cadence, a membership including a clinical representative and the person responsible for records, and a mandate holding the clinical half of `G1` and `G2`. Then name the Information Officer: volume 1 chapters 5 and 18 establish that this is the head of the organisation unless the role is delegated and that the holder must be registered with the Regulator, and `O10` and `O11` hold no such role.
6. Do not add more than one body. A governance layer with seven bodies and three verdicts is worse than one with five.

Done when. Every `Outstanding` decision carries a `Required by` date traceable to a dependency, and every gate in `O58` has a named holder for both its halves.

Goes wrong when. The new body is given a mandate that overlaps `GB-01`'s, at which point the first contested verdict goes to `GB-04` and stays there.

Form 10-D — Governance decision record

One row per decision. Feeds `O60 Governance Decision`. **`Required by` and `Decided on` are columns this repository does not have; adding them is part of step 10.8.** A first pass needs seven rows, one per existing decision. Signed by `GB-04 Executive Committee`.

Ref	Decision	Body	Status	Required by	Decided on
DEC-04	EXAMPLE — delete this row. Designate data owners	GB-03	Outstanding	Before TA-1 exit	

10.9 — Put an expiry on every dispensation, and a decision date on `CR-G05`

Sheet `O61 Governance Change Request` · **Input** `O57`, `O60`, `O30` · **Output** an `Expires` value on every row that needs one · **Signed by** `GB-01 Architecture Board` · **Time** two hours

1. `O61` holds five rows and its sixth column is `Expires`. Only `CR-G01` carries a date, 2029-06. Four rows are blank.
2. `GB-01`'s mandate on `O57` says it "approves dispensations with an expiry". The file's own governance body has a rule its own change-request sheet breaks on four rows out of five.
3. Give each row what it needs. `CR-G01 Dispensation for point-to-point file transfer` has its date; check it against the work package that removes the need for it. `CR-G02` is a scope change, approved — a scope change does not expire, so record `Not applicable` rather than leaving it blank, because a blank cell is not a decision. `CR-G05 Refuse or re-baseline the integration package` is `Board decision required` with **no date by which the board must decide**, against a work package the file marks `Delivered`. That is the row to take to the next meeting.
4. `CR-G03 Raise unfunded building blocks as gaps` is a correction against the repository itself, still only `Raised` — the file asking to have `O29 Gap`'s two rows brought up to the thirty-five building blocks nothing addresses. Approve it or close it.
5. Cross-check each dispensation against the control it suspends. `CR-G01` permits point-to-point file transfer while `SEC-05 Interface logging` is `Partial`, thirteen interfaces are unmanaged and ten carry no logging: a dispensation to 2029-06 against an already-partial control is a four-year hole with a date on it.

Done when. No row in `O61` has an empty `Expires` cell, and every `Board decision required` row has a date by which the board must decide.

Goes wrong when. The expiry is set to the end of the programme. A dispensation that outlives the thing it was granted for has become the architecture.

Form 10-E — Dispensation and change request

One row per request. Feeds `O61 Governance Change Request`. Every row needs a value in `Expires`, including `Not applicable` where a request does not expire. A first pass needs five rows, one per existing request. Signed by `GB-01 Architecture Board`.

Ref	Change request	Against	Type	Status	Expires
CR-G05	EXAMPLE — delete this row. Refuse or re-baseline the integration package	WP-03	Compliance	Board decision required	

10.10 — Gate on the clock: no measured retrieval time, no pass

Sheet `O59` · **Input** `O72 Retention Clock`, `O49 Resilience Tier`, `O20`, volume 1 chapters 13 and 19 · **Output** a retrieval verdict per record class · **Signed by** `GB-06 Clinical Safety and Records Group` · **Time** one day

1. This is where the clock this volume has been carrying since the Preliminary Phase is tested. Preliminary set it, Phase C recorded it, Phase D measured it. **Phase G refuses it.**
2. The rule is one sentence, and it goes in the evidence standard from step 10.1: **a record class with no measured retrieval time does not pass any gate.** Not an estimated time. Not a target. A time somebody measured, on a date, with a result.
3. Run it per record class, not per application. Take `O72 Retention Clock`'s `Record class` column and ask the gate's three questions of each: what was measured, by whom, against what. The *against what* is a stopwatch and a named request.
4. Record the verdict in `O59` against the building block that owns the class, with the measured time in `Evidence`. `Non-compliant` with evidence "not measured" is a complete row; an empty cell is not.
5. Have the reason to hand, because this is the gate most often argued down. The Auditor-General's 2026 health-sector findings record fifteen of twenty-seven facilities with missing or lost records, seventeen not tracking records after issue, and **three of nine paid medico-legal claims conceded because the record could not be found** [12]. A conceded claim is a measured price for an unmeasured retrieval time, and it is the only number in this argument that an executive will not dispute.
6. Check `O49 Resilience Tier` too, because it is the same question asked twice. `O49`'s tier label carries no information at all — `Tier 1` holds a 24-hour objective twice, four hours twice, one hour once and fifteen minutes once, so the loosest objective in the file and the tightest are both on it, while `RES-01 EHR / EPR` is `Tier 1` at 24 hours and `Never tested` — and nine of twelve applications including both `Critical` ones have never been tested. An untested recovery objective is an estimate, and an estimate does not pass either.

7. Note what the gate cannot do. Volume 1 chapter 13 shows that the retention period itself is unsettled: the guideline gives at least six years from dormancy in its 2016 edition and ideally indefinite, particularly if electronic, in its September 2022 review draft [4][5], and the duty is subject to the archives and access regime [1]. The gate does not resolve the period. It refuses a class with no instrument named against it, which is a different and achievable test.

Done when. `O72` has a row per record class, every row has a measured retrieval time with a date, and `O59` carries the verdict.

Goes wrong when. A target is accepted as a measurement, which happens whenever the gate is run by the team that owns the system.

10.11 — Enforce the classification rule where the data is read

Sheet `O64 Security Policy` and `O65 Enforcement Point` · **Input** `O20`, `O37`, `R17`, `O48` · **Output** a dated position per policy and a verdict per enforcement point · **Signed by** `ROL-17 Security Architect` · **Time** half a day

1. All four policies on `O64` read `Breached` in the `Position` column, and the sheet has no date column at all. A policy that has been breached since nobody knows when is not a policy. **Add `Last assessed`.**
2. Read `O65`'s `Position` column. Every one of the five enforcement points records its own defect: `EP-01 Design review gate` is `Newly established`; `EP-02 Identity provider` "enforces authentication, not attribution"; `EP-03 API gateway` records `Product not yet selected`; `EP-04 Data platform` records `Classification not applied at read`; `EP-05 Non-production provisioning` records "Most refreshes are unmasked".
3. `EP-04` is the one that matters most here, because it is where this volume's second overlay is enforced or not. Classification inherits downward: an attribute may be tighter than its entity and may never be looser. `O37` breaks that on **twenty-six of sixty attributes**, including `ATR-001`, the patient `Identifier`, at `Public` beneath a `Restricted` entity — against the file's own principle `PRN-14` at `O05` row 18. With classification not applied at read, none of those twenty-six is enforced anywhere, so correcting `O37` changes nothing until `EP-04` does.
4. So the gate takes two verdicts, not one: the classification values are right, and the point that reads them applies them. Write both into `O59`, against the data building blocks.
5. `EP-03` blocks `POL-03`: the API gateway is the enforcement point for logging material activity and the product has not been selected, which is `DEC-02`, which is step 10.8. Record the chain in one row, so the board sees one decision rather than three findings.
6. Finally, read `O48 Environment` against `POL-01`: the file records production as holding no personal data while development and pre-production do. Either production is mislabelled or the environments are, and `POL-01` cannot be assessed until somebody says which.

Done when. Every policy on `O64` has a `Last assessed` date, and every enforcement point on `O65` has a verdict in `O59` separate from the policy's.

Goes wrong when. The classification work is signed off on `O37` alone. The sheet is the claim; the read path is the control.

10.12 — Rebuild the three reports a regulator would actually be shown

Sheet `RPT09`, `RPT11`, `RPT12` · **Input** `O63`, `O41`, `O32`, `R18`, `O59` · **Output** three reports that compute something · **Signed by** `ROL-08 Chief Enterprise Architect` · **Time** one day

1. `RPT09 Regulatory Coverage` is `O63` reprinted with a footnote. Give it the three columns the obligation conversation needs and `O63` now has: `Instrument`, `Last assessed`, and the `O59` rows that evidence it. Then make the footnote state its test, because at present three sheets label the same figure three ways.
2. `RPT11 Security Position` is `O41` reprinted. Its footnote — "3 controls are absent. A control that is absent is not a control." — is correct, and the best sentence in the file. Keep it and add the missing column: which control objective each control meets, from `R16` as re-pointed in step 10.6. `SEC-08` stays blank until that step runs, which is the point of the column.
3. `RPT12 Standards Conformance` is `O32` reprinted, and its `Conformance` column is one application's verdict presented as a portfolio position: `R18` gives each of the sixteen standards exactly one application, cycling the six odd-numbered ones, and `O32`'s value equals `R18`'s in all sixteen rows. Six applications are governed by no standard at all. Replace the column with two: applications assessed, and applications conformant.
4. Settle the two standards that are both `Deprecated` and `Breach` — `STD-15 Records management standard` and `STD-16 Supplier security standard`. A breach of a withdrawn standard is actionable by nobody. `STD-15` being the records management standard makes it the one to settle first.
5. Put a reference date and a generation date in every report header. None of the thirteen reports carries either, which is why wrong figures have survived in three of them.

Done when. None of the three reports is a single object sheet with a column dropped, and each footnote states the test it applied.

Goes wrong when. You improve the reports and leave `RPT01`'s dashboard reading 843. The dashboard is the only page most readers see.

What the two running examples do with this

WORKED ANSWERS — THE SAME GATE, TWO ORGANISATIONS

Driehoek Health Group — nine acute hospitals, two day clinics, about 1,750 registered beds. Its first full `G1` produced ninety-one verdicts, fifty-three of them `Non-compliant` with the evidence value "not tested". The board accepted all fifty-three rather than argue them, because the alternative was three verdicts and no record. Its step 10.3 reconciliation found `R07` wrong on ten of twelve rows, which changed the resilience programme more than the gate did.

Rietvlei District Health Service — one district hospital, three community health centres, 22 fixed clinics, a mobile service. Rietvlei could not run step 10.10 at all on first pass, because no record class had a measured retrieval time and the paper archive had no index. Its gate output was therefore a single row: `Non-compliant`, evidence "retrieval not measurable; no index exists", tested against a named request on a named date. That one row funded the index, which three years of policy had not.

BEFORE YOU LEAVE PHASE G

Seven questions. Each answer is a document reference, a count or a date, or the gate is not running.

For the last gate you ran: what was reconciled, by whom, and against what?

How many verdicts are recorded in O59, out of how many the gate regime requires?

Which instrument, with which provision, stands behind each of the seven obligations?

Which body holds the clinical half of G1 and G2, and when did it last meet?

Name every live dispensation, the control it suspends and the date it expires.

For each record class, what is its measured retrieval time and the date it was measured?

On what date was each O63 and O64 position last assessed, and by whom?

THE CHAPTER IN TEN LINES

- Eight relationship sheets resolve perfectly and mean nothing, so the gate cannot ask whether a reference resolves.
- The Phase G question is what was reconciled, by whom, against what. Three answers, and a missing third answer is itself a verdict.
- O59 holds three rows against the hundred and forty-eight verdicts four gates over thirty-seven building blocks require.
- All three sit at G2, all three read Non-compliant and all three record that no evidence was produced. They are the most honest cells in the file.
- O33's compliance status is O30's delivery column recoded: Delivered became Not yet assessable on five of five rows and Past end date became Non-compliant on three of three.
- Four work packages worth 21.0 of the 61.3 million dollars have no architecture contract, and GT-0 tests that a contract exists before work starts.
- All seven control objectives in O63 restate their own obligation, so none of them can fail. Name the instrument and the provision instead.
- There is no South African instrument for the clinical safety of IT. An obligation whose instrument does not exist is a different finding from one whose control is absent.
- R16 gives one control per objective in row order and copies the objective's position into the control's, contradicting O41 on four rows.
- Three checks on ZZ Checks are loosely labelled. Print every reading: 2, 6 and 7 of 7 for evidence; 5 and 8 of 14 for lineage; 0 for roles with ten entities owned by a title that does not exist.
- No body in O57 holds clinical safety or the records duty, and no role in O11 is the Information Officer.
- Four of five change requests have no expiry, against a board whose own mandate is to approve dispensations with one.
- A record class with no measured retrieval time does not pass any gate. Not a target, not an estimate: a time somebody measured, on a date.
- Correcting the classification values on O37 changes nothing while O65 records that classification is not applied at read.

SOURCES FOR THIS CHAPTER

- **1.** National Health Act 61 of 2003, consolidated text (section 13 records duty, subject to the National Archives and Record Service of South Africa Act 43 of 1996 and the Promotion of Access to Information Act; section 14 confidentiality and written consent to disclosure; section 17 offences) https://www.saflii.org/za/legis/consol_act/nha2003147.pdf
- **2.** Protection of Personal Information Act 4 of 2013, section 26, prohibition on processing of special personal information https://popia.co.za/section-26-prohibition-on-processing-of-special-personal-information/
- **3.** Protection of Personal Information Act 4 of 2013, section 32, authorisation concerning a data subject's health or sex life https://popia.co.za/section-32-authorisation-concerning-data-subjects-health-or-sex-life/
- **4.** HPCSA Booklet 9, Guidelines on the Keeping of Patient Records, September 2016 edition https://www.hpcsa-blogs.co.za/wp-content/uploads/2022/08/Booklet-9-Guidelines-on-Patient-Records.pdf
- **5.** HPCSA Booklet 9, Keeping of Patient Records, review draft marked September 2022 https://www.hpcsa-blogs.co.za/wp-content/uploads/2022/11/Booklet-9-Keeping-of-Patient-Records_Review-Draft_-FINAL_Sept-2022.pdf
- **6.** SAHPRA, SAHPGL-MD-04 version 5, Guideline for the Classification of Medical Devices and IVDs, effective 27 February 2025 https://www.sahpra.org.za/wp-content/uploads/2025/03/SAHPGL-MD-04_v5-Guideline-for-Classification-of-MD-and-IVDs.pdf
- **7.** SAHPRA, Communication to industry MD08-2025/2026, Regulatory Requirements of Artificial Intelligence and Machine Learning Enabled Medical Devices, 26 September 2025 https://www.sahpra.org.za/wp-content/uploads/2025/09/MD08-20252026_-SAHPRA-Communication-to-Industry-AI-Medical-devices_Acknowledgements.pdf
- **8.** IEC 80001-1:2021, Application of risk management for IT networks incorporating medical devices, part 1. Catalogue entry; the standard's content is described, never reproduced https://www.iso.org/standard/72026.html
- **9.** Office of Health Standards Compliance, 2023/24 Annual Performance Report presentation, 16 October 2024 (inspection and certification of health establishments, and the complaints route) https://static.pmg.org.za/241016Updated_OHSCs_Presentation_-_16_Oct_2024.pdf
- **10.** HPCSA, Adhering to scope of practice and scope of profession (scope of profession set by statute for a profession; scope of practice set for one named individual) https://www.hpcsa-blogs.co.za/adhering-to-scope-of-practice-and-scope-of-profession/
- **11.** South African Nursing Council, eRegister (the register as the authority of record for registration status) https://www.sanc.co.za/erHelp.aspx
- **12.** Auditor-General findings on the health sector, via MedicalBrief, 14 October 2026 (15 of 27 facilities with missing or lost records, 17 not tracking records after issue, three of nine paid medico-legal claims conceded because the record could not be found) https://www.medicalbrief.co.za/health-sector-crippled-by-recurring-systemic-weaknesses-ag-warns/

ITEM 11

Architecture Change Management and the retention clock

The phase that produces nothing of its own, and the only one that runs twice: six queries against six date columns, and the counts that drift the moment anybody repairs anything.

The only phase that runs twice

Every phase before this one produced something for the first time. Phase H produces nothing of its own. Its content is **re-reading what the other phases wrote, on a cadence, and deciding what to do when the reading has changed.**

That makes it the phase clients skip and the one this industry cannot afford to skip, because five of the six sheets this volume creates carry a date column. A retention period set against an edition of a guideline. A consent with an expiry. A code system binding with an effective window. A device with a calibration date. A retrieval time measured on a Tuesday in November. **Every one is true on the day it is written and false later**, and nothing in the repository will tell you when.

Three jobs: **the change request**, how a change enters the architecture at all; **the trigger test**, what makes a change an architecture change rather than an implementation change, decided in advance because decided in the moment it is decided by whoever is loudest; and **the periodic re-reading**, six queries, each one line, each returning rows somebody has to act on.

Overlay 1 closes here. Preliminary set the retention clock; Phase C recorded it against the entities; Phase D measured retrieval and got a number; Phase G refused anything without one. **Phase H re-runs the measurement and re-reads the clock**, and if it does not, the number on the page is a number from a date that has passed and the organisation is quoting history as a position.

What Phase H has to work with, and why it hangs off nothing

Sheet	Rows	What it should do here	What it does
`061 Governance Change Request`	5	Carry every change in, with a raiser and a date	An `Against` column taking two kinds of value, and no raiser or date columns
`060 Governance Decision`	7	Record what was decided, by whom, when	Four `Outstanding`, three `Approved`, no date column of any kind
`066 Architecture Increment`	6	Say what has landed and what has not	Six increments on a relative timebase with no recorded origin
`R23`	6	Join an increment to the project that lands it	A positional join, re-derived in item 9 to seven rows
`058 Governance Gate`	4	Give the re-read a gate to report into	Four well-formed gates, one of which has ever been run
`067` to `072`	new	Hold the six things that go stale	Created in items 2, 5 and 7; this item is the first to re-read them

The structural problem is finding 7, and it bites hardest here. **`060` and `061` appear in no relationship sheet at all**, nor does `057 Governance Body`, nor `058`, nor `005 Principle`. Twenty of the sixty-five object sheets — 206 records, every principle, constraint, governance body, gate, decision and change request among them — are referenced by nothing. The governance layer hangs off the rest of this repository by `061`'s `Against` column and `060`'s free-text `Body` column, and nothing else. **Change management cannot structurally reach the thing it changes**, so this item starts with wiring.

> **WHAT MAKES PHASE H DIFFERENT FROM THE OTHER NINE ITEMS**
>
> Every other item tells you to write something down. This one tells you to **read something back** — and the test of whether the phase is running is not whether a document exists but whether a query was run on a date and returned a number somebody acted on. A Phase H with minutes and no row counts is not running.

The change request, and the column that can only point at two things

`061`'s six columns are `Ref`, `Change request`, `Against`, `Type`, `Status`, `Expires`, over five rows. Read the third column and the wiring problem is visible.

Ref	Against	Type	Status
`CR-G01`	`WP-04`	Dispensation	Granted, expires 2029-06
`CR-G02`	`WP-03`	Scope	Approved
`CR-G03`	`Repository`	Correction	Raised
`CR-G04`	`Repository`	Correction	Approved
`CR-G05`	`WP-03`	Compliance	Board decision required

Three rows point at a work package. Two point at the literal string `Repository`. **There is no third kind of value**, so there is no way to raise a change against a principle, a standard, a data entity, a classification, a building block, a recovery objective, a retention period, a consent scope or a code system edition. Everything this volume has spent eleven items building has nowhere to be changed from.

That is also a trap for anyone working from a findings summary. `CR-G03` is widely described as raised against `O29 Gap`. It is not — **its `Against` value is the literal string `Repository`**, and an architect sent to `O29` finds two rows about capability uplift and no change request.

`CR-G03` asks for a derivation the file cannot perform

`CR-G03` reads "Raise unfunded building blocks as gaps" and has sat at `Raised` since handover. It is quoted as the moment the file notices its own emptiest sheet, and it is that. It is also **unperformable on the file's own terms.**

`O55` holds 37 building blocks. `O30` holds 12 work packages and costs every one, twelve figures summing to the `$61.3m` the file agrees on in three places. `R13` maps packages to blocks and reaches **all 37**. **No sheet in this file records funding against a building block** — no cost column on `O55`, none on `O35`, `O44`, `O45` or `O46`, no money on `R13`. On the file's own terms the set of unfunded blocks is empty, and the derivation is unavailable rather than hard. The request is really asking for a column, and what to take from it is not "find the unfunded blocks" but "this repository cannot cost a building block, and somebody noticed". While costs are in view: **every cost here is in US dollars with no rate, no rate date and no base year** — `O30`, `O54`, `O56`, `R14` — in a repository whose banner names a South African provider. Quote `$61.3m` and `$46m` as the file spells them and never convert silently.

`CR-G04` is quieter and worse. "Record benefit realisation dates per package", `Against` `Repository`, `Status` **`Approved`** — no date of approval, no owner, nothing done, and `GT-3 G3 Benefit` still testing "Stated benefit confirmed against a date" over twelve packages that have none. An approved correction nobody performed is worse than a raised one, because the sheet says it is finished.

11.1 — Extend `O61`, then clear all five rows

Sheet `O61 Governance Change Request` · **Input** `O55`, `O30`, `R13`, `O29`, `O67` to `O72` · **Output** an `Against` convention, four new columns, a dated disposition on five rows · **Signed by** `ROL-08 Chief Enterprise Architect` · **Time** a day

1. Keep the five existing rows and their references exactly; two are quoted in the governance record, and renumbering is how a change log loses its history.
2. Write the `Against` convention: either an object reference that resolves on its own sheet — `PRN-14`, `STD-04`, `E01`, `BBB-01`, `RTN-02`, `CSY-03`, `DEV-01` — or the literal `Repository` for a change to the file's structure. Nothing else; free text here is how a change request stops being findable. Add four columns as recorded additions — `Raised by`, `Raised on`, `Required by`, `Decided on` — and one `Type` value, `Lapse`, for an expired consent, a closed code system edition, an overdue calibration or a superseded instrument.
3. Close `CR-G03` with its arithmetic on the row: no sheet costs a building block; `R13` reaches all 37 from 12 costed packages, so on the file's own terms none is unfunded; the request converts to a schema change adding `Allocated cost` to `O55`. Then date it. Do **not** use `CR-G03` as the route to the gap set — item 8 derived that from the reports and the derivation stands alone.
4. Dispose of the other four. `CR-G04`: an owner and a date, or withdraw the approval. `CR-G05`: the next agenda with a `Required by` date. `CR-G02`: a `Lapse`. `CR-G01`: check its 2029-06 expiry against the package that removes the need for it.

Done when. Every `Against` value resolves or reads `Repository`, every row carries a raiser and a raised date, and no row is older than one cycle without a dated disposition.

Goes wrong when. `CR-G03` is quietly re-raised each cycle so the sheet looks active. A change request raised four times has been refused four times.

Form 11-A — Governance change request, part one: identify

Feeds `O61 Governance Change Request` · **Rows for a first pass** 8 to 14, the five existing rows plus whatever the first re-read returns · **Signed by** `ROL-08 Chief Enterprise Architect` · **Ratified by** `GB-01`

The sheet's six real columns in sheet order; all six go straight into Excel. Form 11-B carries the four added ones, keyed on `Ref`.

Ref	Change request	Against	Type	Status	Expires
CR-G06	EXAMPLE — delete this row. Consent CNS-04 lapsed and not withdrawn	CNS-04	Lapse	Raised	Not applicable

Form 11-B — Governance change request, part two: who raised it and who decides

Feeds `O61 Governance Change Request` · **Rows for a first pass** the same 8 to 14 rows as form 11-A · **Signed by** `ROL-08 Chief Enterprise Architect` · **Ratified by** `GB-01`

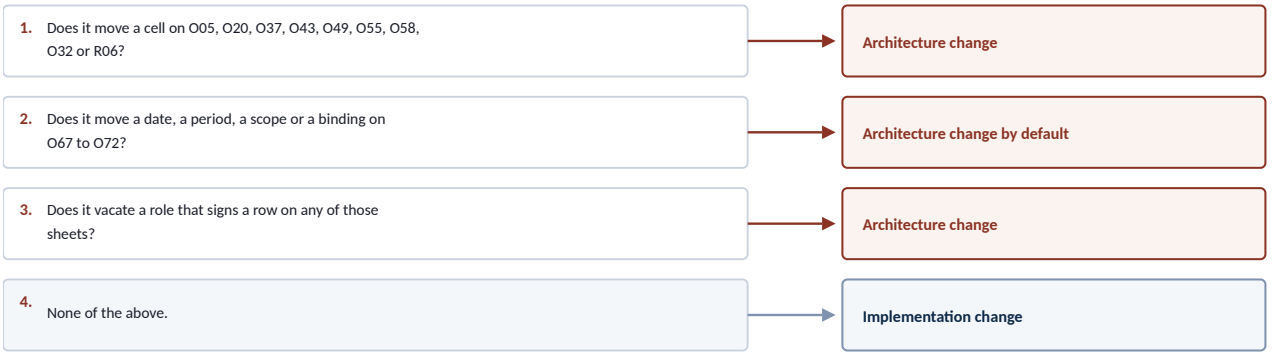
`Raised by`, `Raised on`, `Required by` and `Decided on` are four columns this repository does not have; step 11.1 adds them. Until it runs they exist only in a meeting minute, which is why four of five rows have no expiry and nobody can date `CR-G03`.

Ref	Change request	Raised by	Raised on	Required by	Decided on
CR-G03	EXAMPLE — delete this row. Raise unfunded building blocks as gaps	unknown — asked 2026-11-19	unknown		

Architecture change or implementation change

The decision Phase H exists to make, and the one most often made by accident. Wrong one way and every version upgrade in the estate arrives at the Architecture Board, which stops holding a board within two cycles. Wrong the other and a code system edition changes under a coded field and nobody is told. Write the test down **before** the first contested change, and make it answerable by somebody who is not an architect.

Does this change go on O61, or into the ordinary change process?



Test 2 is default-deny: the raiser may argue it down, and the argument is minuted.

Tests run in order. The first yes stops the test. A no on all four is the only route to the ordinary change process, which is where most changes belong.

Figure 11.1 The trigger test, in the order it is applied. Four questions, the first yes stopping the test, and only a no on all four routing the change away from O61. Test 2 exists because a date on one of the six new sheets drives a disposal, a disclosure refusal or a device withdrawal.

Architecture change — the sheet and column it moves	Implementation change
A principle on `005` ; a standard's status on `032`	A report layout, a field label, a meeting date
A classification on `020` or `037`	A server, a container image, a certificate
The system an entity is mastered in, `020` or `043`	A software version that moves no risk class
An attribute-to-code-system binding, `069` `Where bound`	A code system edition where the binding holds
A period, trigger, hold rule or authority, `072`	A storage tier or backup target
A consent type's scope, `067` ; a device's risk class or segment, `068`	A consent form's wording at unchanged scope; a device's asset tag
A recovery objective, `049` ; a measure's numerator, denominator, exclusions or boundary rule, `070`	A monitoring threshold; a dashboard's colour scale
A gate criterion on `058` ; a block's acceptance criterion on `055` ; an interface becoming or ceasing to be managed, `R06` ; a signing role becoming vacant	A retry count on a managed interface; a staffing change that vacates no signing role

Two boundary cases are worth settling in writing, because both come up and both are usually got wrong.

A code system edition. An implementation change **if** the binding stays put and the point-in-time join still resolves historical records against the edition in force when they were created. It becomes an architecture change the moment the new edition discontinues a code the organisation still reports on, because the measure definition reading it has changed meaning without changing text. Volume 1 chapter 12 has the mechanism: the national diagnosis reference table carries start and end dates per code, a status of added, modified or discontinued, and a revision history, and updates biennially on 1 July [1].

A device software version. An implementation change, until the manufacturer's own change plan says otherwise. For a device whose algorithm updates, the regulator's 2025 communication to industry locks the version at licensing and requires an approved change plan with a stated trigger before the version moves [2]. A version moving outside that plan is a regulatory event.

The last row of the table is the one nobody classifies as a change at all. `011` `Role` holds eighteen roles and `R21` records all eighteen filled, which is why `ZZ` `Checks` reports zero vacant architecture roles — silent on the eight data-ownership titles in `020` 's `Owner` column that exist nowhere in the file. When the person signing a retention class leaves, the row's signature has lapsed and nothing recomputes.

11.2 — Write the trigger test, and give it to the change process that will use it

Sheet `058` `Governance Gate` , `061` , `005` `Principle` · **Input** the two lists above and the organisation's existing change process · **Output** a one-page test and a routing rule · **Signed by** `ROL-08` `Chief Enterprise Architect` · **Time** half a day

1. Write the table above out on one page in the organisation's own words, keeping the sheet and column against each line. A test that names a column is applied; a test that names a concept is argued. Keep the right-hand column at least as long as the left, because a test that says only what is in scope is read as asserting scope over everything. Settle the two boundary cases in writing, with the instrument named.
2. Then route it. The existing change process raises the change; this test decides whether `061` gets a row. Give that process one question — does this alter anything on the left-hand list — and the name of the person to ask when the answer is unclear. Do not ask a change manager to read a principle.
3. Add one rule the lists do not cover: **a change that moves a date on any of `067` to `072` is an architecture change by default**, and the raiser may argue it down. Then record what the test does not decide — urgency, funding, whether the change is a good idea. It decides which register it lands in.

Done when. The test is one page, minuted, and the change process has the routing question and the named person.

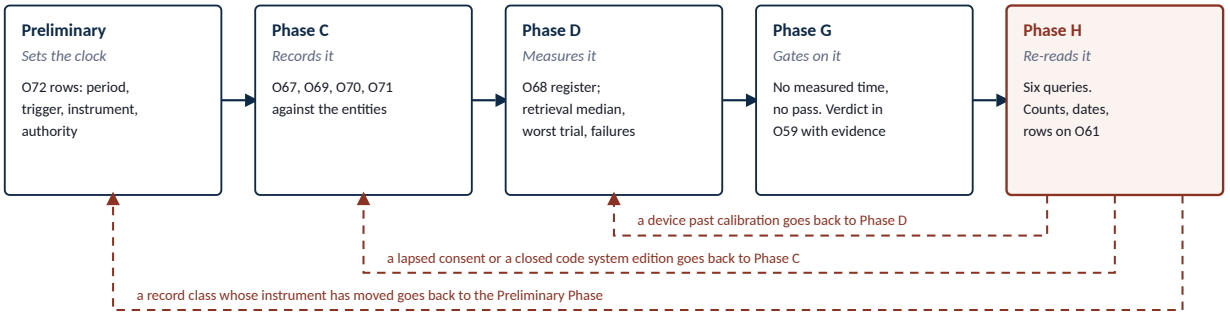
Goes wrong when. It is written as a severity matrix. Severity is about consequence; this test is about kind.

The six re-reads

Six queries, each a single condition against a date. Run them on a cadence, write the count down, raise what they return on `061` .

Sheet	Column read	The query	What a row means
`072` `Retention Clock`	`Instrument`	The edition named is no longer current	The schedule was built against superseded guidance
`067` `Consent`	`Expires` , `Withdrawn`	`Expires` in the past and `Withdrawn` empty	A consent has lapsed and nobody has been told
`069` `Code System`	`Effective to` , `Where bound`	`Effective to` in the past and `Where bound` not empty	A live field is bound to a closed edition
`068` `Medical Device`	`Calibration due`	The date is in the past	A device is producing readings nobody has validated
`070` `Measure Definition`	`Definition version` , `Period`	`Version` changed inside an unclosed `Period`	A published figure is not comparable with itself
`071` `Disclosure Event`	`Date` , `Consent reference`	Dated after its consent expired or was withdrawn	A disclosure that may have had no authority

Overlay 1, one full turn: what each phase writes, and what Phase H sends back



Everything before Phase H is a commitment. Phase H is where it becomes a number with a date on it, twice.

Figure 11.2 Overlay 1 closing. Preliminary sets the clock, Phase C records it, Phase D measures retrieval, Phase G refuses anything without a measurement, and Phase H re-reads all of it and hands each finding back to the phase that owns the object.

11.3 — Re-read `O72 Retention Clock` and re-date the instrument

Sheet `O72 Retention Clock` · **Input** the current edition of the professional guidance, obtained from the council; `O63` ; `O32` · **Output** a dated re-read with an instrument check per row · **Signed by** `ROL-07 Chief Compliance Officer` · **Time** half a day

1. Take the `Instrument` column first. Every row names what sets its period and the edition it was built against, because item 2 made that a rule. The query is one line: is that edition still in force?
2. This is not theoretical here. **There is no South African retention instrument.** Every period in circulation traces to professional guidance made under the Health Professions Act 56 of 1974 — not a statute, not a regulation — and it disagrees with itself across versions: at least six years from dormancy in the September 2016 edition, ideally indefinite and particularly if electronic in the revision marked September 2022 [3][4]. Which is in force could not be established for volume 1. A re-read finding the organisation built against the 2016 text has turned the general adult period from a schedule into an archive strategy.
3. Read `Period` against `Clock starts at` : a period with no trigger cannot be computed, and dormancy needs a definition the system of record can evaluate. Then read `Hold rule` and act on it first — a class whose period has run while its hold rule is engaged, by litigation, a complaint, an inspection or an access request, is not a disposal candidate, and a schedule that deletes it has destroyed the record that was about to be called for.
4. Re-read `Destruction authority` last and hardest, because it is likeliest to have moved: the records duty sits on the person in charge of the establishment and is expressly subject to the archives and access-to-information regime [5], which is why a provincial facility cannot set its own schedule and a private group can. Then automate no deletion. Item 2 deferred automation to here on two conditions, a measured retrieval time and a recorded gate verdict; automate disposal only where `Instrument` , `Hold rule` and `Destruction authority` are all non-empty, retrieval has been measured twice, and the register of destroyed records exists.

Done when. Every row carries the edition it was built against, a re-read date, and a disposition of hold, dispose or not yet.

Goes wrong when. The schedule is re-read from a secondary account of the guidance rather than the guidance. A retention schedule built on a cited summary is a compliance artefact built on sand.

11.4 — Find the consents that have expired and not been withdrawn

Sheet `O67 Consent` and `O71 Disclosure Event` · **Input** the consent capture system and the disclosure log · **Output** three row counts and a stop list · **Signed by** `ROL-07 Chief Compliance Officer` · **Time** half a day

1. Run the query on `O67` : `Expires` earlier than today **and** `Withdrawn` empty. Those rows are what this phase exists for, and they are invisible in any repository that models consent as a flag.
2. Count three populations and never merge them, because the obligations differ. **Lapsed** — expired, not withdrawn: the authority ran out and the next use needs re-consent. **Withdrawn** — a dated act by the subject, which stops future disclosure and does not unsend what was sent, so it needs the disclosure log read backwards. **Open-ended** — `Expires` empty, needing a scope decision rather than a date, and the population most often relied on wrongly.
3. Then cross `O67` against `O71` on `Consent reference` . Any `O71` row dated after the cited consent's `Expires` , or after its `Withdrawn` , is a disclosure made without a live authority. **This is the most valuable query in the phase**, and the file as handed over cannot run it: `E13 Consent` appears on `O20` , two rows of `R05` and the `RPT05` copy and nowhere else — no attributes, no mastering rule, no building block, no mapping, no entity relationship, and a lineage of `Full` for an entity with no lineage path anywhere.
4. Produce the stop list as the output, not the count. "Stop sending this class of record to this recipient, from this date, because the consent cited lapsed on that date" is actionable by an integration team. A count is a slide.

Done when. The three populations are counted and dated, and every disclosure found without a live authority has a row on `O61` with `Type Lapse` .

Goes wrong when. Clinical access is gated on the consent flag. The lawful basis for processing for care is a statutory authorisation, not consent [6]; consent governs disclosure outside the care relationship, and research.

11.5 — Find the code systems whose window has closed, and the measures that read them

Sheet `O69 Code System` and `O70 Measure Definition` · **Input** the publishers' release notes, the funder's rejection report, the figures published last cycle · **Output** live bindings against closed editions, and a comparability statement per measure · **Signed by** the named data owner of the entity the system is bound to · **Time** one day

1. Run two queries on `O69`. `Effective to` earlier than today **and** `Where bound` not empty — a live field bound to a closed edition. And more than one row for the same `Code system` with `Effective to` empty — two editions both claiming to be current, which item 2 set as this sheet's defining defect.
2. A closed-edition binding is not automatically wrong. It is wrong when the field is still written to against that edition, right when the field holds historical records that must resolve against the edition in force at the time of care. Record which: that one word is the difference between a defect and a correctly dated archive. Re-read `Licence position` too — an expired licence on a code set bound to a live field is a commercial exposure no clinical check will find.
3. Then run `O70`'s query: a `Definition version` that changed while a `Period` was still open. Any measure in that state has published two figures under one name. A measure whose numerator counts coded events inherits the code system's edition, so a discontinued code silently moves the numerator — which is why these two sheets are read together. Re-read `Exclusions` and `Boundary rule` hardest, because they move without anybody calling it a change.
4. Publish a comparability statement, one line per measure with its version and period. The reason this needs a sheet rather than a convention is in volume 1 chapter 2: four defensible theatre-utilisation figures come out of one list, raw utilisation at 80% against touch-time utilisation at 60% on the same list, and **there is no conversion between them** [8]. Raise what you find against the entity rather than the code system — the entity is what moved.

Done when. Every `Code system` value has exactly one row with an empty `Effective to`, every closed-edition row with a live binding is marked historical or raised, and every figure published last cycle names the `O70` row and version it was computed under.

Goes wrong when. The edition is treated as reference-data housekeeping, or the definition changes to improve the trend and the version is not incremented. The second is not a reporting error; it is a misstatement with a date on it.

11.6 — Find the devices past calibration, and the versions that moved

Sheet `O68 Medical Device` · **Input** the maintenance management system, the manufacturer's release notes, `O27` · **Output** four counts and a withdrawal list · **Signed by** `ROL-14 Head of Operations` · **Time** one day

1. Run four queries: `Calibration due` earlier than today; `Maintenance due` earlier than today; `Risk class` still reading `Not established`; `Software version` different from the version recorded at the last re-read.
2. State the first count clinically rather than in compliance terms, because that is what gets it acted on. A device past calibration is producing readings a clinician acts on without re-checking them, and those readings land in `E03 Clinical Observation`. The device record is `Public` on `O20` and the observations are `Confidential` — overlay 2 broken in the technology layer, and item 7 set the check: no row of `O68` may be classified more openly than the entity its readings land in.
3. Treat the fourth count as the regulatory one: a version that moved outside the manufacturer's approved change plan is a regulatory event [2], and the device is now a different device from the one that was assessed. Argue the register's existence from the file, because it will be questioned — the extension sheet declares the type as "a clinical device with calibration, maintenance and risk class", and **the word "calibration" appears twice in the whole file**, in that definition and in `E14`'s.
4. Produce a withdrawal list, not a report. A device past calibration either comes out of clinical use or carries a dated, named, time-boxed acceptance of the risk — and the second is a dispensation on `O61` with an expiry. Check `Network segment` against the segment list item 7 produced: a free-text value not on that list is a blank.

Done when. Four counts are recorded with a date, and every overdue device is either withdrawn or carries a dated dispensation.

Goes wrong when. The query is run against the maintenance system instead of `O68`. The maintenance system knows about the devices somebody entered into it; the register is the architecture's statement of what exists, and the gap between the two is itself a finding.

11.7 — Re-measure the retrieval time, with the same method and a different week

Sheet `O49 Resilience Tier`, `O26`, and the measurement record · **Input** item 7's method and its first result · **Output** a second dated measurement and a trend · **Signed by** `ROL-03 Chief Operating Officer` · **Time** one afternoon per site

1. Use item 7's method unchanged — ten trials on the live read-only record with the primary unreachable, patients picked at random from the admitted list, the clock on a named clinician getting a named patient's medication and allergy list into their hand on paper. **Changing the method between cycles destroys the only thing a second measurement gives you.** Record the same five figures: median, 95th percentile, worst trial, trials that produced no record at all, and the age of the oldest record in the copy at the moment of the trial.
2. Then record the two Phase D could not produce: the change since the last measurement, and the number of cycles since a measurement was taken. The second catches a lapsed practice, and it is the only metric in this item that gets worse by itself. Run it unannounced, with clinicians

who have not used the copy, in a different week and a different shift.

3. Keep the reason to hand, because this is the measurement most often argued down to an annual formality. The Auditor-General's 2026 health-sector findings record fifteen of twenty-seven facilities with missing or lost records, seventeen not tracking records after issue, and **three of nine paid medico-legal claims conceded because the record could not be found** [7].
4. Re-read `O49`'s `Tested` column on the same pass. Nine of twelve applications have never been tested, including both the file marks `Critical` and the electronic patient record, at `Medium` criticality, `Tier 1`, a 24-hour objective, `Never tested`. And read the tier labels for what they are worth: `Tier 1` holds a 24-hour objective twice and a 15-minute objective once, `Tier 2` spans 15 minutes to 4 hours, `Tier 3` holds 15 minutes twice. **The loosest objective in the file and the tightest are both on Tier 1, so the tier label carries no information at all** — sharper than calling the tiering inverted, and it survives somebody checking.

Done when. A second measurement exists with the same method, a date, and a stated change against the first.

Goes wrong when. The measurement is delegated to the team that runs the primary system. A drill run by the people who built the thing measures the team, not the estate.

Form 11-C — The Phase H re-read register

This form feeds no sheet. The repository has none for it and this volume creates none, because the register is the minute of a cycle rather than an architecture object. Keep it with the governance minutes and carry the counts into `O61`. **Rows for a first pass** one per query per cycle, so six to nine · **Signed by** `ROL-08 Chief Enterprise Architect` · **Ratified by** `GB-01`

Sheet	What was read	Rows returned	Raised on O61	Re-read by	Re-read on
O67	EXAMPLE — delete this row. Expires past, Withdrawn empty	11 of 34	CR-G06 to CR-G09	ROL-07	2026-11-19

The counts that drift the moment anybody repairs anything

Four counts, three places each, and nothing in the file recomputes them

Count	00 Index	RPT01 Dashboard	ZZ Checks	Other	What the sheets hold
Records	843	843	843	—	820
Relationship instances	430	430	430	—	431 after R23
Work packages past end date	—	3	3	3 on RPT08	0
Components past support	—	2	2	2 on RPT10, "today"	1
Date the count was taken	—	—	—	—	absent everywhere

Red is a figure the file prints. Navy on the right is what counting the sheets returns. A wrong figure printed in three places is corrected in three places, and the dashboard is the one an executive reads.

Figure 11.3 The four counts Phase H recomputes, against the places the file prints them. Every red figure is hand-keyed from another sheet, which is why a single wrong number survives in three places and why none of them carries the date it was taken.

Phase H inherits a bookkeeping problem no earlier phase could fix, because the numbers only go wrong once the repairs start. The file as handed over says it holds **843 records** in three places — `00 Index` row 2, `RPT01`'s dashboard row and `ZZ Checks` row 6. The sheets hold **820**, and the Index's own sixty-five per-sheet figures sum to 820, contradicting the total printed directly above them. Item 1 carries that correction and item 10 the `ZZ Checks` ones. Neither can carry what happens next.

Six new sheets arrive, each with rows, placed after `O66` and before the extension sheet with six new `00 Index` rows in the Index's own format. Every re-derivation under this volume's convention copies a positional join to a new tab — `R09 RD`, `R22 RD`, `R23 RD` — adding evidence, confirmer and date

columns to the right of the original headers. And item 9's re-derivation of `R23` **cannot stay one-to-one**: `INC-3` needs two projects, so the sheet grows from six rows to seven and the Index's relationship-instance total of 430 moves to 431.

After one pass the file has more object sheets than the Index lists, more tabs than the Index has a format for, and a relationship-instance count wrong in the opposite direction from the one it started wrong in. **Nothing in the file recomputes any of it.** The reports are hand-keyed copies of object sheets and `ZZ Checks` a hand-keyed copy of the reports, which is why a wrong 3 for work packages past end date survives in three places and a wrong 2 for components past support in three — one stating it with the word "today", a hard error rather than a matter of reading.

11.8 — Recount the repository, and re-read `O66` against what landed

Sheet `00 Index`, `RPT01 Dashboard`, `ZZ Checks`, `O66 Architecture Increment`, `R23` as re-derived · **Input** every object and relationship sheet, `O56`, `O53`, `O60` · **Output** one dated recount and a landed verdict per increment · **Signed by** `ROL-08 Chief Enterprise Architect` · **Time** one day

1. Count from the sheets, never from the Index. The Index's per-sheet figures were right when the total above them was wrong, which tells you which of the two is derived. Produce four numbers with the date beside each: object sheets, object records, relationship sheets including re-derived tabs, and relationship instances.
2. Add the Index rows the conventions require: six for `O67` to `O72` in the Index's own `<n> records · <domain>` format, and one per re-derived tab — the Index has no format for a re-derived tab, so decide one and apply it everywhere. Correct row 2 in the same edit and propagate to `RPT01` and `ZZ Checks`. Add the check the sheet lacks, date of last recount, and a generation date and reference date in the header of all thirteen reports.
3. Then turn to the increments, starting with the decision, because without it nothing on `O66` can be read. Item 9 raised a row on `O60` to set the programme baseline date. `O66`'s `Lands` column holds Month 10, 16, 22, 30, 42 and 56, and **the file contains no baseline date anywhere**. If the decision is still outstanding, the first line of the Phase H report is that every increment date, transition exit date and gate date has been relative for another cycle.
4. Run the ordering check that needs no baseline, again each cycle, because re-deriving `R23` moves it. On the file as handed over, `INC-1` lands at month 10 and is delivered by the latest-starting of the six named projects, 2028-11; `INC-3` lands at month 22 and is delivered by the earliest-starting, 2027-03. Read `INC-3` closely, because it is this volume's increment: "Compliant by mechanism — Retention, masking, consent, glossary" is where the retention clock and the consent object become mechanisms rather than intentions, and a clock set in the Preliminary Phase waits the better part of two years for it.
5. Give each increment a dated verdict with its evidence — landed, partly landed with the part named, or not landed — tested against `O53`'s eighteen exit criteria, counted on the sheet and written down nowhere in the file until item 9 wrote them. Do not re-date `Lands`: converting a relative month to a calendar date without a decision is how an unrecorded assumption becomes a commitment.

Done when. The four counts agree across three sheets and carry a date, every tab has an Index row, and every increment has a dated landed verdict.

Goes wrong when. Somebody updates the Index and not the dashboard. `RPT01` is the only page most readers ever see.

The cadence, and where it is held

A re-read with no cadence is a one-off; a cadence with no holder is a calendar invitation. Both decay inside two quarters, so the last step makes the other eight repeat.

What is re-read	Cadence	Signed by	Tested against
`O67` consents, lapsed and withdrawn	Monthly	`ROL-07`	The consent capture system and `O71`
`O68` calibration and version	Monthly	`ROL-14`	The maintenance system and the manufacturer's notes
`O69` editions and licences	Quarterly, and on any publisher release	the named data owner	Release notes and the funder's rejection report
`O70` definitions against published figures	Each reporting period	`ACT-05`	The figures actually published
`O72` instrument, holds and authority	Half-yearly, and on any reorganisation	`ROL-07`	The guidance as obtained from the council
The retrieval measurement	Half-yearly, and after every change to the primary	`ROL-03`	A stopwatch and a named request
The repository recount	Every cycle	`ROL-08`	The sheets themselves

11.9 — Put the cadence where it is held, and give the phase a gate to report into

Sheet `O57 Governance Body`, `O58 Governance Gate`, `O60 Governance Decision` · **Input** the cadence table above and `O61` as extended · **Output** a mandated cadence and one gate criterion · **Signed by** `ROL-08 Chief Enterprise Architect` · **Time** half a day

1. Put each re-read in a body's mandate on `O57`, not in a plan. `GB-01` holds `G1` and `G2`, `GB-02` funding and benefit, `GB-03` data owners, `GB-04` escalation, `GB-05` the repository. Item 10 added `GB-06` for clinical safety and the records duty, and the consent, device and retrieval re-reads belong to it.
2. Give `GT-3 G3 Benefit` the criterion it is missing. It tests "Stated benefit confirmed against a date" and nothing in the file supplies the date — which is what `CR-G04` asked for and never got. Add: the six re-reads have been run within their cadence and their counts are recorded. Record

the cadence itself as a decision on `O60`, using the two columns step 11.1 added, so a cadence nobody has met shows up as an overdue decision rather than a forgotten intention.

3. Make the output a standing board item of six numbers and nothing else: lapsed consents, bindings on closed editions, devices overdue, measures without a comparability statement, record classes without a current instrument, and the retrieval median with its worst trial — each dated, each beside the previous cycle's value.
4. Then set the rule that keeps the phase honest. **A re-read that was not run is reported as not run, with the number of cycles since the last one.** It is the only line in the pack that gets worse without anybody doing anything, which is why it books the next cycle. Name a deputy for every signatory: a stopped re-read on a consent sheet is a disclosure made without authority.

Done when. Every row of the cadence table sits in a body's mandate on `O57`, and `GT-3` tests the re-read.

Goes wrong when. The cadence is set to annual because that is the audit cycle. The audit cycle is when somebody checks; the cadence is when the fact changes.

WORKED ANSWERS — THE FIRST PHASE H CYCLE, TWO ORGANISATIONS

Driehoek Health Group — nine acute hospitals, two day clinics, about 1,750 registered beds. Its first cycle ran all six queries, and the one that changed behaviour was not the expected one. The consent re-read returned eleven lapsed rows of thirty-four, which the governance group treated as a backlog. The cross-read against the disclosure log returned **three disclosures to a research partner dated after the consent cited had expired**, which it treated as an incident, and which produced a stop list the integration team acted on the same afternoon. The retrieval re-measurement came in nine minutes worse than Phase D's median at two of nine sites, both of which had replaced a workstation image since the first measurement.

Rietvlei District Health Service — one district hospital, three community health centres, 22 fixed clinics, a mobile service. Rietvlei could run two of the six queries. `O68` returned four devices past calibration out of the four on the register — a complete answer to a small register, and a better deliverable than a partial answer to a large one. `O72`'s re-read found every row built against the 2016 edition of the guidance, and the cycle's single output was a letter to the council asking which edition is in force, still unanswered at the second cycle and recorded as unanswered with the date of asking on every row. Its retention clock has not moved, and the file now says why.

BEFORE YOU CLOSE A PHASE H CYCLE

Seven questions. Each answer is a count, a date or a document reference, or the cycle has not run.

Which of the six re-reads ran this cycle, and how many cycles since each of the others?

How many consents have expired without being withdrawn, and what stopped as a result?

Which live fields are bound to a code system edition that has closed?

How many devices are past calibration, and which are still in clinical use?

What is the retrieval median and worst trial this cycle, and the change since last?

Which record classes name an instrument edition that is no longer current?

What are the four counts, on what date were they taken, and do all three sheets print them?

WHAT PHASE H HANDS BACK, AND TO WHICH PHASE

A lapsed consent and a closed code system edition go back to Phase C, because both objects are Phase C artefacts. A device past calibration goes back to Phase D. A record class whose instrument has moved goes back to the Preliminary Phase, the only phase that can decide what the organisation builds against. A failed retrieval measurement goes to Phase G, which refuses the gate. And a count nobody can reproduce goes nowhere except onto `O61`, with a date and a name, because an unreproducible count is a change to the architecture's own statement of itself.

THE CHAPTER IN TEN LINES

- Phase H produces nothing of its own. Its content is re-reading what the other phases wrote, on a cadence, and acting on what has changed.
- Five of the six sheets this volume creates carry a date column, so every one of them is true on the day it is written and false later.
- O61's Against column takes only a work package reference or the literal string Repository, so nothing in the architecture can be changed from it until the convention is extended.
- CR-G03 is raised against Repository, not against O29. An architect sent to O29 will not find it.
- CR-G03 asks for unfunded building blocks to be raised as gaps, and no sheet in the file costs a building block, so on the file's own terms none is unfunded and the derivation is unavailable.
- CR-G04 is an approved correction that nobody performed, which is worse than a raised one because the sheet says it is finished.
- The trigger test is written down before the first contested change, and a change that moves a date on O67 to O72 is an architecture change by default.
- The six re-reads are one line each: a superseded instrument, a lapsed consent, a closed code system edition, an overdue calibration, a version that moved inside a period, and a disclosure with no live authority.
- Crossing O67 against O71 on the consent reference finds disclosures made after the consent cited had expired. It is the most valuable query in the phase and the file as handed over cannot run it.
- Re-measure retrieval with item 7's method unchanged, and record the number of cycles since the last measurement. It is the only metric here that gets worse by itself.
- The tier label on O49 carries no information at all: the loosest objective in the file and the tightest are both on Tier 1.
- Six new sheets and every re-derived tab move the counts, and re-deriving R23 takes the relationship-instance total from 430 to 431. Nothing in the file recomputes any of it.
- A re-read that was not run is reported as not run, with the number of cycles since the last one.
- A role becoming vacant is an architecture change. When the person who signs a retention class leaves, the row's signature has lapsed and nothing recomputes.

SOURCES FOR THIS CHAPTER

- **1.** User guide to the ICD-10 Master Industry Table, January 2014, National Department of Health (South African start and end dates per code, status of added, modified or discontinued, and the revision history) https://www.health.gov.za/wp-content/uploads/2021/02/cd-10mit_userguide_january2014.pdf
- **2.** SAHPRA, Communication to industry MD08-2025/2026, Regulatory Requirements of Artificial Intelligence and Machine Learning Enabled Medical Devices, 26 September 2025 https://www.sahpra.org.za/wp-content/uploads/2025/09/MD08-20252026_-SAHPRA-Communication-to-Industry-AI-Medical-devices_Acknowledgements.pdf
- **3.** HPCSA Booklet 9, Guidelines on the Keeping of Patient Records, September 2016 edition https://www.hpcsa-blogs.co.za/wp-content/uploads/2022/08/Booklet-9-Guidelines-on-Patient-Records.pdf
- **4.** HPCSA Booklet 9, Keeping of Patient Records, review draft marked September 2022 https://www.hpcsa-blogs.co.za/wp-content/uploads/2022/11/Booklet-9-Keeping-of-Patient-Records_Review-Draft_-FINAL_Sept-2022.pdf
- **5.** National Health Act 61 of 2003, consolidated text (section 13, the record-keeping duty on the person in charge of the establishment, subject to the National Archives and Record Service of South Africa Act 43 of 1996 and the Promotion of Access to Information Act) https://www.saflii.org/za/legis/consol_act/nha2003147.pdf
- **6.** Protection of Personal Information Act 4 of 2013, section 32, authorisation concerning a data subject's health or sex life https://popia.co.za/section-32-authorisation-concerning-data-subjects-health-or-sex-life/
- **7.** Auditor-General findings on the health sector, via MedicalBrief, 14 October 2026 (15 of 27 facilities with missing or lost records, 17 not tracking records after issue, three of nine paid medico-legal claims conceded because the record could not be found) https://www.medicalbrief.co.za/health-sector-crippled-by-recurring-systemic-weaknesses-ag-warns/
- **8.** Zhang and Pandit, British Journal of Anaesthesia 2023;131(1):130-134, theatre utilisation calculation methods https://www.sciencedirect.com/science/article/pii/S0007091223001794

ITEM 12

Requirements Management

A traceability link is an assertion somebody made, on a date, against evidence, or it is decoration — and in this file all thirty-two of them are decoration that resolves perfectly.

The discipline that is not a phase

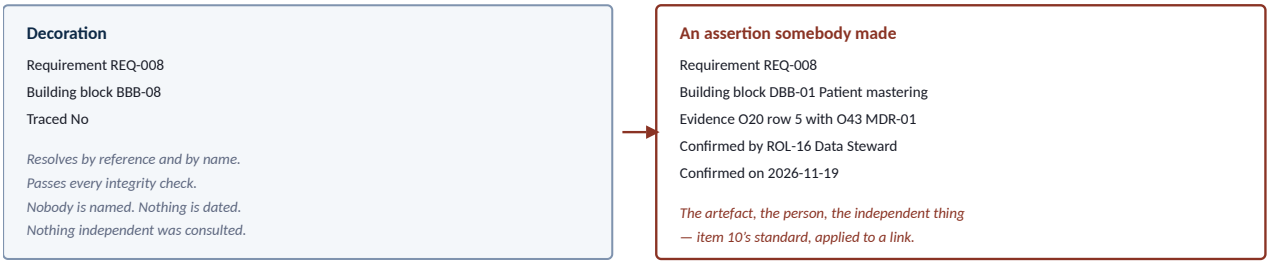
Requirements Management sits in the middle of the cycle because it runs through all of it. Every other item in this volume is opened at the phase you are in. This one is open the whole time, and its single product is a set of links from what the organisation needs to what somebody will build.

This repository has those links. All of them resolve. `R09` **gives every one of the thirty-two requirements a building block, by reference and by name, and not one of the thirty-two pairings is about the requirement it names.** That is the finding item 1 opened the volume on, and this is the item that would have caught it — because a traceability link is not a row in a join table. **It is an assertion that somebody made, on a date, against evidence, or it is decoration.**

So this item is about the discipline that would have caught the other nine positional joins before any of them reached a board paper. Not a tool, not a check, not referential integrity — which in this file is perfect and always was. A rule about what a link has to carry before it counts as one.

> **THE QUESTION IS ITEM 10'S, AND IT IS NOT REINVENTED HERE**
>
> Item 10 settled the Phase G question: **what was reconciled, by whom, against what?** Three answers, with a three-part evidence standard behind them — the artefact, the person, the independent thing.
>
> Requirements Management asks the same question of a **link** rather than of a verdict, and the three parts translate exactly. The **artefact** is the row and the document behind it. The **person** is a named individual holding a role that exists in `O11 Role`. The **independent thing** is something the claimant did not produce — the entity list for a mastering requirement, `R06` for an interface requirement, `O26` for a technology adoption requirement. Where there is no independent thing, the link records that, and that sentence is worth more than the link.

The same link, as decoration and as an assertion



The left-hand row is one of the thirty-two this file holds. The right-hand row is the same link re-derived onto R09 RD, with three columns added to the right of the original headers and the original left untouched as the record of what was inherited.

Figure 12.1 One link, twice. The left-hand version resolves by reference and by name and passes every integrity check a tool can run. The right-hand version is the same link with the three things item 10's evidence standard requires, which is the whole difference between a traceability record and a join table.

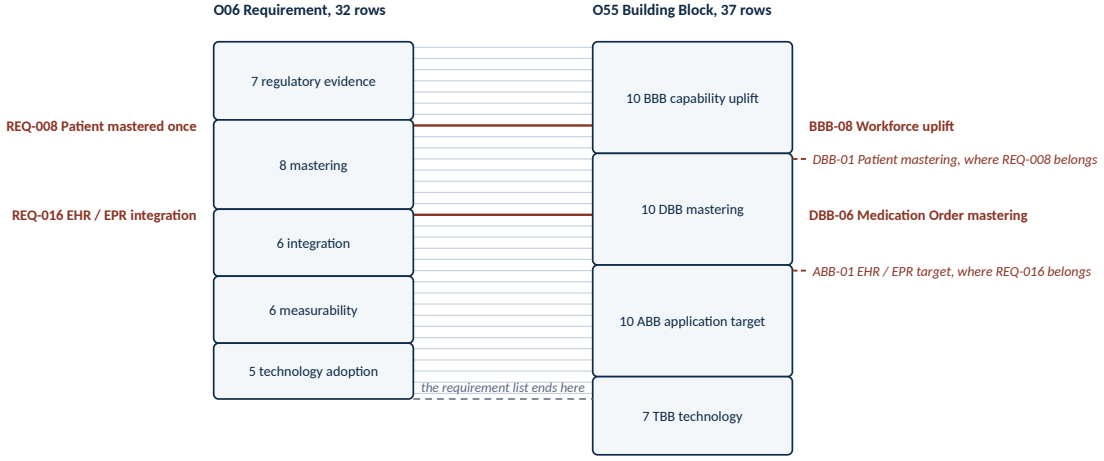
What Requirements Management has to work with

Sheet	Rows	What it should hold	What it holds
`O06 Requirement`	32	Statements that can be verified and can fail	27 template fills and 5 statements copied from another sheet
`O07 Constraint`	8	Constraints that bind named things	Eight rows that appear in no relationship sheet at all
`O29 Gap`	2	What is missing, including from the requirement set	Two rows, both business capability uplift
`R08`	32	A derivation of each requirement from a driver	`O06`'s own `Driver` column reprinted with the name looked up
`R09`	32	A requirement reaching the block that delivers it	Requirement `n` paired with building block `n` in `O55`'s row order
`RPT06 Requirements Traceability`	32	Which requirements reach nothing	`O06` reprinted, plus `R09`'s `Traced` flag relabelled
`O31 DLV-20`	1	The Requirements Catalogue, approved	Status `Draft`, phase `RM`, the only deliverable nobody has signed

Seven things, and only two of them hold facts nothing else holds: `O06`'s statements and `O07`'s constraints. `R08` and `RPT06` are derived views presented as content, and `R09` is a join whose rows resolve and mean nothing.

`R09` , read cell by cell

R09 pairs requirement n with the nth building block, and the groups do not line up



Every join line is horizontal, because the join is identity on row number. The five TBB blocks below the dashed line reach no requirement at all: identity resolution, decision support, device integration, zero-downtime clinical systems and the secure research enclave. The list runs out five rows short.

Figure 12.2 Why none of the thirty-two pairings is about its own requirement. The join is identity on row number, so every line is horizontal; the requirement list and the building block list are grouped differently, so the subjects never meet. The block each of the two marked requirements should reach is on the same sheet, a few rows away.

The arrangement is simple and you can check it in two minutes. `O55 Building Block` holds thirty-seven blocks in one order: ten `BBB-` capability uplifts, ten `DBB-` mastering blocks, ten `ABB-` application targets, seven `TBB-` technology blocks. `O06` holds thirty-two requirements in a different order: seven regulatory evidence, eight mastering, six integration, six measurability, five technology adoptions. **`R09` pairs requirement `n` with the `n`th block on `O55`.** Nothing else is happening.

R09 row	Requirement	Reaches	Which is
5	`REQ-001` Evidence Health data protection	`BBB-01`	Patient Access & Scheduling capability uplift
12	`REQ-008` Patient mastered once	`BBB-08`	Workforce & Rostering capability uplift
20	`REQ-016` EHR / EPR integration managed	`DBB-06`	Medication Order mastering
25	`REQ-021` Theatre Management integration managed	`ABB-01`	EHR / EPR target
36	`REQ-032` Adopt Medical device integration	`TBB-02`	Terminology services

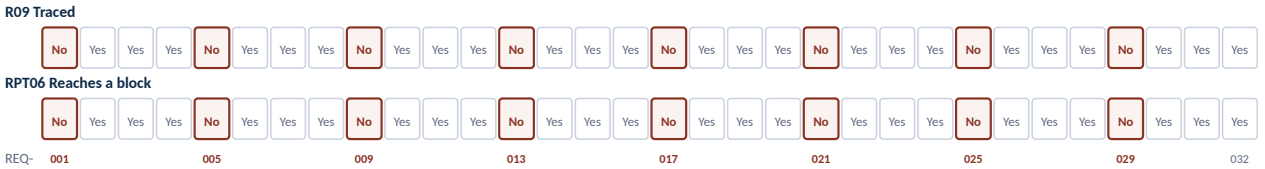
The two middle rows are the ones to take to a client, because the blocks those requirements should reach **exist on the same sheet**. `DBB-01 Patient mastering` is there; `ABB-01 EHR / EPR target` is there; `REQ-008` and `REQ-016` reach neither. A reader who argues the join is "approximately right" has to explain that.

Then count the far end. Thirty-two requirements laid against thirty-seven blocks leaves five satisfying no requirement at all: `TBB-03 Master patient index`, `TBB-04 Clinical decision support`, `TBB-05 Medical device integration`, `TBB-06 Zero-downtime clinical systems`, `TBB-07 De-identification and secure enclave`. That is identity resolution, decision support, device integration, the resilience capability item 7 measured and the research access path — **the five things a provider's technology architecture most needs a requirement behind** — and the list runs out before it reaches them because thirty-two is less than thirty-seven.

The `Traced` column is a stride, not an assessment

`R09`'s fifth column is `Traced`, and it reads `No` on eight of the thirty-two rows: `REQ-001`, `REQ-005`, `REQ-009`, `REQ-013`, `REQ-017`, `REQ-021`, `REQ-025`, `REQ-029`. Write those out and the pattern is immediate. **Every fourth row, starting at the first.** Eight of thirty-two is exactly 25%, and the eight are spaced at a stride of four with no exceptions.

R09 Traced, and RPT06 Reaches a block: eight No values at a stride of four



All 32 rows of R09 carry a building block, the eight marked No included: REQ-001 has BBB-01 and REQ-029 has ABB-09. The flag is asserted at a fixed stride, not computed. RPT06 copies it on all 32 rows and relabels it, and its footnote describes a condition the sheet cannot detect. Two of the eight are regulatory Must requirements.

Figure 12.3 R09's Traced column and the RPT06 column that copies it. Eight No values at a fixed stride of four, on a sheet where all thirty-two rows have a building block — a flag that was typed, not derived, and then relabelled in the report an executive reads.

So the flag is not derived from anything in the repository. Every one of the thirty-two rows has a building block, including all eight marked `No` — `REQ-001` has `BBB-01`, `REQ-029` has `ABB-09`. The column is **asserted, and asserted mechanically.**

`RPT06 Requirements Traceability` then copies it. The report's six columns are `Requirement`, `Name`, `Category`, `Priority`, `Driver` and `Reaches a block`; the first five are `O06` reprinted and the sixth equals `R09`'s `Traced` value on all thirty-two rows. Its footnote, at row 38, reads: *a requirement that reaches no building block is a requirement nothing will deliver.* The sentence is true. **The sheet does not detect the condition it describes**, because on its own data no requirement reaches no block.

And the eight it does flag include **two of the seven regulatory `Must` requirements** — `REQ-001 Evidence Health data protection` and `REQ-005 Evidence Professional registration and scope`. An executive reading `RPT06` is told that two regulatory obligations have no delivery route, by a column that counted to four.

> **THREE FAILURES, AND THEY ARE NOT THE SAME FAILURE**
>
> **One.** The link is wrong on substance: thirty-two of thirty-two. **Two.** The flag that reports on the link is a stride, not a computation: eight rows, mechanically placed. **Three.** The report that carries the flag relabels it, from `Traced` to `Reaches a block`, which are not the same statement.
>
> A repository can be wrong at all three levels at once and pass every check a tool can run. Fixing the first without retiring the second leaves a correct join reported on by a decoration.

`R08` is correct by construction, which makes it evidence of nothing

This is the sheet a reader samples first to reassure themselves the file is sound, and it is worth reading closely for exactly that reason.

`O06` has seven columns, and the sixth is `Driver` — a reference into `O01 Driver`, populated on all thirty-two rows and resolving cleanly on all thirty-two. `R08`'s four columns are `Requirement`, `Requirement name`, `Driver`, `Driver name`. **`R08` is `O06`'s sixth column reprinted with the driver's name looked up**, which makes it a derived view presented as a relationship.

Then read what the derivation is. `REQ-001` to `REQ-007` are named `Evidence Health data protection`, `Evidence Clinical records retention` and so on — **the driver's name with one word in front of it** — and they point at `DRV-01` to `DRV-07` in order. The requirement was generated from the driver, so the join is true by construction, exactly as `R03` is true by construction because its ten services were generated from its ten capabilities. Both sheets are correct, both are the first thing a reader checks, and **neither tells the reader anything** — the most useful single fact about sampling a repository.

The information `R08` does carry is in what is absent from it.

Driver	Requirements derived from it in R08
`DRV-10 Legacy estate`	11 — six integration and five technology adoptions
`DRV-11 Data fragmentation`	8 — the mastering set
`DRV-08 Cost pressure`	6 — the measurability set
`DRV-01` to `DRV-07`, the seven regulatory drivers	1 each
`DRV-09`, `DRV-12`, `DRV-13`	0

`DRV-02 Clinical records retention`, at `Pressure High`, drives exactly one requirement — and that requirement is `REQ-001`'s template sibling, "The architecture must produce evidence that Clinical records retention is met, without manual reconciliation", reaching `BBB-02 Clinical Care Delivery capability uplift`. That is the entire architectural response in this file to the obligation overlay 1 is built on. Three drivers produce nothing, and all seven regulatory drivers are owned by `ACT-07 Chief Compliance Officer`, who heads no organisation unit in `O09`.

`O06` read as text, and the test a requirement has to pass

Read the `Statement` column before you read anything else, because twenty-seven of the thirty-two statements are template fills and the seams are visible in the cells.

Seven read "The architecture must produce evidence that *driver* is met, without manual reconciliation." Eight read "*entity* is mastered in one system and consumed elsewhere through a published interface." Six read "Every interface to or from *application* is managed, published and logged." Six read "*capability* exposes measures that the executive can act on." Three of the first seven are ungrammatical, because the driver name is plural and the frame is not — `REQ-004` reads "that Clinical safety standards **is** met", and `REQ-006` and `REQ-007` do the same.

The remaining five, `REQ-028` to `REQ-032`, are not written either. **Each statement is copied verbatim from `O26`'s `Rationale` cell for the same-named technology component**, and the same seven strings appear again on `O46`, `O47` and `O55`. A requirement whose statement is a component's rationale has not stated a requirement; it has named a product. Four of those five propose adopting components `O26` already records as `Current`, at priority `Could` and status `Proposed`.

> **WHAT A REQUIREMENT HAS TO CARRY BEFORE IT IS ONE**
>
> A subject, a condition and a means of verification — and it must be capable of failing. "The architecture must produce evidence that Clinical records retention is met" names a subject and no condition, and cannot fail, because nothing says what evidence, over which record classes, retrieved in what time, authorised by whom.
>
> Rewritten so it can fail: "Every record class in `O72 Retention Clock` carries a period, a named instrument with its edition, a destruction authority and a retrieval time measured within the last six months." That statement fails on the file as handed over, which is what makes it a requirement.

`O07` and `O29`: the two ends nothing reaches

`O07 Constraint` holds eight rows — `Ref`, `Constraint`, `Description`, `Type` — and **appears in no relationship sheet in the file.** A constraint that constrains nothing named is a sentence in a document. Two of the eight bear directly on everything this volume has built: `CON-05 Data residency`, "Certain data may not leave the jurisdiction", which governs every offshore component in the estate; and `CON-06 Change freeze windows`, "Peak periods are closed to change", which governs the cadence item 11 set. Neither is attached to anything. A third, `CON-08 Union and works council consultation`, is generic language that has not been written for a South African provider, and `CON-06`'s "peak periods" is retail phrasing — a hospital has winter pressure.

`O29 Gap` holds two records, and item 8 owns the derivation of the real gap set. What belongs here is narrower and nobody has it: **there is no relationship anywhere in this repository from a requirement to a gap.** `R12` closes gaps with work packages; `R09` reaches building blocks; nothing records that a requirement nobody will deliver is itself a gap. So the requirements discipline cannot raise one, which is why `O29` has two rows while `RPT06` prints eight `No` values.

12.1 — Adopt item 10's standard, and write what it means for a link

Sheet `R09`, `R08`, `O11 Role` · **Input** item 10's evidence standard as minuted · **Output** a one-page link standard and a `Confirmer` convention · **Signed by** `ROL-08 Chief Enterprise Architect` · **Time** half a day

1. Take item 10's three parts as written and do not restate them in different words. The artefact, the person, the independent thing. A second standard in the same repository is how two teams come to disagree about what evidence is.
2. Write the one page that says what each part is for a link. **Artefact**: the sheet row plus the document, count or minute behind it. **Person**: a named individual holding a role that exists in `O11 Role` — not "the team", not `Architecture Board`, which is `GB-01` on `O57` and is a body, so it ratifies and does not confirm. **Independent thing**: for a mastering link, `O20` and `O43`; for an integration link, `R06`; for a technology adoption, `O26` and `O47`; for a regulatory link, the instrument and its provision.
3. Set the rule that makes the standard enforceable: **a link with no confirmer and no date is retired, not deleted**, and the original stays as the record of what was inherited. Then say what the standard is not. It is not a demand that every link be right; it is a demand that every link say who thinks so and why — a question a person answers in thirty seconds and a tool cannot answer at all.

Done when. The standard is one page, minuted, and names the independent thing for each of the five categories in `O06`.

Goes wrong when. Somebody writes a second evidence standard because the first was written for Phase G. There is one standard; the gate and the link are two places it is applied.

12.2 — Test `R09` for meaning, sheet-wide, and publish three counts

Sheet `R09` · **Input** `O06`, `O55`, the link standard from 12.1 · **Output** three counts and a per-row verdict · **Signed by** `ROL-10 Domain Architect, Data` · **Time** one day

1. Run the test the file cannot run, one row at a time, read aloud: **does the subject of this requirement appear anywhere in the name or the acceptance criterion of the block it reaches?** It is a string comparison a person can do in an hour over thirty-two rows.
2. Record three counts, never one: rows checked, rows that survive the reading, rows re-pointed. A reconciliation with one number in it is a row count, and the row count was never in doubt — the sheet has had thirty-two rows since handover and every reference resolves.
3. Expect zero survivors and say so plainly, because the arrangement is mechanical: requirement `n` against block `n`. The one case that needs a judgement rather than a reading is `REQ-022` reaching `ABB-02 PAS target` — a measurability requirement about patient access reaching the patient administration system's target disposition, which is nearly plausible and is still a coincidence of row numbers.
4. Then test the far end and record it as a fourth number: building blocks reaching no requirement. On this file it is five, all technology, and three of the five are the components `REQ-030`, `REQ-031` and `REQ-032` propose adopting — so the requirement exists, the block exists, and the join puts them in different rows.

Done when. Every row has a verdict, and four counts are published with the date they were taken.

Goes wrong when. The test is run on a sample. A sample of a positional join returns the join's accuracy, which is zero, but it cannot tell you the arrangement, and the arrangement is the finding.

12.3 — Re-derive `R09`, and retire the `Traced` column

Sheet `R09 RD`, new · **Input** `O06`, `O55`, step 12.2's verdicts · **Output** a re-derived join with three added columns, and a retired flag · **Signed by** `ROL-10 Domain Architect, Data` · **Time** two days

1. Copy `R09` to a new tab, `R09 RD`, keeping the original row-4 headers exactly and in order: `Requirement`, `Requirement name`, `Building block`, `Block name`, `Traced`. Leave the original untouched as the record of what was inherited.
2. Add three columns to the right and nothing else: `Evidence`, `Confirmed by`, `Confirmed on`. Those three are the link standard made into a schema, and they are the whole repair. A re-derived join with right answers and no confirmer is the same artefact as before with better guesses in it.
3. Re-point on subject, allowing many-to-many, because a requirement may need more than one block and a block may serve more than one requirement. `REQ-008 Patient mastered once` needs `DBB-01` and, if identity resolution is in scope, `TBB-03 Master patient index`. The seven regulatory requirements need the blocks that actually produce the evidence, which for `REQ-002` is nothing currently on `O55` — record the absence and raise it, do not invent a pairing.
4. Retire `Traced` rather than correcting it. Set every cell to `Retired — asserted, not computed`, dated, and derive the replacement: a requirement reaches a block when `R09 RD` holds a row for it with a non-empty `Evidence` and `Confirmed by`. That value is computed, so it moves when the file moves, which the old one never did.

Done when. `R09 RD` exists, every row carries evidence, a confirmer and a date, and `Traced` is dated and retired on both tabs.

Goes wrong when. The original `R09` is edited in place. Six months later nobody can show what was inherited, and the first auditor to ask will be told the file was always like that.

Form 12-A — The traceability assertion, feeding `R09 RD`

Feeds `R09 RD`, a new tab this volume's conventions create; the repository has `R09` only. **`Evidence`, `Confirmed by` and `Confirmed on` are three columns `R09` does not have**; step 12.3 adds them to the right of its original headers. `Requirement name` and `Block name` are typed straight into Excel from `O06` and `O55`. **Rows for a first pass** 32 to 48, one per requirement plus the second block some requirements need · **Signed by** `ROL-10 Domain Architect, Data` · **Ratified by** `GB-01`

Requirement	Building block	Block name	Evidence	Confirmed by	Confirmed on
REQ-008	EXAMPLE — delete this row. DBB-01	Patient mastering	O20 row 5 and O43 MDR-01, read together 19 Nov	ROL-16	2026-11-19

12.4 — Rewrite the thirty-two statements so that each one can fail

Sheet `O06 Requirement` · **Input** `O01`, `O26`, `O47`, `O63`, the link standard · **Output** thirty-two verifiable statements and two added columns · **Signed by** `ROL-09 Domain Architect, Business` · **Time** three days

1. Keep the references. `REQ-001` to `REQ-032` are cited in `R08`, `R09` and `RPT06`, and renumbering gains nothing. Rewrite the `Statement` cell and leave `Ref` alone.
2. Add `Verification` and `Verified on`, and record them as columns the repository does not have. A requirement with no means of verification is a preference, and the file's seven regulatory requirements all say "without manual reconciliation" — which is a method, not a verification.
3. Take the five `REQ-028` to `REQ-032` rows first, because they are the quickest win. Each statement is `O26`'s rationale for the same-named component, so each is a product choice masquerading as a need. Rewrite each as the capability the organisation needs, and note that four of the five propose adopting something `O26` already records as `Current` while nothing in the file records any of the seven in production.

4. Then rewrite the seven regulatory statements against the instrument, using the `Instrument` column item 10 added to `O63 Control Objective`: `REQ-002` becomes testable against `O72`, `REQ-003` against `O68` and the device classification guideline [1], `REQ-001` against the consent and disclosure sheets and the authorisation that lifts the prohibition on processing health information [2][3]. Fix the three ungrammatical cells while you are in them — a client will notice `REQ-004` before `R09`.

Done when. Every statement names a subject, a condition and a verification, and a reader can say what would make it fail.

Goes wrong when. The statements are rewritten and the categories left alone. `Category` and `Priority` are the two columns a steering committee reads, and five `Could` rows at the bottom of the sheet are where the technology architecture went.

Form 12-B — `O06 Requirement`, part one: identify

Feeds `O06 Requirement` · **Rows for a first pass** 32, one per existing requirement · **Signed by** `ROL-09 Domain Architect, Business` · **Ratified by** `GB-01`

These are six of the sheet's seven real columns, in sheet order. Form 12-C carries the `Statement` column and the two added ones, keyed on `Ref`. All six here go straight into Excel.

Ref	Requirement	Category	Priority	Driver	Status
REQ-002	EXAMPLE — delete this row. Clinical records retention is mechanised	Regulatory	Must	DRV-02	Approved

Form 12-C — `O06 Requirement`, part two: state it so it can fail

Feeds `O06 Requirement` · **Rows for a first pass** the same 32 rows as form 12-B · **Signed by** `ROL-09 Domain Architect, Business` · **Ratified by** `GB-01`

`Verification` and `Verified on` are two columns this repository does not have; step 12.4 adds them. Until then the only place a verification exists is the covering note, which is why twenty-seven of thirty-two statements could be template fills without anybody noticing.

Ref	Statement	Verification	Verified on
REQ-002	EXAMPLE — delete this row. Every record class in O72 carries a period, a named instrument with its edition, a destruction authority and a retrieval time measured in the last six months	Count O72 rows failing any of the four tests; target zero	2026-11-19

12.5 — Re-derive `R08`, and record the drivers that drive nothing

Sheet `R08 RD`, new, and `O01 Driver` · **Input** `O06`, `O01`, `O63` · **Output** a derivation per requirement with a confirmer, and a list of drivers with no requirement · **Signed by** `ROL-09 Domain Architect, Business` · **Time** one day

1. Say in one sentence why the sheet is being re-derived at all, because somebody will point out that `R08` is correct. It is correct **because the requirements were generated from the drivers**, which makes it a restatement of `O06`'s own `Driver` column and not a derivation anybody performed. Copy it to `R08 RD` and add the same three columns: `Evidence`, `Confirmed by`, `Confirmed on`.
2. Confirm each derivation against the driver's `Description` rather than its name. `DRV-11 Data fragmentation` reads "The same fact is held differently in several systems", which supports the eight mastering requirements; `DRV-08 Cost pressure` reads as margin pressure, which does not obviously support six requirements about executive-facing measures.
3. List the drivers that reach no requirement and raise each: `DRV-09`, `DRV-12`, `DRV-13`. Then list the ones that reach too few, which matters more. **`DRV-02 Clinical records retention` is a `High`-pressure regulatory driver with one template-generated requirement behind it**, against a control objective whose evidence reads `No mechanism`, a standard at `Emerging` and `Not assessed`, and a metadata element whose coverage is `Absent`. Four sheets record that retention is unmanaged and one requirement addresses it. Write the new requirements that follow, keep them few, continue the series at `REQ-033` and do not renumber.

Done when. Every row of `R08 RD` has a confirmer and a date, and every driver either reaches a requirement or carries a dated note saying why it does not.

Goes wrong when. The sheet is left alone because it is correct. A correct join with no confirmer is indistinguishable from a lucky one, and this file contains both.

12.6 — Make `O07`'s constraints bind something

Sheet `O07 Constraint` · **Input** `O06`, `O55`, `O48`, `O50`, `O72`, item 11's cadence · **Output** a binding per constraint and two added columns · **Signed by** `ROL-08 Chief Enterprise Architect` · **Time** half a day

1. `O07`'s four columns hold no reference to anything, and the sheet appears in no relationship sheet. Add `Binds` and `Assessed on`, and record them as additions. `Binds` takes object references under the same convention item 11 set for `O61`'s `Against` column.
2. Bind the two that matter most to this volume. `CON-05 Data residency` binds every environment on `O48`, the research extract classes on `O72`, and any offshore component — and volume 1 chapter 18 establishes that an offshore component needs the transborder conditions met **and** prior authorisation, which is a sequencing constraint rather than a design preference. `CON-06 Change freeze windows` binds the cadence item 11 set, and a monthly consent re-read that falls inside a freeze still has to run, because reading is not changing.
3. Rewrite the two that are not about this industry. `CON-06`'s "peak periods" is retail phrasing: a provider has winter pressure, a strike, a water interruption and a mass-casualty diversion, and those are different windows with different lengths. `CON-08 Union and works council consultation` is generic language that has not been written for a South African provider.
4. Then check each constraint against the requirements it should have blocked. A constraint that has never blocked a requirement, a design or a work package has not been applied, and the honest entry in `Assessed on` is the date you first looked.

Done when. Every constraint names what it binds, and `O07` appears in at least one relationship.

Goes wrong when. `Binds` is filled with free text. The column is only useful if a reader can follow it to a row.

Form 12-D — `007 Constraint`, bound to something

Feeds `007 Constraint` · **Rows for a first pass** 8 to 12, the eight existing rows plus any the re-read adds · **Signed by** `ROL-08 Chief Enterprise Architect` · **Ratified by** `GB-01`

`Binds` and `Assessed on` are two columns this repository does not have; step 12.6 adds them. The first four columns are the sheet's real ones, in sheet order, and go straight into Excel.

Ref	Constraint	Description	Type	Binds	Assessed on
CON-05	EXAMPLE — delete this row. Data residency	Certain data may not leave the jurisdiction	Regulatory	ENV-01 to ENV-04, RTN-07, TBB-07	2026-11-19

12.7 — Raise the requirements that reach nothing as gaps, and join the two sheets

Sheet `O29 Gap` and a new relationship · **Input** `R09 RD`, `O06` as rewritten, item 8's derived gap set · **Output** a gap per unserved requirement and one new join · **Signed by** `ROL-08 Chief Enterprise Architect` · **Time** one day

1. Take the output of step 12.3: every requirement with no `R09 RD` row carrying evidence and a confirmer. On the file as handed over that is all thirty-two; after a re-derivation it is the set that genuinely has nothing to deliver it.
2. Raise each as a row on `O29`, continuing at `GAP-03`, with `Domain` taken from the requirement's `Category` and `Severity` from its `Priority` — a `Must` requirement with nothing behind it is `Critical` and nothing else. Item 8 owns the rest of the gap set; this step adds only the gaps the requirement set produces, and says so on each row.
3. Then create the join nothing in the file has: **requirement to gap.** It is the missing edge that makes the discipline operable, because without it a requirement nobody will deliver has no route onto a sheet anybody reads. Three columns are enough — requirement, gap, and the date raised — placed after the existing relationship sheets with its own `00 Index` row.
4. Check the arithmetic before you publish. `O29` holds two rows while `RPT03` shows all ten level-1 capabilities with a maturity gap, `O32` six standards in breach, `O41` three absent controls, `O63` obligations without evidence and `O27` a component past support. Adding requirement-derived gaps to a sheet already two orders of magnitude short produces a number that frightens people, so publish the derivation beside the count.

Done when. Every requirement with nothing behind it has a gap row and a dated join, and `O29` says where each row came from.

Goes wrong when. The gaps are raised and `CR-G03` is left open as though it were the same work. It is not: that change request asks for unfunded building blocks, which item 11 shows the file cannot identify.

12.8 — Add the requirements this industry needs and this file has no row for

Sheet `O06`, continuing at `REQ-033` · **Input** `O67` to `O72`, volume 1 chapters 12, 13, 18 and 19 · **Output** six to nine new requirements, each verifiable · **Signed by** `ROL-07 Chief Compliance Officer` · **Time** two days

1. This is the step that makes Requirements Management worth running in a provider rather than anywhere else. Six sheets exist in this volume because the obligations behind them are stated in the file and the structure to meet them is not. Each needs a requirement, and the requirement is what funds it.
2. Write one per sheet, each naming its instrument and its verification. **Retention**: every record class in `O72` carries a period, an instrument with its edition, a destruction authority and a retrieval time measured in the last six months — the records duty sits on the person in charge of the establishment and is subject to the archives and access regime [4], and the periods come from guidance that disagrees with itself across its own editions [5][6]. **Consent**: every consent in `O67` carries a scope, a grant date, an expiry and a withdrawal field, and no disclosure in `O71` postdates the expiry of the consent it cites [2][3]. **Devices**: every device in `O68` carries a risk class, a software version and a calibration date, the class being the manufacturer's read with the regulator's determination [1].
3. Then the three that look optional and are not. **Code systems**: every coded attribute names the system and the edition in force on the date of care, with a point-in-time join — the national diagnosis reference table carries start and end dates per code and a revision history, and updates biennially [7]. **Measure definitions**: every published figure names the `O70` row and version it was computed under, because four defensible

theatre-utilisation figures come out of one list with no conversion between them [8]. **Disclosure**: the organisation can tell a patient what was disclosed about them, to whom and under what authority.

4. Give each a `Category`, a `Priority` and a `Driver` — the drivers already exist, `DRV-01` to `DRV-07` waiting for requirements that could be verified. Then put each new requirement through step 12.3, so it reaches a block with evidence and a confirmer on the day it is written.

Done when. Each of the six new sheets has at least one requirement behind it, naming an instrument and a verification, and each reaches a building block with a confirmer.

Goes wrong when. The new requirements are written at `Should`. Each one of them traces to a stated regulatory driver at `High` pressure, and the priority column is the only part a funding committee reads.

12.9 — Rebuild `RPT06` so that it computes something

Sheet `RPT06 Requirements Traceability` · **Input** `R09 RD`, `O06`, `O29`, the new join from 12.7 · **Output** a report that derives its own columns · **Signed by** `ROL-08 Chief Enterprise Architect` · **Time** half a day

1. `RPT06`'s first five columns are `O06` reprinted and its sixth is `R09`'s retired flag relabelled. Replace the sixth with three derived columns: blocks reached with evidence, the date of the most recent confirmation, and the gap reference where the count is zero.
2. Rewrite the footnote so it states the test it applies, which the present one does not. "A requirement reaches no building block" was never true on this data; "a requirement has no link carrying evidence and a confirmer" is, on all thirty-two rows as handed over.
3. Add the two counts that make the report readable in a board pack: requirements with at least one evidenced link, out of thirty-two; and building blocks with at least one requirement behind them, out of thirty-seven. Both are computable, both move when the file changes, and both read as zero and thirty-two respectively on the file as handed over.
4. Put a generation date and a reference date in the header. None of the thirteen reports carries either, which is item 11's finding and this report's problem too: the eight `No` values have been on that page since handover with no date against them.

Done when. Every column in `RPT06` is derived from a sheet rather than copied from one, and the footnote states its test.

Goes wrong when. The report is rebuilt and `RPT01`'s dashboard is not. Nothing on the dashboard mentions requirements at all, which is its own finding.

12.10 — Sign `DLV-20`, and set the cadence that keeps the catalogue true

Sheet `O31 Deliverable`, `O58 Governance Gate`, `O60 Governance Decision` · **Input** everything from 12.1 to 12.9 · **Output** one signed deliverable and a standing cadence · **Signed by** `ROL-08 Chief Enterprise Architect` · **Time** one board meeting

1. `DLV-20 Requirements Catalogue`, phase `RM`, is `Draft` — and of the twenty deliverables on `O31`, thirteen are `Approved`, including every Phase A to D artefact, while the catalogue the other thirteen were supposed to satisfy has never been signed. Say that in one sentence at the top of the board paper.
2. Do not sign it on the rewritten statements alone. The condition for signing is the link standard applied: thirty-two statements that can fail, thirty-two or more evidenced links, a confirmer and a date on each, the drivers that reach nothing recorded, and the constraints bound. A catalogue signed on its own contents is what produced the file in front of you.
3. Set the cadence, the same shape as item 11's. Every requirement's link is re-confirmed when the block it reaches changes, when the driver behind it changes, or annually, whichever comes first. A link confirmed three years ago against a block since re-scoped is a decoration with a date on it, which is worse than one without. Then put the catalogue into `GT-0 G0 Charter`, which already tests that scope, acceptance criteria and a contract exist before work starts: add that every requirement the package is said to satisfy has an evidenced link. On the file as handed over no package would pass, and that is the number to take to the board rather than the method.

Done when. `DLV-20` is `Approved` with a date and a named signatory, and `GT-0` tests the link.

Goes wrong when. The catalogue is signed to close the phase. A signed catalogue of template fills is the one artefact in this repository that would survive an audit and should not.

WORKED ANSWERS — THE SAME TEST, TWO ORGANISATIONS

Driehoek Health Group — nine acute hospitals, two day clinics, about 1,750 registered beds. Its step 12.2 pass over its own equivalent of `R09` took a data architect four hours and returned 41 rows checked, 6 surviving and 35 re-pointed. The six survivors were all in one category, because that category's links had been built by the team that owned the data rather than by the programme office. The finding that changed the budget was not the re-pointing: it was that nine `Must` requirements had no block at all once the coincidental pairings were removed, and five of the nine were regulatory.

Rietvlei District Health Service — one district hospital, three community health centres, 22 fixed clinics, a mobile service. Rietvlei had eleven requirements and no relationship sheet, so step 12.2 was unnecessary and step 12.8 was the whole engagement. Six new requirements, one per new sheet, each naming an instrument and a verification, each at `Must`, each reaching a block that did not previously exist. Its requirements catalogue is four pages and every line of it can fail, which is more than the thirty-two-row version can say.

BEFORE YOU SIGN THE REQUIREMENTS CATALOGUE

Seven questions. Each answer is a count, a date or a document reference, or the catalogue is not ready.

For your traceability links: what was reconciled, by whom, and against what?

How many requirements have at least one link carrying evidence, a confirmer and a date, out of how many?

Which requirements would fail if the architecture were as it is today, and which could not fail at all?

Which drivers reach no requirement, and which reach only a template fill?

Which building blocks have no requirement behind them, and which of those are technology blocks?

What does each of your eight constraints bind, and when was each last assessed?

On what date was each link last confirmed, and what is the oldest confirmation still relied on?

THE CHAPTER IN TEN LINES

- Requirements Management is not a phase. It runs through all of them, and its product is a set of links from what the organisation needs to what somebody will build.
- A traceability link is an assertion somebody made, on a date, against evidence, or it is decoration.
- The question is item 10's and is not reinvented here: what was reconciled, by whom, against what, with the artefact, the person and the independent thing behind it.
- R09 pairs requirement n with the nth building block on O55. The requirement list and the block list are grouped differently, so no pairing is about its own requirement.
- REQ-008 Patient mastered once reaches BBB-08 Workforce & Rostering, while DBB-01 Patient mastering sits three rows away on the same sheet.
- Five building blocks satisfy no requirement at all, and they are identity resolution, decision support, device integration, zero-downtime clinical systems and the secure research enclave.
- R09's Traced column reads No on eight rows at a stride of exactly four, and every one of the thirty-two rows has a building block, the eight included.
- RPT06 copies that flag, relabels it Reaches a block, and carries a footnote describing a condition the sheet cannot detect. Two of the eight it flags are regulatory Must requirements.
- R08 is O06's own Driver column reprinted with the name looked up, and the seven regulatory requirements were generated from the drivers, so the join is true by construction and evidence of nothing.
- Three drivers reach no requirement, and Clinical records retention, a High-pressure regulatory driver, reaches exactly one template fill.
- Twenty-seven of the thirty-two statements are template fills, three are ungrammatical, and the remaining five are copied verbatim from O26's rationale for a technology component.
- A requirement needs a subject, a condition and a means of verification, and it must be capable of failing.
- O07's eight constraints appear in no relationship sheet, and nothing anywhere in the repository joins a requirement to a gap, so the discipline cannot raise one.
- Re-derive to R09 RD with evidence, confirmer and date; retire Traced rather than correcting it; and sign DLV-20 only once the links carry all three.

SOURCES FOR THIS CHAPTER

- **1.** SAHPRA, SAHPGL-MD-04 version 5, Guideline for the Classification of Medical Devices and IVDs, effective 27 February 2025 https://www.sahpra.org.za/wp-content/uploads/2025/03/SAHPGL-MD-04_v5-Guideline-for-Classification-of-MD-and-IVDs.pdf
- **2.** Protection of Personal Information Act 4 of 2013, section 26, prohibition on processing of special personal information https://popia.co.za/section-26-prohibition-on-processing-of-special-personal-information/
- **3.** Protection of Personal Information Act 4 of 2013, section 32, authorisation concerning a data subject's health or sex life https://popia.co.za/section-32-authorisation-concerning-data-subjects-health-or-sex-life/
- **4.** National Health Act 61 of 2003, consolidated text (section 13, the record-keeping duty on the person in charge of the establishment, subject to the National Archives and Record Service of South Africa Act 43 of 1996 and the Promotion of Access to Information Act) https://www.saflii.org/za/legis/consol_act/nha2003147.pdf
- **5.** HPCSA Booklet 9, Guidelines on the Keeping of Patient Records, September 2016 edition https://www.hpcsa-blogs.co.za/wp-content/uploads/2022/08/Booklet-9-Guidelines-on-Patient-Records.pdf
- **6.** HPCSA Booklet 9, Keeping of Patient Records, review draft marked September 2022 https://www.hpcsa-blogs.co.za/wp-content/uploads/2022/11/Booklet-9-Keeping-of-Patient-Records_Review-Draft_-FINAL_Sept-2022.pdf
- **7.** User guide to the ICD-10 Master Industry Table, January 2014, National Department of Health (South African start and end dates per code, status of added, modified or discontinued, and the revision history) https://www.health.gov.za/wp-content/uploads/2021/02/cd-10mit_userguide_january2014.pdf
- **8.** Zhang and Pandit, British Journal of Anaesthesia 2023;131(1):130-134, theatre utilisation calculation methods https://www.sciencedirect.com/science/article/pii/S0007091223001794

APPENDIX A

The sheet-by-sheet reference

All 111 sheets — what each holds, what one row of it is, who fills it, and what is wrong with the one you were given.

How to read a row

One hundred and eleven sheets: the 105 the repository shipped with and the six this volume creates. Keep it open beside the spreadsheet. Nothing here teaches; every line is a lookup.

Every sheet has the same shape. Row 1 title, row 2 banner, row 3 blank, row 4 headers, data from row 5. The first column of every object sheet is `Ref`; on a relationship sheet it is the left-hand end's reference instead, except on `R06` and `R19`, which carry a `Ref` of their own.

Each group comes in two tables. Part one says what the sheet holds, so you know you are on the right tab. Part two says which phase fills it, which item tells you how, and the shortest true statement of what is wrong with it. Where a defect is a matter of reading rather than counting, both readings are in the cell and neither is picked. Part two keys on the bare reference, which is how the rest of this volume names a sheet.

`Rows` is the count on 25 October 2026, and it agrees with the sheet's own banner and with `00 Index`'s per-sheet figure in all sixty-five cases. The only count that agrees with nothing is the Index's own total: row 2 says 843, the sixty-five figures beneath it sum to 820, and the difference of 23 is exactly `R06`'s row count.

The numbering runs `O01` to `O66` with **`O34` and `O51` absent**. Nothing accounts for the two missing slots and there is no note about them. They are not hidden and not corrupt. Do not go looking.

Four traps in the tab bar

One. Twenty-three tabs are exactly 31 characters, Excel's limit, and twenty-two of them lost a word. `RPT06 Requirements Traceability` is 31 and complete by coincidence, so it is not one of the twenty-two.

What matters is where the truncation went next. It reached the sheet's own row-1 title on **two sheets only** — `O26`, whose row 1 reads "Logical Technology Componen", and `O27`, "Physical Technology Compone". `00 Index` rows 40 and 41 reproduce both. All twenty truncated relationship tabs carry a **complete** row 1: `R02`'s tab stops at `Capabilit` and its row 1 reads "Value stream uses Capability"; `R09`'s stops at `Bu` and its row 1 reads "Requirement satisfied by Building Block". So read the sheet's name off row 1, not off the tab — and know that on `O26` and `O27` row 1 *is* the tab. Anything that addresses sheets by name addresses them by the tab string, truncation included.

Figure 13.1 Twenty-three tab names are exactly 31 characters and twenty-two lost a word, but the truncation reached the sheet's own row-1 title on only two — O26 and O27. All twenty truncated relationship tabs carry a complete row 1. R04's cut fell after a word boundary, leaving a trailing space in the tab name.

Two. One tab name ends in a trailing space. The cut on `R04` fell after a word boundary, so the last character of the tab name is a space and a lookup written against the name you can see will not find the sheet. Quote it as `R04` and give the re-derived tab a name with no trailing space.

Three. `046` has two columns headed `Ref` — the only duplicate header in the supplied file. Column 1 holds `TBB-01` to `TBB-07`; column 3 holds `T01` to `T07`, which resolve nowhere in the 105 sheets. Two readings, both carried: either a mis-keyed pointer to `026`'s `LTC-01` to `LTC-07`, which match the seven rows name for name and rationale for rationale in the same order, or a sheet numbered `T01` to `T07` was dropped and nothing closed the hole. The cells support the first. Either way seven of seven rows point at nothing, and anything written against `046` addresses the two columns by position.

Four. Four sheets are named by reference and never by their header text. `015`'s row 1 and column 2, `019`'s column 2, `R03`'s column 3 and `R04`'s column 1 and row 1 all name an element ArchiMate 4 no longer carries. This volume names them `015`, `019`, `R03` and `R04` and carries no form for `015` or `019`. The Excel transcription workbook keeps the file's real spellings on its cards, which is correct: those cards are typed into the spreadsheet and have to match it.

Where the six new sheets sit

`067` to `072`, and the numbering is closed: nothing else may claim those slots and there is no seventh. Item 2 defines `069` and `072`; item 5 defines `067`, `070` and `071`; item 7 defines `068`.

Place them after `066` and before `EXT`, add six `00 Index` rows in the Index's own `<n> records · <domain>` format, and correct row 2 to 820 in the same edit. One caution if you work from the shipped empty template rather than the populated file: that template seats the six after `EXT` instead. The two disagree on the position of those tabs and on nothing else — same references, same columns, same order.

What is counted twice

Before you fill a sheet, check whether the file already holds its content under another name.

Sixty-seven of the 820 records are a second copy of something already in the file

Before you fill a sheet, check whether the file already holds it under another name. Four of these pairs are byte-for-byte.

Sheet	Rows	Already in the file as	How exact
O35 Business Building Block	10	= O55 rows 5–14	byte-for-byte
O44 Data Building Block	10	= O55 rows 15–24	byte-for-byte
O45 Application Building Block	10	= O55 rows 25–34	byte-for-byte
O46 Technology Building Block	7	= O55 rows 35–41	plus a second Ref column that resolves nowhere
O36 Data Entity Relationship	11	= R19	byte-for-byte; one model counted as 11 records and 11 instances
O59 Compliance Assessment	3	= R24	byte-for-byte; three verdicts counted twice
O56 Project	12	= O30	same cost, start and end on all twelve rows; one column added
O47 Reference Model Entry	7 of 11	= O26	same seven components, a different status on four of them

Add the nine reports that reprint one sheet and the count of distinct facts in this repository falls again — see item 14.

None of this is caught by anything the file checks: ZZ Checks counts rows, and a duplicated row counts twice.

Figure 13.2 Sixty-seven of the 820 object records are a second copy of content already in the file, four of the pairs byte-for-byte. ZZ Checks counts rows, so a duplicated row counts twice and nothing in the file catches it.

The consequence for Phase E is that there are not thirty-seven building blocks plus ten plus ten plus ten plus seven. There are thirty-seven, and `035`, `044`, `045` and `046` are `055` cut four ways. Fill `055` and let the four be views of it, or the reverse — but say which, in writing, and never edit both.

How far each sheet reaches, and what that decides

What decides whether a sheet can be repaired is not whether it is full. It is whether anything in the model reaches it.

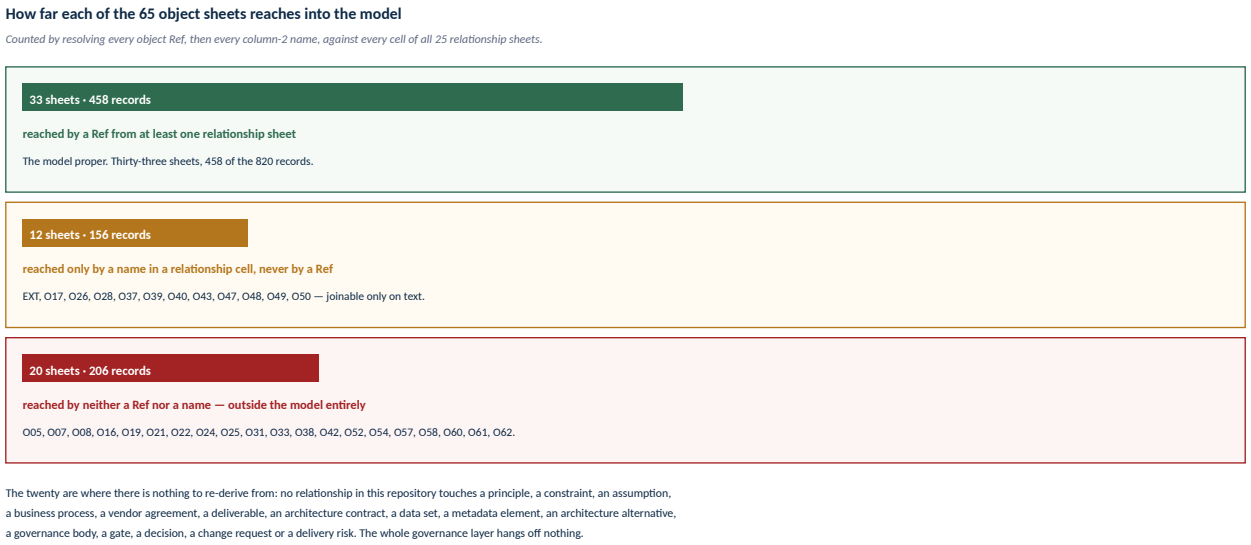


Figure 13.3 Every object Ref and then every column-2 name resolved against all 430 relationship rows. Thirty-three sheets are reached by a reference, twelve by a name alone, and twenty — 206 records, the whole governance layer among them — by neither.

The twenty unreached sheets have nothing to re-derive **from**. On the other forty-five you can test a row against the thing at the far end of the join and decide whether the pair belongs together; on the twenty you cannot, because there is no far end. They need connecting, which is a Phase G and Phase H job rather than a repair.

The nine sheets whose joins resolve and mean nothing — `R04`, `R05`'s consumer half, `R07`, `R09`, `R13`, `R20`, `R22`, `R23`, `R25` — are re-derived, never repaired in place: a new tab named with the sheet's reference plus `RD`, the original row-4 headers exactly and in order, three columns added to the right for the evidence, the confirmer and the date, and the original left untouched.

The index and the motivation layer

Part one — the sheet.

Sheet	Rows	What it holds
`00 Index`	—	The inventory: one sheet per row, with its record count and domain.
`001 Driver`	13	Seven regulatory drivers and six business, each with an origin, owner and pressure.
`002 Goal`	6	Six goals, all `Active`, each with an owner.
`003 Objective`	14	Objectives under the goals, each with a `Horizon` and a status.
`004 Measure`	16	Sixteen measures, each with a unit, baseline figure and cadence.
`005 Principle`	24	The generic principle set, four still `Draft`.
`006 Requirement`	32	Requirements, each with a statement, priority, driver and status.
`007 Constraint`	8	Eight constraints, each typed.
`008 Assumption`	6	Six assumptions, `Open` or `At risk`.

Part two — who fills it, and what is wrong with it.

Sheet	Phase, item	What is wrong in the supplied file
`00 Index`	Preliminary; every close · 1, 13	Row 2 says 843; the sixty-five figures beneath it sum to 820. Rows 40 and 41 carry `O26` 's and `O27` 's truncation; row 79 spells `EXT` with a lower-case e.
`O01`	A · 3	All seven regulatory drivers are owned by `ACT-07` , who heads no unit. Three drivers reach no requirement.
`O02`	A · 3	`GOL-03` grows revenue from existing customers — retail, unrewritten.
`O03`	A · 3	`Horizon` is a relative axis with no origin. Two rows are retail objectives.
`O04`	A; defined in C · 3, 5	No numerator, denominator, exclusion or boundary rule for any of the sixteen. `MSR-08` 's baseline is 25% where ten `Managed` of twenty-three is 43%.
`O05`	Preliminary · 2, 19	Not one is about clinical safety, consent, the record or the patient. All are owned by "Architecture Board", a governance body and not an actor.
`O06`	RM · 12	Twenty-seven of thirty-two statements are template fills, three ungrammatical. Stops part-way down every list it derives from.
`O07`	A · 3	Peak windows and works-council consultation are generic residue. No owner, no expiry.
`O08`	A · 3	No owner, no test, no date for resolution.

The business layer

Part one — the sheet.

Sheet	Rows	What it holds
`O09 Organization Unit`	12	Twelve units, each with a head's title and a headcount.
`O10 Actor`	18	Eighteen posts, every one `Filled` .
`O11 Role`	18	Eighteen roles, each `Performed by` a `Person` .
`O12 Location`	8	Eight sites, each typed and given a `Scale` in people.
`O13 Business Capability`	100	Ten level-1 capabilities to level 3, with criticality, maturity and a disposition.
`O14 Value Stream`	5	Five clinical value streams, each with a stage count and a capability list.
`O15`	10	Ten services, one per level-1 capability, with an exposure and a criticality.
`O16 Business Process`	20	Four numbered stages per value stream, with an automation state.
`O17 Business Function`	10	Ten functions, one per level-1 capability.
`O18 Product`	5	Five care products, each typed.
`O19`	10	Ten vendor agreements, one per application, with a counterparty and expiry month.

Part two — who fills it, and what is wrong with it.

Sheet	Phase, item	What is wrong in the supplied file
`O09`	B · 4	No clinical unit at all. Headcount sums to 1,607 against `O12` 's 1,769; six of twelve heads are not actors on `O10` .
`O10`	B · 2, 4	No clinician, pharmacist, safety officer or records manager. `ACT-07` owns seven drivers and heads no unit.
`O11`	B · 2, 4	A one-for-one mirror of `O10` . The eight data-ownership titles `O20` uses appear here nowhere.
`O12`	B, D · 4, 7	`Scale` is an estates figure: no ward, clinic, theatre or bed. Sums to 1,769 against `O09` 's 1,607.
`O13`	B · 4, 20	Ninety of the hundred names and descriptions are generated. Thirty-six of ninety children outrank their parent; `Disposition` restates the maturity gap exactly.
`O14`	B · 4	Declares twenty-four stages against `O16` 's twenty and contradicts `R02` on two rows. Against chapter 9 the shortfall is sixteen.
`O15`	B · 4	Row 1 and column 2 name an element ArchiMate 4 no longer carries, so it takes no form here. Six of ten set `Exposure` to `Internal` under an "externally consumable" description.
`O16`	B · 4	No name beyond a stream and a stage number, no performer, no system. Contradicts `O14` for three of five.
`O17`	B · 4	All ten descriptions are the identical string and the name is the capability's verbatim.
`O18`	B · 4	Three of five are care settings, not products. Eight of twelve applications support none.
`O19`	B, G · 4, 10	Column 2's header names an element ArchiMate 4 no longer carries, so it takes no form here. `Counterparty` holds the system's own name on all ten rows.

Data, application and technology

Part one — the sheet.

Sheet	Rows	What it holds
`020 Data Entity`	14	The fourteen entities, each with a grading, owner, master system and lineage state.
`021 Logical Data Component`	12	A logical store per entity.
`022 Physical Data Component`	12	A physical store per application, with a storage type.
`023 Application Component`	12	Twelve applications, each with a disposition, criticality, health, in-service year, users and owning unit.
`024 Information System Service`	12	A service per application, with a style and a published state.
`025 Technology Service`	12	Twelve platform-level services, each categorised.
`026`	7	Seven logical technology components, each with a rationale and a status.
`027`	12	Twelve physical components, each with a support-end month, position and site.
`028 Platform Service`	8	Eight platform services.

Part two — who fills it, and what is wrong with it.

Sheet	Phase, item	What is wrong in the supplied file
`020`	C · 5	Not one entity is mastered in a system that could own it. Six patient-identifying or clinical entities are `Public` or `Internal`; ten of fourteen owners are titles the file does not define.
`021`	C · 5	Stops at `E12`, so Consent and Medical Device have none. Adds a `Ref` and the word "store".
`022`	C · 5	A one-per-application cycle: no size, location, owner or retention.
`023`	C · 6	The most real content in the file. `Owner` cycles the unit list — imaging to Human Resources, theatre scheduling to Internal Audit — and `A01` is `Medium` while billing is `Critical`.
`024`	C · 6	Seven of twelve are `Unpublished` against `OBJ-08`; the four `Published` are styled `File`.
`025`	D · 7	`Category` contradicts the service in seven of twelve: the data platform is `Security`.
`026`	D · 2, 7	One of two sheets whose row 1 carries the truncation. Differs from `047` on four of the same seven; its `Rationale` appears verbatim five times.
`027`	D · 7	The second and last sheet whose row 1 carries the truncation. `Position` is typed, not computed — two rows read `Past support` and only `PTC-04` is.
`028`	D · 7	No `Ref` resolves to it; the only cross-references are free text on `041` and `065`.

Gaps, packages and building blocks

Part one — the sheet.

Sheet	Rows	What it holds
`029 Gap`	2	The gap register: building block, severity, status, due month.
`030 Work Package`	12	Twelve packages, each with a cost, dates, approval state and delivery state.
`031 Deliverable`	20	The twenty standard artefacts, by phase, with a status.
`032 Standard`	16	Sixteen standards, each with a status and a conformance verdict.
`033 Architecture Contract`	8	A contract per work package, with a compliance status.
`035 Business Building Block`	10	The ten business blocks, each with a criterion.
`036 Data Entity Relationship`	11	The conceptual model: two entities, a verb, a cardinality.
`037 Data Attribute`	60	Six generic attributes for each of ten entities, with a key flag and a grading.
`038 Data Set`	6	Six data sets.
`039 Data Migration`	6	Six entity migrations, each with a source, target and status.
`040 Source To Target Mapping`	12	Twelve field mappings, each with a rule.

Part two — who fills it, and what is wrong with it.

Sheet	Phase, item	What is wrong in the supplied file
O29	E · 8	Two rows against ten maturity gaps, six standards in breach, three absent controls and seven unevidenced obligations. Thirty-five of thirty-seven blocks have no gap record.
O30	E; dated in F · 8, 9	Seven read Delivered and three Past end date before any has reached its start date. Dollars with no rate, rate date or base year.
O31	every phase · 1, 10	No owner, no date, and no artefact this industry needs — no safety case, no retention position, no measured retrieval time.
O32	Preliminary, D · 7, 19	Conformance is one application's verdict shown as the portfolio position. Six in Breach, four Deprecated, two both at once.
O33	G · 10	Compliance status is O30's Delivery recoded in both directions, with no third value either side. No assessment took place.
O35	E · 8	Byte-for-byte O55 rows 5 to 14, so every business block is counted twice.
O36	C · 5	Byte-for-byte identical to R19. Walks the entity list in a circle: a protocol composed of a clinician, a ward with one funder claim.
O37	C · 5	The same six attributes for every entity, stopping at E10. Twenty-six of sixty are graded above their parent, ATR-001 the patient identifier among them.
O38	C · 5	Four of six are not healthcare, one of them a risk set holding "exposure and position data". No clinical or safety set.
O39	C, F · 5, 9	In six of six neither the source nor the target is the entity's master per O20. No date, no volume, no reconciliation.
O40	C · 5	All twenty-four field cells are placeholders. No real field name appears anywhere in the file.

Security, metadata, mastering and the estate

Part one — the sheet.

Sheet	Rows	What it holds
O41 Security Control	8	Eight controls, each with a position and an enforcement point.
O42 Metadata Element	14	Fourteen metadata elements, each with a coverage state.
O43 Master Data Rule	10	A mastering rule per entity, with a rule type and a golden source.
O44 Data Building Block	10	The ten data blocks.
O45 Application Building Block	10	The ten application blocks.
O46	7	The seven technology blocks, each with a second reference and a criterion.
O47 Reference Model Entry	11	Eleven reference-model entries, each with an adoption status.
O48 Environment	4	Four environments, each flagged for personal data.
O49 Resilience Tier	12	A tier, recovery objective and test state per application.
O50 Network Link	10	Ten site-to-site links, with a bandwidth and a diverse-path flag.

Part two — who fills it, and what is wrong with it.

Sheet	Phase, item	What is wrong in the supplied file
O41	D, G · 7, 10	Three are Absent, attributable access among them. R16 reports two In place controls as Partial and In breach.
O42	C · 5	A catalogue that records its own absence. MTD-02 contradicts O20, MTD-10 contradicts O37, MTD-08 records lineage as Full.
O43	C · 5	Names a different golden source from O20 for eight of ten; MDR-01's type is literally Golden source. Stops at E10.
O44	E · 8	A copy of O55 rows 15 to 24, stopping at E10, so four entities have no block.
O45	E · 8	A copy of O55 rows 25 to 34, stopping at A10, so two applications have no block.
O46	E · 8	Two columns headed Ref, and column 3's T01 – T07 resolve nowhere. Carry both readings; address the two columns by position.
O47	Preliminary · 2	Its first seven rows are O26's components with a different status on four. Nothing records any of the seven in production.
O48	D · 7	ENV-01 Production reads No while two non-production rows read Yes — and ENV-03 Test also reads No. No masking state, no refresh date.
O49	D · 7	The tier label carries no information: the loosest and tightest objectives in the file are both Tier 1. Nine of twelve never tested.
O50	D · 7	Two location pairs are carried twice with different figures. The two slowest links with no diverse path carry imaging and the data-centre pair.

Sequencing, alternatives and governance

Part one — the sheet.

Sheet	Rows	What it holds
052 Implementation Factor	8	Eight sequencing factors, each with a deduction.
053 Transition Architecture	3	Three transition states, each with its packages and an exit-criteria count.
054 Architecture Alternative	5	Five costed alternatives, each with a recommendation.
055 Building Block	37	The whole block set, each with a criterion.
056 Project	12	A project per work package, with a cost, dates and a lead.
057 Governance Body	5	Five bodies, each with a cadence, membership and mandate.
058 Governance Gate	4	The four-gate regime: when each applies and what it tests.
059 Compliance Assessment	3	The verdicts reached: block, criterion, gate, verdict, evidence.
060 Governance Decision	7	Seven decisions, each with a body and a status.
061 Governance Change Request	5	Five requests, each with a target, type, status, expiry.

Part two — who fills it, and what is wrong with it.

Sheet	Phase, item	What is wrong in the supplied file
052	E · 8	IF-03 's peak trading freeze is retail residue. No owner, no date, no resolution path.
053	F · 9	Counts eighteen exit criteria and writes none down. Covers seven of twelve packages, and TA-3 begins before TA-2 .
054	E · 8	ALT-B is Recommended at $46m against a plan of $61.3m , and no sheet reconciles the 33% gap.
055	E · 8	The union of 035 , 044 , 045 and 046 . Thirty-four of thirty-seven are never assessed, and nothing in the file costs a block.
056	F · 9	A one-to-one restatement of 030 . Six leads do not match the work; six projects deliver no increment.
057	Preliminary, G · 2, 10	Membership is prose, not references, and no body is named as ratifier of anything on another sheet.
058	G · 10	Four gates defined, one used: every row of 059 is at G2 . GT-0 tests for a contract and four packages have none.
059	G · 10	Three rows of the 148 verdicts a four-gate regime over thirty-seven blocks requires. No assessor, no date.
060	G · 10	No date column at all, so nothing here can be late and no check can say otherwise. Four of seven are Outstanding .
061	G, H · 10, 11	Expires is blank on four of five. CR-G03 's Against value is the literal string Repository , not 029 .

Risk, obligations and the extension declaration

Part one — the sheet.

Sheet	Rows	What it holds
062 Delivery Risk	8	Eight programme risks, each with a likelihood, impact, owner.
063 Control Objective	7	Seven obligations, each with a source text and an evidence state.
064 Security Policy	4	Four policies, each with a scope and a position.
065 Enforcement Point	5	Five enforcement points, each with what it acts on and a position.
066 Architecture Increment	6	Six increments, each with the month it lands.
EXT Industry Extension	6	The six types this sector needs beyond the core.

Part two — who fills it, and what is wrong with it.

Sheet	Phase, item	What is wrong in the supplied file
062	F · 9	Not one clinical or patient-safety risk. No mitigation, no date, no review.
063	Preliminary, G · 2, 10	All seven Control objective cells are one generated sentence, and not one obligation is evidenced. No instrument column.
064	D, G · 7, 10	All four read Breached . No owner, no date, no exception register.
065	D, G · 7, 10	All five record a defect in Position . EP-05 enforces no policy although POL-01 's scope text names it.
066	F · 9	Lands is a relative axis with no origin. INC-1 lands at month 10 and is delivered by the latest-starting of the six projects.
EXT	Preliminary, C, D · 1, 2, 5, 7	A declaration with no mechanism: nothing resolves to X01 – X06 , and Medical Device and Consent exist as one row of 020 and nothing more.

The six sheets this volume creates

Part one — the sheet.

Sheet	Rows	What it holds
`067 Consent`	new	A consent: type, subject, scope, grant and expiry dates, withdrawal, evidence, system of record.
`068 Medical Device`	new	A device: model, risk class, software version, calibration and maintenance dates, network segment, owner.
`069 Code System`	new	A code system: edition, effective window, licence position, where it is bound, owner.
`070 Measure Definition`	new	A definition: numerator, denominator, exclusions, boundary rule, period, version, owner.
`071 Disclosure Event`	new	A disclosure: record subject, recipient, purpose, authority, date, consent reference, where logged.
`072 Retention Clock`	new	A record class: period, clock start, instrument, system of record, hold rule, destruction authority.

Part two — who fills it, and what is wrong with it.

Sheet	Phase, item	What is wrong in the supplied file
`067`	C · 5	Does not exist. `E13` is one attributeless row of `020` whose lineage reads `Full` on an entity no mapping carries.
`068`	D · 7	Does not exist. "Calibration" appears twice in the whole file and "risk class" once, each time in definition text.
`069`	Preliminary · 2	Does not exist. `E04` and `E05` are defined as coded and neither has a code, code-system or version attribute.
`070`	C · 5	Does not exist. `004` carries sixteen baselines and a definition for none. The shipped template gives this sheet two columns headed `Version`.
`071`	C · 5	Does not exist. With no disclosure log, a withdrawal of consent cannot be acted on and an access request cannot be answered.
`072`	Preliminary · 2, 11	Does not exist. Four sheets record retention as unmanaged and none can hold a period.

The relationship sheets

Twenty-five sheets, 430 instances, and that count is right: the sum across `R01` to `R25` is 430 and every sheet's banner matches its own row count. All 430 rows resolve on both ends, by `Ref` and by name, with no exceptions — the one promise `00 Index` makes and keeps, and the file's blind spot, because nothing in it tests whether the two ends of a resolved link belong together.

The relationship sheets, `R01` to `R13`

Part one — the sheet.

Sheet	Rows	What it holds
`R01 Capability decomposition`	90	Every parent-child pair from `013`, with the child's level.
`R02`	15	Three capabilities per value stream.
`R03`	10	Each level-1 capability to its one service.
`R04`	10	Each service to one application.
`R05`	28	Which application masters and which consumes each entity.
`R06`	23	Every interface: two ends, transport type, frequency, managed and logging state.
`R07`	12	Each application to the component it runs on.
`R08`	32	Each requirement to its driver.
`R09`	32	Each requirement to the block said to satisfy it, plus a `Traced` flag.
`R10`	14	Each objective to its measure, with the unit.
`R11 Goal realised by Objective`	14	Each objective under its goal.
`R12 Gap closed by Work Package`	2	The two gaps and the two packages that close them.
`R13`	37	Each block to the package that delivers it, with the block's domain.

Part two — who fills it, and what is wrong with it.

Sheet	Phase, item	What is wrong in the supplied file
R01	B · 4	The cleanest arithmetic in the file, and nothing in it is a decision anybody took.
R02	B · 4	Contradicts O14 's own capability list for two of five; C08 , C09 and C10 appear in no value stream either way.
R03	B · 4	A positional join correct by construction, because the ten services were generated from the ten capabilities — so it tells you nothing.
R04	B · 4	A positional join: six of ten plainly wrong. Its tab name ends in a trailing space. Re-derive as R04 RD .
R05	C · 5, 6	Keyed by name on rows 5–18 and by Ref on rows 19–32, so a lookup on column 1 returns fourteen of twenty-eight. The consumer half is the ninth positional join.
R06	C · 6	The one relationship sheet with real content. Fifteen rows contradict themselves on four declared tests, seventeen on five — so declare the test, then publish the count.
R07	D · 7	A positional join: ten of twelve wrong, and the component named "Integration engine servers" hosts the imaging system.
R08	RM · 12	Agrees with O06 's own Driver column on all thirty-two rows. Six of thirteen drivers are reached by nothing.
R09	RM · 12	The worst of the nine positional joins: not one of thirty-two reaches a block whose subject it names, and Traced is derivable from nothing.
R10	A · 3	A clean one-to-one join. MSR-15 and MSR-16 measure no objective.
R11	A · 3	Agrees with O03 's own Goal column on all fourteen rows. Nothing wrong and nothing decided.
R12	E · 8	Two rows, so ten of twelve packages close no gap. Coverage reads 100% only because the register holds two rows.
R13	E, F · 8, 9	A positional join cycling twelve packages through O55 , so no package delivers the blocks its own name describes.

The relationship sheets, R14 to R25

Part one — the sheet.

Sheet	Rows	What it holds
R14	12	Each project to its package, with the cost.
R15	7	Which packages sit in which transition state.
R16	7	Each control objective to the control said to meet it.
R17	4	Each policy to its enforcement point.
R18	16	Each standard to one application, with a verdict.
R19	11	The conceptual data model, again.
R20	12	Each application to a site.
R21 Role performed by Actor	18	Each role to its actor, with a status.
R22 Org Unit owns Capability	10	Which unit owns which level-1 capability.
R23	6	Each increment to the project that delivers it.
R24	3	Each assessment to the block it tested.
R25	5	Each product to the application that supports it.

Part two — who fills it, and what is wrong with it.

Sheet	Phase, item	What is wrong in the supplied file
R14	F · 9	A one-to-one restatement of O56 . Sums to $61.3m , agreeing with O30 and O56 and with nothing on O54 .
R15	F · 9	Seven rows for twelve packages, so five sit in no state at all — the most expensive among them.
R16	G · 10	Position is O63 's Evidence copied verbatim, so it reports the objective and not the control, contradicting O41 on four rows.
R17	G · 10	Four rows for five points: EP-05 enforces no policy although POL-01 's scope text names it.
R18	D, G · 7, 10	Cycles the six odd-numbered applications, so six are governed by no standard and O32 's Conformance is one application's verdict.
R19	C · 5	Byte-for-byte identical to O36 , and counted twice. Carries Ref .
R20	D · 7	A positional join that wraps, contradicting R07 and O27 's own Location column for eleven of twelve applications.
R21	B · 2, 4	Eighteen titles joined to the same eighteen, all Filled — true by construction, and silent on the eight data-ownership titles with no holder.
R22	B · 4	A positional join, and not one of ten is right: Human Resources owns medicines management. Re-derive as R22 RD .
R23	F · 9	A positional join to the odd-numbered projects. Re-deriving it cannot stay one-to-one, so it grows to seven rows and the file's relationship total moves.
R24	G · 10	The same three rows as O59 under a second name, so the file's three verdicts are counted twice.
R25	B · 4	A positional join striding the applications in fours. Eight of twelve support no product.

The reports and the checks

Part one — the sheet.

Sheet	Rows	What it holds
RPT01 Dashboard	18	The file's own headline figures.
RPT02 Application Portfolio	12	O23 without its Description column.
RPT03 Capability Heatmap	10	O13 's ten level-1 rows plus a computed Gap .
RPT04 Interface Catalogue	23	R06 with the reference columns dropped.
RPT05 Data Ownership	14	O20 without its Definition column.
RPT06 Requirements Traceability	32	O06 plus R09 's flag as Reaches a block .
RPT07 Gap Analysis	2	O29 with R12 's package added.
RPT08 Roadmap	12	O30 without its Domains column.
RPT09 Regulatory Coverage	7	O63 , column for column.
RPT10 Technology Currency	12	O27 , column for column.
RPT11 Security Position	8	O41 , column for column.
RPT12 Standards Conformance	16	O32 , column for column.
RPT13 Orphan Analysis	2	What nothing else references, with a finding.
ZZ Checks	15	Fifteen typed figures, each with a threshold.

Part two — who fills it, and what is wrong with it.

Sheet	Phase, item	What is wrong in the supplied file
`RPT01`	every close · 14	Three of eighteen are wrong: 843 records, three packages past end date, two components past support.
`RPT02`	C · 14, 6	A reprint. Computes nothing and carries no closing line.
`RPT03`	B · 14, 4	The only report that computes anything, and the column beside the computation restates it in words.
`RPT04`	C · 14, 6	A reprint of a relationship sheet; footnote correct.
`RPT05`	C · 14, 5	A reprint; footnote correct on the stated test, silent on three `Partial` rows.
`RPT06`	RM · 14, 12	Its tab is 31 characters and complete. The flag is asserted, not computed, and the footnote describes a condition the sheet does not detect.
`RPT07`	E · 14, 8	Two rows, no closing line.
`RPT08`	E, F · 14, 9	A reprint, and its own row-18 footnote prints the wrong three.
`RPT09`	G · 14, 10	A reprint, and its footnote is the only place the file states the evidence test.
`RPT10`	D · 14, 7	A reprint, and its row-18 footer states the wrong two with the word "today".
`RPT11`	D, G · 14, 7	A reprint. Footnote correct.
`RPT12`	D, G · 14, 7	A reprint, and the only report with nothing added at all.
`RPT13`	every phase · 14	Two rows, both found by one test against one relationship sheet, and each is referenced by three or four others.
`ZZ Checks`	every close · 14, 1	Three results wrong, three labels carrying more than one reading, eight of fifteen failing their threshold of zero.

What this appendix does not tell you

It does not say what a row should say: the defect column says what is wrong, the item column says where the method is. It does not resolve the two genuine ambiguities — `O46`'s duplicate `Ref` with its unresolvable `T01` to `T07`, and `O66`'s `Month N` against `O56`'s absolute dates, two time axes with no baseline anywhere in the file. Carry both readings into the client's document and do not invent a programme start to make the second go away. It does not carry the figures the file prints about itself, which is item 14. And it does not say how many rows a first pass needs: item 18 reprints every form with that figure on its face.

THE CHAPTER IN TEN LINES

- One hundred and eleven sheets: the 105 the repository ships with and the six this volume creates.
- Row 1 is the title, row 2 the banner, row 4 the headers and row 5 the first data row, on every sheet.
- Twenty-three tabs are exactly 31 characters, twenty-two lost a word, and the loss reached row 1 only on O26 and O27.
- All twenty truncated relationship tabs carry a complete row-1 title, so read the name off row 1.
- R04's tab name ends in a trailing space, so quote it as R04 and expect a lookup on the visible name to fail.
- O34 and O51 do not exist and nothing in the file accounts for them.
- O46 carries two columns headed Ref and its T01 to T07 resolve nowhere; carry both readings, address the columns by position.
- Correct 00 Index row 2 from 843 to 820 in the same edit that adds the six new Index rows.
- Sixty-seven of the 820 records are a second copy of something already in the file, four pairs byte-for-byte.
- Thirty-three object sheets are reached by a reference, twelve by a name alone, and twenty by neither.
- The twenty unreached sheets, 206 records, have nothing to re-derive from; they need connecting, not repairing.
- The nine positional joins are re-derived to a new RD tab, never repaired in place.
- O67 to O72 go after O66 and before EXT; the shipped empty template seats them after EXT instead.

APPENDIX B

The report catalogue

Thirteen reports and a sheet of checks: what each computes, what it gets wrong, and what the set would have to add to be worth running.

Thirteen reports, and what they are for

A report earns its place by computing something the sheets do not already say. Judge the thirteen on that test and the set has three members, one of which gets its one test wrong.

Eight of the thirteen are a column-for-column reprint of a single sheet: `RPT02` prints `O23` with the `Description` column dropped, `RPT05` prints `O20` without `Definition`, `RPT08` prints `O30` without `Domains`, and `RPT09`, `RPT10`, `RPT11` and `RPT12` print `O63`, `O27`, `O41` and `O32` column for column. `RPT04` is the same thing against a relationship sheet: `R06` with the two reference columns dropped and the names left in. Two more reprint one sheet and lift a single column from a second — `RPT06` is `O06` with `R09`'s `Traced` flag relabelled, and `RPT07` is `O29` with `R12`'s work package. So ten of the thirteen compute nothing.

What each of the fourteen computed sheets actually does

Ten of the thirteen reports print a sheet you already have. Three compute something, and one of those three computes it wrong.

Sheet	Reads	What it does with it	Kind
RPT02 Application Portfolio	O23	prints it, one column dropped	reprint
RPT04 Interface Catalogue	R06	prints it, two Ref columns dropped	reprint
RPT05 Data Ownership	O20	prints it, one column dropped	reprint
RPT08 Roadmap	O30	prints it, one column dropped	reprint
RPT09 Regulatory Coverage	O63	prints it, column for column	reprint
RPT10 Technology Currency	O27	prints it, column for column	reprint
RPT11 Security Position	O41	prints it, column for column	reprint
RPT12 Standards Conformance	O32	prints it, column for column	reprint
RPT06 Requirements Traceability	O06 + R09	prints one sheet, lifts one column	reprint + 1
RPT07 Gap Analysis	O29 + R12	prints one sheet, lifts one column	reprint + 1
RPT03 Capability Heatmap	O13, level 1	Target minus Current, right on 10 of 10	computes
RPT13 Orphan Analysis	R04 only	one test, on applications only	computes
RPT01 Dashboard	the whole file	18 typed measures, 3 of them wrong	asserts
ZZ Checks	the whole file	15 typed figures, 3 of them wrong	asserts

Eight reports are a column-for-column reprint of one sheet. Two more reprint one sheet and lift a single column from a second. So ten of the thirteen compute nothing at all, and of the three that do, RPT13 runs a test far narrower than the one its own row 2 claims.

Figure 14.1 Eight reports are a column-for-column reprint of one sheet and two more reprint one sheet and lift a single column from a second, so ten of the thirteen compute nothing. RPT03 subtracts, RPT13 runs one narrow test, and RPT01 and ZZ Checks type their figures rather than deriving them.

That leaves `RPT03`, which subtracts; `RPT13`, which runs one test; and `RPT01`, which is not a report at all but eighteen figures typed onto a page. `ZZ Checks` is the same thing again, fifteen figures with a threshold beside each.

The practical consequence is that printing a report costs you nothing and tells you nothing. You cannot take `RPT05` to a data owner and ask them to confirm it, because `RPT05` is `O20` and the owner has already seen `O20`. What you take to the data owner is the thing the report set does not produce: the `O20` row beside the `O43` row that contradicts it for eight of ten entities, and the `O23` description of the system `O20` says masters it.

The eight reprints, and what to do with each

Leave them. They cost nothing to keep and a client will expect the shape they were sold. But do not describe them as reports in the Phase A document, do not put one in front of a steering committee as evidence of anything, and do not let a reviewer treat agreement between a report and its source as corroboration. `RPT10` agreeing with `O27` is not two sheets agreeing. It is one sheet printed twice.

Report	Prints	Changed	Closing line
RPT02	O23, 12 rows	drops Description	none
RPT04	R06, 23 rows	drops both reference columns, keeps the names	correct: 13 of 23 unmanaged, 10 unlogged
RPT05	O20, 14 rows	drops Definition	correct on the stated test; silent on three Partial
RPT08	O30, 12 rows	drops Domains	**wrong**: "3 work packages are past their end date"
RPT09	O63, 7 rows	nothing	correct, and the only statement of the evidence test anywhere
RPT10	O27, 12 rows	nothing	**wrong**: "2 components are running past vendor support today"
RPT11	O41, 8 rows	nothing	correct: three controls absent
RPT12	O32, 16 rows	nothing	none at all

Two things in that table are worth more than the rest of it. RPT09's closing line is the only place in the whole file that says what "without evidence" means, which is the only reason the number six is defensible. And RPT12 is the one report that adds nothing whatever — no dropped column, no footnote, no gloss — while reporting that six of sixteen standards are in Breach and four are Deprecated, two of them both at once. A report that prints a breach and says nothing about it is worse than no report.

The three that do something

RPT03 Capability Heatmap takes O13's ten level-1 rows and adds a Gap column of Target minus Current. The arithmetic is right in all ten rows. It is the only computation in the report set, and the column beside it undoes the point: Disposition reads Sustain on every row with a gap of 1 and Build on every row with a gap of 2, with no exceptions in ten rows here or in the hundred rows of O13. The one derived figure in the set is restated in words immediately to its right, so a reader who covers the Gap column loses nothing. **What to do with it:** keep the subtraction, delete Disposition, and replace it with the thing a disposition is supposed to carry — what the organisation has decided to do and who decided it.

RPT06 Requirements Traceability prints O06 and relabels R09's Traced column as Reaches a block. Its closing line reads that a requirement reaching no building block is a requirement nothing will deliver, which is true and which this sheet cannot detect: all thirty-two requirements reach a block, including the eight the flag marks No. The flag is asserted, not derived, and two of the eight are regulatory Must requirements. **What to do with it:** the column to compute is not whether a requirement reaches a block but whether the block it reaches names the requirement's subject. On that test the answer is nought of thirty-two, and item 12 owns the method.

RPT07 Gap Analysis prints O29's two rows and adds the package from R12. Both gaps are closed by a package and coverage reads 100%, which is true and worthless, because the register holds two rows against thirty-five building blocks with no gap record. **What to do with it:** item 8 rebuilds the gap set first; the report is then worth printing, and not before.

RPT01, the dashboard, line by line

Eighteen measures on rows 5 to 22, and a closing line on row 24 which reads that every red figure is computed from the workbook and not asserted. That closing line is the only outright false statement in the report set. The figures are asserted. Three of them are wrong, and they are wrong precisely because they were typed rather than derived.

Measure	Value	Verdict
Object types populated	65	correct; `029` carries two rows, so it is a count and not a quality
Records	843	**wrong**: 820
Relationship types	25	correct
Relationship instances	430	correct; every `R` sheet's banner matches its own row count
Reports	13	correct
Applications	12	correct
Interfaces	23	correct
Unmanaged interfaces	13	correct
Interfaces with no logging	10	correct
Direct database reads	3	correct; two of the three are labelled `Managed` on `R06`, against this row's own gloss
Platform components past vendor support	2	**wrong today**: 1
Security controls absent	3	correct
Control objectives with no evidence	6	2 / 6 / 7 of 7
Data entities with no lineage	5	5 `Absent` / 8 without full
Vacant architecture roles	0	0 of 18, silent on 8
Gaps overdue	0	correct, and true only because the register holds two rows
Work packages past end date	3	**wrong**: 0
Level 1 capabilities	10	correct, and the decomposition adds up: 10 + 30 + 60

Eleven are right, three are wrong, three carry a label that does not say which test produced them, and one — gaps overdue — is right for a reason that makes it useless, because a gap register of two rows cannot produce an overdue gap. Four figures are about interfaces and all four are right, which is what you would expect, because `R06` is the one sheet in the file with real content to count.

What to do with it. This is the page an executive sees, so it is the page to fix first and the page to date. Replace the three wrong figures, add the test beside the three loose ones, delete the row-24 claim that the figures are computed, and add the four measures the sheet has no row for: work packages delivered before they start, which is seven; building blocks never assessed, which is thirty-four of thirty-seven; entities whose master system and golden source disagree, which is eight of ten; and applications never recovery-tested, which is nine of twelve. Every one of those is derivable from the sheets in front of you and every one is worse than anything the dashboard currently reports.

The three wrong numbers, and the nine places they are printed

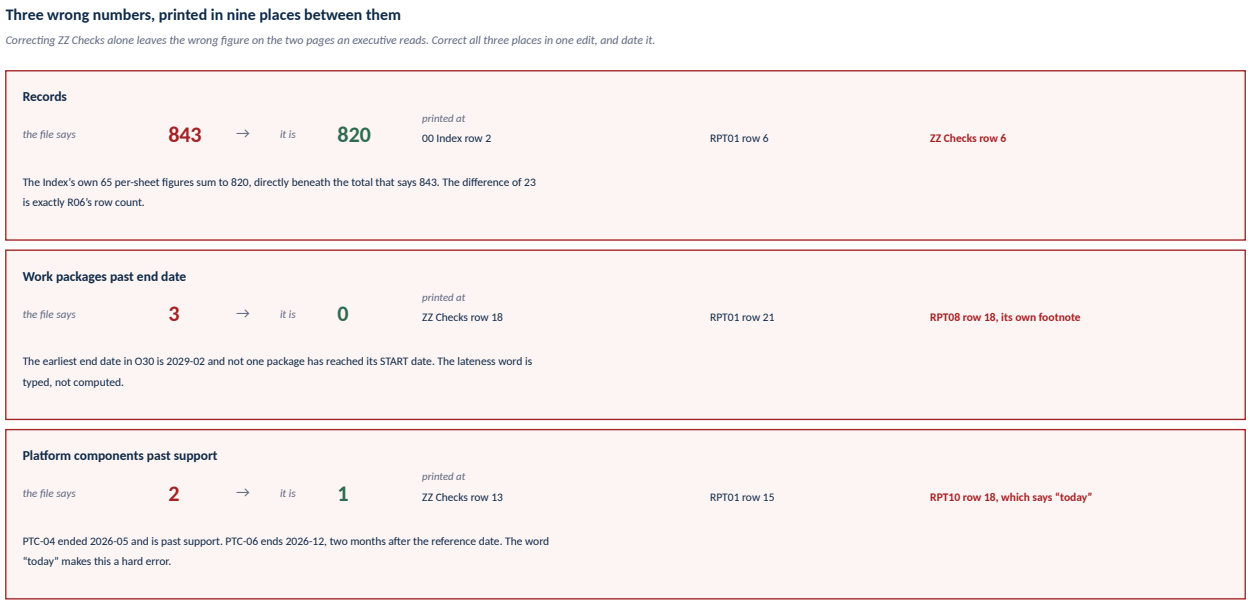


Figure 14.2 Each wrong figure is printed in three places, and in two of the three cases the third place is the report's own footnote. RPT10's states its figure with the word today, which makes it a hard error rather than a matter of reading.

The record count is wrong by 23 and the overstatement equals `R06`'s row count exactly. The lateness count is wrong because `O30`'s `Delivery` column is typed rather than computed: the earliest end date on the sheet is February 2029 and not one package has reached its start date, so on 25 October 2026 the figure is nought. The support count is wrong because `O27`'s `Position` column is typed too: `PTC-04`'s support ended in May 2026 and `PTC-06`'s ends in December 2026, so one component is past support and the other becomes so on 1 January 2027.

Correct all three in one edit, and correct every place each appears. This is the part that gets missed. An item that corrects `ZZ Checks` alone leaves 843 printed on the Index's front page and on the dashboard; leaves "3 work packages are past their end date" printed as `RPT08`'s own footnote, which is the page a delivery board reads; and leaves "2 components are running past vendor support today" printed as `RPT10`'s row-18 footer. That last one is not a matter of reading. It states a condition at a named time, and at that time it is false for one of the two.

Date the correction when you make it. Two of the three numbers are true of a date: the support figure changes on 1 January 2027 and the lateness figure changes in 2029. A corrected figure with no date against it will be wrong again, and the next reader will have no way of knowing when it was last right.

`RPT04`, the reprint worth keeping, and what it should carry

`RPT04` is a reprint like the other seven, but its source is the one relationship sheet in the file with real content, so it is the only one where widening the report is worth the afternoon. `R06` gives twenty-three interfaces with a transport type, a frequency, a managed state and a logging state, and the report prints all of it with the references dropped.

Three columns it does not have and should. **The entity that crosses.** Not one of the twenty-three rows says what data moves, which means no interface in this repository can be assessed against a classification, a retention period or a consent. **The standard it conforms to.** `R18` gives each standard exactly one application and no standard reaches an interface at all. **The hazard.** Nineteen of the twenty-three bypass the integration engine, every interface into or out of the clinical record among them, and no sheet in the file records a clinical consequence of any of them failing.

And one column it has that needs a test printed beside it. `Managed`, `Logged` and `Type` contradict each other on fifteen of the twenty-three rows against four declared tests — a direct store read marked `Managed`, a `Managed` row carrying no logging, an `Unmanaged` row producing correlated logs, and a transport type at odds with its own frequency. `INT-010` and `INT-020` fail two tests each and eight rows survive all four. A defensible fifth test takes the count to seventeen with six survivors. Which is why the count goes under the test and never alone: item 6 owns the method and this report is where the result belongs.

The three checks whose label does not say which test produced them

None of these three is wrong. Each is defensible under a test that is stated somewhere other than beside the number, or nowhere at all. Carry every reading; do not pick one.

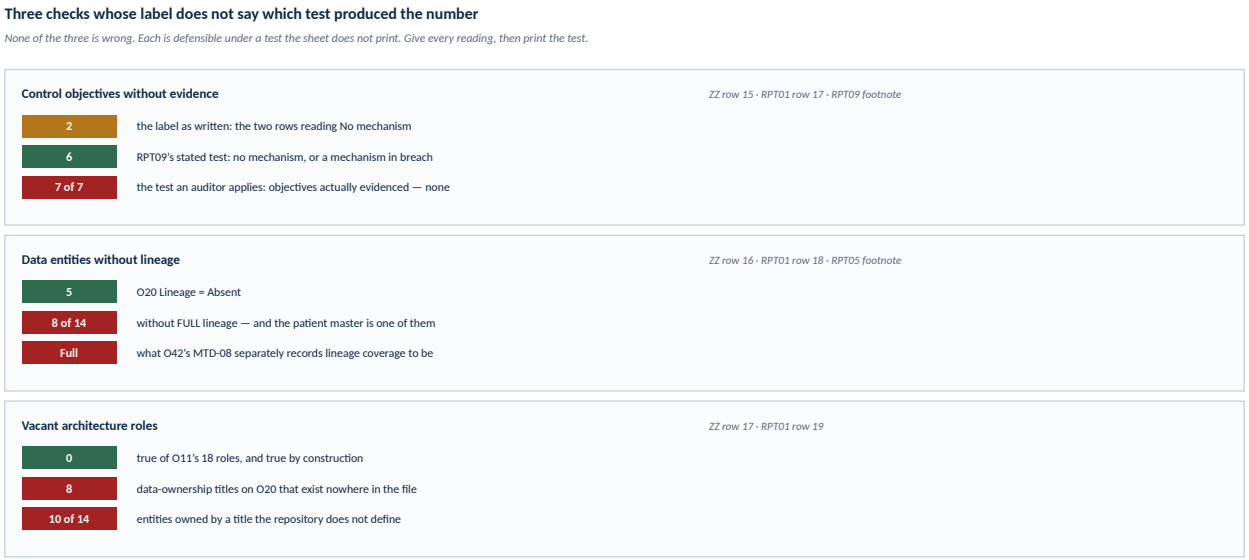


Figure 14.3 Three checks whose number is defensible under a test stated somewhere other than beside it. Carry every reading rather than picking one, and print the test with the count.

Control objectives without evidence, printed as 6. Under the label exactly as written it is 2 — the two rows of `O63` whose `Evidence` reads `No mechanism`. Under `RPT09`'s own closing line, which is the only definition the file offers, it is 6: no mechanism, or a mechanism in breach. Under the test an auditor would apply — objectives that are actually evidenced — it is 7 of 7, because not one of the seven is evidenced and the best any of them manages is `Partial`. Report all three and lead with the last, because it is the one that changes what anybody does.

Data entities without lineage, printed as 5. Correct for `Lineage` = `Absent`. Eight of fourteen lack *full* lineage once the three `Partial` rows are counted, and the patient master is one of the three. Add the contradiction the check cannot see: `O42`'s `MTD-08 Lineage` records coverage as `Full`, which cannot be true alongside either figure. And `MSR-11`, the measure that would track this, carries a baseline of 23% that reconciles to neither.

Vacant architecture roles, printed as 0. True of `O11`'s eighteen roles, and true by construction, because `R21` joins the same eighteen titles to themselves one for one and every row reads `Filled`. The check is silent on the eight data-ownership titles `O20` uses — Head of Patient Access, Head of Care Coordination, Head of Estates, Head of Revenue Cycle, Head of Clinical Care Delivery, Head of Clinical Governance, Head of Population Health, Head

of Diagnostics — none of which exists anywhere else in the file. Ten of fourteen data entities are therefore owned by a title the repository does not define, on a file whose own principle `PRN-13` says an owner is a person and not a committee.

`RPT13`, and the test it did not run

`RPT13` holds two data rows. Both name an application, both are correct on the narrow test its `Finding` column states — neither `A11` nor `A12` is reached by any row of `R04` — and neither is an orphan on the much wider test the sheet's own row 2 announces, which is objects that nothing else references. `A11` is named on four relationship sheets across eight rows and `A12` on three sheets across five. The sheet the Index calls the one that finds what nothing references names two objects that four and three sheets respectively reference.

Run the wide test and the answer is of a different order. Resolve every object `Ref` against every cell of all 430 relationship rows and **thirty-two of the sixty-five object sheets — 362 records — are reached by no reference at all.** Tighten it to the version a reader can check without arguing about leaf types, resolving the column-2 name as well as the reference, and **twenty sheets and 206 records are reached by neither**: `005`, `007`, `008`, `016`, `019`, `021`, `022`, `024`, `025`, `031`, `033`, `038`, `042`, `052`, `054`, `057`, `058`, `060`, `061`, `062`. Every principle, every constraint, every assumption, every business process, every vendor agreement, every deliverable, every architecture contract, every governance body, gate, decision and change request, and every delivery risk. The whole governance layer hangs off nothing.

That is still not everything `RPT13` should have found. On the file's own data: eight of twelve applications support no product, six are governed by no standard, five of twelve work packages sit in no transition state, six of twelve projects deliver no increment, ninety of a hundred capability records have nothing beneath them, five building blocks satisfy no requirement, and thirty-four of thirty-seven are never assessed.

What to do with it. Keep the sheet and widen the test, one layer at a time, and write the test in column 4 beside each finding rather than in the row-2 gloss. An orphan is not always wrong — an attribute points at its entity and the entity does not point back — which is exactly why the test has to be printed with the count. The useful version of this report is two columns: the test that was run, and what failed it.

`ZZ Checks`, line by line

Fifteen checks on rows 5 to 19. Row 21 is a closing sentence, not a check, which is why the sheet is sometimes counted as sixteen.

Check	Result	Threshold	Verdict
Object types populated	65	a sentence	literally true; `029` carries two rows
Records	843	—	**wrong**: 820
Relationship types	25	—	correct
Relationship instances	430	—	correct
Reports	13	13	correct, and met
Interfaces unmanaged	13	0	correct, and failed
Interfaces with no logging	10	0	correct, and failed
Direct database reads	3	0	correct, and failed; two of the three are labelled `Managed`
Platform components past support	2	0	**wrong**: 1 on this date
Security controls absent	3	0	correct, and failed
Control objectives without evidence	6	0	2 / 6 / 7 of 7
Data entities without lineage	5	0	5 `Absent` / 8 without full
Vacant architecture roles	0	0	0 of 18, silent on 8
Work packages past end date	3	0	**wrong**: 0
Records validated against a real organisation	0	0	correct, and the only threshold met honestly

Eight of the fifteen fail a stated threshold of nought. Three carry no threshold at all. Three meet one, and two of those three meet it by reporting nought of something rather than by anything being in order. The last line is the one to cite whenever a figure from this file is quoted outside it.

Which of them to print, and when

A report set is worth running on a cadence or not at all. On this one the cadence is four points, and three of the thirteen carry the weight.

Before Phase A closes, print `RPT01` and `ZZ Checks` and keep the copy. They are the record of what you inherited and they are the baseline every later figure is measured against. Correct the three wrong numbers in that same sitting and date the correction, so the Phase A document shows both what the file claimed and what it held.

At the design gate, `RPT12` and `RPT11`. Six standards in breach and three controls absent are the two facts a design review has to work against, and both are one sheet printed twice, so the reviewer is reading the source. Print them anyway, because the governance record needs the artefact.

Before a build gate, `RPT06` and `RPT03`, and neither of them as they stand. `RPT06` with the re-derived traceability from item 12 beside it, so the gate sees thirty-two requirements against the blocks that actually name their subject. `RPT03` with `Disposition` replaced by a decision and a decider.

After operation, `RPT08` and `RPT10`, both with their footnotes corrected and dated, and `RPT13` widened one layer. The orphan report is the only one in the set whose value grows as the repository fills, because every new sheet creates a new way for something to be unreferenced.

`RPT02`, `RPT05`, `RPT07` and `RPT09` are not gate artefacts at all. They are working sheets, and a working sheet that is a reprint of another working sheet should be opened at the source.

What this report set would have to add to be worth running

Six things, and the shipped empty template does all six, which is the shortest demonstration that they are achievable in a spreadsheet rather than a tool.

One, compute instead of type. Every number on `ZZ Checks` and `RPT01` is typed. Three of them are wrong because the underlying column — `O30`'s `Delivery`, `O27`'s `Position` — is also typed. A lateness figure derived from an end date against today cannot be wrong; a lateness word someone keyed in is wrong the moment the date passes or does not.

Two, test the typing of a join. Read both ends of a relationship row against the object sheet each reference resolves to, and count the rows where the reference and the name beside it do not belong to the same record. That check reads nought here, which is the point: every join in this file resolves. Run it to catch the typing, then read the sheets to catch the sense.

Three, test the sense of a join, which no formula can do. The reconciliation is the work and it is a person's judgement. What a check can do is demand the record of it, which is item 10's formulation: what was reconciled, by whom, against what.

Four, check the four registers this sector needs and this file has no place for. Record classes with no retention clock, consents with no expiry, devices with no calibration date, measures with no measure definition. Those are `O72`, `O67`, `O68` and `O70`, and the checks that read them cannot run against the 105 sheets as delivered because the sheets do not exist. That is not a gap in the checks; it is the finding.

Five, check coverage, not just violation. Applications with no resilience tier, entities on `O20` with no row on `O43`, entities where `O20`'s master system and `O43`'s golden source disagree, acceptance criteria with no assessment. Every one of those reads a figure this file would fail, and not one is on `ZZ Checks`.

Six, print the test beside the number. Three of the fifteen checks here carry a label that does not say which test produced the figure, and the only definition the file offers for any of them sits in a footnote on a different sheet.

The shipped empty template computes **seventy checks against the fifteen this file prints**, every result a formula over the sheets of the workbook rather than a typed figure, in six sections — shape; the record and whether it can be produced; consent and disclosure; the device, the code system and the measure; the estate and the joins that resolve and lie; and the template itself. **Twenty-nine of the seventy cannot be computed over the 105 sheets as delivered**, because the columns and sheets they read do not exist there: no consent register, no device register, no code-system register, no measure definition, no disclosure log, no retention clock, no entity named on an interface, no masking state on an environment, no instrument on a control objective, no date on a governance decision, no benefit measure on a work package.

Run it twice in every phase: once before you start, so you know what you inherited, and once at the end, so the difference is attributable to you. The workbook holds no history, so keep the copy and date it.

THE CHAPTER IN TEN LINES

- Eight of the thirteen reports are a column-for-column reprint of one sheet, and two more reprint one and lift a column from a second.
- Ten of the thirteen compute nothing, so a report agreeing with its source is one sheet printed twice.
- RPT03 is the only computation in the set, and the column beside it restates the result in words.
- RPT06 prints a flag that is asserted rather than derived, and its footnote describes a condition the sheet cannot detect.
- 843 records is printed on 00 Index row 2, RPT01 row 6 and ZZ Checks row 6; the figure is 820.
- Three work packages past end date is printed on ZZ Checks row 18, RPT01 row 21 and RPT08's own footnote; the figure is 0.
- Two components past support is printed on ZZ Checks row 13, RPT01 row 15 and RPT10's row-18 footer, which says today; the figure is 1.
- Control objectives without evidence reads 2 on the label, 6 on RPT09's stated test and 7 of 7 on the test an auditor applies.
- Data entities without lineage reads 5 for Absent and 8 of 14 without full lineage, while O42 records lineage coverage as Full.
- Vacant architecture roles reads 0 of O11's eighteen and is silent on eight data-ownership titles that exist nowhere.
- RPT13 names two orphans that four and three relationship sheets respectively reference, against 206 records on twenty sheets reached by nothing.
- Eight of the fifteen checks fail their own threshold of zero, and the only one met honestly is the line saying nothing was validated against a real organisation.
- The shipped empty template computes seventy checks against the fifteen this file prints, and twenty-nine cannot run over the 105 sheets as delivered.
- Run the checks before and after each phase, and keep the copy, because the workbook holds no history.

APPENDIX C

Workshop packs, checklists and the ninety-day calendar

Seven packs, seven sittings, thirteen weeks, and three Fridays on which somebody signs.

Seven packs, and they are not this item's invention

The seven workshop packs here are the seven capture sheets of the Excel workbook that ships with this volume, `Healthcare_EA_Workshop_Capture.xlsx`. Same seven, same order, same columns. Do not retype them and do not improve them in the room: the capture sheet is the artefact that goes into the repository afterwards, and a facilitator who rebuilds it on a flip chart has turned the paste step into a transcription with no audit trail.

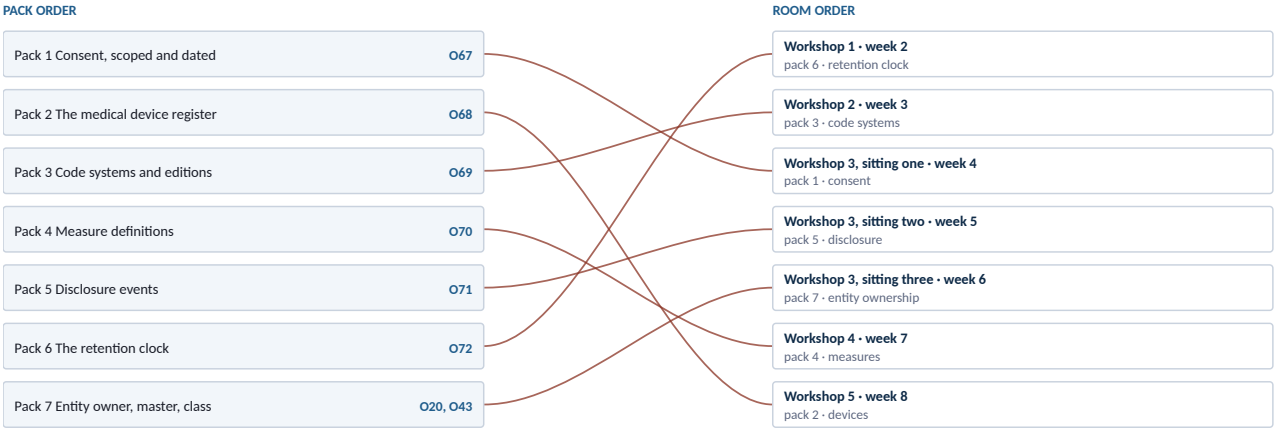
Each capture tab has a fixed shape, and it is not the repository's. Row 1 is the pack title. Row 2 names the sitting and the sheet the pack fills. Rows 3 to 7 are five briefing lines — the length and the room, what to arrange in advance, the row rule, the two columns the sheet exists for, and what leaves the room signed. **Row 8 is the header row, row 9 is a marked example row, and rows 10 to 129 are the hundred and twenty blank capture rows.** Delete the example before you paste.

Pack	Tab	Fills	Sitting	Length	Serves step
1 Consent, scoped and dated	`P1`	`O67 Consent`	Workshop 3, first	three hours	5.9
2 The medical device register	`P2`	`O68 Medical Device`	Workshop 5	three hours	7.4, 7.5
3 Code systems and editions	`P3`	`O69 Code System`	Workshop 2	two and a half hours	2.5, 2.6
4 Measure definitions	`P4`	`O70 Measure Definition`	Workshop 4	three hours	5.11
5 Disclosure events	`P5`	`O71 Disclosure Event`	Workshop 3, second	two hours	5.10
6 The retention clock	`P6`	`O72 Retention Clock`	Workshop 1	three hours	2.7, 2.8
7 Entity owner, master, class	`P7`	`O20` , `O43`	Workshop 3, third	three hours	5.2, 5.3, 5.5

Read the fourth column against the first, because the rooms do not run in pack order. The packs are numbered by the sheet each fills. The rooms run retention clock first, because it needs nothing modelled and the person who controls the off-site store needs a fortnight's notice; code systems second, because `O69` is a precondition for the measure definitions and for every coded attribute after them; then the long third workshop in three sittings; then measures; then devices last, because that register fails on access to an asset list rather than on argument. Pack 1 alone needs a return visit of two hours a fortnight later, and the reason is below.

The packs are numbered by the sheet they fill; the rooms run in another order

Seven capture tabs, seven sittings, five numbered workshops



Pack 1 alone returns for a second sitting in week 6, to close the expiry column.

Figure 15.1 The seven capture tabs on the left, in the order the Excel workbook numbers them; the seven sittings on the right, in the order they run. The retention clock goes first because it needs nothing modelled; the device register goes last because it fails on access to an asset list rather than on argument.

The three columns that never leave the capture workbook

Every capture tab carries more columns than the sheet it fills, and the extra ones stay behind.

Three **check columns** sit immediately right of the sheet's own columns — on pack 1 `Scope check`, `Clock check` and `Capacity check`; on pack 7 `Inheritance check`, `Mastering check` and `Owner check`. They are the facilitator's record that the three things most often skipped were put to the room. Then an **evidence column**, naming the document and date the row rests on, and a **`Signed by`** column, naming the role and the countersignature. The repository has nowhere to put any of the five, and that is the finding rather than a shortcoming of the workbook: a sheet holding a consent's scope

but not who asserted it, against what, on what date, cannot be defended a year later. `O59 Compliance Assessment`'s `Evidence` is the only evidence column in the whole file — three rows of it, all reading that none was produced.

So the paste rule is mechanical. `Ref` across to the last of the sheet's own columns goes into the repository from row 5. Everything right of that stays in the capture workbook.

One further tab, `ZZ Lists`, is hidden and should stay hidden. It holds eighteen named ranges, one per drop-down, and two are load-bearing in ways that are not obvious. `L_Classification` runs loosest to tightest and pack 7's inheritance check reads its **position**, so re-ordering it silently inverts the check. `L_Code_Role` holds what a code system is being used *as* rather than product names. Add a value at the bottom of a column and extend the named range; never rename a column or a defined name.

Form 15-A — Workshop readiness, one row per pack

Feeds no sheet in the repository — the engagement pack carries it · **Rows for a first pass** 7, one per pack · **Signed by** the Chief Enterprise Architect (`ROL-08`) · **Ratified by** `GB-05`

No sheet in the supplied file records that a workshop happened, who attended or what it produced. `O60 Governance Decision` is the nearest and has four columns, none of them a date.

Pack	Room date	Read access	Extract frozen	People confirmed	Facilitator
6 Retention clock	wk 2 Tue — example	n/a	store list, 14 Feb	5 of 5	lead architect

15.1 — Fix the seven sitting dates and the three signing dates before you book a room

Sheet no sheet; the engagement calendar · **Input** the pack table above and a named start Monday · **Output** seven dated sittings and three dated signings, circulated · **Signed by** the Chief Enterprise Architect (`ROL-08`) · **Time** two hours

1. Pick the start Monday and write it down. Everything below is relative to it, and the calendar is thirteen weeks because ninety days is thirteen weeks and a day.
2. Place workshop 1 in week 2, workshop 2 in week 3, the three sittings of workshop 3 in weeks 4, 5 and 6, workshop 4 in week 7, workshop 5 in week 8, and pack 1's return visit in week 6.
3. Place the three signings on the Friday of weeks 2, 6 and 12.
4. Send all ten dates in one message. Sending them separately is how workshop 5 ends up without clinical engineering in the room.
5. Name a facilitator and a separate scribe per sitting.

15.2 — Check the capture workbook against the repository before the first room

Sheet `O20` , `O37` , `O42` , `O43` , `00 Index` · **Input** the capture workbook and read access · **Output** a one-page note of every difference, and pack 7 column J filled · **Signed by** the Chief Enterprise Architect (`ROL-08`) · **Time** half a day

1. Read row 2 of each capture tab against the sheet it names. Where the repository has changed since the pack was written, the note records it; do not edit the pack.
2. Fill pack 7's `Loosest attribute classification` from `O37` before the room sits. It is the only column in the seven packs that must be pre-filled, and the room cannot derive it live.
3. Confirm the six new tabs exist, are empty, and already carry row 4 headers. Step 1.4 did this; check it, because a pack pasted into a sheet with no header row lands at row 1.
4. Confirm `00 Index` row 2 reads 820 and not 843. Rows added to a wrong total compound it.
5. Delete the example row from every working tab, so the room never sees a filled row to copy.

The seven packs

Each section gives the room, what you bring, the running order, the questions that produce rows, what a bad answer sounds like, and what leaves the room signed. Rows 3 to 7 of each capture tab are the short form of the same thing.

Pack 1 — Consent, scoped and dated

In the room: the information officer, the head of health records (`ROL-20`), one admitting clinician, one theatre sister, the person who answers subject-access requests, and whoever configures the consent screen in `A01 EHR / EPR` . The last is the one most often left out and the one who knows what the field actually accepts. Bring `O20` row 17 and `O42` row 11 on a screen, and nothing else.

Four beats. Open on one cell: `O20` row 17 grades `E13 Consent` lineage `Full` , the strongest value on the sheet, for an entity that appears in no mapping, no data set and no migration — then `MTD-07 Consent basis` at `O42` row 11, coverage `Partial` , a metadata element *about* consent rather than an instance of one, referenced by nothing. Second, set the row rule and hold it: **one row per consent type per subject class, not one row per patient.** Volume 1, chapter 4 holds consent to treatment, consent to disclosure and consent to research as three objects with three lifecycles, and the room discovers halfway through that it has been holding them as one. Eight to fourteen rows; one row means the objects have not been separated. Third, work `Scope` and the two dates. Fourth, paste and sign.

PACK 1 — THE QUESTIONS THAT PRODUCE ROWS

For this consent type, what is the purpose, the class of recipient and the class of record?

What ends this consent: a date, an event, or nothing? If nothing, who decided that and when?

Who may give this consent for a child over twelve, and where is the maturity assessment recorded?

A bad answer is `Scope` = "all care". Nine rows reading that is the flag you came to replace with an object. The second is a tick in `Withdrawn` : it is a date or it is blank, never a tick, because a withdrawal with no date cannot be acted on. Capacity turns on age and on an assessed maturity that is itself a recorded clinical act — section 129 of the Children's Act permits a child over twelve with sufficient maturity to consent to their own treatment [1] — and section 14(2)(a) of the National Health Act requires consent to disclosure **in writing** [2], so form is a required property.

Signed out: eight to fourteen `CNS-` rows, by the information officer, countersigned by the clinician who holds the treatment side, plus the list of rows the room could not close. The return visit exists for the expiry column: the room agrees the scope in three hours and will not agree the expiry, because nobody has been asked to put a date on it before.

Pack 2 — The medical device register

In the room: clinical engineering, who hold the asset list; the networks engineer, who knows which segment each device sits on; the theatre and imaging leads; whoever signs the vendor service contracts; and the information officer, because a device holding patient data is a processing location. Bring the clinical engineering asset list and one network inventory, both frozen and dated.

Arrange read access to both extracts a week before the room, not on the day. This is the step that fails and it fails in advance. Open on the arithmetic: the word calibration appears twice in the whole workbook, in `E14` 's definition and in `X05` 's, and the phrase risk class appears once. `O27` is the nearest structure and its `Class` values are Application, Storage, Network, Database, End user and Compute, which is why `PTC-07 Medical device gateways` is filed as infrastructure. Then the row rule: **one row per device type per segment, not one row per serial number**, twelve to thirty rows. Then `Risk class` and `Software version` , the two columns the room wants to leave blank.

PACK 2 — THE QUESTIONS THAT PRODUCE ROWS

What risk class has the licence holder assigned to this device, and where is that in writing?

What software version runs on it today, and who patches it by name?

Which network segment is it on, and can it reach the clinical record from there?

A bad answer is a register of serial numbers with no software version, which is the asset list you already had. `Unknown` is a true value and a finding with a date on it; blank reads as not asked. `Owner` is a named person with a role reference — never a department, never the vendor, because the vendor holds the contract and not the duty. The risk class is assigned by the licence holder under the regulator's classification guideline [3] and is not a judgement for the room to make; the room records what was assigned and where it is written.

Signed out: twelve to thirty `DEV-` rows, by the head of clinical engineering, countersigned by the clinical safety officer (`ROL-19`) — a role the supplied file has no row for and that item 2 creates.

Pack 3 — Code systems, and the edition each is bound to

In the room: the clinical coding manager or senior coder, the laboratory information manager, the pharmacy systems lead, the claims manager, and one clinician who codes their own diagnoses. The coder and the biller will disagree about what a code is for, and that disagreement is the output of the room rather than a problem in it.

Open on the file: the only appearance of any terminology in the whole workbook is one string repeated verbatim five times — at `O06 REQ-029` , `O26 LTC-02` , `O46 TBB-02` , `O47 RME-02` and `O55 TBB-02` — and it is a component name, not a code system. `E04 Diagnosis` is defined as a coded condition attributed to a patient and carries no code, code system or version attribute on any of its six `O37` rows. Then read volume 1, chapter 4's three-way distinction out loud — a terminology so a clinician can record what was observed, a classification so cases can be counted, a payable-unit nomenclature naming a unit of work that is not a clinical act — and ask which of the three each system in the room is being used as. Then the row rule: **one row per code system per edition in use**, six to fourteen rows. Two live editions in two systems is two rows, and the difference is the finding.

PACK 3 — THE QUESTIONS THAT PRODUCE ROWS

Which of the three roles is this system being used as here, and who decided that?

Which edition is live in this system today, and from what date to what date?

What binds this edition — which interface, screen or report? If nothing binds it, say so.

A bad answer is a list of code system names with no edition and no binding, which is what the five verbatim strings already amount to. `Effective from` and `Effective to` are the columns the sheet exists for. `Licence position` is a fact about the organisation rather than the code system, so get it from whoever signs. Use the `L_Code_Role` values rather than naming a product where the role is the fact that matters.

Signed out: six to fourteen `CSY-` rows, by the data owner for coded clinical data, countersigned by the clinical coding manager, with a note on the same page that the decision anticipates a national terminology position that does not yet exist — the gazetted normative standards framework names neither of the two terminologies most architects expect, and lists the diagnostic classification in its acronym list only [4].

Pack 4 — Measure definitions, so a figure can be reproduced

In the room: the head of information or whoever builds the board pack, the finance analyst who builds the other version of the same figure, the theatre or bed manager depending which figure you picked, and one clinician who will say the number does not match the ward. Bring the three figures the executive argues about **with the query behind each**. A figure without its query is a discussion, not a row.

The room exists to settle one thing: which of the defensible figures was filed. Volume 1, chapter 2 shows four defensible theatre-utilisation figures computed from one list with no arithmetic relationship between them — raw utilisation at eighty per cent and touch-time at sixty on the same session, on one specialty in one trust over fifteen weeks [5]. Carry that denominator with the finding. Then the boundary rule, the column an architect from another industry skips and the one that decides the figure: pick from `L_Boundary` , name the rule, and compute from the lowest-level timestamped facts rather than a pre-cut aggregate. Then exclusions, then version and period. `004 Measure` already holds sixteen rows in six columns — `Ref` , `Measure` , `Unit` ,

`Domain`, `Baseline`, `Cadence` — with no numerator, denominator, exclusions, boundary rule or version, which is why `MSR-11` carries a baseline reconciling with no reading of `O20` and `MSR-13` a count of 43 against twelve possible subjects.

PACK 4 — THE QUESTIONS THAT PRODUCE ROWS

What exactly is the numerator, and from which timestamped fact is it counted?

What is excluded from the denominator — day cases, transfers in, deaths, leave of absence?

Which boundary rule applies, and who wrote it down?

A bad answer is nine definitions each reading "as per the dashboard". The second is a definition edited in place: a change is a new version with a date, because the old figures were computed under the old rule and remain correct under it. Signed out: `MDF-` rows for the three disputed figures at minimum, by the measure owner, countersigned by whoever presents the figure, with the matching `MSR-` reference in `Measure` so `O04` and `O70` join.

Pack 5 — Disclosure events, one row per release

In the room: the information officer, the head of health records (`ROL-20`), the person who answers subject-access requests, the claims manager, and whoever sends the statutory and programmatic returns. Those last two send most of the volume and are the two least likely to think of themselves as disclosing anything. Two hours, the shortest room in the pack.

Start from the outward-flow register rather than from the systems — volume 1, chapter 22 asks for every statutory and programmatic report sent out, with its recipient and its clock, on day one for exactly this reason. Then the row rule: **one row per class of disclosure on a first pass, not one row per release.** The operational sheet holds one row per release once a system writes to it, and that is the target state rather than the workshop output. Ten to twenty rows. `Record subject` takes a class of subject: **no patient identifier goes on this card or into this sheet.** Then `Authority` and `Consent reference`, a pair and the point of the room. Where the authority is a patient consent, `Consent reference` takes the `CNS-` key from pack 1 and the two sheets join; where it is a statutory duty or a court order, it is deliberately blank and `Authority` carries the weight.

PACK 5 — THE QUESTIONS THAT PRODUCE ROWS

What class of record left the organisation, to whom, and for what purpose?

Under what authority — a consent, a statutory duty, a court order or an agreement?

If a patient withdrew consent tomorrow, could you list who already has their record?

A bad answer is a register holding only the disclosures somebody already felt uneasy about, which is the opposite of the sample you need. Nothing in the 105 sheets records what left the organisation, to whom, or under what authority: `SEC-05 Interface logging` reads `Partial` with ten interfaces carrying no logging, `EP-02` enforces authentication and not attribution, and `PRN-21 Attributable Access` is `Approved` with nothing behind it. Signed out: ten to twenty `DSC-` rows, by the information officer.

Pack 6 — The retention clock, per record class

In the room: the head of health records (`ROL-20`), the information officer, the medico-legal lead, the clinician who owns the longest-running record class you hold, and the person who physically controls the off-site store. The last knows which boxes exist and is never invited. Three hours, held in the first fortnight and before anything is modelled.

Open on the arithmetic of absence: four sheets independently record that retention is unmanaged — `DRV-02` states the obligation, `CO-02` reads `No mechanism`, `STD-04` reads `Emerging` and `Not assessed`, `MTD-06` reads coverage `Absent` — and no sheet has anywhere to hold a period. `O20` has no retention column and `O37` no retention attribute. Then the governing sentence: **the period, the instrument and the destruction authority are three different facts and the repository holds none of them.** Then the row rule: one row per record class, fourteen to twenty on a first pass. `Clock starts at` is the column that does the work, because a period with no event is not a clock.

PACK 6 — THE QUESTIONS THAT PRODUCE ROWS

What is the period for this record class, and which instrument states it?

From what event does the clock start — last entry, discharge, dormancy, death, majority?

Who has authority to destroy a record of this class, and under what instrument?

A bad answer is a single corporate period applied to everything, which is unlawfully short for the longest class and needlessly expensive for all the rest. Four start events produce four destruction dates for the same box. `Destruction authority` takes an instrument or a named approval, never a guideline; where the honest value is that none has been identified, record it. South Africa has no retention instrument and every period in circulation traces to professional guidance that disagrees with its own editions [6][7], while the National Health Act makes record-keeping expressly subject to the National Archives and Record Service of South Africa Act 43 of 1996 and to the Promotion of Access to Information Act 2 of 2000, which is where the destruction authority sits [2]. The Auditor-General's audit of 27 facilities across nine provinces found 15 reporting missing or lost records, 17 not tracking records after issue, and three of nine paid medico-legal claims conceded because the record could not be found [8].

Signed out: fourteen to twenty `RTN-` rows, by the head of health records, countersigned by the information officer for every class an instrument reaches. The measured retrieval time is added at step 7.8, not here — a retention period with no measured retrieval time is a promise.

Pack 7 — Entity owner, master system and classification

In the room: the clinical coding manager, the head of health records, the laboratory information manager, the pharmacy systems lead, and the one role in the repository that already resolves honestly. Three hours, and do not let it run to four, because the room will want to design a data model.

Open on one cell. `E02 Encounter` is `Public`, and so are `E05 Procedure`, `E12 Claim` and `E14 Medical Device`; `E04 Diagnosis` and `E07 Diagnostic Order / Result` are `Internal`. Read `E04`'s own definition aloud — a coded condition attributed to a patient — then ask on what basis it is graded `Internal`. Then the comparison that lands: `E09 Clinical Protocol`, which identifies nobody and which the organisation publishes, is `Restricted` — as tightly as `E01 Patient`, and those two are the only `Restricted` rows of fourteen. Then the rule: **classification inherits downward and never loosens**, broken in 26 of 60 `O37` rows including `ATR-001`, the patient identifier, `Public` beneath a `Restricted` entity, against the file's own `PRN-14` at `O05` row 18. Then mastering: do not leave the room until `Master system` and `Golden source` agree for every entity.

PACK 7 — THE QUESTIONS THAT PRODUCE ROWS

On what basis is this entity classified at this grade, and who graded it?

Which single system is authoritative, and which system does the organisation actually read?

Who owns it, by name and with a role reference — not a title and not a committee?

A bad answer is a `Head of` title. Eight sit in `O20` today and resolve to nobody, because the posts they name have no unit in `O09` to sit in. They are not discarded: they are the only evidence in the file that somebody knew the industry, and item 4 turns them into roles.

Signed out: fourteen rows pasted into `O20` and the matching `O43` rows in the same sitting, or the three sheets disagree again by Friday. Signed by the data owner for each entity, countersigned by the information officer for every entity graded `Confidential` or `Restricted`.

15.3 — Run the fifteen minutes before and the fifteen minutes after

Sheet the capture tab for the pack · **Input** Form 15-A for this pack · **Output** a dated capture tab and a list of rows the room could not close · **Signed by** the facilitator · **Time** hours

1. Before: delete the example row, project the tab, read row 2 aloud, and name the signatory and countersignatory. A room that does not know who signs produces rows nobody will defend.
2. In the room: the scribe types into the capture tab. No flip charts and no separate notes.
3. After: fill the three check columns yourself, from what was said, before you leave the building.

4. After: list every row the room could not close, with the reason and the name of whoever would close it. That list is the output nobody expects and the one the next sitting runs on.
5. After: save with the date in the filename and circulate the same day.

15.4 — Paste a pack without losing the check columns

Sheet the sheet the pack fills, plus `00 Index` · **Input** a signed capture tab · **Output** the sheet populated from row 5 with its `00 Index` count corrected · **Signed by** the sheet's signatory · **Time** hours

1. Select `Ref` across to the last of the sheet's own columns — on pack 1 `System of record`, on pack 6 `Destruction authority`, on pack 7 `Lineage` into `O20` and `Golden source` plus `Rule type` into `O43`.
2. Paste as values into row 5 downward. Do not paste formatting; row 4's header style is what a later reader matches on.
3. Update row 2 of the sheet and the matching `00 Index` row in the Index's own `<n> records · <domain>` format.
4. Leave the check, evidence and `Signed by` columns behind, and record in the change note which capture file, by filename and date, the rows came from.
5. Where a row was retired in a later sitting, date it rather than deleting it.

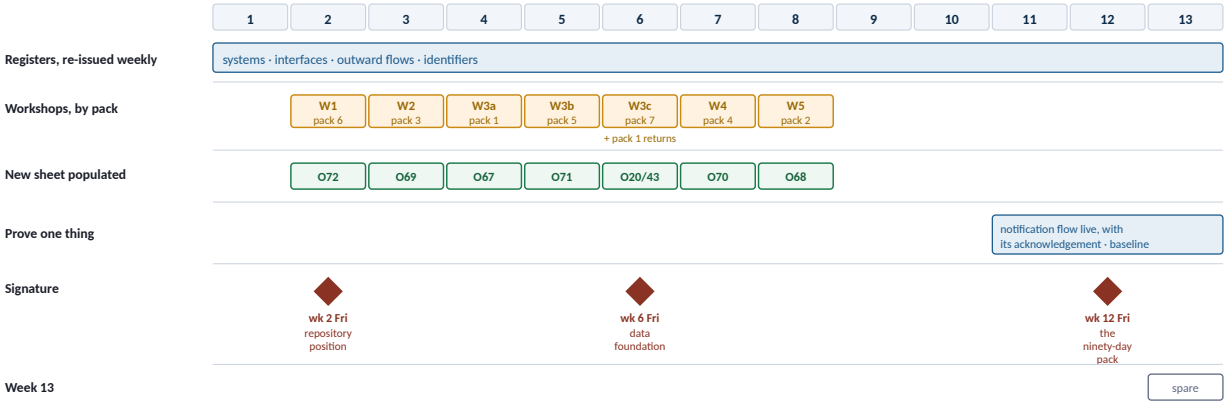
The ninety-day calendar

Volume 1, chapter 22 gives the decision order — identity, terminology and integration strategy settled before the application strategy, with event-driven clinical notification as the highest-return first capability — reads the Western Cape Provincial Health Data Centre as a worked reference design with its own five published compromises [9], and names eight artefacts with the role that signs each. None of that is restated. What follows is that chapter turned into dated weeks: which pack runs when, which step produces which artefact, and the three dates on which somebody signs.

Thirteen weeks, because ninety days is thirteen weeks and a day. Call the first Monday week 1 and write the date on the register.

Thirteen weeks, seven sittings, three signatures

Everything is relative to a named start Monday. Week 13 is deliberately empty.



W3a, W3b and W3c are the three sittings of one workshop. Nothing in weeks 1 and 2 needs a budget, a vendor or a committee.

Figure 15.2 The ninety days as thirteen weeks against a named start Monday. Six of the seven sheets the repository does not have are populated by week eight. The three diamonds are the only dates on which anybody signs anything.

Weeks 1 and 2 — count, collect, and sign the repository position

Week	What runs	Artefact	Sheet	Signed by
1 Mon–Tue	Steps 1.2, 1.1	count sheet, three corrections; meaning rate on two sheets	`00 Index`, `ZZ Checks`	ROL-08
1 Wed	Step 1.3	mapping note, or the documents that do not exist	—	ROL-08
1 Thu	Step 1.4	six empty tabs, six Index rows, row 2 to 820	`00 Index`	ROL-08
1 Fri	Step 15.1; registers opened	ten dates; four registers	—	ROL-08
2 Mon	Steps 2.1, 2.2	five bodies staffed; `ROL-19` and `ROL-20` created	`O57`, `O11`	ROL-08
2 Tue–Thu	**Workshop 1, pack 6**; steps 2.7, 2.8, 15.4	`O72` fixed, populated, pasted	`O72`	ROL-20
2 Fri	**Signing point 1**	the repository position pack	—	below

Week 2, Friday. One pack, five pages, one signature page. The corrected count — 820 against the 843 printed in three places. Two `ZZ Checks` corrections: work packages past end date 3 to 0, and components past support 2 to 1 — with the note that the wrong 3 also appears in `RPT01` and in `RPT08`'s own footnote, and the wrong 2 in `RPT01` and in `RPT10`'s row-18 footer, which states it with the word "today". The meaning rate from two relationship sheets, beside the integrity rate of 430 of 430. The six reserved tab numbers. The two new posts, with what each signs. And `O72 Retention Clock` populated.

Signed by the Chief Enterprise Architect (ROL-08) for the corrections and the Head of Health Records (ROL-20) for the clock, countersigned by the Chief Compliance Officer (ROL-07) for every record class an instrument reaches, and ratified by GB-01 .

Nothing in that week required a budget, a vendor or a steering committee.

Weeks 3 to 6 — name the boundaries, and sign the data foundation

Week	What runs	Artefact	Sheet	Signed by
3	**Workshop 2, pack 3**; steps 2.5, 2.6, 2.3	069 populated; principles re-owned and re-statused	069 , 005	data owner
4	**Workshop 3, sitting one, pack 1**; steps 5.1, 5.9, 2.4	067 populated; Phase C deliverables back to draft	067 , 031	information officer
5	**Workshop 3, sitting two, pack 5**; steps 5.10, 5.6, 5.7	071 populated; identifier rows; merge and unmerge	071 , 037	ROL-05
6 Tue	**Workshop 3, sitting three, pack 7**; steps 5.2, 5.3, 5.5	fourteen entities regraded, re-seated, re-owned	020 , 043	data owner
6 Wed	pack 1's return visit; steps 5.4, 2.9	every CNS- expiry closed; 26 breaches closed	067 , 037	ROL-07
6 Fri	**Signing point 2**	the data foundation pack	—	below

Week 6, Friday. Chapter 22's three documents — the identity position paper, the vocabulary note and the measure definitions for the figures already in dispute — go in this pack whether or not the measure room has run. Beside them: 069 and 072 populated and effective-dated; 067 with an expiry on every row or a written reason for its absence; 071 as the baseline sample; the fourteen entities with a grade, an owner, a master system and a lineage value that three sheets now agree on; and the twenty-six attribute breaches closed, split by method — how many closed by tightening the attribute and how many by re-grading the entity. Signed by the Chief Data Officer (ROL-05) for classification, mastering and lineage, the Chief Compliance Officer (ROL-07) for the clock and the instruments, and the named data owner for each entity. Ratified by GB-03 for the owner designations and by GB-01 for the Preliminary refusals on the same page.

Weeks 7 to 12 — measure, prove one thing, and sign the quarter

Week	What runs	Artefact	Sheet	Signed by
7	**Workshop 4, pack 4**; step 5.11	070 for the three disputed figures, joined to 004	070 , 004	measure owner
8	**Workshop 5, pack 2**; steps 7.4, 7.5	068 populated; network segments written into it	068	ROL-19
9	Item 6 on R06 ; steps 5.13, 5.14	interface position on a declared test; 039 re-derived	R06 , 039	ROL-08
10	Steps 7.6, 7.7, 7.2	048 repaired, reading stated; objectives re-derived clinically	048 , 049	ROL-19
11	Step 7.8; the notification flow	retrieval time published; one live flow, with its acknowledgement	072	ROL-20
12 Mon–Thu	Twelve measures baselined; step 15.8	first measurement, method per figure; register re-dated	070 , 004	measure owner
12 Fri	**Signing point 3**	the ninety-day pack	—	below

Week 12, Friday. Six sheets that did not exist in week 1, populated and signed. The measured retrieval time, the only number in overlay 1 that can be measured and the one the medico-legal argument runs on. Recovery objectives re-derived on clinical grounds, with the finding that the tier label in the supplied file carries no information at all — 049 's Tier 1 holds both the loosest objective in the file and the tightest, and nine of twelve applications including both Critical ones have never been tested. The device register with a risk class and a software version per row, or Unknown with a date. One notification flow live, with the proportion never acknowledged. The first measurement of the twelve benchmark measures, with the method beside each figure and "not measurable" recorded where that is true. And the refusal register, re-dated.

Signed by the Chief Enterprise Architect (ROL-08) for the pack, the Clinical Safety Officer (ROL-19) for the recovery objectives and device risk classes, the Head of Health Records (ROL-20) for the retrieval time and record classes, and the Chief Data Officer (ROL-05) for the entity set. Ratified by GB-01 and reported to GB-04 .

Week 13 is deliberately empty. It is where the signature that slipped goes, and in three engagements out of four something does.

What is signed on three Fridays, and by whom

One pack each. A signature with no artefact behind it is a meeting.

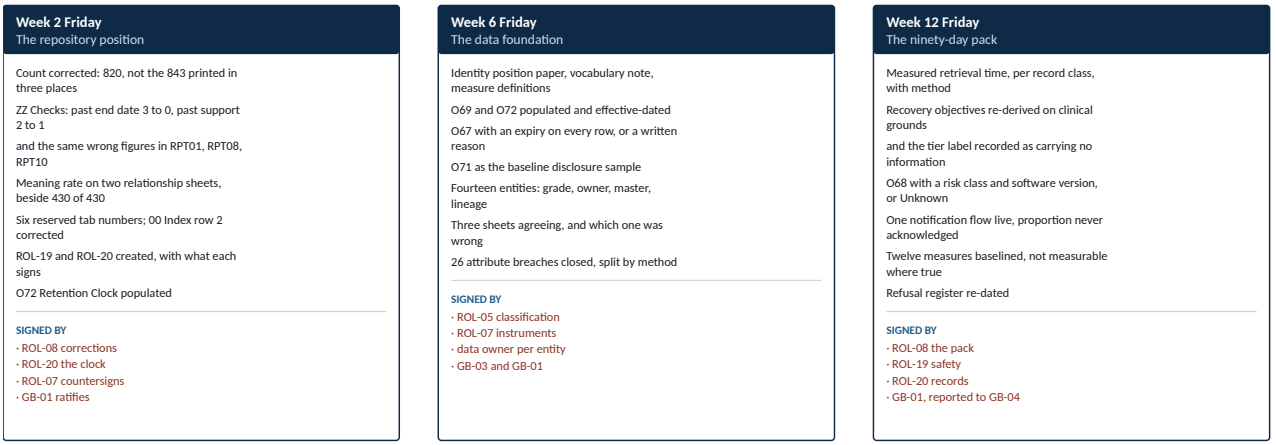


Figure 15.3 The three signature packs, page by page, with the role that signs each part and the body that ratifies it. Nothing on the week 2 page required a budget, a vendor or a steering committee.

> **`O60 Governance Decision`**
>
> `Ref`, `Decision`, `Body`, `Status`. Seven rows. `Body` holds a body name, not a `Ref`, so a lookup against `O57` matches on name. `Status` takes `Outstanding` and `Approved`. **There is no date column and no column for who signed**, which is why Form 15-B carries both and says so.

Form 15-B — The ninety-day signature register

Feeds `O60 Governance Decision` · **Rows for a first pass** 12, one per signed artefact across the three signings · **Signed by** the Chief Enterprise Architect (`ROL-08`) · **Ratified by** `GB-01`

`O60` has no date column and no signatory column. Columns 1 to 4 are `O60`'s own and are typed straight into Excel; columns 5 and 6 have nowhere to go in the repository.

Ref	Decision	Body	Status	Date	Signed by
DEC-08	Record count corrected to 820 — example	Architecture Board	Approved	wk 2 Fri	ROL-08

What is deliberately not decided

Refusal needs a form of words, because "we need more information" reads as a stall. The form that holds is: *this decision has a named prerequisite, here it is, here is when we expect it, and here is what runs meanwhile.* Chapter 22 gives six refusals that belong to the industry. These seven belong to the repository, and they are the ones a client presses on between week 2 and week 12.

Refuse to decide	The named prerequisite	What runs meanwhile
The target application estate	The identity, terminology and integration decisions, in that order	Publish the four registers and the three week-6 documents
A programme start date	A board-minuted baseline date	Carry `O66` 's relative months, `O03` 's `Horizon` and `O30` 's absolute dates as three axes, two with no origin
Which of `046` 's two readings is right	The file's version history, or its author	Carry both: a mis-keyed pointer to `026` 's seven rows, or a dropped `T` -numbered sheet
The programme cost in rand	A stated rate on a stated date, and a base year	Quote the file's own strings — `$61.3m` on `O30` , `O56` and `R14` , `$46m` for `ALT-B` — never convert silently
Whether a device is software as a medical device	The licence holder's classification, in writing [3]	Record the class as `Unknown` with the date it was asked for
The national terminology binding	A designated release centre, not yet operating	Store the code system and edition on every coded value, which `O69` now does
The retention periods themselves	The current edition of the professional guidance, whose versions disagree [6][7]	Keep `O72` as data, `Destruction authority` reading "None identified" where true

Two are worth more than the rest. The cost one, because every cost in the file is in US dollars with no rate, no rate date and no base year, in a repository whose own banner is a South African provider — so restating the column in rand at a stated rate on a stated date is a precondition of taking the programme cost anywhere. And the retention one, because the honest position is more useful to an accounting officer than a confident six-year rule that is wrong twice over.

Form 15-C — The refusal register

Feeds no sheet · **Rows for a first pass** 13, the six from volume 1 chapter 22 and the seven above · **Signed by** the Chief Enterprise Architect (`ROL-08`) · **Ratified by** `GB-01`

No sheet in the repository holds a refusal. `O61 Governance Change Request` is the nearest and records requests, not refusals; `O08 Assumption` has three columns and no date.

Refusal	Prerequisite	Expected	Meanwhile	Re-dated	Open
Programme start date — example	board-minuted baseline	wk 6	three axes carried	wk 6 Fri	yes

15.5 — Hold the week 2 signing

Sheet `00 Index` , `ZZ Checks` , `RPT01` , `O72` · **Input** steps 1.2, 1.4, 2.2 and 2.8 complete · **Output** a five-page pack and one signature page · **Signed by** the Chief Enterprise Architect (`ROL-08`) · **Time** half a day

1. Put the corrected count first and the meaning rate second. The count establishes that you check; the meaning rate establishes what checking is for.
2. Name all three places each wrong `ZZ Checks` figure is printed. A pack that corrects one leaves the wrong number on the page an executive reads.
3. Show the six reserved tab numbers and the two new posts on one page, with what each post signs.
4. Put `O72` in the same pack. It is the only populated new sheet at this point and it is what makes the pack a deliverable rather than a critique, and record the refusals on the signature page.

15.6 — Hold the week 6 signing

Sheet `O20` , `O37` , `O43` , `O67` , `O69` , `O71` , `O72` · **Input** workshops 1 to 3 complete and pasted · **Output** the data foundation pack, signed per entity · **Signed by** the Chief Data Officer (`ROL-05`) · **Time** a day

1. Produce one page per entity: grade, owner by name, master system, lineage value, and the loosest attribute classification beneath it. Fourteen pages, and they are read.
2. Show the twenty-six attribute closures split by method. The split is the evidence that a rule was applied rather than a column overwritten.
3. Show `O20`, `O43` and `R05` agreeing, and say which was wrong. Two sheets agreeing is not corroboration when one was copied from the other.
4. Put `O67`'s open expiries on the page with a named person against each, and have `GB-03` ratify the owner designations in the same meeting.

15.7 — Hold the week 12 signing

Sheet the six new sheets, plus `O48`, `O49`, `O68` · **Input** workshops 4 and 5 complete, the retrieval measured, the notification flow live · **Output** the ninety-day pack and a baseline · **Signed by** the Clinical Safety Officer (`ROL-19`) and the Head of Health Records (`ROL-20`) · **Time** two days

1. Open with the measured retrieval time, per record class, with the method. It is the one number in the pack nobody can argue with and nobody had before.
2. Put the recovery objectives next, with the clinical grounds for each and the signature of the role that set them. State that the tier label carried no information and has been replaced.
3. Show the baseline of the twelve benchmark measures with the method beside each figure, and record "not measurable" where that is the answer. Several will be, and that is a finding, not an apology.
4. Show the notification flow and the proportion never acknowledged. One working thing with a number on it moves the next budget conversation further than the other eleven pages.
5. In a client file: where there is no `O57` equivalent, have the accounting officer's delegate sign and record that no body exists. Do not invent a committee.

15.8 — Publish the refusal register, and re-date it at every signing

Sheet no sheet; `O61` is the nearest and records requests rather than refusals · **Input** Form 15-C · **Output** a dated refusal register, re-issued three times · **Signed by** the Chief Enterprise Architect (`ROL-08`) · **Time** hours

1. Write each refusal in the four-part form: the decision, the prerequisite, the expected date, what runs meanwhile.
2. Circulate it in week 1, not in week 12. A refusal published after somebody has asked reads as a defence.
3. Re-date it at each signing and name what moved. A register with the same thirteen rows at week 12 that it had at week 2 is a register nobody worked.
4. Where a prerequisite arrived and the decision was still not taken, name who holds it.

THE FOUR QUESTIONS A SPONSOR ASKS AT WEEK TWELVE

What did you change in the repository, and who signed each change?

Which of the six new sheets is populated, and which is a header row with nothing under it?

What is the measured retrieval time, and for which record classes is it still unmeasured?

If I replaced you on Monday, which artefacts would your successor find, and whose names are on them?

THE CHAPTER IN TEN LINES

- The seven workshop packs are the seven capture tabs of the shipped Excel workbook, in its order, with its columns. Do not retype them and do not improve them in the room.
- Each capture tab puts the header on row 8, a marked example on row 9 and a hundred and twenty blank rows below. Delete the example before you paste.
- The packs are numbered by the sheet each fills; the rooms run retention clock, code systems, the three sittings of workshop 3, measures, devices.
- Three check columns, an evidence column and a signatory column stay in the capture workbook, because no sheet in the repository can hold any of the five.
- ZZ Lists is hidden for a reason: the classification list is ordered loosest to tightest and pack 7 reads its position, so re-ordering it inverts the check.
- Every pack has a row rule, and holding it is most of the facilitation: one row per consent type per subject class, per device type per segment, per code system per edition, per record class, per class of disclosure.
- A bad answer has a shape per pack: scope reading all care, a register of serial numbers, a code system with no edition, a definition reading as per the dashboard, a single corporate retention period, a Head of title.
- Ninety days is thirteen weeks and a day, and the thirteenth week is deliberately empty.
- Week 2 Friday signs the repository position: the corrected count, both ZZ Checks corrections in all three places each is printed, the meaning rate, the six reserved tabs, the two new posts, and the retention clock populated.
- Week 6 Friday signs the data foundation: the three documents from chapter 22, the code system and clock sheets effective-dated, consent and disclosure populated, and fourteen entities three sheets now agree on.
- Week 12 Friday signs the ninety-day pack: the measured retrieval time, recovery objectives re-derived clinically, the device register, one live notification flow with its acknowledgement, and twelve measures baselined.
- Seven refusals belong to the repository rather than to the industry, and the cost one is the sharpest: every figure in the file is in US dollars with no rate, no rate date and no base year.
- Publish the refusal register in week 1, not in week 12, and re-date it at each signing.

SOURCES FOR THIS CHAPTER

- **1.** Children's Act 38 of 2005, section 129, consent to medical treatment and surgical operation (a child over 12 of sufficient maturity may consent to their own treatment) https://www.acts.co.za/childrens/129_consent_to_medical_treatment_and_surgical_operation.php
- **2.** National Health Act 61 of 2003, consolidated text (s13 record-keeping duty, expressly subject to the National Archives and Record Service of South Africa Act 43 of 1996 and the Promotion of Access to Information Act 2 of 2000; s14(2)(a) consent to disclosure in writing; s16(1)(b) research authorisations; s17(2) offences) https://www.saflii.org/za/legis/consol_act/nha2003147.pdf
- **3.** SAHPRA, SAHPGL-MD-04 version 5, Guideline for the Classification of Medical Devices and IVDs, effective 27 February 2025 (the four risk classes and the classification responsibility; described, not reproduced) https://www.sahpra.org.za/wp-content/uploads/2025/03/SAHPGL-MD-04_v5-Guideline-for-Classification-of-MD-and-IVDs.pdf
- **4.** Health Normative Standards Framework 2021, Notice 2667, Government Gazette 47337, 21 October 2022, published under National Health Act s74(1) (names neither of the two terminologies most architects expect; the diagnostic classification appears in the acronym list only) https://www.health.gov.za/wp-content/uploads/2022/10/HNSF_Gazette_21_October_2022.pdf
- **5.** Zhang and Pandit, British Journal of Anaesthesia 2023;131(1):130-134, four theatre-utilisation calculation methods on one specialty in one trust over fifteen weeks, with no simple mathematical relationship between them https://www.sciencedirect.com/science/article/pii/S0007091223001794
- **6.** HPCSA Booklet 9, Guidelines on the Keeping of Patient Records, September 2016 edition (retention at least six years from dormancy; minors to the 21st birthday; cl. 9.5 authorisation to destroy) https://www.hpcsa-blogs.co.za/wp-content/uploads/2022/08/Booklet-9-Guidelines-on-Patient-Records.pdf
- **7.** HPCSA Booklet 9, Keeping of Patient Records, review draft marked Revised September 2022 (general records retained ideally indefinitely, particularly if electronic; which edition is in force could not be established) https://www.hpcsa-blogs.co.za/wp-content/uploads/2022/11/Booklet-9-Keeping-of-Patient-Records_Review-Draft_-FINAL_Sept-2022.pdf
- **8.** Auditor-General findings on the health sector, reported via MedicalBrief, 14 October 2026 (27 facilities across nine provinces: 15 reporting missing or lost records, 17 not tracking records after issue, 25 with inadequate storage; three of nine paid medico-legal claims conceded for want of the record) - headline source, secondary https://www.medicalbrief.co.za/health-sector-crippled-by-recurring-systemic-weaknesses-ag-warns/
- **9.** Coetzee and others, data resource profile of the Provincial Health Data Centre of the Western Cape, International Journal of Population Data Science (the reference design and its five published compromises) https://ijpds.org/article/view/1143

APPENDIX D

The repository in ArchiMate 4

Forty-two elements, eleven relationships, and nineteen of the seventy object sheets named after something neither of them carries.

What this item is, and what chapter 20 already did

Volume 1, chapter 20 owns the modelling patterns. It fixes the element names, gives the six limits of the notation, and draws the seven patterns a provider engagement needs. Do not restate it. When a reviewer asks why a retention period is a property rather than a box, send them chapter 20 and get on with the import.

This item does the thing chapter 20 deliberately did not: the sheet-by-sheet mapping. One hundred and eleven sheets — the 105 supplied plus the six this volume tells you to create — against forty-two elements and eleven relationships, with the places where the mapping fails and what to write instead.

The sheets sort into four kinds, and only two of them become model content:

- **Seventy object sheets** that hold things: `O01` to `O66` with no `O34` and no `O51`, which is 64, plus `O67` to `O72`. These become elements, or do not.
- **Twenty-five relationship sheets**, `R01` to `R25`, which become relationships.
- **Fifteen sheets that are queries**: `00 Index`, the thirteen `RPT` reports and `ZZ Checks`. A report is a view, and importing one as elements is how a repository acquires a second copy of its own content.
- **One extension sheet**, `EXT`, whose six declared types are specialisations rather than elements. The notation lets you specialise; it does not let you add.

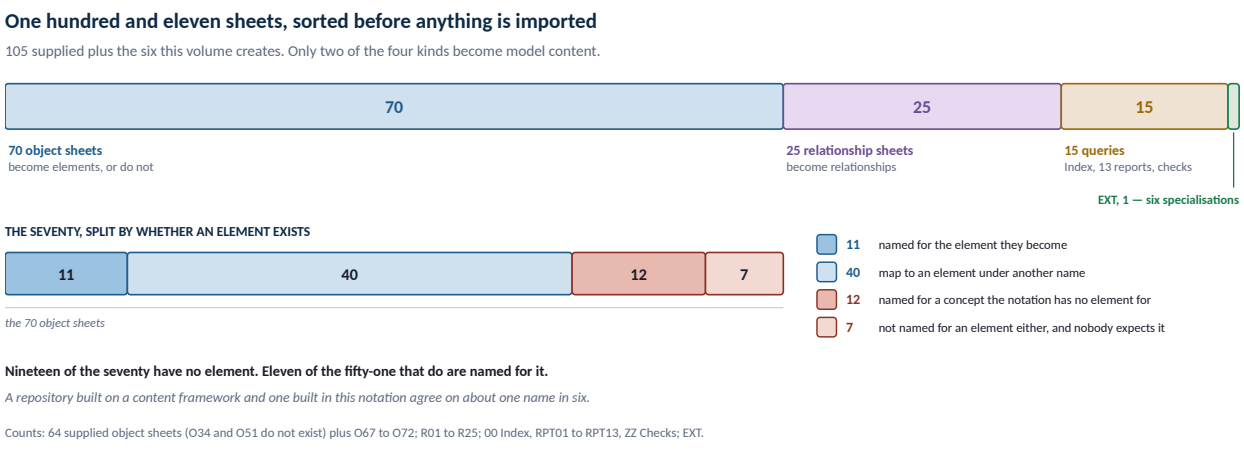


Figure 16.1 The 111 sheets by kind, and the seventy object sheets by whether the notation has an element for what each is named after. Nineteen do not: twelve where the absence is a surprise and seven where it is not.

The arithmetic, before the tables

Forty-two elements. The seventy object sheets reach **thirty-one** and never reach eleven. Eleven relationships. The twenty-five relationship sheets reach **nine** and never reach two. Both counts are checkable against the tables below, and both are more interesting than the mapping itself.

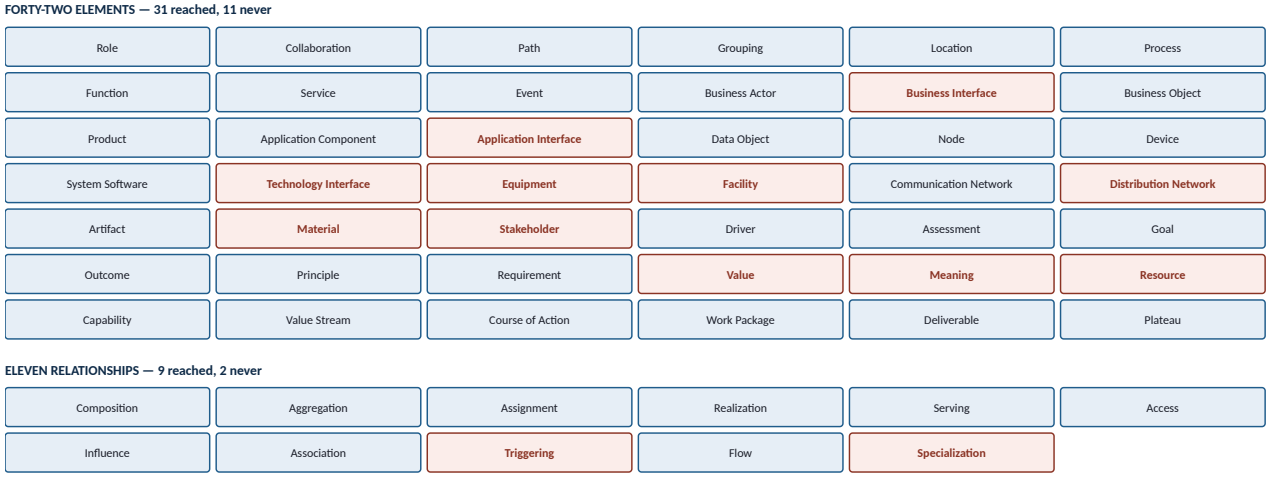
The eleven elements no sheet reaches, with what each would hold in a provider:

Element	What it would hold here	Why its absence shows
`Business Interface`	The enquiry counter, the referral fax line, the ward telephone	Every handover in the file is drawn application to application
`Application Interface`	The endpoint one system publishes and another calls	`R06` holds 23 interfaces and no sheet holds an interface
`Technology Interface`	The device port, the segment boundary	`O65`'s five enforcement points are the nearest thing
`Equipment`	The autoclave, the theatre table, the specimen analyser	`O68` is the sheet that needs it; `O27` files devices as compute
`Facility`	The hospital, the clinic, the laboratory, the mortuary	`O12`'s eight rows are places with headcounts, not buildings
`Material`	Medicines, consumables, blood products, specimens	`C04 Medicines Management` exists and nothing it moves does
`Distribution Network`	The specimen courier round, the pneumatic tube, the pharmacy run	`O50`'s ten links carry bandwidth and move nothing physical
`Stakeholder`	The regulator, the funder, the patient, the accounting officer	`O10` and `O11` are internal; nothing outside has a concern
`Value`	Why any of it is worth doing, in words	`O02`'s six goals are the nearest and they are statements of intent
`Meaning`	The agreed sense of a term	`O42`'s `MTD-01 Business definition` records that it should exist
`Resource`	Beds, theatre sessions, staffed hours, funded capacity	`O09`'s `Headcount` and `O12`'s `Scale` are the only resource facts

Three of those eleven matter more than the rest. **No interface element of any kind appears on any sheet**, in a repository whose own `R06` is twenty-three rows of interfaces and whose `MSR-08` measure is about managed interfaces. **No `Facility`**, in a provider. And **no `Material`**, in a file with a medicines capability, a pharmacy application and a diagnostics value stream.

What the repository reaches, and what it never does

Forty-two elements and eleven relationships, against all 111 sheets.



Nothing records that one thing triggers another, and nothing records that one thing is a kind of another.

No interface element of any kind appears on any sheet, in a file whose R06 holds twenty-three interfaces.

Figure 16.2 The forty-two elements and eleven relationships, with the ones no sheet in the repository reaches marked. The eleven unreached elements are what a provider has and this model does not; the two unreached relationships are sequence and specialisation.

The object sheets, mapped

Four columns: the sheet, its record count, the element to create, and what the import carries as a property or loses. Where the element column reads **none**, the section after the tables says what to write instead.

Motivation

Sheet	Rows	Element	What the import carries or loses
`001 Driver`	13	`Driver`	`Type`, `Origin`, `Pressure` as properties; `Owner` as an `Association` to a `Role`
`002 Goal`	6	`Goal`	`Status` as a property
`003 Objective`	14	**none** — use `Outcome`	`Horizon` is a relative timebase with no origin; carry it as a string, never as a date
`004 Measure`	16	**none**	There is no metric element and `Value` is not a number; chapter 20 is explicit. `070` is where a measure becomes an object
`005 Principle`	24	`Principle`	`Statement` into documentation; `Category` and `Status` as properties
`006 Requirement`	32	`Requirement`	`Priority` and `Status` as properties; `Driver` becomes the `Influence` on `R08`
`007 Constraint`	8	**none** — use `Requirement`	Carry a property marking it restrictive, or the eight rows become indistinguishable from `006`'s 32
`008 Assumption`	6	**none**	An `Assessment` marked unverified is the closest, and it changes the meaning

Business

Sheet	Rows	Element	What the import carries or loses
`009 Organization Unit`	12	`Business Actor`	`Head` as an `Assignment` to a `Role`; `Headcount` as a property. Six of twelve heads are not actors in `010`
`010 Actor`	18	`Business Actor`	`Type` and `Status` as properties
`011 Role`	18	`Role`	`Performed by` becomes the `Assignment` on `R21`
`012 Location`	8	`Location`	`Scale` holds a headcount, not a size. A building is a `Facility` and no sheet holds one
`013 Business Capability`	100	`Capability`	`Level` and `Parent` become the `Composition` on `R01`; `Current`, `Target`, `Criticality`, `Disposition` as properties
`014 Value Stream`	5	`Value Stream`	`Stages` holds a count, not stages. The stages themselves exist only on `016`, and twenty of thirty-six
`015`	10	`Service`, business layer	The sheet's own name is a pre-version-4 element name. Import by sheet number and name the element for what the row is
`016 Business Process`	20	`Process`	`Automation` as a property. No performer, no system and no trigger on the sheet
`017 Business Function`	10	`Function`	`Capability` becomes an `Association`; the file has no relationship sheet for it
`018 Product`	5	`Product`	The five rows are care settings rather than products; `Location` is arguably the honest element and item 4 settles it
`019`	10	**none** — `Business Object`	Ten vendor agreements. The counterparty is a `Business Actor`, the obligation a `Requirement`. Import by sheet number

Data

Sheet	Rows	Element	What the import carries or loses
`020 Data Entity`	14	`Business Object`, realised by a `Data Object` per holding system	`Classification`, `Owner`, `Master system` and `Lineage` as properties. The retention class attaches here
`021 Logical Data Component`	12	`Data Object`	`Entity` becomes a `Realization` back to the `Business Object`
`022 Physical Data Component`	12	`Artifact`	`Application` becomes a `Composition` from the `Application Component`
`036 Data Entity Relationship`	11	not an element — `Association`	`Cardinality` has no expression in the notation. All eleven values are lost on import
`037 Data Attribute`	60	**none**	Attributes are properties. Twenty-six carry a classification looser than their entity, and a property set is the first thing an export drops
`038 Data Set`	6	`Data Object`, or a `Grouping` of them	Four of the six are not healthcare data sets at all
`039 Data Migration`	6	`Work Package` plus a `Flow`	`Status` as a property; the two systems become the flow's ends
`040 Source To Target Mapping`	12	`Artifact` holding the mapping	`Rule` as a property. The mapping is not itself a relationship between elements
`041 Security Control`	8	**none**	`Requirement` for what it must do, realised by the `Node` named in `Enforced at`
`042 Metadata Element`	14	**none**	A property schema, not content. `MTD-01` is the nearest thing to a `Meaning` in the file
`043 Master Data Rule`	10	**none**	A `Requirement` naming the authoritative system. `Rule type` as a property
`044 Data Building Block`	10	`Grouping` over `Requirement`	`Acceptance criteria` into documentation

Application and technology

Sheet	Rows	Element	What the import carries or loses
`023 Application Component`	12	`Application Component`	`Disposition`, `Criticality`, `Health`, `In service`, `Users` as properties; `Owner` as an `Association`
`024 Information System Service`	12	`Service`, application layer	`Style` and `Published` as properties. All twelve descriptions are generated from the application name
`025 Technology Service`	12	`Service`, technology layer	`Category` cannot be used as a discriminator: storage is categorised Security and security operations Platform
`026`	7	`Node`, logical	Row 1 is truncated on this sheet. Name the element from neither the tab nor row 1
`027`	12	`Node`; `Device` where `Class` reads End user; `Communication Network` where it reads Network	Row 1 is truncated here too. `Support end`, `Position`, `Vendor` as properties
`028 Platform Service`	8	`Service`, technology layer	Three columns only; nothing to lose
`045 Application Building Block`	10	`Grouping` over `Requirement`	One per application, generated from the application name
`046 Technology Building Block`	7	`Grouping` over `Requirement`	Two columns headed `Ref`. The second carries `T01` to `T07`, which exist on no sheet; carry both readings
`048 Environment`	4	`Grouping`	`Personal data present` as a property, and it is wrong on three of four rows
`049 Resilience Tier`	12	**none**	`Requirement` for the recovery objective, `Assessment` for `Tested`. The `Tier` value carries no information
`050 Network Link`	10	`Path` between two `Location` elements	Its ends are locations, not nodes, so it is a `Path` and not a network. The only sheet reaching `Path`

Realization, governance and security

Sheet	Rows	Element	What the import carries or loses
`029 Gap`	2	**none**	A gap is the difference between two `Plateau` elements, not a thing. Model it as a `Requirement` with an `Assessment`
`030 Work Package`	12	`Work Package`	`Cost`, `Start`, `End`, `Approval`, `Delivery` as properties. Every cost is in US dollars with no rate
`031 Deliverable`	20	`Deliverable`	`Phase` and `Status` as properties
`032 Standard`	16	**none**	`Requirement` for what it demands, `Business Object` for the document plus its version, `Assessment` for `Conformance`
`033 Architecture Contract`	8	**none** — `Business Object`	`Compliance status` is `030`'s `Delivery` column recoded, in both directions, on every row
`035 Business Building Block`	10	`Grouping` over `Requirement`	`Acceptance criteria` into documentation
`047 Reference Model Entry`	11	**none**	A `Grouping`, or an `Artifact` holding the reference. Seven of eleven duplicate `026` name for name
`052 Implementation Factor`	8	`Assessment`	`Category` holds five different kinds of thing, three of which have no element of their own
`053 Transition Architecture`	3	`Plateau`	`Exit criteria` holds a count, not criteria
`054 Architecture Alternative`	5	`Course of Action`	`Cost` and `Recommendation` as properties
`055 Building Block`	37	`Grouping` over `Requirement`	The parent of `035`, `044`, `045` and `046`, which are its four domain slices
`056 Project`	12	**none** — a second `Work Package`	A project composes work packages, which is exactly what `R14` asserts
`057 Governance Body`	5	`Collaboration` of `Business Actor`	`Cadence`, `Membership`, `Mandate` as properties. The only sheet that reaches `Collaboration`
`058 Governance Gate`	4	`Event`, with a `Requirement` for what it tests	The only sheet that reaches `Event`, in a file with twenty processes
`059 Compliance Assessment`	3	`Assessment`	`Verdict` and `Evidence` as properties. Three rows against the 148 a four-gate regime over 37 blocks needs
`060 Governance Decision`	7	**none**	A `Business Object` for the minute. `Course of Action` is a chosen option, not a decision record
`061 Governance Change Request`	5	**none**	A `Business Object` for the request, a `Requirement` for what it asks
`062 Delivery Risk`	8	**none**	`Assessment` for the finding, `Driver` for what makes it matter. `Likelihood` and `Impact` as properties
`063 Control Objective`	7	**none**	`Requirement` for the obligation, `Assessment` for `Evidence`, `Driver` for the instrument behind `Source`
`064 Security Policy`	4	**none**	A `Grouping` of `Requirement` elements, or a `Principle` where the row is genuinely a principle
`065 Enforcement Point`	5	`Node`	`Acts on` and `Position` as properties. A `Technology Interface` is the better element where the point is a boundary
`066 Architecture Increment`	6	`Plateau`	`Lands` holds a relative month against no origin. The second plateau sheet in the file

The six sheets this volume creates

Sheet	Element	What the import carries
`O67 Consent`	`Business Object`, one per consent type	`Scope`, `Granted`, `Expires`, `Withdrawn` as properties. Chapter 20 draws it as three objects with three owners
`O68 Medical Device`	`Node` composed of a `Device` and a `System Software`	`Risk class` is assigned by the licence holder under the regulator's guideline [1], not by the modeller. `Equipment` is the right element where there is no software
`O69 Code System`	`Business Object`	`Edition or version`, `Effective from`, `Effective to` as properties; `Where bound` as an `Association` to the interface
`O70 Measure Definition`	`Business Object`	The sheet exists because the notation has no metric element. Every quantity in the model points here
`O71 Disclosure Event`	`Business Object`, accessed by the `Process` that disclosed	`Authority` as an `Association` to a `Driver` or to a `CNS-` consent
`O72 Retention Clock`	`Requirement`, with a `Business Object` for the schedule	`Instrument` becomes a `Driver`; the destruction authority sits under the National Archives instrument the records duty is made subject to [2]

Twelve sheets named after something the notation does not have

Twelve of the supplied sheets carry a title that an architect expects to be an element — because the framework names it, or because a pre-version-4 version of the notation had it — and that version 4 does not carry. The absence is a surprise in all twelve, which is why they are listed together.

Sheet	What to create instead	The thing you lose if you do not
`O03 Objective`	`Outcome`, with the horizon as a property	The dated commitment; an `Outcome` has no date either
`O04 Measure`	A `Business Object` per measure definition, on `O70`	Everything: no quantity is expressible as an element
`O07 Constraint`	`Requirement` with a restrictive statement	The distinction between a demand and a prohibition
`O08 Assumption`	`Assessment` marked unverified	That nobody tested it
`O19`	`Business Object` plus a `Business Actor` and a `Requirement`	The counterparty and the obligation, which are the parts that bind
`O29 Gap`	`Requirement` plus an `Assessment`, between two `Plateau` elements	That a gap is a difference and not a thing
`O32 Standard`	`Requirement`, `Business Object` for the document and version, `Assessment` for conformance	The version, which is what a standard actually is
`O33 Architecture Contract`	`Business Object` plus the `Requirement` set it imposes	The obligations, which is the whole content
`O41 Security Control`	`Requirement` realised by the `Node` that enforces it	Where it is enforced, which is `O41`'s own last column
`O56 Project`	A second `Work Package`, composing the first	Nothing, as long as the composition is drawn
`O62 Delivery Risk`	`Assessment` plus a `Driver`	Likelihood and impact, which become properties nobody prints
`O63 Control Objective`	`Requirement`, `Assessment`, and a `Driver` for the instrument	The instrument, and therefore the authority

Four of those twelve are the same shape, and it is worth seeing once: an obligation arrives from outside, a requirement states what must be true, something realises it, and an assessment records whether it does. `O63`, `O41`, `O64` and `O65` are four sheets of one chain with no element of their own between them, and the two relationship sheets that join them do it in row order. `R16` pairs `CO-01` to `CO-07` with `SEC-01` to `SEC-07` one for one, so `SEC-02 Design review` is recorded as the control meeting `CO-02 Clinical records retention`. `R17` pairs `POL-01` to `POL-04` with `EP-01` to `EP-04` the same way — against `O64`'s own text, which says `POL-04` is enforced at the design gate, which is `EP-01`. `R17` has four rows for five enforcement points, and `EP-05` appears on none of them.

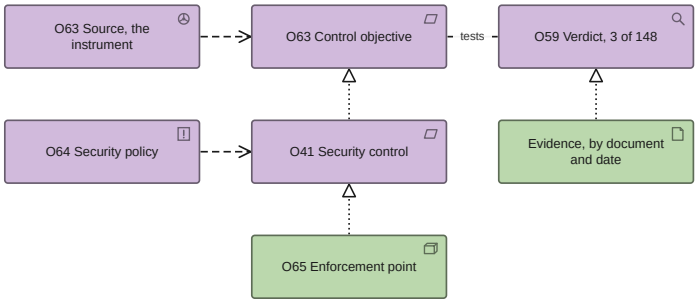


Figure 16.3 Four sheets with no element of their own, modelled as one chain. The obligation behind O63's Source becomes a Driver, the control objective and the security control become two Requirement elements, the enforcement point realises the control and the compliance assessment tests the objective. The file joins them only in row order, on R16 and R17.

Seven more that are not elements either

Seven further sheets have no element behind their name, and nobody expects them to, because the thing they are named after is a property, a rule, a category or a record. They are listed here so a reader counting absences gets the same number twice running: `O37 Data Attribute` and `O42 Metadata Element` are properties; `O43 Master Data Rule` is a `Requirement` about mastering; `O47 Reference Model Entry` is a `Grouping` or an `Artifact`; `O49 Resilience Tier` is a `Requirement` plus an `Assessment` with a label in front of it; `O60 Governance Decision` and `O61 Governance Change Request` are `Business Object` records of what a body did.

Add the twelve and the seven and nineteen of the seventy sheets have no element. Fifty-one do. Of those fifty-one, exactly eleven are named for the element they become — `O01`, `O02`, `O05`, `O06`, `O11`, `O12`, `O14`, `O18`, `O23`, `O30` and `O31` — and the other forty are named for something else. Which is the real lesson of the mapping table: a repository built on a content framework and a repository built in this notation agree on about one name in six.

Four sheets, two elements

Two pairs collapse, and both collapses are worth drawing because they change what the import looks like rather than just what it is called.

`O30 Work Package` and `O56 Project` both become `Work Package`. There is no project element, and there does not need to be: `R14` already says a project delivers work packages, which is a `Composition` between two `Work Package` elements. The twelve projects compose the twelve work packages one to one, which is the tell that the sheet was generated rather than planned.

`O53 Transition Architecture` and `O66 Architecture Increment` both become `Plateau`. Three rows and six rows, on two incompatible time axes — `O53` has no dates at all and `O66` dates its six in relative months against no origin anywhere in the file, while `O30` and `O56` carry absolute dates and `O03` carries a third relative scale. Importing both plateau sheets without reconciling them produces a roadmap view with nine states on it and no way to order them.

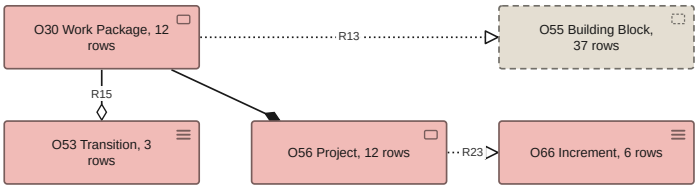


Figure 16.4 Four sheets, two elements. There is no project element, so O56 becomes a second Work Package composing the first, which is what R14 already asserts. O53 and O66 are both Plateau, on two time axes the file never reconciles.

The relationship sheets, mapped

Eleven relationships. Nine are reachable from the twenty-five sheets; `Triggering` and `Specialization` are not reached by any of them. That is the sharpest single thing in this item. **Nothing in the repository records that one thing triggers another** — twenty business processes, five value streams, twenty-four declared stages, and no sequence anywhere. And **nothing records that one thing is a kind of another**, in a file whose `EXT` sheet declares six industry extension types that are precisely specialisations and records them as six rows of text.

Sheet	Instances	Relationship	Reads in the right order
R01 capability decomposition	90	Composition , parent to child	yes
R02 value stream to capability	15	Serving , capability to stream	no
R03 capability to one of O15 's rows	10	Realization , service to capability	no
R04 one of O15 's rows to an application	10	Serving , application to service	no
R05 application manages data entity	28	Access , with Role as the access type	yes
R06 application interfaces with application	23	Flow for file, batch and event; Serving for API	yes
R07 application deployed on technology	12	Assignment , node to component	no
R08 requirement derives from driver	32	Influence , driver to requirement	no
R09 requirement satisfied by building block	32	Realization , block to requirement	no
R10 objective measured by measure	14	Association only — the far end has no element	undirected
R11 goal realised by objective	14	Realization , outcome to goal	no
R12 gap closed by work package	2	Association only — the near end has no element	no
R13 work package delivers building block	37	Realization , package to block	yes
R14 project delivers work package	12	Composition , between two Work Package elements	yes
R15 transition contains work package	7	Aggregation , plateau to package	yes
R16 control objective met by control	7	Realization between two Requirement elements — neither end has an element of its own	no
R17 policy enforced at enforcement point	4	Realization , node to requirement — the near end has no element	no
R18 standard applies to application	16	Realization , application to requirement — the near end has no element	no
R19 data entity relates to data entity	11	Association ; cardinality lost	undirected
R20 application located at location	12	Aggregation , location to application	no
R21 role performed by actor	18	Assignment , actor to role	no
R22 org unit owns capability	10	Assignment , actor to capability	yes
R23 increment delivered by project	6	Realization , package to plateau	no
R24 compliance assessment tests building block	3	Association — an assessment has no structural relationship to a grouping	undirected
R25 application supports product	5	Serving , application to product	no

Three things fall out of that table and all three go in the import note.

Six sheets name an end that has no element. R10 's measure, R12 's gap, R14 's project, both of R16 's ends, R17 's policy and R18 's standard. A relationship whose end has to be invented on import is not a relationship the repository recorded; it is one the importer wrote.

Four sheets reduce to Association, which asserts that two things are connected and nothing else. R10 , R19 , R24 and R12 . R19 is the only one of the four where Association is the right answer rather than the residue.

Fifteen of the twenty-two directed sheets name their ends in the reverse order. Only R01 , R05 , R06 , R13 , R14 , R15 and R22 read source first. That is not an error in the file — a sheet titled "x satisfied by y" is readable English — but an importer that takes column 1 as the source and column 3 as the target will build fifteen sheets backwards, and every one of them will still resolve.

One thing to carry out of this table and into item 1's test rather than into the import. Three sheets outside item 1's positional set behave exactly like it and deserve the same twenty-row sample: R16 and R17 , above, and R18 , which cycles A01 , A03 , A05 , A07 , A09 and A11 across all sixteen standards on the identical stride R25 uses — so the records management standard applies to the prescribing system, and six of the twelve applications are governed by no standard at all.

Form 16-A — The element map

Feeds no sheet in the repository; the modelling convention note in the engagement pack · **Rows for a first pass** 70, one per object sheet including the six created · **Signed by** the Chief Enterprise Architect (`ROL-08`) · **Ratified by** `GB-01`

No sheet holds the mapping. `00 Index` holds a domain per sheet and nothing about elements. Keep this with the repository mapping note from step 1.3 and version it from the first week.

Sheet	Rows	Element	Layer property	Properties carried	Lost
O49 Resilience Tier	12	none — Requirement + Assessment — example	technology	tier, objective, tested	nothing

Form 16-B — The relationship map

Feeds no sheet in the repository · **Rows for a first pass** 25, one per relationship sheet · **Signed by** the Chief Enterprise Architect (`ROL-08`) · **Ratified by** `GB-01`

The `Reversed` column is the one that earns the form. Fifteen of the twenty-five read in the opposite order to the relationship that carries them, and an importer that misses it builds a model that resolves perfectly and points the wrong way.

Sheet	Instances	Relationship	Source end	Reversed	Invented end
R18	16	Realization — example	application	yes	the standard, which has no element

The truncation, head-on, because it decides what you name things

Twenty-three tab names are exactly 31 characters, Excel's limit. One of them — `RPT06 Requirements Traceability` — is 31 by coincidence and lost nothing, so **twenty-two tabs are genuinely truncated**: `O26` , `O27` , and twenty of the twenty-five relationship sheets.

The truncation propagated into the sheet's own row-1 title on **two** of the twenty-two, and both are object sheets: `O26` row 1 is missing its final letter and `O27` row 1 is missing its final two. `00 Index` rows 40 and 41 reproduce both. **Every one of the twenty truncated relationship sheets carries a full row-1 title.** Their row 1 reads the whole sentence the tab name could not hold, which is why the import rule below works at all.

One tab, `R04` , was cut after a word boundary, so its thirty-first character is a space. A formula or a script written against the name without the trailing space will not find the sheet, and nothing on screen will show you why. Match on the `R04` prefix.

So the import rule is three lines. **Read the element name from row 1 of the sheet, not from the tab** — for twenty of the twenty-five relationship sheets that is the difference between a readable relationship and a fragment. **For `O26` and `O27` , read it from neither**, because both are wrong; name those two elements for what their rows are. **And match every sheet on its prefix**, never on its full name.

16.1 — Decide the layer property before you import anything

Sheet `00 Index`, as the list of sources · **Input** the mapping tables above · **Output** a layer property on every unprefixed element type, and the validation rule that enforces it · **Signed by** the Chief Enterprise Architect (`ROL-08`) · **Time** two hours

1. `Service`, `Process`, `Function`, `Event`, `Role` and `Collaboration` carry no layer in their names. The layer exists only as a position in a view and a fill colour, and an export loses both.
2. Define a layer property, set it on creation, and validate it. Four sheets import to `Service` — `015`, `024`, `025` and `028` — and without the property they are one undifferentiated set of 42.
3. Record the decision in the modelling convention note, because the second architect will not infer it.
4. In a client file: ask the tool whether it holds the layer as a first-class attribute. Several do not, and that answer decides how much of this is manual.

16.2 — Map every object sheet to one element, and record the nineteen with none

Sheet all 70 object sheets · **Input** Form 16-A · **Output** 70 rows, each with an element or the word none and what replaces it · **Signed by** the Chief Enterprise Architect (`ROL-08`) · **Time** two days

1. Work in sheet order and do not skip. The sheets with no element are not clustered; they run from `003` to `063`.
2. For each, write the element, the layer, the properties that carry across and what is lost. The lost column is the one a reviewer reads.
3. Where the element is `none`, write the substitution as elements and relationships, not as prose. `063` becomes four things, and a reader who sees "model it as a requirement" will build one.
4. Count your own nineteen. A different number means your rule differs from this item's, and the note should say what your rule is.
5. In a client file: a tool that ships sixty object types is shipping a content framework, not the notation, and the same nineteen will be in it.

16.3 — Map the twenty-five relationship sheets, and get the direction right

Sheet `R01` to `R25` · **Input** Form 16-B · **Output** 25 rows with relationship, source end and a reversed flag · **Signed by** the Chief Enterprise Architect (`ROL-08`) · **Time** a day

1. For each sheet, read row 1 — not the tab — and write the relationship it implies.
2. Write which column is the source *in the notation*, which is not always column 1. It is column 3 on fifteen of the twenty-two directed sheets.
3. Flag the six sheets with an end that has no element, and name what you will create for it. That element is yours, not the file's, and the note must say so.
4. Flag the four that reduce to `Association` and say for each whether `Association` is the answer or the residue.
5. Record that `Triggering` and `Specialization` are reached by nothing, and what that means: no sequence anywhere, and six declared extension types that are specialisations recorded as text.

16.4 — Rename on import, or do not import

Sheet `015`, `019`, `024`, `026`, `027`, `045`, `046`, `047` · **Input** the two maps · **Output** a rename list, or a decision not to import a sheet · **Signed by** the Chief Enterprise Architect (`ROL-08`) · **Time** half a day

1. Import the element name from row 1 of the sheet. For `026` and `027`, from neither row 1 nor the tab.
2. Name an element for what it is, not for the sheet it came from. A row whose description reads "externally consumable service realising clinical care delivery" is a capability name dropped into a sentence; promoting it preserves the emptiness and gives it a reference number.
3. Where two sheets hold the same rows under different numbers — `026`, `046` and `047` share seven names — import once and relate, or you will have three copies in three layers.
4. Where a sheet's content is generated from another sheet's names, mark it and do not import it until somebody rewrites it.

16.5 — Write the one page that says what the model does not carry

Sheet no sheet · **Input** steps 16.1 to 16.4 · **Output** one page at the front of every architecture document · **Signed by** the Chief Enterprise Architect (`ROL-08`) · **Time** hours

1. List what is held as a property rather than drawn: classification, retention period, code system edition, device risk class, recovery objective, effective dates.
2. List what has no expression in the notation at all — time on an element, cardinality, state, quantity, the type and instance distinction — and name where each is maintained instead.
3. List the eleven elements nothing in this repository reaches, and say which of them the organisation actually has. A provider with no `Facility` and no `Material` is modelling a software estate.

4. Name the nineteen sheets with no element and the six relationship ends you invented, and date it. Re-issue it whenever the maps change; it is the cheapest credibility a modelling deliverable buys.

16.6 — Check the views deck against the repository it claims to draw

Sheet the repository; the deck is the input · **Input** the 27-slide views deck · **Output** a caption-by-caption note · **Signed by** the Chief Enterprise Architect (`ROL-08`) · **Time** two hours

1. The deck draws one view per viewpoint, twenty-five of them. Chapter 20 says draw six, each with the question it settles; a catalogue is read once. Do not expand the deck, reduce it.
2. Read every caption against the sheet it claims to draw. Five of the twenty-five name an organisation, a value stream, a process or a count that the repository does not hold — one says twelve level-one capabilities where `013` holds ten.
3. Where a caption names an organisation you have not heard of, that is a copy from another industry's set that escaped review. Report it; do not edit the supplied file silently.
4. Record which of the six views chapter 20 asks for the deck already contains, and build the rest.

FIVE QUESTIONS TO SETTLE BEFORE THE FIRST ELEMENT IS CREATED

Which tool are we importing into, and does it hold the layer as a first-class attribute?

Who signs the nineteen substitutions, and where is the list kept?

Which of the six invented relationship ends are we prepared to create, and which do we leave out?

What do we do about the twenty-three tab names at exactly 31 characters, including the one ending in a space?

Which six views are we drawing, and what question does each settle?

What this mapping is not

It is not a licence to import the file. Every finding in this volume survives the mapping intact: the positional relationship sheets import as perfectly resolving relationships that mean nothing, the twenty-six loose attribute classifications import as properties nobody prints, and `033`'s recoded compliance column imports as a property on eight `Business Object` elements. The notation has no opinion about any of it. Chapter 20 puts it in one sentence worth repeating: a model can satisfy every derivation rule the specification carries and still pair two objects that have nothing to do with each other, because the error is semantic and the tool only tests structure.

What the mapping gives you is narrower. It tells you, before anybody opens a tool, which seventy sheets become elements, which nineteen of them need something invented, which twenty-five relationship sheets become nine relationships, which fifteen point backwards, and which eleven elements the organisation will have to add itself. That is a day's work and it saves a rebuild.

THE CHAPTER IN TEN LINES

- Chapter 20 owns the patterns and the element names. This item owns the mapping of 111 sheets, and cites chapter 20 rather than repeating it.
- Seventy object sheets become elements or do not; twenty-five relationship sheets become relationships; fifteen are queries and become nothing; EXT declares six specialisations.
- The seventy sheets reach thirty-one of the forty-two elements and never reach eleven.
- The eleven never reached are the three interface elements, Equipment, Facility, Material, Distribution Network, Stakeholder, Value, Meaning and Resource.
- No interface element of any kind appears on any sheet, in a repository whose own R06 holds twenty-three interfaces and whose MSR-08 measure is about managed ones.
- Twelve sheets are named after a concept the notation has no element for: O03, O04, O07, O08, O19, O29, O32, O33, O41, O56, O62 and O63.
- Seven more have no element either and nobody expects them to: O37, O42, O43, O47, O49, O60 and O61. Nineteen of seventy in all.
- Of the fifty-one that do map, exactly eleven are named for the element they become.
- Four sheets collapse into two elements: O30 and O56 both become Work Package, O53 and O66 both become Plateau on two unreconciled time axes.
- The twenty-five relationship sheets reach nine of the eleven relationships. Triggering and Specialization are reached by nothing.
- Six relationship sheets name an end that has no element, so the importer invents it: R10, R12, R14, R16, R17 and R18.
- Fifteen of the twenty-two directed relationship sheets name their ends in the reverse order, and every one of them still resolves.
- Twenty-three tabs are exactly 31 characters and twenty-two are genuinely truncated, but only O26 and O27 carry the truncation into row 1. Every truncated relationship sheet has a full row-1 title.
- Read the element name from row 1, not from the tab; for O26 and O27 read it from neither; and match every sheet on its prefix, because one tab name ends in a space.

SOURCES FOR THIS CHAPTER

- **1.** SAHPRA, SAHPGL-MD-04 version 5, Guideline for the Classification of Medical Devices and IVDs, effective 27 February 2025 (the risk class is assigned by the licence holder; described, not reproduced) https://www.sahpra.org.za/wp-content/uploads/2025/03/SAHPGL-MD-04_v5-Guideline-for-Classification-of-MD-and-IVDs.pdf
- **2.** National Health Act 61 of 2003, consolidated text (s13 record-keeping duty, expressly subject to the National Archives and Record Service of South Africa Act 43 of 1996 and the Promotion of Access to Information Act 2 of 2000) https://www.saflii.org/za/legis/consol_act/nha2003147.pdf

APPENDIX E

Where to look in volume 1

Twenty-five items, 259 pages, and one question you will actually ask: which chapter settles this, now, while the workshop is still in the room.

Volume 1 is 259 pages: twenty-two chapters and three appendices. You will not read it again. You will, several times a week, hit a question in the middle of a phase and need the one chapter that settles it. This item is the index for that, built backwards: the question first, the chapter second, and the workbook item that put the question in front of you third.

It is organised by the reader's question rather than by chapter order, because chapter order is teaching order and you are not being taught. Two rules make it usable. Cite by chapter number and move on; a workbook step that restates volume 1 has stopped being a step. And when the chapter does not answer the question, the answer is a local decision with an owner, a version and an effective date, which is what `O69`, `O70` and `O72` exist to hold.

Definitions: chapter 4 or appendix B

Volume 1 holds the vocabulary in two places and they do different jobs. Chapter 4 settles sixty-two terms and those definitions are binding on this volume. Appendix B is a glossary of one hundred and eighty-two entries: the sixty-two carried unchanged in substance, plus everything chapter 4 did not reach.

Use chapter 4 when you are about to write a definition into the repository. The `Definition` column on `O20`, the boundary rule on a `O70` row, the `Record class` on `O72`, the scope text on an `O67` row: all of those are definitions you are publishing, and publishing one that contradicts chapter 4 means two parts of the same engagement disagree. Chapter 4's rules bite here. Admission is an event and never a stay, so a column that holds a duration is not called admission. Claim is never bare, so an `O20` row reads *funder claim* or *medico-legal claim*. A member pays a contribution. Spell and consultant episode are English definitions adopted for want of South African ones, so a deliverable that presents them as South African is wrong on its face.

Use appendix B when you have met a word in somebody else's document and need to know what it could mean. It prints both senses where a term has two legitimate ones, names the chapter that settles this volume's usage, and for about a dozen entries states that the research could not establish an edition, a version or a status and names what would settle it. That last group is the useful part in a workshop: it tells you the argument in the room is not resolvable from the literature.

Appendix A's map one is the third place, and it answers a different question: which repository sheet corresponds to which ArchiMate 4 element, including the sheets named after elements the notation does not contain. Appendix C answers whether you can obtain a standard at all, at what price, and which eleven items nobody could retrieve.

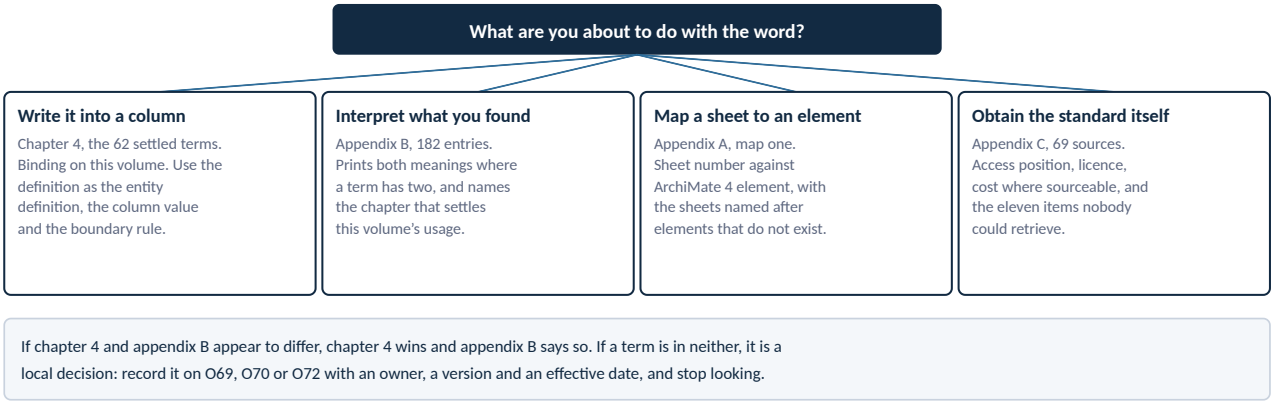


Figure 17.1 The four reference places in volume 1, and the question each one answers. Chapter 4 is binding; appendix B interprets; appendix A maps; appendix C tells you whether you can obtain the thing at all.

Before you start, and the Preliminary Phase

The question in front of you	Volume 1	Raised in
Which of my instincts from another industry will mislead me here	1	item 1
Why does this repository resolve perfectly and say almost nothing	21	items 1, 10, 12
Two official figures for the same thing. Which is right	2	items 1, 3, 20
What does this word mean before I type it into a column	4, then B	every item
Who sets policy, who holds the budget, who employs the staff	3	items 2, 3
Which body may decide this, and who is allowed to sign it	5	items 2, 10
Is there a national specification for this, or do I have to make one	12	items 2, 19
Can I get the standard, and what does it cost	C	items 2, 19
Where does the retention period come from	18, then 13	items 2, 11
Who may authorise destruction of a record	18	items 2, 10, 11
Why is the new funding statute not the reason for the work	7	items 3, 8, 12

Chapter 12 is the one to read before the Preliminary workshop rather than during it. Its subject is the nine decisions nobody has made nationally, and the whole of item 2 is the apparatus for making them locally: two fixed points, nine absences, and a register with an owner and an effective date against each answer.

Phase A

The question in front of you	Volume 1	Raised in
Where is the business case, in numbers an accounting officer accepts	13, then 1	item 3
Which measure, on whose basis, as at what date	2	items 3, 5, 20
What does this organisation actually look like, against its own chart	3, 5	items 3, 4
Which three figures will the executive argue about	2, 8	items 3, 5, 15
What does a defensible first ninety days contain	22	items 3, 15
What may I refuse to decide, and how do I say so	22	items 3, 12

Chapter 13 opens on the arithmetic rather than on the record, and that order is deliberate: the medico-legal exposure is the only part of a records programme that is already published, already audited and already in the accounting officer's own report. Item 3's exposure worksheet is that chapter turned into five measured episodes and four exposure lines.

Phase B

The question in front of you	Volume 1	Raised in
What are the five journeys called, and what are their stages	9	items 4, 15
Who performs this stage, and under what authority	5	items 4, 10
Where does a referral actually fail	9, 3	items 4, 6
What cadence does this run on, and which clock governs it	8	items 4, 11
Why can a facility master not derive the category from the bed count	3	items 4, 13
What leaves the organisation, to whom, on what clock	8, A map four	items 4, 11
Why is there no single owner for this entity	5, A map three	items 4, 5, 13

Chapter 9 names stages the repository only numbers, and the counts do not agree: chapter 9's own stage names run to thirty-six, `014` declares twenty-four and `016` holds twenty. Item 4 carries the arithmetic. Read chapter 9's five "where it fails" sections before the workshop; they are the handoff column of item 4's stage form, already filled in for a generic provider.

Phase C, the data half

The question in front of you	Volume 1	Raised in
What are the entities, and which ones are missing	10	items 5, 13
What identifies a person, and what must never be the key	11, 10	item 5
How tight should this classification be	18, 13	items 5, 10
Is this merge reversible, and what makes it so	11	item 5
Which code system, which edition, effective from when	12	items 2, 5
What is a Measure Definition and why does nobody ask for one	13, 10, 2	items 5, 20
How many objects is a consent	10, 13, 20	item 5
What is a disclosure event for	13, 18	items 5, 11
How do I state a cardinality the notation cannot hold	20, 10	items 5, 16

Chapter 11 is the one chapter whose numbers you should quote verbatim rather than paraphrase, because South Africa has a published, quantified measurement of linkage failure and almost nobody uses it. Chapter 10's last three sections — the sixteen omitted entities, the two that exist only as a row, and the structural absence after them — are the argument for three of the six sheets this volume creates.

Phase C, the application half

The question in front of you	Volume 1	Raised in
What survives a change of vendor	14	items 6, 8
Is this boundary real, or an artefact of who sold what first	14	item 6
What is actually in the estate, system by system	15	items 6, 13
Why does every hospital have an integration engine	14, 16	item 6
What governs this seam, and what is the actual contract	16	items 6, 19
Do I integrate with the practice system or with the switch	15, 6	item 6
Which version of the exchange standard, and is that the right question	16	items 6, 19
Which six categories does a landscape review usually omit	15	items 6, 13

Chapter 16 is the longest chapter in volume 1 and the one most often read for the wrong reason. Its finding is not which standard to pick; it is that most seams are governed by two or three standards at different layers at once, plus a profile, plus a local interface specification, and only the last of those is the contract. Item 6's seam work and item 19's standards set both rest on that sentence.

Phase D

The question in front of you	Volume 1	Raised in
How long before a patient is harmed, and what is the manual substitute	19	items 7, 11
What does a read-only record have to be able to do	19, 1	item 7
Which shared services stop clinical work, and who owns them	1, 15, 19	items 7, 19
How do I model a regulated device without losing the regulation	20, 18	items 7, 16
What does the products regulator require of software	18	items 7, 19
How long does retrieval actually take, measured	13	items 7, 11
Which maturity instrument fits a provider, and which does not	19	items 7, 20

Chapter 19's seven incidents are the only part of volume 1 written as case notes, and they carry the planning assumption: weeks of degraded operation, not hours, and three of the six sourced attacks were on a supplier rather than on the organisation. Item 7 turns the chapter's read-only record from a principle into a Phase D deliverable with a measured retrieval time against it.

Phases E and F

The question in front of you	Volume 1	Raised in
How do I price this properly, and what is the instance set	17	items 8, 9
What should I not accept from a supplier at face value	17	items 8, 19
Why can this supplier's share claim not be checked	17	item 8
Which release windows are closed, and why	8	items 8, 9
Which annual boundaries cut the plan	8	items 9, 11
What does donor-funded running cost do to year four	17	items 8, 9
What does a decision order look like, against a teaching order	8, 22	items 8, 9, 15

Chapter 17's eight claims not to accept at face value are the shortest useful thing in volume 1 for Phase E, and chapter 8's two calendar sections are the shortest useful thing for Phase F. Neither needs re-deriving; both need copying into the plan as constraints with a source.

Phases G and H, and Requirements Management

The question in front of you	Volume 1	Raised in
What does the audit and accreditation cycle demand, and when	8, 5	items 10, 11
What distinguishes a lawfully destroyed record from a lost one	13, 18	items 10, 11
Which offence is the strongest argument for access auditing	13, 1	items 10, 19
What can the privacy regulator actually do	18	items 10, 19
Which law wins where two of them govern the same record	18	items 10, 12
What has to be re-measured, how often, and by whom	19, 2	items 11, 20
Where does this requirement come from, in an instrument in force	18, 12, 7	items 12, 19
What has ArchiMate 4 no element for	20	items 12, 16
Which sheet corresponds to which element	A map one	items 13, 16
Which views, in what order, and why not twenty-five	20	item 16
What is wrong with the supplied repository, and in what order	21	items 1, 13, 14, 20

The four chapters you will open more than once

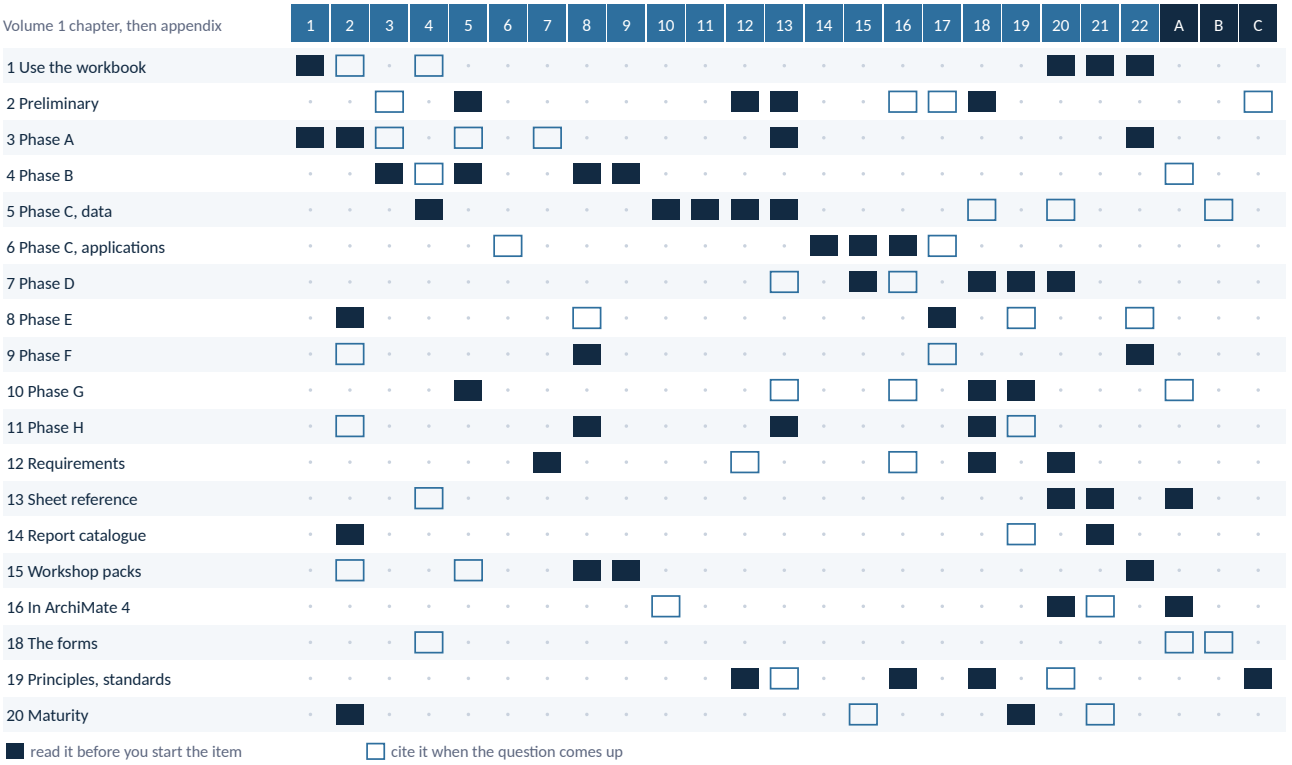


Figure 17.2 Which volume 1 chapters each workbook item reaches into. A filled square is a chapter to read before starting the item; an outlined square is one to cite when the question comes up. Item 17 is omitted because it reaches into all of them.

Chapter 2 is the measurement chapter and every phase hits it, because every phase produces a figure somebody will dispute. Its eight questions — the denominator, the as-at date, the publishing institution, thing against register, measured against derived, aggregate against patient-level, the second official value, and who benefits — are the interrogation to run on any number before it goes into a deliverable, including the numbers you generate yourself.

Chapter 4 is the vocabulary and it is binding. Chapter 20 is the notation, and the section that earns its keep is the one on what the notation will not do: no attributes, no cardinality, no validity period on an element, no type-instance distinction, no states and no swimlanes. Say that at the front of any model document and half the review comments disappear.

Chapter 21 is the repository read end to end, and it is the chapter to read before the client workshop rather than after it, because every defect this volume works on is described there once, in order, with the arithmetic shown. The chapter's closing sentence is the one to quote: a repository can pass every integrity check a tool can run and still describe an architecture that would harm patients.

Where volume 1 disagrees with itself, and what to do about it

Two places, both of which you will meet.

The outward-flow register appears twice at two different lengths. Chapter 8's table holds twenty-two rows and says so; appendix A's map four holds twenty-one and says "at minimum twenty-one". They are not the same twenty-one. Eighteen flows appear in both. Chapter 8 carries three patient-safety flows and a foreign safety issue that appendix A does not; appendix A carries a device-incident flow that chapter 8 does not, and splits the programme indicators into two and the perinatal and child death flow into two. Neither table is wrong about itself and neither number is the register. Build it from both — at chapter 8's grain that is twenty-three rows, at appendix A's finer grain twenty-five — declare the grain you counted at, and then publish the count. Item 4 starts the register and item 11 re-runs it.

Retention is the second. Chapter 13 and chapter 18 both set out the professional guideline, and the guideline disagrees with itself between its 2016 and 2022 editions: at least six years from dormancy in one, ideally indefinite and particularly if electronic in the other [1][2]. Volume 1 does not resolve it and nor does this volume, because it is not resolvable from the literature. What is resolvable is the destruction authority, which the records duty puts under the archives and access-to-information regime rather than under the guideline [3]. Carry both editions on the `O72` row, in the `Instrument` column, and name the authority separately.

What volume 1 does not answer, and where this volume does

Volume 1 teaches the industry. It does not tell you which tab to type a thing into, who signs it, how long the step takes, or what the paper looks like in the room. Six questions fall entirely to this volume.

- Which sheet, by its real tab name, and which columns, spelled as the file spells them. Item 13 is the sheet-by-sheet reference and item 14 the report catalogue.

- Who signs and who ratifies. A role signs; a body ratifies. Items 2 and 10 settle the bodies.
- What the paper looks like. Item 18 reprints all fifty-six forms from items 1 to 16 as one photocopiable set.
- Which sheets the repository does not have. Six, numbered `O67` to `O72`, defined in items 2, 5 and 7 and referenced everywhere else.
- A healthcare principle set, because the twenty-four generic principles on `O05` carry none of this industry's rules. Item 19.
- How to score a maturity assessment against sheets that are missing as well as sheets that are weak. Item 20.

Three things volume 1 tells you to re-check before you rely on it

Volume 1 shipped on 31 October 2026 and names three of its own facts as perishable. The Constitutional Court heard the two public-participation challenges to the funding statute from 5 to 7 May 2026 and judgment was still reserved at that date [4]. The national release centre for the clinical terminology South Africa joined in May 2026 was not yet operating, so a platform bought from mid-2026 must be able to take a terminology binding it does not yet have. And which edition of the retention guideline is in force could not be established.

Re-check all three before a deliverable leaves the building, and record the date you checked on the face of the document. A dated check is a defensible position; an undated assertion from a twelve-month-old book is not.

THE CHAPTER IN TEN LINES

- Volume 1 is 259 pages, twenty-two chapters and three appendices, and this item indexes it by the question rather than by the chapter.
- Cite volume 1 by chapter number. A workbook step that restates a chapter has stopped being a step.
- Chapter 4 settles sixty-two terms and they are binding: use it when you are writing a definition into a column.
- Appendix B holds one hundred and eighty-two entries for a word met in somebody else's document; appendix A map one maps sheet to element and appendix C gives the access position for a standard.
- Chapter 2, chapter 4, chapter 20 and chapter 21 are the four you will open more than once.
- Chapter 2's eight questions are the interrogation to run on any figure, including the figures you produce yourself.
- Chapter 12 is read before the Preliminary workshop, not during it: nine decisions nobody has made nationally.
- Chapter 13 opens on the medico-legal arithmetic because that is the part of the business case already published and already audited.
- Chapter 19 sets the planning assumption at weeks of degraded operation, and names supplier concentration as the dominant failure mode.
- Chapter 21 is read before the client workshop, not after: every defect this volume works on is described there once, in order.
- Volume 1 disagrees with itself on the outward-flow count, twenty-two in chapter 8 against twenty-one in appendix A, and neither is the union.
- The retention guideline disagrees with itself across two editions and neither volume resolves it; carry both editions on the O72 row and name the destruction authority separately.
- Three facts in volume 1 are perishable: the reserved judgment, the terminology release centre, and which retention edition is in force. Re-check and date the check.

SOURCES FOR THIS CHAPTER

- **1.** HPCSA Booklet 9, Guidelines on the Keeping of Patient Records, September 2016 edition (general records retained at least six years from the date they became dormant) https://www.hpcsa-blogs.co.za/wp-content/uploads/2022/08/Booklet-9-Guidelines-on-Patient-Records.pdf
- **2.** HPCSA Booklet 9, Keeping of Patient Records, revision marked September 2022 (general records retained ideally indefinitely, particularly if electronic; which edition is in force could not be established) https://www.hpcsa-blogs.co.za/wp-content/uploads/2022/11/Booklet-9-Keeping-of-Patient-Records_Review-Draft_-FINAL_Sept-2022.pdf
- **3.** National Health Act 61 of 2003, consolidated text (section 13 record-keeping duty, expressly subject to the National Archives and Record Service of South Africa Act 43 of 1996 and the Promotion of Access to Information Act 2 of 2000) https://www.saflii.org/za/legis/consol_act/nha2003147.pdf
- **4.** Constitutional Court of South Africa, hearing of the National Health Insurance Act public-participation challenges, 5 to 7 May 2026, judgment reserved https://www.concourt.org.za/index.php/judgements/cases-heard

APPENDIX F

The forms

Every form in the volume, reprinted once, in order, with the example row still on it and nothing renumbered.

Everything in items 1 to 16 that has to be written on, in one place. Photocopy from here; work from here in the room. Nothing in this item is new — if a form raises a question, the item that printed it first is the one that answers it, and the contents table below names it against every form.

There are 56 forms. Each is reproduced exactly as its own item prints it: the same label, the same header line naming what it feeds and how many rows a first pass needs and who signs and who ratifies, the same single worked example row, and the same blank ruled rows beneath it. Do not renumber anything. A form's label is how the signed paper is cited in a minute, in a gate pack and in a `072` row, and a label that moves between the item and this appendix destroys the audit trail.

How to work the set

Six rules, and the last two are the ones people break.

Strike out the example row first, in ink, before anybody writes. It is there so the person filling the form can see the grain of a row. It is also the commonest thing to find transcribed into a repository six months later, sitting among the real rows with nothing to mark it.

Write the date and your name at the top. No form has a field for either, because a field for them would need a seventh column. A form without them cannot be told from a form filled at a different engagement.

Six columns is the maximum. A 170mm text column takes six hand-written columns and no more. Where a sheet is wider, the form is split into a part one that identifies and a part two that assesses, keyed on `Ref`, and the two halves are printed adjacent here as they are in the source item. Five repository sheets are wider than six — `023` at ten, `013` at nine, `020` at eight, `030` at eight and `027` at seven — and four of the six new sheets are nine columns wide.

Never hand-write a computed value. A hand-written derived value is a hand-written error. `013`'s level is derivable from the dots in its `Ref`, `027`'s support position from its support end date and today, and the classification inheritance check from `020` against `037`. None of the three appears on a form.

No patient identifier goes on a form. The consent and disclosure forms take a class of subject, not a person. The device form takes a device type, not a serial number. A printed form left on a meeting-room table is a disclosure, and `071` has no row for it.

Transcribe the same day, and initial the paper when you do. A form typed up a week after the workshop has lost the hedging, and the hedging is usually the finding. The paper is the audit trail for the sheet it feeds: keep it for as long as the matching `072 Retention Clock` row says to keep the record it describes, and not one day less.

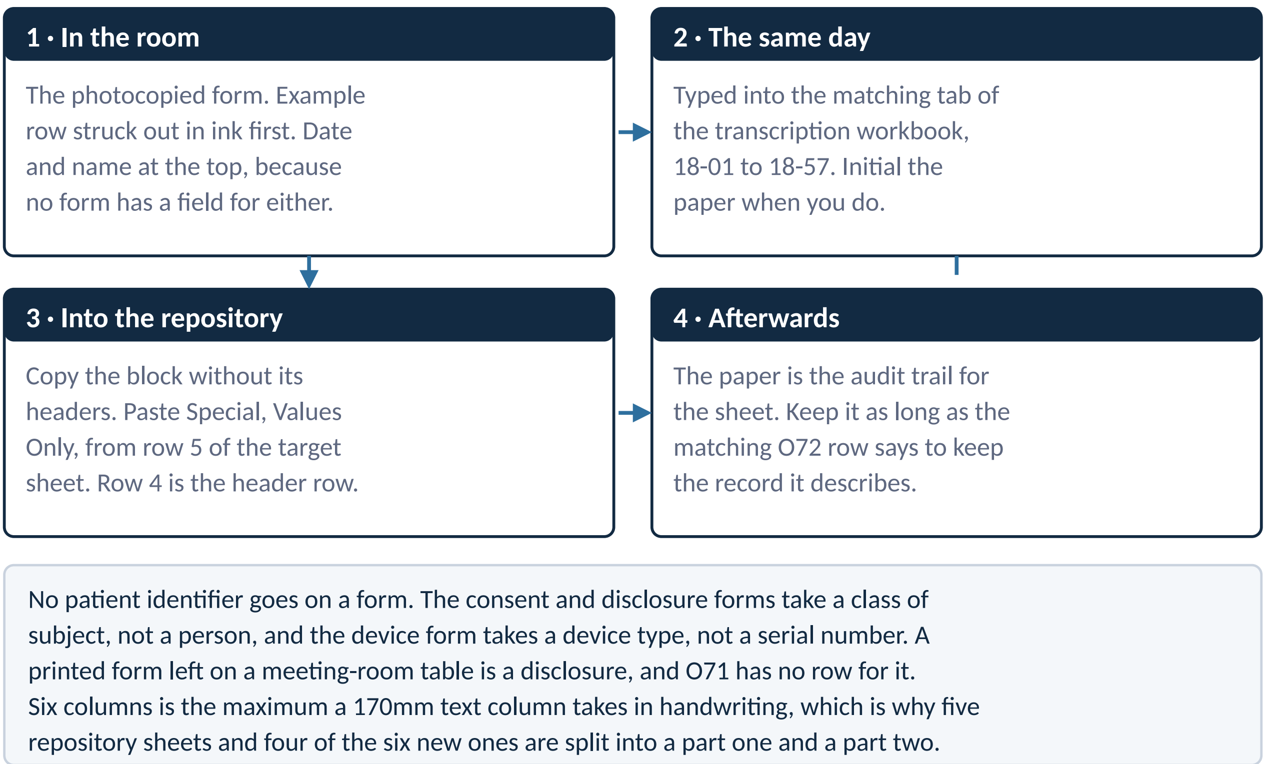


Figure 18.1 The path a form takes, and the four rules that travel with it. The paper is the audit trail for the sheet, which is why it is kept for as long as the record it describes.

Contents

Every form in the volume, the sheet it feeds, and the item that explains it. A form that feeds no sheet says so, and the reason is always the same: the repository has no column for the thing, and the form is the record instead.

Form, items 1 to 3	The sheet it feeds	Explained in
Form 1-A	R22, plus a reconciliation note no sheet holds	item 1
Form 1-B	00 Index row 2, RPT01 row 6, ZZ Checks row 6	item 1
Form 1-C	The engagement pack. No sheet holds it	item 1
Form 2-A	O57 Governance Body	item 2
Form 2-B	The sourcing register. No sheet holds a route	item 2
Form 2-C	O69 Code System, new	item 2
Form 2-D	O69 Code System, new	item 2
Form 2-E	O72 Retention Clock, new	item 2
Form 2-F	O72 Retention Clock, new	item 2
Form 3-A	O09 Organization Unit	item 3
Form 3-B	No sheet. O31 has no approver and no date column	item 3
Form 3-C	No sheet. O04 has no target column and O62 no financial column	item 3

Form, items 4 to 5	The sheet it feeds	Explained in
Form 4-A	O14 Value Stream column 4, and O16 Business Process	item 4
Form 4-B	O16 Business Process	item 4
Form 4-C	No sheet. Nothing joins a process to a role or an application	item 4
Form 5-A	O20 Data Entity	item 5
Form 5-B	O37 Data Attribute	item 5
Form 5-C	O67 Consent, new	item 5
Form 5-D	O67 Consent, new	item 5
Form 5-E	O71 Disclosure Event, new	item 5
Form 5-F	O71 Disclosure Event, new	item 5
Form 5-G	O70 Measure Definition, new	item 5
Form 5-H	O70 Measure Definition, new	item 5

Form, items 6 to 8	The sheet it feeds	Explained in
Form 6-A	O23 Application Component	item 6
Form 6-B	R06	item 6
Form 6-C	O24 Information System Service	item 6
Form 6-D	R07	item 6
Form 7-A	R07	item 7
Form 7-B	O68 Medical Device, new	item 7
Form 7-C	O68 Medical Device, new	item 7
Form 7-D	O48 Environment	item 7
Form 7-E	O49 Resilience Tier	item 7
Form 8-A	O29 Gap	item 8
Form 8-B	No sheet. The covering note to O54	item 8
Form 8-C	O52 Implementation Factor	item 8

Form, items 9 to 11	The sheet it feeds	Explained in
Form 9-A	O30 Work Package	item 9
Form 9-B	O53 Transition Architecture	item 9
Form 9-C	R23	item 9
Form 9-D	O31 Deliverable	item 9
Form 10-A	O59 Compliance Assessment, columns one to four	item 10
Form 10-B	O59 Compliance Assessment, columns five to seven	item 10
Form 10-C	O63 Control Objective	item 10
Form 10-D	O60 Governance Decision	item 10
Form 10-E	O61 Governance Change Request	item 10
Form 11-A	O61 Governance Change Request	item 11
Form 11-B	O61 Governance Change Request	item 11
Form 11-C	No sheet in the repository holds it	item 11

Form, items 12 to 16	The sheet it feeds	Explained in
Form 12-A	R09 RD, a new tab this volume's conventions create; the repository has R09 only	item 12
Form 12-B	O06 Requirement	item 12
Form 12-C	O06 Requirement	item 12
Form 12-D	O07 Constraint	item 12
Form 15-A	No sheet in the repository — the engagement pack carries it	item 15
Form 15-B	O60 Governance Decision	item 15
Form 15-C	No sheet in the repository holds it	item 15
Form 16-A	No sheet in the repository; the modelling convention note in the engagement pack	item 16
Form 16-B	No sheet in the repository	item 16

The transcription map

`Healthcare_EA_Transcription_Forms.xlsx` holds 58 sheets: a `00 Contents` sheet and 57 cards numbered `18-01` to `18-57`, one per card, each carrying a single block of six columns or fewer under a header that names the repository sheet it pastes into. Fill the paper form in the room, type it into the matching card, then copy the block without its headers and Paste Special, Values Only, into the target sheet from row 5. Row 1 is the sheet title, row 2 the organisation and record count, row 3 blank, row 4 the headers. Every one of the 105 sheets has that shape.

Five cards carry exactly one column the repository sheet does not have, and each says so on its own face: `18-07 Standard card` adds a conformance test to `O32`, `18-11 Role register` adds a registration basis to `O11`, `18-12 Location register` adds the care setting to `O12`, `18-18 Agreement register` adds a renewal-window date to `O19`, and `18-24 Physical data comp` adds the entity to `O22`. Add the column to the sheet and say in the covering note that you added it.

The map below is form to card, not form to sheet. Where a form splits a sheet the cards split it in the same place, so a part one and a part two land on the same card or on the two cards that pair. Where the column says None, there is no card and the form is typed into the sheet directly. The first column drops the word Form to keep the label on one line: every entry reads Form followed by what is printed there.

Form, items 1 to 3	Tab in the transcription workbook	What it leaves you to type straight in
1-A	None	No card. The re-derivation goes straight onto a new tab R22 RD, original headers kept and three columns added.
1-B	None	No card. Correct the three places in one edit; the counted figure replaces 843 everywhere it appears.
1-C	None	No card. It is the mapping note the client keeps, not a repository row.
2-A	18-48 Governance body	Nothing. Columns 5 and 6 of the form are working columns and stay on the paper.
2-B	None	No card. O23 has no sourcing column and this volume does not add one.
2-C	18-54 Code system card	The card carries these five and Where bound; Licence position and Owner are typed in Excel.
2-D	18-54 Code system card	Where bound is on the card. Licence position and Owner are typed straight into Excel.
2-E	18-57 Retention clock	The card carries these five and Destruction authority.
2-F	18-57 Retention clock	Destruction authority is on the card. System of record and Hold rule are typed straight into Excel.
3-A	18-09 Organisation unit reg	Nothing. Five columns, all O09's own.
3-B	None	No card. It is the decision record, kept with the vision and carried into Phase G.
3-C	None	No card. The accounting officer signs the paper; the figures go into the Phase A pack.

Form, items 4 to 5	Tab in the transcription workbook	What it leaves you to type straight in
4-A	18-17 Process register	Stage counts go onto 18-15 Value stream register; the stage rows themselves onto 18-17.
4-B	18-17 Process register	Nothing. Five columns, all O16's own.
4-C	None	No card. Four answers per stage, carried on paper into Phase C and Phase D.
5-A	18-19 Entity classify and 18-20 Entity owner master	Classification goes on 18-19; Owner, Master system and Lineage on 18-20. Class and Definition are already right.
5-B	18-21 Data attribute card	Nothing. The inheritance check is computed on the workshop capture workbook, never written by hand.
5-C	18-52 Consent card	The card carries these five and Granted and Expires.
5-D	18-52 Consent card	Granted and Expires are on the card. Withdrawn, Evidence and System of record are typed straight into Excel.
5-E	18-56 Disclosure event	The card carries these five and Authority.
5-F	18-56 Disclosure event	Authority is on the card. Consent reference and Logged in are typed straight into Excel.
5-G	18-55 Measure definition	The card carries Measure and these four rule columns.
5-H	18-55 Measure definition	Measure is on the card. Period, Version and Owner are typed straight into Excel.

Form, items 6 to 8	Tab in the transcription workbook	What it leaves you to type straight in
6-A	18-28 Application identify and 18-29 Application assess	Disposition goes on 18-28; Criticality and Health on 18-29. Owner is typed straight into Excel.
6-B	18-31 Interface identify and 18-32 Interface assess	Source and Target go on 18-31; Type, Managed and Logged on 18-32. The two queue columns are added in Excel.
6-C	18-30 Info system service	Nothing. All six columns are O24's own.
6-D	None	No card. R07 is a relationship sheet with no Ref column; the re-derivation goes onto R07 RD.
7-A	None	No card. Application name and Component name are lookups and are never keyed by hand.
7-B	18-53 Device card	The card carries Ref, Device, Model, Risk class, Software version and Calibration due.
7-C	18-53 Device card	Software version and Calibration due are on the card. Maintenance due, Network segment and Owner are typed straight into Excel.
7-D	18-36 Environment register	Ref, Environment and Description already exist and are not retyped. The fifth column stays off the sheet.
7-E	18-37 Resilience tier	Ref and Application already exist. Add a Date tested column rather than writing Tested in words. The sixth column goes in the covering note.
8-A	None	No card, and the contents sheet does not say there is none. Building block is a lookup from O55 after the row exists.
8-B	None	No card, deliberately. Four lines and a signature; the moment it is a formula somebody changes an input.
8-C	18-40 Implementation factor	Ref only. The other three are keyed from the signed form.

Form, items 9 to 11	Tab in the transcription workbook	What it leaves you to type straight in
9-A	18-42 Work package dates	Domains and Cost are on 18-41 Work package scope and are typed straight into Excel from the existing sheet.
9-B	18-43 Transition state	Nothing. The exit criteria cell takes the numbered list, not a count.
9-C	None	No card. R23 grows from six rows to seven, and the Index relationship total moves with it.
9-D	None	No card, and the contents sheet does not say there is none. Owner and Due are columns you add to O31.
10-A	None	No card. The contents sheet says O59 is filled from a card in another item; this is that card.
10-B	None	No card. Block name is a lookup from O55. Tested against and Assessed by are added to R24.
10-C	18-51 Control objective	Source is typed straight into Excel from the existing sheet. Instrument and Last assessed are columns you add.
10-D	None	No card. Required by and Decided on are columns you add to O60.
10-E	None	No card. Every row needs a value in Expires, including Not applicable.
11-A	None	No card for O61 in the transcription workbook. Type it into the sheet directly.
11-B	None	No card for O61 in the transcription workbook. Type it into the sheet directly.
11-C	None	No card. The form is the record; nothing in the repository holds it.

Form, items 12 to 16	Tab in the transcription workbook	What it leaves you to type straight in
12-A	None	No card for R09 in the transcription workbook. Type it into the sheet directly.
12-B	18-08 Requirement register	Six columns on the form; what the sheet holds beyond them is typed straight into Excel.
12-C	18-08 Requirement register	Six columns on the form; what the sheet holds beyond them is typed straight into Excel.
12-D	18-05 Constraint register	Six columns on the form; what the sheet holds beyond them is typed straight into Excel.
15-A	None	No card. The form is the record; nothing in the repository holds it.
15-B	None	No card for O60 in the transcription workbook. Type it into the sheet directly.
15-C	None	No card. The form is the record; nothing in the repository holds it.
16-A	None	No card. The form is the record; nothing in the repository holds it.
16-B	None	No card. The form is the record; nothing in the repository holds it.

Two of the gaps matter and the workbook's own contents sheet does not name them. `O29 Gap` and `O31 Deliverable` have no card, and they are the only two object sheets that are neither carded nor listed among the sheets the contents sheet says deliberately have none. Both carry a form in this volume. Type them into the sheet directly, and note on the covering sheet that you did, because a figure that reached a sheet without passing through a card has no paper behind it.

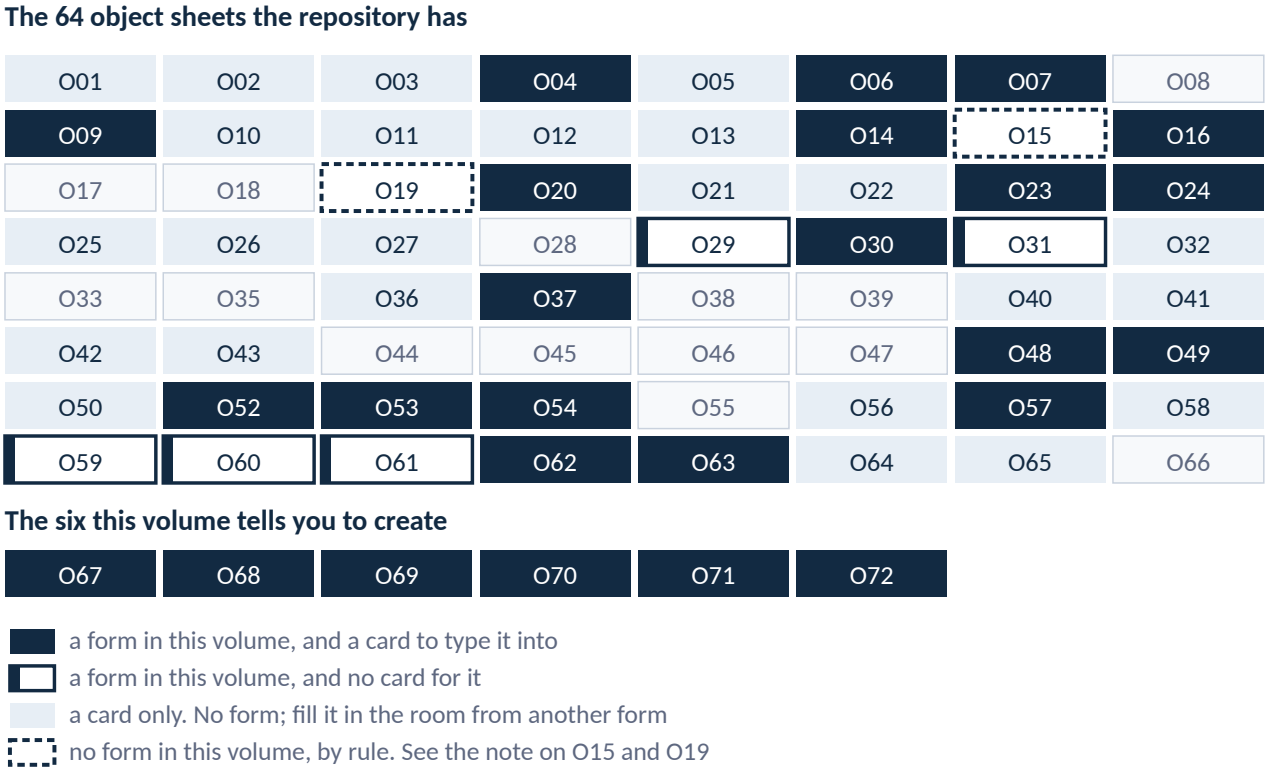


Figure 18.2 Form and card coverage across the repository. An outlined sheet has a form in this volume and no card in the transcription workbook; a dashed sheet carries no form by rule.

The forms that feed sheets the repository does not have

Six sheets do not exist in the supplied repository and this volume tells the reader to create them, after `O66` and before `EXT`, with six new `00 Index` rows in the Index's own `<n> records · <domain>` format. Every form that feeds one of the six carries that statement in bold on its face, and every one of them is split in two, because four of the six are nine columns wide and two are eight.

New sheet	Prefix	The forms that feed it	Defined in
`O67 Consent`	`CNS-`	Form 5-C, Form 5-D	item 5
`O68 Medical Device`	`DEV-`	Form 7-B, Form 7-C	item 7
`O69 Code System`	`CSY-`	Form 2-C, Form 2-D	item 2
`O70 Measure Definition`	`MDF-`	Form 5-G, Form 5-H	item 5
`O71 Disclosure Event`	`DSC-`	Form 5-E, Form 5-F	item 5
`O72 Retention Clock`	`RTN-`	Form 2-E, Form 2-F	item 2

The numbering is fixed and nothing else may claim `O67` to `O72`. The transcription workbook already carries all six, as cards `18-52` to `18-57`, each flagged in its own contents row as a sheet to build before the paste. Build the sheet first. A paste into a tab that does not exist creates one with the wrong name, and a sheet named with a trailing space is a defect you will find in Phase H and not before.

Twelve forms feed the six. Nine of the twelve carry the missing-sheet statement in bold on their own face; the three that do not are the part-two halves in item 5, which are keyed on their part one and inherit it. If you photocopy a part two on its own, write the warning on it by hand before it goes into the room.

Separately, 24 forms in the set feed a sheet the repository does have, or no sheet at all, and carry a statement about a column that does not exist: `O31` has no owner and no due date, `O60` has no required-by or decided-on column, `O63` has no instrument and no last-assessed date, `O49` records whether a recovery objective was tested but not when, `O48` has no refresh or masking column, `O29` has no place for the source the gap was derived from, `O31` again has no approver and no date for a scope decision, and `R07` has no `Ref` column at all. Each of those forms keeps the statement on its face, in bold or in its header line, in the same place as the missing-sheet warning, and says what carries the column instead.

The two sheets that carry no form at all

`O15` and `O19` carry no form in this volume, and the reason is a collision between two rules rather than anything wrong with either sheet.

The first rule is that a form's columns are the real columns of the real sheet, in sheet order, spelled exactly as the file spells them. A form that renames a column has stopped being a transcription instrument: somebody types the renamed heading into the sheet and the next reader has two columns for one thing.

The second rule is the volume's banned-string list, which is matched without regard to case. `015`'s second column, and the sheet's own row-1 title, name an element type that does not exist in ArchiMate 4 and that this volume may not write. `019` is the same problem one column over: its second column repeats the sheet's own name, and that name is a thing the notation does not have either — volume 1, chapter 20 settles both, and tells you to handle them by sheet number.

The two rules cannot both be kept, so the form is dropped and the sheet is named by reference. `015` holds ten rows, each one an externally consumable service realising a capability on `013`, with a capability reference, a description, an exposure value and a criticality. `019` holds ten rows, each one an agreement with a counterparty, with a type and an expiry date, and the renewal window that would make the expiry actionable is the column the transcription card adds. Work both sheets from the cards — `18-16` and `18-18`, which keep the real spellings, which is correct, because the validator runs on this volume's text and not on the spreadsheets.

This is worth one more sentence, because it will come up at a gate. Not writing a string is not the same as not modelling the thing. The ten rows on `015` and the ten on `019` are real and both sheets carry work: `019`'s expiry dates are the only contractual clock in the repository, and the integration product the whole Phase F plan depends on has no row on it at all.

What this set does not carry

Four things, and none of them is an oversight. Items 1 to 16 print 27 ruled question blocks as well as the forms; a question block has no sheet behind it, no example row and no signature line, and its answers go into a covering note rather than into a column, so photocopy those from the item that prints them. No form here carries a column a formula can fill, which is why the set is narrower than the repository is wide. `015` and `019` carry no form at all, for the reason set out above. And the forms in items 19 and 20 stay with their items, because the principles starter set and the maturity assessment are each a single exercise rather than part of the phase-by-phase run, and the maturity sheet feeds no repository sheet at all — no maturity sheet exists.

The example rows, and which organisation each belongs to

Every form carries exactly one worked example row, marked as an example, and the organisation it is drawn from is not interchangeable. Three run through this volume and they are kept apart on purpose.

Driehoek Health Group is the private worked answer: nine acute hospitals, two day clinics, about 1,750 registered beds. It carries the example rows on the technology and delivery forms, where the question is what a private group with its own estate would write — Form 7-A, Form 7-B, Form 7-E and Form 8-C.

Rietvlei District Health Service is the public worked answer: one district hospital, three community health centres, 22 fixed clinics and a mobile service. It carries the example rows where the answer differs by sector — the scope decision on Form 3-B, the consent register on Form 5-C, and the environment position on Form 7-D.

Ravensmere Health Network is neither. It is the repository the reader actually opens, and every sheet reference, every count and every defect in this volume is Ravensmere. Where an example row is read straight off a sheet rather than imagined — Form 1-A, Form 1-B, Form 8-A and Form 8-B — it is Ravensmere, and the row is a real row of the supplied file.

Do not mix them on one form. A form whose example row names a private group and whose first real row describes a district hospital teaches the reader that the distinction does not matter, and in this industry the distinction is most of the architecture.

The split forms, and keeping the halves together

Where a sheet is wider than six columns the form is in two parts and the two parts are keyed on `Ref`. Part one identifies: the reference, the name, and the facts that do not change. Part two assesses: the judgements, the dates and the states. Three rules keep them together.

Photocopy them as a pair and staple them. A part two on its own has a `Ref` column and nothing else to tell a reader what it is about, and a part two that reaches the repository without its part one produces rows whose key resolves to nothing.

Fill part one for every row before starting part two. The reverse order is how two rows end up sharing a reference, and nothing in a spreadsheet will tell you that happened.

Sign them separately. In most of the splits the two halves have different signatories, because identifying a thing and assessing it are different acts with different authority — on the device forms the estates function signs the identification and the clinical engineer signs the operating state, with the network segment countersigned by the Security Architect. A single signature across both halves is a sign that one of the two was filled in by the other person's assistant.

The forms

In item order, then in the order each item prints them. A split sheet keeps both halves and keeps them adjacent.

Item 1 — How to use this workbook

Three forms, Form 1-A, Form 1-B and Form 1-C.

Form 1-A — Relationship meaning test

Feeds `R22`, plus the reconciliation note that `ZZ Checks` has no row for · **Rows for a first pass** 10, one per `R22` row · **Signed by** the Chief Enterprise Architect · **Ratified by** `GB-01`

Columns 1 to 4 are `R22`'s own columns and are typed straight back into Excel. Columns 5 and 6 are working columns; the repository has nowhere to put them, which is the point of the exercise.

Org unit	Unit name	Capability	Capability name	Verdict	Who said so, and when
ORG-04	Human Resources	C04	Medicines Management	Wrong — example row	Chief Pharmacist, 4 Mar

Form 1-B — Repository count sheet

Feeds `00 Index` row 2, `RPT01` row 6 and `ZZ Checks` row 6 · **Rows for a first pass** 16, one per `ZZ Checks` row · **Signed by** the Chief Enterprise Architect · **Ratified by** `GB-01`

Columns 1 to 3 are `ZZ Checks`' own columns and are typed straight back into Excel. Columns 4 to 6 are working columns.

Check	Result	Threshold	Counted	Correct figure	Where you counted it
Records	843	—	820 — example row	820	rows 5 down, 65 sheets

Form 1-C — Repository mapping

Feeds the repository mapping note in the engagement pack; no sheet in this workbook holds it · **Rows for a first pass** 65, one per object type on `00 Index` · **Signed by** the Chief Enterprise Architect · **Ratified by** `GB-01`

Sheet	Records	Client object type	Columns this volume needs	Present	Note
O20 Data Entity	14	Data Object — example row	Classification, Owner, Master system, Lineage	3 of 4	no master system field

Item 2 — Standing the practice up where nothing is specified nationally

Six forms, Form 2-A, Form 2-B, Form 2-C, Form 2-D, Form 2-E and Form 2-F.

Form 2-A — Governance body membership

Feeds `057` · **Rows for a first pass** 5, one per governance body · **Signed by** the Chief Enterprise Architect · **Ratified by** `GB-04`

Columns 1 to 4 are `057`'s own columns and are typed straight back into Excel. Columns 5 and 6 are working columns.

Ref	Governance body	Cadence	Membership	Named holders	First meeting
GB-03	Data Governance Council	Monthly	CDO, business data owners, stewards	example row — 1 named, 4 vacant	18 Mar

Form 2-B — Sourcing route and shared-service dependency

Feeds the sourcing register in the Preliminary pack; no sheet holds a route · **Rows for a first pass** 12 applications plus the shared services · **Signed by** the Chief Enterprise Architect · **Ratified by** `GB-02`

Ref	Application	Type	Disposition	Sourcing route	What stops when it stops
A11	Integration Engine	Platform	Consolidate	bought — example row; no O19 row	19 of 23 interfaces bypass it already

Form 2-C — O69 Code System, part one: identify

Feeds `O69`, which the supplied repository does not have · **Rows for a first pass** 8 to 14, one per code system per edition · **Signed by** the named data owner · **Ratified by** `GB-03`

All five columns go straight into Excel; form 2-D carries the other four, keyed on `Ref`.

Ref	Code system	Edition or version	Effective from	Effective to
CSY-03	Laboratory local test catalogue — example row	as at 2026-07 release	2026-07-01	

Form 2-D — O69 Code System, part two: assess

Feeds `O69` , which the supplied repository does not have · **Rows for a first pass** the same 8 to 14 rows as form 2-C · **Signed by** the named data owner · **Ratified by** `GB-03`

`Ref` and `Code system` repeat from form 2-C as the key; the last three go straight into Excel.

Ref	Code system	Licence position	Where bound	Owner
CSY-03	Laboratory local test catalogue	none needed — example row	E07, result code	Chief Pathologist

Form 2-E — O72 Retention Clock, part one: identify

Feeds `O72` , which the supplied repository does not have · **Rows for a first pass** 10 to 16, one per record class · **Signed by** the Chief Compliance Officer · **Ratified by** `GB-01`

All five columns go straight into Excel; form 2-F carries the other four, keyed on `Ref` .

Ref	Record class	Period	Clock starts at	Instrument
RTN-02	Minors, including obstetric — example row	to 21st birthday	date of birth	professional guidance, cl. 9.3.1

Form 2-F — O72 Retention Clock, part two: assess

Feeds `O72`, which the supplied repository does not have · **Rows for a first pass** the same 10 to 16 rows as form 2-E · **Signed by** the Head of Health Records · **Ratified by** `GB-01`

`Ref` and `Record class` repeat from form 2-E as the key; the last three go straight into Excel.

Ref	Record class	System of record	Hold rule	Destruction authority
RTN-02	Minors, including obstetric	A01 plus paper — example row	any open claim or complaint	Deputy Director-General

Item 3 — Architecture Vision

Three forms, Form 3-A, Form 3-B and Form 3-C.

Form 3-A — The clinical organisation units the repository does not have

Feeds `O09 Organization Unit` · **Rows a first pass needs** six to ten · **Signed by** the chief executive

Ref	Organisation unit	Type	Head	Headcount
ORG-13 (example)	Medical Directorate	Clinical	Clinical Manager	214

Form 3-B — Scope decision record

Feeds no sheet: there is none for it, because `031` has no approver and no date column. This form is the record of the decision; keep it with the vision and carry it into Phase G. **Rows a first pass needs** one per scope line · **Signed by** the sponsor

Scope line	In scope	Out of scope	Decided by	Date
Establishments (example)	Rietvlei District Hospital; three community health centres	22 fixed clinics; mobile service	Head of Department	2027-02-15

Form 3-C — Medico-legal exposure and retrieval worksheet

Feeds no sheet: there is none for it, because `004` has no target or definition column and `062` no financial column. It is the Phase A instrument the accounting officer signs. **Rows a first pass needs** five measured episodes plus four exposure lines · **Signed by** the chief financial officer and the accounting officer

Line	What is measured	Source and date	Figure	Who confirmed
Retrieval, episode 1 (example)	Hours from request to complete record	Records office log, 2027-02-09	61 hours; chart missing	Records Manager

Item 4 — Business Architecture

Three forms, Form 4-A, Form 4-B and Form 4-C.

Form 4-A — Value stream stage register

Feeds `014 Value Stream` column 4 and `016 Business Process` · **Rows a first pass needs** thirty-six, one per stage · **Signed by** the value stream owner

Ref	Business process	Value stream	Description	Automation
BPR-06 (example)	Collect and label	VS02	Identity verified at the patient; specimen type and collection time recorded	Manual

Form 4-B — Stage definition, part A: identify

Feeds `016 Business Process`. Columns 1 to 5 are typed straight into the sheet; part B's four answers do not fit `016` and are carried in this form to Phase C. **Rows a first pass needs** thirty-six, one per stage · **Signed by** the value stream owner

Ref	Business process	Value stream	Description	Automation
BPR-24 (example)	Acknowledge and act	VS02	Named recipient records that the result was seen and what was done	Manual

Form 4-C — Stage definition, part B: assess

Feeds no sheet. The repository has no relationship sheet joining a process to a role or to an application, and no column for a handoff, so this form is the Phase B record and the input to Phase C and Phase D. **Rows a first pass needs** thirty-six, one per stage · **Signed by** the operational manager who runs the step

Ref	Performer, by role	System touched	Where the handoff fails	What the data must carry
BPR-24 (example)	Responsible clinician, or whoever covers	None in most deployments	No acknowledgement event exists	An actor and a timestamp

Item 5 — Data Architecture

Eight forms, Form 5-A, Form 5-B, Form 5-C, Form 5-D, Form 5-E, Form 5-F, Form 5-G and Form 5-H.

Form 5-A — Regrade, re-seat and re-own the fourteen entities

Real `O20` columns in sheet order, less `Class` and `Definition`, which are already right and are typed straight into Excel. One row per entity; a first pass needs all fourteen. Signed by the entity `Owner` and `GB-03`.

Ref	Data entity	Classification	Owner	Master system	Lineage
E12	Claim — EXAMPLE, delete this row	Confidential		Revenue Cycle / Claims	Absent

Form 5-B — The attribute inheritance sweep

All six real `037` columns in sheet order. Work one entity at a time, writing the entity's settled grade at the head of each block. A first pass needs twenty-six rows, one per breach, plus the identifier and code-system rows from 5.4 and 5.6. Signed by the Chief Compliance Officer.

Ref	Entity	Attribute	Type	Key	Classification
ATR-001	Patient — EXAMPLE, delete this row	Identifier	String	Yes	Restricted

Form 5-C — `067 Consent`, identify

`067 Consent` does not exist in the repository. This form creates it. Five of the nine fixed `067` columns, in sheet order; form 5-D carries the lifecycle four. One row per consent type per care setting — at Rietvlei District Health Service, about twenty rows. Signed by the Chief Compliance Officer.

Ref	Consent type	Subject	Scope	System of record
CNS-001	Disclosure — EXAMPLE, delete this row	Patient, E01	Discharge summary / referring practitioner / continuity of care / 12 months	

Form 5-D — `067 Consent`, lifecycle

The remaining four `067` columns, in sheet order, keyed on form 5-C's `Ref`. `Withdrawn` is a date and an actor, never a tick. Signed by the Chief Compliance Officer; `Evidence` countersigned by the records function.

Ref	Granted	Expires	Withdrawn	Evidence
CNS-001	2026-03-14 — EXAMPLE, delete this row	2027-03-13		Signed form, registry file, retrieval measured at 4 days

Form 5-E — `071 Disclosure Event`, the disclosure

`071 Disclosure Event` does not exist in the repository. This form creates it. Five of the eight fixed `071` columns, in sheet order. One row per disclosure; a first pass covers twelve months. Signed by the records function. Form 5-F carries the authority.

Ref	Record subject	Disclosed to	Purpose	Date
DSC-001	Patient, E01 — EXAMPLE, delete this row	Named scheme administrator	Funder claim substantiation	2026-04-02

Form 5-F — `071 Disclosure Event`, the authority

The remaining three `071` columns plus `Ref`, in sheet order. A row with no `Authority` is a finding, not a blank. Signed by the Chief Compliance Officer.

Ref	Authority	Consent reference	Logged in
DSC-001	Written consent, NHA s14 — EXAMPLE, delete this row	CNS-001	Revenue Cycle / Claims

Form 5-G — `O70 Measure Definition`, the rule

`O70 Measure Definition` does not exist in the repository. This form creates it. `Ref` plus the four rule columns, in sheet order; form 5-H carries the identity four. Sixteen rows, one per `O04` measure. Signed by the measure `Owner`.

Ref	Numerator	Denominator	Exclusions	Boundary rule
MDF-10	O20 rows whose Owner resolves to an O10 Actor — EXAMPLE, delete this row	14 O20 rows	None	Not applicable, entity count

Form 5-H — `O70 Measure Definition`, the identity

`Ref` plus the four identity columns, in sheet order, keyed on form 5-G's `Ref`. `Owner` is a person. `Version` increments whenever any column on 5-G changes, and the old version is kept. Signed by `GB-03`.

Ref	Measure	Period	Version	Owner
MDF-10	Entities with a named owner — EXAMPLE, delete this row	Quarterly	1.0, from 2026-11-01	

Item 6 — Application Architecture

Four forms, Form 6-A, Form 6-B, Form 6-C and Form 6-D.

Form 6-A — The portfolio re-grade

Six of the ten real `023 Application Component` columns, in sheet order. `Type`, `Description`, `In service` and `Users` are confirmed from the systems and typed straight into Excel. Twelve rows, one per application. Signed by `GB-02 Investment Committee`; `Owner` countersigned by `GB-04`.

Ref	Application	Disposition	Criticality	Health	Owner
A09	Incident & Risk — EXAMPLE, delete this row	Invest	Critical	Poor	

Form 6-B — The interface register, one row per interface

Six of the nine real `R06 Application interfaces with` columns, in sheet order. `Source name`, `Target name` and `Frequency` resolve from the refs and are typed straight into Excel. Twenty-three rows; write the test you declared at the head of the sheet before the first row. Signed by the Domain Architect, Application.

Ref	Source	Target	Type	Managed	Logged
INT-010	A07 — EXAMPLE, delete this row	A01	Direct DB read	Unmanaged	No

Form 6-C — The service catalogue

All six real `O24 Information System Service` columns, in sheet order. Expect more than twelve rows when you are finished, and fewer applications. `Published` means a consumer can find and call it without asking an engineer. Signed by the Domain Architect, Application.

Ref	Information system service	Application	Description	Style	Published
ISS-01	EHR / EPR Service — EXAMPLE, delete this row	A01		File	Unpublished

Form 6-D — Re-deriving a positional join

All five real `R07 Application deployed on Tec` columns, in sheet order. **This sheet has no `Ref` column.** Use the same method for `R04`, `R20` and `R25`, whose columns differ: name the evidence for every row, and leave the sheet short rather than filling it. Signed by the Domain Architect, Technology.

Application	Application name	Platform component	Component name	Environment
A04	RIS / PACS — EXAMPLE, delete this row	PTC-02	PACS image store	Production

Item 7 — Technology Architecture, devices and the downtime path

Five forms, Form 7-A, Form 7-B, Form 7-C, Form 7-D and Form 7-E.

Form 7-A — Deployment record, feeding `R07`

Twelve rows for a first pass, one per application on `O23`. Row 1 is the worked example — Driehoek Health Group. Signed by the Head of Service Management. Columns are `R07`'s own, spelled as the sheet spells them; `R07` is a relationship sheet and has no `Ref` column.

Application	Application name	Platform component	Component name	Environment
A01	EHR / EPR	PTC-01	EHR application tier	Production

Typed straight into Excel: `Application name` and `Component name`, because both are lookups from `O23` and `O27` and should never be keyed by hand. Key the three that carry meaning.

Form 7-B — Device register, identify. Feeding `O68 Medical Device`

This form feeds a sheet the repository does not have. Create `O68` before you type it in. Ten to twelve rows for a first pass. Row 1 is the worked example — Driehoek Health Group, theatre complex. Signed by the Head of Estates.

Ref	Device	Model	Risk class	Owner
DEV-01	Volumetric infusion pump, general ward	as the plate reads	C, manufacturer classified	named clinical engineer

Typed straight into Excel: nothing on this form. Every cell is read off a device or out of a maintenance system by a person standing next to it, which is why the form is on paper.

Form 7-C — Device register, operating state. Feeding `068 Medical Device`

This form feeds a sheet the repository does not have. The same rows as form 7-B, joined on `Ref`. Ten to twelve rows for a first pass. Row 1 is the worked example. Signed by the clinical engineer, with the network segment countersigned by the Security Architect.

Ref	Software version	Calibration due	Maintenance due	Network segment
DEV-01	as the manufacturer states it	2027-03	2027-01	unpatchable clinical devices

Typed straight into Excel: `Calibration due` and `Maintenance due`, where a maintenance management system already holds them and can export. Everything else is keyed from the form.

Form 7-D — Environment position, feeding `048 Environment`

Four rows go into `048`, which has four; the rules below them are for the second pass, when an environment is added or a shared one is split out and has to be assessed on its own. Row 1 is the worked example — Rietvlei District Health Service. Signed by the Chief Data Officer. The fifth column is not on `048` and is the reason you can defend the fourth; keep it on the form and in the covering note.

Ref	Environment	Description	Personal data present	Date last refreshed, and masked or not
ENV-01	Production	Live service	Yes	n/a — production is not refreshed from anywhere

Typed straight into Excel: `Ref`, `Environment` and `Description`, which already exist and should not be retyped. Key only `Personal data present`, and keep the fifth column out of the sheet.

Form 7-E — Recovery objectives, feeding `049 Resilience Tier`

Twelve rows for a first pass against the current sheet, then one row per clinical function once the decomposition in step 6 is agreed. Row 1 is the worked example — Driehoek Health Group. Signed by the Chief Operating Officer on clinical advice. The sixth column is not on `049`; it is the derivation, and without it the fourth column is an assertion.

Ref	Application	Tier	Recovery objective	Tested	How long before a patient is harmed
RES-01	EHR / EPR	Tier 1	15 min read, 4 hours write	2027-02-11, partial	minutes for allergy and medication retrieval

Typed straight into Excel: `Ref` and `Application`, which already exist. `Tier` is keyed last, after the objective, and the sixth column goes into the covering note rather than the sheet.

Item 8 — Opportunities and Solutions

Three forms, Form 8-A, Form 8-B and Form 8-C.

Form 8-A — The derived gap set, feeding `029 Gap`

Twenty-six rows minimum for a first pass, one per candidate from 8.1, before 8.2's additions. Row 1 is the worked example — Ravensmere, read straight off `RPT03`. Signed by the Chief Enterprise Architect. The sheet has seven columns and this form carries six; `Building block` is the seventh and is typed in Excel from `055` after the row exists.

Ref	Gap	Domain	Severity	Status	Due
GAP-03	Estates & Medical Devices at maturity 1 against target 3	Business	Critical	Open	

Typed straight into Excel: `Building block`, which is a lookup from `055`, and the source reference, which `029` has no column for and which goes in the covering note keyed by `Ref`. Key the six above by hand, because each one is a judgement about severity that a lookup cannot make.

Form 8-B — Cost reconciliation, feeding the covering note to `054`

Four lines at the top level, and the sheet has no column for any of them — **this form feeds no sheet the repository has**, and the reconciliation lives in the covering note to `054` with a row reference. The rules below the four are for the sub-lines each one breaks into, because "what is in the packages and not in the alternative" is never one line. Row 1 is the worked example — Ravensmere, read off `054` row 6. Signed by the Investment Committee.

Line	Amount, with currency and rate date	Source
The recommended alternative as costed	$46m, no rate, no date	`054` row 6, `ALT-B`

Typed straight into Excel: nothing. The reconciliation is four lines and a signature, and it is the one Phase E artefact that should never be in a spreadsheet, because the second it is a formula somebody will change an input.

Form 8-C — Implementation factors, feeding `052 Implementation Factor`

Eight rows to replace the eight that are there, plus the additions from step 5. Row 1 is the worked example — Driehoek Health Group, replacing `IF-03`. Signed by the Chief Operating Officer. All four columns are `052`'s own.

Ref	Implementation factor	Category	Deduction
IF-03	Winter respiratory pressure, June to August	Constraint	No clinical-system release in the window; device work deferred

Typed straight into Excel: `Ref` only. The other three are the workshop's output and should be keyed from the signed form so that there is a paper original of what was agreed.

Item 9 — Migration Planning

Four forms, Form 9-A, Form 9-B, Form 9-C and Form 9-D.

Form 9-A — Work package, re-dated and re-stated

One row per work package. Feeds `030 Work Package`. The `Domains` and `Cost` columns are typed straight into Excel from the existing sheet and are not on this form. A first pass needs twelve rows, one per existing package. Signed by `GB-02 Investment Committee`.

Ref	Work package	Start	End	Approval	Delivery
WP-05	EXAMPLE — delete this row. Access and security	2027-03	2030-03	Not started	Not started

Form 9-B — Transition architecture and its exit criteria

One row per state. Feeds `053 Transition Architecture`; the `Exit criteria` cell takes the numbered list, not a count. A first pass needs five rows — the three states the file holds plus the two step 9.6 adds. Signed by `GB-01 Architecture Board`.

Ref	Transition architecture	Description	Work packages	Exit criteria
TA-2	EXAMPLE — delete this row. Integrable	Managed integration carrying production traffic	WP-03	1. Zero unmanaged interfaces on R06. 2. …

Form 9-C — Increment, re-pointed to the project that delivers it

One row per increment-to-project pair; an increment may take more than one row. Feeds `R23`. `Lands` is copied from `066` unchanged, including its relative form. A first pass needs seven rows. Signed by `ROL-13 Head of Delivery` and `GB-01 Architecture Board`.

Increment	Increment name	Lands	Project	Project name
INC-5	EXAMPLE — delete this row. Analytics and evidence	Month 42 (relative)	PRJ-06	Analytics and reporting

Form 9-D — Deliverable, owned and dated

One row per deliverable. Feeds `031 Deliverable`. **`Owner` and `Due` are columns this repository does not have; adding them is part of the step.** A first pass needs twenty rows — or seven, if you start with the drafts. Signed by `ROL-08 Chief Enterprise Architect`.

Ref	Deliverable	Phase	Status	Owner	Due
DLV-16	EXAMPLE — delete this row. Migration Plan	F	Draft	ROL-08	

Item 10 — Implementation Governance

Five forms, Form 10-A, Form 10-B, Form 10-C, Form 10-D and Form 10-E.

Form 10-A — Compliance assessment A, identify

One row per building block per gate. Feeds `059 Compliance Assessment`, columns one to four. The remaining three columns are on form B. A first pass needs one row per block at the gate you are running — thirty-seven for a full gate. Signed by `GB-05 Architecture Function`.

Ref	Building block	Block name	Acceptance criterion
CA-04	EXAMPLE — delete this row. ABB-01	EHR / EPR target	No unmanaged interface to or from EHR / EPR.

Form 10-B — Compliance assessment B, assess

The same rows as form 10-A, carried forward by `Ref`. Feeds `059 Compliance Assessment`, columns five to seven. **`Tested against` and `Assessed by` are columns this repository does not have**; step 10.2 adds them to `R24`. `Block name` is typed straight into Excel from `055`. A first pass needs the same count as form A. Signed by `GB-01 Architecture Board`.

Ref	Gate	Verdict	Evidence	Tested against	Assessed by
CA-04	EXAMPLE — delete this row. G1	Non-compliant	19 of 23 interfaces bypass the engine	R06, counted 14 Nov	ROL-11

Form 10-C — Control objective, with the instrument behind it

One row per obligation. Feeds `O63 Control Objective`. **`Instrument` and `Last assessed` are columns this repository does not have; adding them is part of step 10.5.** The `Source` column is typed straight into Excel from the existing sheet. A first pass needs seven rows, one per existing obligation. Signed by `ROL-07 Chief Compliance Officer`.

Ref	Obligation	Control objective	Instrument	Evidence	Last assessed
CO-02	EXAMPLE — delete this row. Clinical records retention	Every record class has a period, an authority and a measured retrieval time in O72	NHA s13, read with the archives regime	No mechanism	

Form 10-D — Governance decision record

One row per decision. Feeds `O60 Governance Decision`. **`Required by` and `Decided on` are columns this repository does not have; adding them is part of step 10.8.** A first pass needs seven rows, one per existing decision. Signed by `GB-04 Executive Committee`.

Ref	Decision	Body	Status	Required by	Decided on
DEC-04	EXAMPLE — delete this row. Designate data owners	GB-03	Outstanding	Before TA-1 exit	

Form 10-E — Dispensation and change request

One row per request. Feeds `O61 Governance Change Request`. Every row needs a value in `Expires`, including `Not applicable` where a request does not expire. A first pass needs five rows, one per existing request. Signed by `GB-01 Architecture Board`.

Ref	Change request	Against	Type	Status	Expires
CR-G05	EXAMPLE — delete this row. Refuse or re-baseline the integration package	WP-03	Compliance	Board decision required	

Item 11 — Architecture Change Management and the retention clock

Three forms, Form 11-A, Form 11-B and Form 11-C.

Form 11-A — Governance change request, part one: identify

Feeds `O61 Governance Change Request` · **Rows for a first pass** 8 to 14, the five existing rows plus whatever the first re-read returns · **Signed by** `ROL-08 Chief Enterprise Architect` · **Ratified by** `GB-01`

The sheet's six real columns in sheet order; all six go straight into Excel. Form 11-B carries the four added ones, keyed on `Ref`.

Ref	Change request	Against	Type	Status	Expires
CR-G06	EXAMPLE — delete this row. Consent CNS-04 lapsed and not withdrawn	CNS-04	Lapse	Raised	Not applicable

Form 11-B — Governance change request, part two: who raised it and who decides

Feeds `O61 Governance Change Request` · **Rows for a first pass** the same 8 to 14 rows as form 11-A · **Signed by** `ROL-08 Chief Enterprise Architect` · **Ratified by** `GB-01`

`Raised by`, `Raised on`, `Required by` and `Decided on` are four columns this repository does not have; step 11.1 adds them. Until it runs they exist only in a meeting minute, which is why four of five rows have no expiry and nobody can date `CR-G03`.

Ref	Change request	Raised by	Raised on	Required by	Decided on
CR-G03	EXAMPLE — delete this row. Raise unfunded building blocks as gaps	unknown — asked 2026-11-19	unknown		

Form 11-C — The Phase H re-read register

This form feeds no sheet. The repository has none for it and this volume creates none, because the register is the minute of a cycle rather than an architecture object. Keep it with the governance minutes and carry the counts into `O61`. **Rows for a first pass** one per query per cycle, so six to nine · **Signed by** `ROL-08 Chief Enterprise Architect` · **Ratified by** `GB-01`

Sheet	What was read	Rows returned	Raised on O61	Re-read by	Re-read on
O67	EXAMPLE — delete this row. Expires past, Withdrawn empty	11 of 34	CR-G06 to CR-G09	ROL-07	2026-11-19

Item 12 — Requirements Management

Four forms, Form 12-A, Form 12-B, Form 12-C and Form 12-D.

Form 12-A — The traceability assertion, feeding `R09 RD`

Feeds `R09 RD` , a new tab this volume's conventions create; the repository has `R09` only. **`Evidence` , `Confirmed by` and `Confirmed on` are three columns `R09` does not have**; step 12.3 adds them to the right of its original headers. `Requirement name` and `Block name` are typed straight into Excel from `O06` and `O55` . **Rows for a first pass** 32 to 48, one per requirement plus the second block some requirements need · **Signed by** `ROL-10 Domain Architect, Data` · **Ratified by** `GB-01`

Requirement	Building block	Block name	Evidence	Confirmed by	Confirmed on
REQ-008	EXAMPLE — delete this row. DBB-01	Patient mastering	O20 row 5 and O43 MDR-01, read together 19 Nov	ROL-16	2026-11-19

Form 12-B — `O06 Requirement` , part one: identify

Feeds `O06 Requirement` · **Rows for a first pass** 32, one per existing requirement · **Signed by** `ROL-09 Domain Architect, Business` · **Ratified by** `GB-01`

These are six of the sheet's seven real columns, in sheet order. Form 12-C carries the `Statement` column and the two added ones, keyed on `Ref` . All six here go straight into Excel.

Ref	Requirement	Category	Priority	Driver	Status
REQ-002	EXAMPLE — delete this row. Clinical records retention is mechanised	Regulatory	Must	DRV-02	Approved

Form 12-C — `006 Requirement`, part two: state it so it can fail

Feeds `006 Requirement` · **Rows for a first pass** the same 32 rows as form 12-B · **Signed by** `ROL-09 Domain Architect, Business` · **Ratified by** `GB-01`

`Verification` and `Verified on` are two columns this repository does not have; step 12.4 adds them. Until then the only place a verification exists is the covering note, which is why twenty-seven of thirty-two statements could be template fills without anybody noticing.

Ref	Statement	Verification	Verified on
REQ-002	EXAMPLE — delete this row. Every record class in O72 carries a period, a named instrument with its edition, a destruction authority and a retrieval time measured in the last six months	Count O72 rows failing any of the four tests; target zero	2026-11-19

Form 12-D — `007 Constraint`, bound to something

Feeds `007 Constraint` · **Rows for a first pass** 8 to 12, the eight existing rows plus any the re-read adds · **Signed by** `ROL-08 Chief Enterprise Architect` · **Ratified by** `GB-01`

`Binds` and `Assessed on` are two columns this repository does not have; step 12.6 adds them. The first four columns are the sheet's real ones, in sheet order, and go straight into Excel.

Ref	Constraint	Description	Type	Binds	Assessed on
CON-05	EXAMPLE — delete this row. Data residency	Certain data may not leave the jurisdiction	Regulatory	ENV-01 to ENV-04, RTN-07, TBB-07	2026-11-19

Item 15 — Workshop packs, checklists and the ninety-day calendar

Three forms, Form 15-A, Form 15-B and Form 15-C.

Form 15-A — Workshop readiness, one row per pack

Feeds no sheet in the repository — the engagement pack carries it · **Rows for a first pass** 7, one per pack · **Signed by** the Chief Enterprise Architect (`ROL-08`) · **Ratified by** `GB-05`

No sheet in the supplied file records that a workshop happened, who attended or what it produced. `060 Governance Decision` is the nearest and has four columns, none of them a date.

Pack	Room date	Read access	Extract frozen	People confirmed	Facilitator
6 Retention clock	wk 2 Tue — example	n/a	store list, 14 Feb	5 of 5	lead architect

Form 15-B — The ninety-day signature register

Feeds `060 Governance Decision` · **Rows for a first pass** 12, one per signed artefact across the three signings · **Signed by** the Chief Enterprise Architect (`ROL-08`) · **Ratified by** `GB-01`

`060` has no date column and no signatory column. Columns 1 to 4 are `060` 's own and are typed straight into Excel; columns 5 and 6 have nowhere to go in the repository.

Ref	Decision	Body	Status	Date	Signed by
DEC-08	Record count corrected to 820 — example	Architecture Board	Approved	wk 2 Fri	ROL-08

Form 15-C — The refusal register

Feeds no sheet · **Rows for a first pass** 13, the six from volume 1 chapter 22 and the seven above · **Signed by** the Chief Enterprise Architect (`ROL-08`) · **Ratified by** `GB-01`

No sheet in the repository holds a refusal. `O61 Governance Change Request` is the nearest and records requests, not refusals; `O08 Assumption` has three columns and no date.

Refusal	Prerequisite	Expected	Meanwhile	Re-dated	Open
Programme start date — example	board-minuted baseline	wk 6	three axes carried	wk 6 Fri	yes

Item 16 — The repository in ArchiMate 4

Two forms, Form 16-A and Form 16-B.

Form 16-A — The element map

Feeds no sheet in the repository; the modelling convention note in the engagement pack · **Rows for a first pass** 70, one per object sheet including the six created · **Signed by** the Chief Enterprise Architect (`ROL-08`) · **Ratified by** `GB-01`

No sheet holds the mapping. `00 Index` holds a domain per sheet and nothing about elements. Keep this with the repository mapping note from step 1.3 and version it from the first week.

Sheet	Rows	Element	Layer property	Properties carried	Lost
O49 Resilience Tier	12	none — Requirement + Assessment — example	technology	tier, objective, tested	nothing

Form 16-B — The relationship map

Feeds no sheet in the repository · **Rows for a first pass** 25, one per relationship sheet · **Signed by** the Chief Enterprise Architect (`ROL-08`) · **Ratified by** `GB-01`

The `Reversed` column is the one that earns the form. Fifteen of the twenty-five read in the opposite order to the relationship that carries them, and an importer that misses it builds a model that resolves perfectly and points the wrong way.

Sheet	Instances	Relationship	Source end	Reversed	Invented end
R18	16	Realization — example	application	yes	the standard, which has no element

THE CHAPTER IN TEN LINES

- This item reprints every form in items 1 to 16 as one photocopiable set: 56 forms, in item order, labels unchanged.
- A form's label is how the signed paper is cited in a minute, a gate pack and an O72 row, so nothing is renumbered here.
- Strike out the example row in ink before anybody writes, and write the date and your name at the top, because no form has a field for either.
- The example rows belong to three organisations that are never mixed: Driehoek Health Group private, Rietvlei District Health Service public, and Ravensmere for anything read off the file.
- Six columns is the maximum a 170mm text column takes in handwriting, which is why five repository sheets and four of the six new ones are split into a part one and a part two.
- Never hand-write a computed value: O13's level, O27's support position and the classification inheritance check are all computed, never written.
- No patient identifier goes on a form. The consent and disclosure forms take a class of subject and the device form takes a device type.
- Transcribe the same day and initial the paper, because a form typed up a week later has lost the hedging and the hedging is the finding.
- The transcription workbook holds 58 sheets: a contents sheet and 57 cards, each pasting into its target sheet from row 5, because row 4 is the header row.
- Five cards add exactly one column to their sheet and say so: a conformance test, a registration basis, a care setting, a renewal window and an entity reference.
- Twelve forms feed the six sheets the repository does not have, and each carries that statement in bold on its face. Build the sheet before the first paste.
- O29 Gap and O31 Deliverable have no card and the workbook's contents sheet does not say so, yet both carry a form here. Type them into the sheet directly and note that you did.
- O15 and O19 carry no form, because their real column headings are on the banned-string list and a form may not rename a column. Work both from their cards; both hold ten real rows and real work.

APPENDIX G

The principles and standards starter set

Thirty-four principles and twenty-five standards a South African provider can adopt, edit or retire in a day, with the evidence from the file behind every row.

Why you start from a set and not from a blank page

`O05 Principle` ships with twenty-four rows under `Ref`, `Principle`, `Category`, `Statement`, `Owner`, `Status` — the generic set, and not one of them about the clinical record, consent, the patient or clinical safety, in a provider repository.

The instinct is to throw them away and draft your own. Do not. Six executives with a blank page will write forty statements nobody can argue with, in no order and with no test, and the register they produce will refuse nothing for the rest of its life. A starter set turns that week into a day of decisions — adopt, edit, reject or defer, one verdict per row — because **argument against a position is cheaper than argument against a blank page.** Read this item with item 2 and with item 5, which creates three of the sheets the tier-1 principles depend on.

Apply one test to every inherited row

Has this organisation ever refused a design because of it, or would it?

A principle that has never stopped anything is a slogan with a reference number. That test produces four verdicts and nothing else.

What happens to the twenty-four the repository ships, and the ten it does not

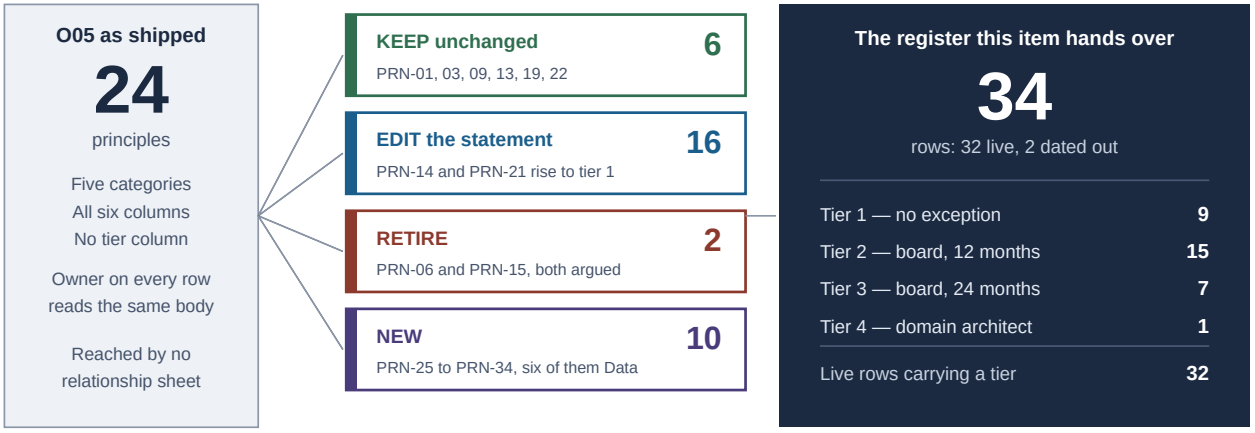


Figure 19.1 The four verdicts applied to the shipped twenty-four, the ten added, and the tier split of the thirty-two that stay live. Counts taken from the register cell for cell.

Verdict	Rows	Which, and why in one line
Keep unchanged	6	PRN-01, PRN-03, PRN-09, PRN-13, PRN-19, PRN-22. The statement survives and the estate contradicts it, which is the most useful shape a principle can have.
Edit the statement	16	True in general and silent on the thing this industry turns on. A carve-out, a direction or a seam is added, and nothing else changes.
Retire	2	PRN-06 Technology Independence and PRN-15 Requirements-Based Change, argued below.
New	10	PRN-25 to PRN-34. Ten statements with no counterpart in what shipped, which is the honest measure of the distance.

Thirty-four rows, thirty-two live. Six of the ten new rows are `Data`, and no new category value is needed — every one fits the five `O05` already ships. Nothing here claims `PRN-35` upward.

Fix the plumbing before you argue about any wording

`O05` is one of the twenty object sheets no relationship sheet reaches, and no report reads it either. **The register governs nothing today, however it is worded** — and that is a one-column repair.

19.1 — Make the register citable before the workshop

Sheet `O06 Requirement`, `O32 Standard`, `O60 Governance Decision` · **Input** the shipped `O05` and `O32` · **Output** four added columns and a dated decision route · **Signed by** `ROL-01` the Chief Enterprise Architect · **Time** a day

1. Add a `Principle` column to `O06 Requirement`, to the right of its seven. Thirty-two rows, one column, half a day, and the cheapest thing available in the file.

2. Point every `O06` row at a `PRN-` reference. A requirement resting on no principle is a preference with a priority code; write that in the cell rather than guessing.
3. Add `Owner` and a conformance test column to `O32 Standard`, in that order — a test with no owner is nobody's job.
4. Add a date column to `O60 Governance Decision`. The sheet has four columns — `Ref`, `Decision`, `Body`, `Status` — so nothing in it can be recorded as decided *when*, and every retirement, reinstatement and re-tiering in this item lands there.
5. In a client file: the equivalent is a requirements register with a free-text justification field. Ask for the three most recent entries and see whether any names a principle.

Until step 19.1 is done, every decision on this register is a decision about wording.

The tier ladder

`O05` has no `Tier` column. Four tiers, one dispensation route each, carried beside the sheet in `Form 19-B`, and the question per row is **who may let this go and for how long** — never how important it is. Importance produces a register where everything is tier 1 and the first dispensation request is granted by whoever happens to be asked.

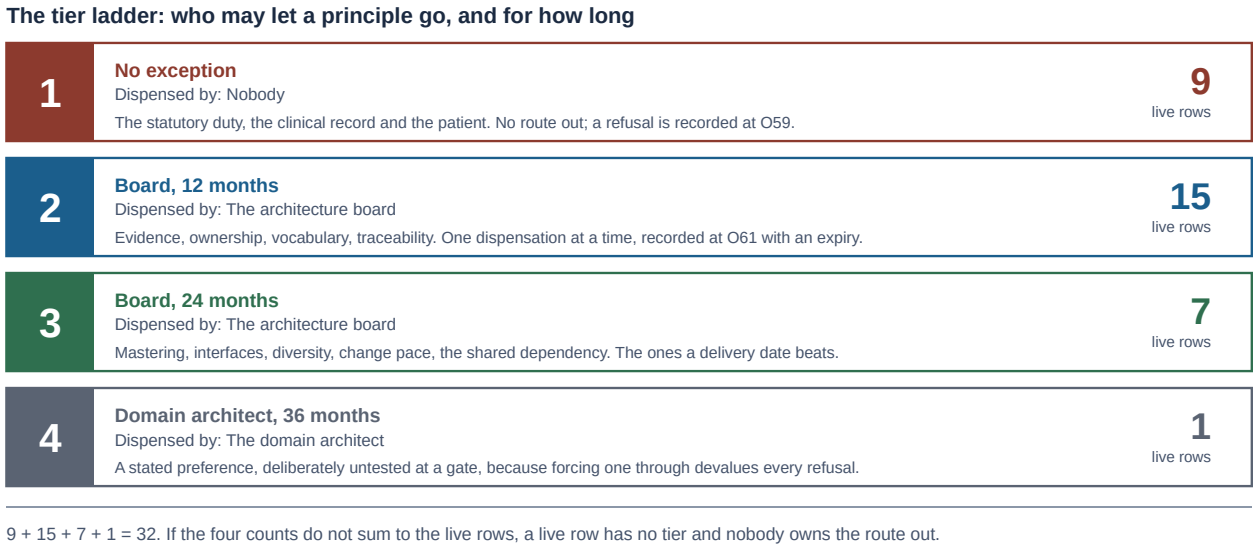


Figure 19.2 Four tiers, one dispensation route each, and the arithmetic a reviewer will test. O05 has no tier column, so the ladder is this volume's and has to be carried beside the sheet.

Nine, fifteen, seven and one. **Tier 2 is the largest band and therefore the one where the ladder does least work** — fifteen rows share one route out, which is tolerable while only one dispensation stands against a principle at a time; a second request is a re-tiering decision. **Tier 4 holds exactly one row and the register says so plainly. Six of the nine tier-1 rows are still `Draft`** — the highest tier and no force, which is the worst combination on the register. All six are new rows, and the workshop either approves them on the day or re-tiers them honestly.

> **WHY FOUR TIERS AND NOT THREE**
>
> Three tiers collapse the difference between a duty that cannot be waived and one that can be deferred against a condition, and that is the difference the board actually needs. Four is also the most a room will hold in its head without a handout.
>
> A principle that *can* be dispensed was not tier 1. Re-tier it rather than granting the dispensation, and record the re-tiering at `O60` with the date column from step 19.1, showing both ladders for the overlapping year. A statement nobody will put a dispensation route against is usually a value, and values belong in the charter rather than on this register.

A dispensation is a record, not a conversation. It lands at `O61 Governance Change Request`, which has `Ref`, `Change request`, `Against`, `Type`, `Status`, `Expires` — and **four of its five shipped rows carry no `Expires` value**, which is exactly what an unrecorded dispensation looks like a year later. Fill that column before granting anything new.

Tier 1 — nine principles, no dispensation route

The statutory duty, the clinical record, the patient, and the things that make the record defensible. No route out and no expiry, because the duty is not the organisation's to waive. A design that breaches one is refused at a gate and recorded at `O59 Compliance Assessment` with a verdict — and `O59` holds three rows against thirty-seven building blocks, so the first application of any of these is the fourth assessment ever written.

PRN-04 Business Continuity (*edited*). Operations continue despite the loss of any single component, **including components the organisation does not own**. The shipped sentence stops at "component" and is false of every provider estate: a shared pathology, administrator or network service that stops, stops care. Downtime in a hospital is a clinical state, not an availability figure, and volume 1 chapter 19 argues the planning assumption is weeks rather

than hours. *Carries:* name every shared service you do not own with the recovery position you can actually obtain, then build the read-only record. *Test:* the date of the last unannounced drill, and the date of the last successful restore from offline backup. Both are dates or they are nothing.

PRN-14 Classification Travels with Data (*edited, raised to tier 1*). Classification is a property of the data and not its location, **and it inherits downward: an attribute, a copy or an extract may be tighter than its parent and may never be looser**. Direction is the whole content of a classification principle. The shipped sentence stops at "location", which is true and refuses nothing, and without direction every downstream copy is free to re-grade. *Carries:* the inheritance check belongs on the Phase C capture sheet and in a gate test, not in a review opinion. *Test:* take any entity graded `Confidential` or `Restricted` and list its attributes with their grades; one looser and the principle is not in force however it is worded. **Twenty-six of `037` 's sixty rows are graded more openly than their parent**, and `ATR-001` , the patient `Identifier` , is `Public` beneath a `Restricted` entity. Item 5 owns the repair.

PRN-21 Attributable Access (*edited, raised to tier 1*). Every access to a clinical record is attributable to a named person **and to the authority they acted under**. Attribution answers who; authority answers why, and here the second question is the one asked afterwards. The National Health Act makes creating, changing or destroying a record without authority, and connecting a user's identifying details to their health status without authority, offences carrying a fine or imprisonment [1]. *Carries:* `071 Disclosure Event` , which item 5 creates. *Test:* pick one record and ask for everyone who read it last month with the authority each acted under — the second half is the half nobody can produce, and `EP-02` records its own position as "Enforces authentication, not attribution".

PRN-25 A Consent is an Object, Not a Flag (*new*). A consent is a record with a type, a subject, a scope, a grant date, an expiry and a withdrawal date; a tick box is not a consent. Consent to treatment, to disclosure and to research are three objects with three lifecycles: withdrawal stops the intervention in the first, stops future disclosure in the second and cannot unsend what has already gone. Written consent is required for a health service, and a child of twelve or older with sufficient maturity may consent for themselves, which makes subject class a column rather than a note [1][2]. *Carries:* `067 Consent` , defined in item 5. *Test:* ask to be shown one consent with its scope and its expiry, then what happens in the record when it is withdrawn. `E13 Consent` is one attributeless row whose `Lineage` is graded `Full` .

PRN-26 The Record is Reconstructable as It Stood (*new*). The clinical record must be reproducible as it stood at a stated point in time, including what had not yet been filed. A screen is not the record and no single system's contents are the record; what the provider must retain is a construct assembled from several systems, and an amendment that silently overwrites makes reconstruction impossible rather than merely difficult. *Carries:* corrections supersede rather than overwrite, on every clinical system and in every migration, which makes a merge reversible by construction. *Test:* pick a record amended in the last year and ask to be shown it as it stood the day before. **The cheapest tier-1 row on the register** — no new sheet, only a supersede rule in each system.

PRN-27 Retention Runs From an Event and Names Its Authority (*new*). Every class of record has a period, the event the period runs from, the instrument behind it and a named destruction authority: four different facts, and the repository holds none of them. South Africa has no retention instrument — every period in circulation traces to professional guidance that disagrees with its own earlier edition, at least six years from dormancy in 2016 against ideally indefinite in the September 2022 revision [3][4] — while the records duty is made expressly subject to the national archives legislation and the access-to-information Act, which is where the destruction authority sits [1]. *Carries:* `072 Retention Clock` , defined in item 2. *Test:* the schedule with the authority cited per period, then the measured time to retrieve a ten-year-old record and the date it was measured. Four sheets record this as unmanaged — `DRV-02` , `CO-02` , `STD-04` and `MTD-06` .

PRN-30 Identity is Resolved Deliberately (*new*). Identity is resolved by a written rule with stated thresholds, a reviewed queue and a reversible merge, **and a patient may decline to be matched**. The asymmetry drives the configuration: a false duplicate splits one person's data, while a false merge puts two people's clinical data in one record and is harder to detect, because a merged record looks complete. The thresholds are a clinical safety decision, not a data-quality setting. *Carries:* `MDR-01` on `043` is typed `Golden source` and must become a `Match rule` ; the queue needs a service-level target and a measured depth. *Test:* the duplicate rate, the false-merge rate, the depth and age of the queue — then who signed the thresholds.

PRN-31 Recovery Objectives are Set Clinically (*new*). A recovery objective is set from the clinical consequence of the system being unavailable, and it is tested on a date. An objective derived from licence cost or a finance classification puts the rostering system above the record. *Carries:* re-derive every tier from clinical consequence so the label predicts the objective, with a named clinical signatory, and put a date against `Tested` . *Test:* the date of the last test of the recovery objective on the clinical record — tested with no date is untested. **`049` 's tier label carries no information at all**: Tier 1 holds four different objectives across its six rows, from fifteen minutes to twenty-four hours, so the loosest objective in the file and the tightest are both on Tier 1. The electronic patient record is Tier 1, twenty-four hours, never tested, and both applications the file marks `Critical` sit below Tier 1. The sheet refutes itself without outside evidence; item 7 owns the repair.

PRN-33 A Disclosure is an Event With a Recipient and a Purpose (*new*). Every release of a record outside the organisation is recorded as an event with its recipient, its purpose, its authority and its date. A withdrawal of consent cannot be acted on and a request for access to one's own information cannot be answered without it, and the outward flows are the only immovable requirements in the estate — they are already happening whether or not anything records them. *Carries:* `071 Disclosure Event` , joined to `067` where the authority is a consent, and starting from the outward-flow register rather than the systems — the statutory and programmatic returns carry most of the volume and nobody thinks of them as disclosures. *Test:* last month's list of disclosures with recipient, purpose and authority.

> **THREE OF THESE NINE WAIT ON A SHEET THAT DOES NOT EXIST**
>
> Consent at `067` , the retention clock at `072` and the disclosure record at `071` ; item 2 and item 5 create them. PRN-26 is the fourth new tier-1 row and needs no new sheet at all, which is why it is the one to start on. **A gate cannot refuse against a sheet that does not exist**, so until the sheet is there the trace defers with a date. Say that out loud when you tier these rows.

Tier 2 — fifteen principles, board dispensation to twelve months

Evidence, ownership, vocabulary and traceability: the things that make a tier-1 duty demonstrable. Each entry gives the statement, the reason the edit was made, what it carries and the test that shows whether it is being followed.

PRN-01 Primacy of Principles (*kept*). Applies to every part of the enterprise without exception. Harmless, and the plumbing beneath it is missing, so step 19.1 is its whole implication. *Test:* name the requirement, the principle it rests on and the gate that tests the pair.

PRN-02 Maximise Benefit to the Enterprise (*edited*). Enterprise benefit outranks local optimisation, **except where the local party is a patient and the trade is clinical**. As shipped the row says a cost saving may be preferred over a clinical consequence, and that is the trade the organisation may not make silently. *Carries:* a named clinical risk owner signs any decision trading a clinical consequence against a financial one, recorded at `O60`. *Test:* find one decision in the last year where a clinical request lost to a cost case, then ask who signed the risk assessment.

PRN-03 Compliance is Designed, Not Added (*kept*). Obligations are architected in, not bolted on. The best-written row on the shipped sheet, disproved on all seven obligations at `O63`: four `In breach`, two `No mechanism`, one `Partial`. The work is attaching it to a gate. *Test:* pick one `O63` row and ask for the last output of the mechanism that evidences it.

PRN-09 Data is an Asset (*kept*). Data has an owner, a definition and a lifecycle — three separate tests, failed three different ways: ten of fourteen entities are owned by a title that exists nowhere else, every definition is one machine-phrased line, and no entity has a lifecycle. *Test:* pick an entity and ask for its owner by name, its definition in one sentence, and the date its oldest instance may be destroyed. The third question is the one nobody can answer.

PRN-11 Data is Accessible (*edited*). Those entitled can reach it without a project, **and every access is recorded and reviewed**. Accessibility without attribution is the finding rather than the fix: emergency access to a record a clinician is not normally entitled to see is a feature, and what makes it safe is that somebody reads the record of it afterwards. *Test:* last month's break-glass count and the proportion reviewed — two numbers and a named reviewer, neither computable from anything in the file.

PRN-12 Common Vocabulary (*edited*). One definition per concept, **and the boundary definitions are declared in writing**. A vocabulary principle here that does not mention the boundary words is decoration: volume 1 chapter 4 gives five words for the boundary of one hospital stay and two unrelated meanings of *claim*. *Carries:* adopt chapter 4's definitions verbatim; spell and consultant episode are English definitions adopted for want of South African ones. *Test:* ask two people in different functions to define an admission, then which answer the board pack used.

PRN-13 Data Ownership is Individual (*kept*). An owner is a person, not a committee — and apply it to `O05` itself, where all twenty-four shipped rows read `Owner = Architecture Board`, the body this principle says an owner may not be. *Test:* read the `Owner` column of any sheet aloud. If it is a body or a title, the row has no owner.

PRN-18 Interoperability (*edited*). Standards are chosen for interoperability, not preference, **and the standard chosen is named at the seam**. The shipped sentence cannot be tested because it names nothing: the whole file contains one terminology string, repeated verbatim five times, as a component name. *Test:* pick a seam and ask which standard governs it and which edition is loaded on each side.

PRN-19 Security is Built In (*kept*). Security is a design input, not a review gate. All four security policies at `O64` read `Breached` and all five enforcement points at `O65` record a defect in their position column — a strong principle with a backlog behind it, headed by attribution rather than authentication. *Test:* which design decision last quarter was changed because of a security input, as opposed to reviewed after the fact.

PRN-20 Least Privilege (*edited*). Access is the minimum the role requires, **and the exception is recorded rather than prevented**. The strict form is unsafe in clinical use: a model that blocks a clinician from a record is defeated by shared logins, and the shared login destroys the attribution PRN-11 and PRN-21 both depend on. *Carries:* break-glass needs a workflow, a log and a reviewer; none of the three exists. *Test:* how a locum reaches a record at two in the morning, and who reads the log.

PRN-23 Models are Governed (*edited*). Analytical and AI models are governed like any other asset, **and a model that informs a clinical decision is governed like a clinical asset**. A model in the clinical path is a hazard with a confidence interval, and the governance question is not accuracy but what the clinician is told about its limits at the moment of the decision. *Test:* the list of models in the clinical path and the clinical owner of each. An empty list is an answer worth having in writing.

PRN-28 A Code System is Effective-Dated (*new*). A code system is named with its edition and its effective window, and the edition travels with every coded value — because a value without its edition cannot be re-interpreted later, which means a trend across an edition change is not a trend. *Carries:* `O69 Code System`, created in item 2; capture in a terminology, aggregate through a classification, bill through a payable-unit nomenclature, all three versioned. *Test:* pick a coded field and ask which edition is loaded today, which two years ago, and what happened to the values in between.

PRN-29 A Derived Figure Travels With Its Measure Definition (*new*). A derived figure is quoted with the versioned definition it was computed from: numerator, denominator, exclusions, boundary rule and period. Volume 1 chapter 2 shows four defensible theatre-utilisation figures computed from one list with no arithmetic relationship between them. *Carries:* `O70 Measure Definition`, created in item 5, and computing from the lowest-level timestamped facts so a definitional change is a re-computation rather than a migration. *Test:* take last month's board figure and ask for its four parts. Four answers, or the figure is an anecdote with a decimal point.

PRN-32 A Shared Service You Do Not Own is Still Your Single Point of Failure (*new*). A dependency outside the organisation's control is named, its recovery position is obtained from the contract, and the fallback is designed. Three of the six sourced attacks in volume 1 chapter 19 were on a supplier rather than on the provider, and the dependency is not reduced by being contractual. *Test:* list the shared services you do not own; for each, the fallback and when it was last exercised. `O19`'s ten agreements record no recovery position at all, and their expiry is `YYYY-MM`, so no renewal window is computable.

PRN-34 A Practitioner's Authority is Live (*new*). Authority to order, prescribe, sign or bill is checked against a live registration and a current privilege, not an attribute copied once. A lapsed practitioner may not practise, their orders should not be actionable and their funder claims should not be payable — and human resources, credentialling, the clinical directorate and billing each own part of the entity and disagree, so the deliverable is a governance decision about who owns which part rather than a data model. *Test:* how the estate knows this morning that a prescriber's registration is current.

Tier 3 — seven principles, board dispensation to twenty-four months

Mastering, interfaces, technology diversity, change pace and the shared dependency: the principles a delivery date beats, predictably, so the ladder gives them a longer route out rather than pretending otherwise. The condition for removal is usually a named piece of work rather than a date.

PRN-05 Common Use Applications (*edited*). Prefer a shared capability to a duplicated one, **except where the duplicate is the record that must survive the shared one**. The read-only clinical record is a deliberate duplicate, and writing the principle without the exception is how an organisation argues itself out of the cheapest resilience it has. *Carries:* the duplicate is named, owned, dated and tested, or it is shadow IT with a good excuse. *Test:* what a clinician reads when the primary record is down, and when it was last read in anger.

PRN-07 Ease of Use (*edited, and deliberately not retired*). A system that is hard to use will be worked around, **and in clinical use a workaround is a safety event**. Every sibling volume drops this row as a design quality no gate tests; it survives here because the workaround is the hazard, and volume 1 chapter 19's twelfth benchmark measure — alerts suppressed, with a documented clinical owner for each suppression — makes it testable. *Test:* the suppression rate and the clinical owner of the three most-suppressed alerts. No owner means the suppression was taken by nobody.

PRN-08 Service Orientation (*edited*). Capability is exposed as a service **with a published specification and a named owner**. As shipped, "with a contract" is satisfied by anything, which is how nineteen of twenty-three interfaces come to bypass the declared integration platform and still pass every integrity check in the file. Five of `O24`'s twelve rows read `Published` and all five are styled `File` — file drops described as published services. *Test:* pick an interface and ask for its specification, its owner, and how deep its exception queue is this morning.

PRN-10 Data is Shared (*edited*). Data is shared through published interfaces, not copies, **and every copy has a named purpose, a refresh cadence and a masking position**. `INT-022` reads the laboratory system's database directly, hourly, and is marked `Managed`. A copy is a second master in waiting and, here, a second classification decision taken by nobody. *Test:* the list of copies of the clinical record and who approved each one.

PRN-16 Responsive Change Management (*edited, and reversed in substance*). The estate changes at **the pace clinical safety allows, which is sometimes slower than the business would like**. As shipped it reads that the estate changes at the pace the business needs, which in a hospital is an argument for shipping a clinical change without a hazard assessment. *Test:* the hazard assessment behind the last clinical system change, and who signed it. The file has nowhere to record either.

PRN-17 Control Technical Diversity (*edited*). Each capability has one approved technology **unless a justification is recorded and dated**. As shipped, "unless justified" has no home, so the justification is a conversation; `O26` and `O47` already give a different status to the same component in four of seven cases, which is what an unrecorded exception looks like a year later. Diversity is not the enemy; undocumented diversity is. *Test:* pick a capability with two technologies behind it and ask for the dated justification.

PRN-22 Defence in Depth (*kept*). No single control is relied on alone. `STD-11` reads `Current` and in `Breach` while clinical devices sit on whatever segment the vendor installed them on, and depth is the only answer where a layer cannot be patched — a device whose software version is unknown is a layer you cannot fix. *Test:* which control would still hold if the device segment were compromised. The device register at `O68`, created in item 7, makes this actionable.

Tier 4 — one principle, domain architect, thirty-six months

PRN-24 Sustainable by Design (*edited*). Energy and resource cost is a design consideration, **named at the site**, because in a hospital group it bites at the site rather than the data centre. No implication beyond stating it in the design document, and no test, because the row is a stated preference and the register says so. If it starts refusing designs, re-tier it rather than quietly enforcing it.

The two retirements, argued

A register where nothing was rejected was not read. Test both in the room rather than nodding them through.

PRN-06 Technology Independence — *applications are independent of the platform beneath them.* Retire it. It is false of the two most expensive things in this estate: an imaging store and a clinical device estate are not independent of the platform beneath them, and `TBB-05 Medical device integration` and `PTC-07 Medical device gateways` are platform-bound by construction. A principle false in the two cases that cost the most is worse than no principle, because it distorts every sourcing decision that cites it. Retiring it is not a concession: the honest version already exists as PRN-17, which says what to do about coupling instead of denying it. *The test it fails:* name one design refused because of it. Nobody can.

PRN-15 Requirements-Based Change — *change is driven by stated requirements.* Retire it from this register. It is a Requirements Management discipline rather than an architecture principle, it is still `Draft` after shipping, and it is contradicted by the file it governs: all thirty-two rows of `R09` were built by pairing two lists in row order, so not one requirement reaches a building block whose subject it names. Move it into item 12. *The test it fails:* read any three `R09` rows against the block they reach. All three will be unrelated.

A retired row is **dated, not deleted**: keep it on the sheet with the decision recorded at `O60`, so a reader holding the original file can see what happened to it.

> **WHY THE CLASSIFICATION REQUIREMENT IS AN EDIT AND NOT A THIRTY-FIFTH ROW**
>
> Eleven things a healthcare principle set needs that the generic twenty-four lack: consent, the reconstructable record, retention from an event, the effective-dated code system, the measure definition, deliberate identity resolution, clinically set recovery objectives, the shared service you do not own, the disclosure event, the live practitioner authority — and classification inheriting downward and never loosening.
>
> Ten become new rows, `PRN-25` to `PRN-34`. The eleventh does not. **`PRN-14` already ships, it is already `Approved`, and the file already breaks it in twenty-six of sixty attributes.** A thirty-fifth principle on the same subject would have left the broken row standing beside the new one, and the first design to cite the easier of the two would cite `PRN-14` as shipped — which refuses nothing. One clause added to the row already in force closes that door.
>
> The general rule: **where the repository already carries a row on the subject, edit it. A new row beside a broken one is two principles in conflict, and the weaker one wins.**

SIX QUESTIONS FOR THE ROOM BEFORE THE REGISTER IS SIGNED

Put these with `005`, `037` and `049` on the screen, and write the answers here in front of them.

Name one design this organisation has refused because of a principle, and say which principle.

Who, by name rather than by body, owns the principle register after today?

Which tier-1 principle will the first dispensation request be made against, and who will refuse it?

On what basis is the published clinical protocol graded as tightly as the patient master?

Which of the nine tier-1 rows can be tested at a gate next month without a new sheet?

If a clinician does not attend the adoption workshop, which rows are deferred and to when?

The standards catalogue

`032 Standard` ships with sixteen rows under `Ref`, `Standard`, `Category`, `Status`, `Conformance`. Five columns: no owner, no conformance test, no end date, no replacement. **Without a test a conformance verdict is an opinion**, which is the state all sixteen shipped rows are in.

This set hands over twenty-five rows: twenty-four live and one recommended for retirement. The shipped sixteen keep their identifiers; the new rows run `STD-17` to `STD-25` and nothing claims `STD-26` upward. **Nine are new**, each naming a mechanism the file asserts an obligation for and does not hold: terminology binding, measure definitions, identity matching, consent recording, record retrievability, device software, downtime and the read-only record, disclosure logging, and the clinical safety case. By category the live twenty-four run seven `Data`, six `Security`, four `Governance`, three `Platform`, three `Delivery` and two `Integration`.

Five of the nine new rows ship with `Conformance` = `Breach`, which looks strange on a standard that did not exist yesterday. It is the honest reading: the duty is already owed and the mechanism is absent, so `Not assessed` would understate the position and `Conformant` would be false. The one new row that ships `Emerging` is the clinical safety case standard, because marking it `Current` would make every clinical change retrospectively non-conformant on day one.

One retirement and one rename. `STD-09 Container image standard` is `Deprecated`, never assessed and governs nothing this estate runs — the only standard whose removal costs nothing, and retiring it makes the remaining deprecated rows visible as a decision rather than as clutter. `STD-12 Coding and review standard` is about source control, and in a healthcare repository every clinician who reads that title will read *clinical* coding, which is a different standard with a different owner. Call it the software development standard and let `STD-17` carry terminology binding.

Status and conformance are independent, and the file does not treat them as independent

Status says whether the standard is in force; conformance says whether the estate meets it, and the pairs they may form are the whole of the lifecycle position.

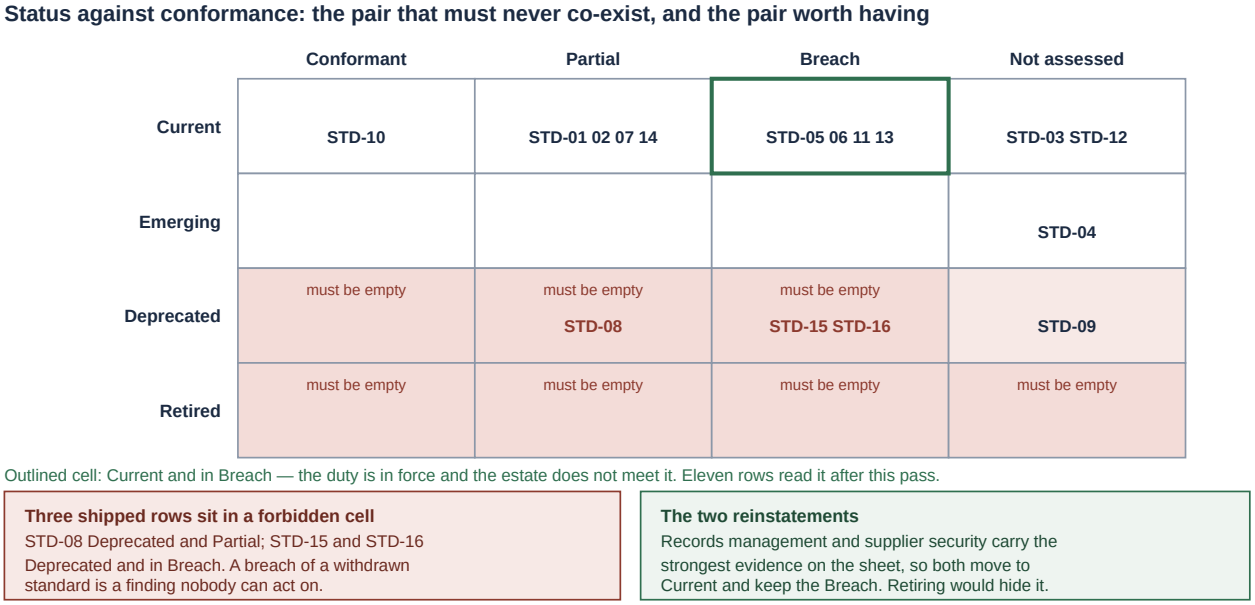


Figure 19.3 The sixteen shipped standards plotted on status against conformance. Deprecated and Retired rows may carry no live verdict; three shipped rows do.

Value	What it means	What it obliges	What it forbids
Current	In force; may be assessed against.	A dated verdict and a named owner.	Nothing. The only status under which a breach verdict is meaningful.
Emerging	Adopted in principle, not in force. New work meets it; existing work does not.	A date on which it becomes current, and what is in scope from then.	A blanket verdict across the estate. Say which work was assessed.
Deprecated	Withdrawn for new work, tolerated in place.	An end date and a named replacement; neither column exists on O32.	A live conformance verdict.
Retired	Out of force. Nothing is assessed against it.	A dated decision at O60, and removal of every citation still in use.	Any verdict at all. Conformance of a retired standard is a category error, not a bad result.

On the conformance side: `Conformant` obliges a scope and a date — `STD-10 Database standard` is the only `Conformant` row, on an estate where three applications read another application's store directly, so the scope is the whole question. `Partial` obliges you to name the part that is not met and a date by which it will be; five shipped rows read `Partial` and none does, which makes `Partial` a synonym for unassessed. `Breach` obliges a dated finding and a decision — fix, dispense or re-scope. `Not assessed` obliges a date by which somebody will look: respectable on a new standard, indefensible on one current for a year. Two shipped rows are `Current` and `Not assessed`, and `STD-03 Data classification standard` is the one that matters.

Current and in Breach is the pair worth having. It says the duty is in force and the estate does not meet it, which is a position somebody can act on inside a quarter; eleven rows read it after this pass, and that is the register working rather than failing. The alternative is `Not assessed`, which hides the same position behind a neutral-sounding word. A high breach count on a healthy register is not an embarrassment; it is the evidence that somebody looked.

The two reinstatements, and why reinstating beats retiring

Four shipped rows read `Deprecated`, and three of those four also carry a live verdict — the pair that must never co-exist, because a withdrawn standard cannot be conformed to. `STD-08 Cloud landing zone standard` is `Deprecated` and `Partial`, and there the decision is a genuine either-or: reinstate and assess, or retire and clear the verdict. Pick one.

The other two are not an either-or, and this is the most serious status error on the sheet. `STD-15 Records management standard` and `STD-16 Supplier security standard` each read `Deprecated` **and** `Breach`.

The temptation is to retire both and clear the contradiction in one edit. Resist it, and the reason is not tidiness. **When a file says a deprecated standard is being conformed to — or breached — the verdict is itself the evidence that somebody is still assessing against it, which means the standard is in force in practice and the status is the thing that is wrong.** Retiring clears the contradiction by deleting the finding; reinstating clears it by correcting the status and keeping the finding, and only one of those two leaves work behind. Retirement is right where the standard governs nothing and has no verdict to lose — `STD-09`, which is `Not assessed`. Where there is a verdict, reinstate.

- `STD-15 Records management` governs creation, custody, tracking, retrieval and destruction of the clinical record. The duty is owed today, and the file has marked the standard carrying it as withdrawn while recording a breach of it. The published evidence is the strongest in the volume: provinces pay in the order of R1,5 billion a year settling medico-legal claims [7], and the Auditor-General's review of twenty-seven facilities found fifteen with missing or lost records, seventeen not tracking records after issue, and **three of nine paid claims conceded because the record could not be located** [6]. Reinstate as `Current`, keep the `Breach`, and test it: a measured retrieval time per record class with the date measured, and a tracking record showing where every record issued last month is now.

- **`STD-16 Supplier security`** governs what is required of a supplier holding or reaching patient data, and what is tested before signature. Three of the six sourced attacks in volume 1 chapter 19 were on a supplier rather than on the provider, so this is the standard with the best evidence behind it on the sheet and it is marked withdrawn. Reinstate as `Current`, keep the `Breach`, and test it against the agreements sheet: every agreement reaching patient data states a recovery position, a breach-notification period and a conformance requirement. None does today.

Two reinstatements, two dated decisions at `O60`. Spend workshop time on these rather than on the retirement.

19.2 — Give every live standard a test, an owner and a scope

Sheet `O32 Standard` with the two columns added in step 19.1 · **Input** the twenty-four live rows · **Output** one test, one owner and one scope statement per row · **Signed by** `ROL-02` the domain architect for the category · **Time** three days

1. Write the test as something a reviewer can ask to be shown, in one sentence, with a count in it wherever one is available. "Every row of the interface register has a published specification reference and no row reads Direct DB read; three do today" is a test. "Interfaces comply with the API design standard" is not.
2. Name the owner as a post, never a body. Then name the assessor, who is neither the author of the standard nor the delivery team being assessed.
3. State the scope the verdict covers. A `Partial` with no scope is unassessed.
4. Mark the five rows whose test names a sheet that does not exist — retention at `O72`, consent at `O67`, devices at `O68`, code systems at `O69`, disclosures at `O71`. Those five are deferred with a date, not assessed, and the date is the date item 2, item 5 and item 7 land.
5. Count the rows with no test. It must read zero before any verdict on this sheet means anything.
6. In a client file: a conformance column with no test column is the normal case. Add the column before the first assessment, not after the first argument.

19.3 — Settle the status column in one sitting

Sheet `O32 Standard`, `O60 Governance Decision` · **Input** the four `Deprecated` rows and the two `Emerging` rows · **Output** six dated decisions · **Signed by** `ROL-01` the Chief Enterprise Architect · **Ratified by** `GB-01` · **Time** half a day

1. Decide status first and verdict second, every time. The reverse order is how a deprecated row acquires a breach.
2. Reinstate `STD-15` and `STD-16` as `Current` and keep both breach verdicts, each recorded at `O60` with the date column from step 19.1.
3. Retire `STD-09` and clear its verdict in the same edit. A retired row carries no verdict at all.
4. Decide `STD-08` either way and record which. Do not leave it `Deprecated` with a `Partial`.
5. Raise `STD-04 Data retention` from `Emerging` to `Current`. `Emerging` means adopted in principle and not in force, which is the wrong status for a duty already owed — four sheets record it as unmanaged. The status is wrong rather than the verdict.
6. Leave `STD-25 Clinical safety case` at `Emerging` and set the date it becomes current. Then count the rows reading `Deprecated` with a live verdict. It must read zero.

The forms

Form 19-A — Principle decision sheet (part one: identify)

Feeds `O05 Principle` · **Rows for a first pass** 34, one per register row · **Signed by** `ROL-01` the Chief Enterprise Architect · **Ratified by** `GB-01`

These six are `O05`'s own columns, in sheet order, typed straight into Excel. Edit `Statement` and nothing else on the first pass, and replace every `Architecture Board` in `Owner` with a named post before the register is signed — all twenty-four shipped rows break `PRN-13` as they stand.

Ref	Principle	Category	Statement	Owner	Status
PRN-14	Classification Travels with Data	Data	Classification is a property of the data and not its location, and it inherits downward — example row	Information officer	Approved

Form 19-B — Tier and dispensation route (part two: assess)

Feeds nothing in the repository; the dispensation it authorises lands at `O61` · **Rows for a first pass** 32, one per live row · **Signed by** `ROL-01` the Chief Enterprise Architect · **Ratified by** `GB-01`

`O05` has no Tier column, no dispensation column and no expiry column. The tier lives in this form and in the principles document; the dispensation is recorded at `O61 Governance Change Request`, which does have `Expires` — and four of its five shipped rows leave it blank. Keyed on `Ref` to part one.

Ref	Tier	Dispensed by	Condition for removal	Expires	Decision note
PRN-31	1	Nobody — no route out			Draft today; approve or re-tier — example row

Form 19-C — Standards catalogue (part one: identify)

Feeds `032 Standard` · **Rows for a first pass** 25, one per catalogue row · **Signed by** `ROL-02` the domain architect for the category · **Ratified by** `GB-01`

These five are `032`'s own columns, in sheet order, and are typed straight into Excel. Decide the status before the conformance value, never the other way round.

Ref	Standard	Category	Status	Conformance
STD-15	Records management standard	Governance	Current — reinstated, example row	Breach

Form 19-D — Conformance test and assessment (part two: assess)

Feeds `032 Standard` once the two columns from step 19.1 exist · **Rows for a first pass** 24, one per live standard · **Signed by** `ROL-02` the domain architect · **Ratified by** `GB-01`

`032` has no Owner column, no conformance test column, no assessor and no date. Step 19.1 adds the first two; the assessor and the date have no home in the sheet and are carried here and in the assessment record at `059`. Keyed on `Ref` to part one.

Ref	Conformance test, in one sentence	Owner post	Scope assessed	Assessed by	Date
STD-21	A measured retrieval time per record class with the date measured — example row	Head of health records	Inpatient records, all nine sites		

Form 19-E — Principle traced to a gate

Feeds `O58 Governance Gate` and `O59 Compliance Assessment` · **Rows for a first pass** 10, the six traces that run today plus the four that wait on a sheet · **Signed by** `ROL-01` the Chief Enterprise Architect · **Ratified by** `GB-01`

`O58` has no column for the body that holds a gate, so the holding body named here has nowhere to land until one is added. The four gates are `G0 Charter`, `G1 Design`, `G2 Build complete` and `G3 Benefit`. PRN-14, PRN-25 and PRN-30 run at G1, PRN-27 at G0, PRN-31 at G2 and PRN-21 before deployment; PRN-26, PRN-28, PRN-29 and PRN-33 defer until their sheet exists.

Ref	Gate	Holding body	What the gate must see	What it does when it does not	Live today?
PRN-14	G1 Design	Data Governance Council	No O37 row graded more openly than its parent, shown as a list — example row	Refuses, and records it at O59 with a verdict	Yes

Run the adoption workshop before anything is modelled

A principle adopted after the design is a rationalisation. One day, and the output is a signed register rather than a discussion.

19.4 — The adoption day

Sheet `O05`, `O32`, `O60`, `O61` · **Input** this item printed, one copy per attendee · **Output** a signed register, a tier per live row, four dated decisions and three actions · **Signed by** `ROL-01` the Chief Enterprise Architect · **Ratified by** `GB-01` · **Time** a day

1. The chair chairs and does not advocate. The chair who argues for a principle cannot rule on the objection to it.
2. Open by reading out the one test, then reading out what is *not* on the agenda: wording. Every row already has a statement; the decision is adopt, edit, reject or defer.
3. Take the inherited twenty-four in `Ref` order at three minutes each — ninety minutes. Most will be adopt or edit. Do not let the room reopen PRN-01: the fix behind it is step 19.1, not a debate.
4. Break, then take the ten new rows at twelve minutes each — two hours. These will be argued, and the argument is the point. **Read the repository evidence before the statement, every time**, because a room that has seen the cell argues about the fix rather than the sentence.
5. After lunch, tier every live row with the board members present. Seventy-five minutes, and the question per row is who may let this go and for how long.
6. Then the two retirements and the two reinstatements, forty-five minutes, each with a dated decision at `O60`.
7. Close by reading back every deferral with the date it returns, then the three actions from step 19.1. Nothing on the register is in force without them.

Who must be in the room. A clinician with clinical governance standing, because five rows name a clinical owner — recovery objectives, merge thresholds, change pace, alert suppression and clinical model governance — and without that person those five are deferred. The information officer, because consent, disclosure, classification and attribution all land on that post. The head of health records, the only person who knows what is in the off-site store and how long it takes to get something out of it, and who is routinely not invited. The clinical coding manager, who will disagree with the billing manager about what a code is for — and that disagreement is the output. **`O10` and `O11` hold no clinician, no pharmacist, no safety officer and no records manager**, so three of those four have no row in the repository until item 2 creates them. Not the vendor.

Four ways the day fails. It becomes a drafting session — if a statement cannot be fixed in one sentence from the floor, mark it deferred with an owner and a date and move on. The clinician does not come — reschedule rather than proceed. Everything is adopted — a register where nothing was rejected was not read. Nothing is tiered — then every principle is tier 1 in practice, and tiering is the half of the day that makes the other half enforceable.

FOUR QUESTIONS TO CLOSE THE WORKSHOP ON

Read these back to the room and write the answers on the register itself, not in the minutes.

Which deferrals are outstanding, and on what date does each return?

Who adds the Principle column to O06 Requirement, and by when?

Which standard will be assessed first, by whom, against what scope?

What is the first design this register will refuse, and at which gate?

Where this register meets a gate

A principle that reaches no gate is decoration. Six of the thirty-two can be traced to a gate today and refuse something; four more need a sheet first and defer with a date. `Form 19-E` carries the ten traces, and three housekeeping actions sit behind all of them. Add `Principle` to `O06`. Add a column to `O58 Governance Gate`, which has `Ref`, `Gate`, `When`, `What it tests` and no column for the body that holds each gate, so every holding body has nowhere to land in the repository. And use the other three gates: `O59` holds three records, all at `G2`, all `Non-compliant`, all with `Evidence` reading "No evidence produced", while `G0`, `G1` and `G3` have never been used and four of the traces need them.

> DO NOT REPRINT A STANDARD'S CONTENT IN THE REGISTER
>
> Every row of this catalogue says what the standard *governs* and what volume 1 cites, and sends the reader to the publisher for the content. No code list, value set, message layout, resource definition or framework decomposition appears anywhere in it, and none should appear in yours. Where an external standard carries its own editions, the row states that the edition is recorded at `O69` with the date you read it, and stops there. The gazetted normative standards framework is worth reading for what it *does* name at a seam and for its conformance-testing position [5]; it is not worth quoting into a register.
>
> The same discipline applies to clinical judgement. Where a principle needs one — recovery objectives, merge thresholds, alert suppression — the row names who signs, and that person is a clinician. The register does not contain the judgement.

THE CHAPTER IN TEN LINES

- Start from the shipped twenty-four rather than a blank page: argument against a position is cheaper than argument against a blank page.
- One test decides every inherited row - has this organisation ever refused a design because of it, or would it.
- Thirty-four rows: six kept unchanged, sixteen edited, two retired and ten new, of which thirty-two stay live.
- Four tiers, nine at tier 1, fifteen at tier 2, seven at tier 3 and one at tier 4, each with one dispensation route.
- A principle that can be dispensed was not tier 1, so re-tier it rather than granting the dispensation.
- The classification requirement is one clause added to PRN-14, not a thirty-fifth row, because the shipped row already breaks it in twenty-six of sixty attributes.
- PRN-06 Technology Independence is retired because it is false of the two most expensive things in the estate; PRN-15 belongs to Requirements Management.
- Nothing in the model points at O05, so the first repair is one column on O06 Requirement naming the principle each requirement rests on.
- Twenty-five standards, twenty-four live, nine new, one retired, and five of the new rows ship in breach because the duty is owed and the mechanism is absent.
- Status says whether a standard is in force and conformance says whether the estate meets it; a deprecated row may carry no live verdict.
- Records management and supplier security are reinstated as current rather than retired, because the breach verdict is the evidence that somebody is still assessing against them.
- Current and in breach is the pair worth having, and eleven rows read it after this pass.
- Run the adoption workshop before anything is modelled, with a clinician, the information officer and the head of health records in the room.

SOURCES FOR THIS CHAPTER

- **1.** National Health Act 61 of 2003, consolidated text (s13 record-keeping duty, expressly subject to the National Archives and Record Service of South Africa Act 43 of 1996 and the Promotion of Access to Information Act 2 of 2000; s14 written consent to a health service; s17 access control and the s17(2) offences) https://www.saflii.org/za/legis/consol_act/nha2003147.pdf
- **2.** Children's Act 38 of 2005, section 129, consent to medical treatment and surgical operation (a child of twelve or older of sufficient maturity may consent) https://www.acts.co.za/childrens/129_consent_to_medical_treatment_and_surgical_operation.php
- **3.** HPCSA Booklet 9, Guidelines on Keeping Patient Records, September 2016 (retention at least six years from the date the record became dormant) https://www.hpcsa-blogs.co.za/wp-content/uploads/2022/08/Booklet-9-Guidelines-on-Patient-Records.pdf
- **4.** HPCSA Booklet 9, Keeping of Patient Records, revised September 2022 (retention ideally indefinite, particularly where records are electronic; attribution of electronic entries and reasons for amendments) https://www.hpcsa-blogs.co.za/wp-content/uploads/2022/11/Booklet-9-Keeping-of-Patient-Records_Review-Draft_-FINAL_Sept-2022.pdf
- **5.** Health Normative Standards Framework 2021, published under National Health Act s74(1), Government Gazette notice of 21 October 2022 (standards named per seam; conformance testing at a department-approved laboratory) https://www.health.gov.za/wp-content/uploads/2022/10/HNSF_Gazette_21_October_2022.pdf
- **6.** Auditor-General findings reported via MedicalBrief, 14 October 2026 (audit of 27 facilities across nine provinces; 15 with missing or lost records; 17 not tracking records after issue; three of nine paid claims conceded for want of a record) https://www.medicalbrief.co.za/health-sector-crippled-by-recurring-systemic-weaknesses-ag-warns/
- **7.** National Treasury, 2026 Budget Review, Chapter 6 (provinces spend about R1,5 billion a year settling medico-legal claims) https://archive.treasury.gov.za/documents/National%20Budget/2026/review/Chapter%206.pdf

APPENDIX H

Running the maturity assessment

Eleven domains, twenty-nine questions and ten named absences, scored from what the file does not have as well as from what it does.

Run it in month six, not in month three

A maturity score in a first-quarter pack is a number the room will argue about instead of arguing about the findings behind it, and the findings are what change anything. Volume 1 chapter 22 is explicit that the statement an architect can defend after ninety days is a single level with one named binding constraint behind it, not a scorecard. So this instrument waits until there is a repository to read, a handful of interviews behind you and at least one artefact you have asked for and not been given.

When you do run it, it produces four things.

- **A score per domain**, each attached to a named piece of evidence you have seen.
- **A candidate gap row for every domain scoring below 2**, pre-drafted with its first seven fields shaped to paste into `O29 Gap`, which holds two records today.
- **A mean** you can write into an `O04 Measure` row with a target and a date, so the next run has a comparison — and it needs an `O70 Measure Definition` row of its own, because a maturity mean is exactly the kind of derived figure that cannot be reproduced six months later without one.
- **A second, harsher number**: the overall level on the lowest-domain rule.

What it is not. Not a stage model: the acute-hospital adoption stage models measure adoption and pervasiveness rather than outcomes, are partly self-assessed, and contain no sourced resilience requirement [3]. Not a certification — nothing here maps to anybody's accreditation. And not a substitute for volume 1 chapter 19's twelve benchmark measures, which are cheaper, harder to game and the thing to baseline inside ninety days. **Six of those twelve appear directly as questions here**: the exception-queue count and the age of the oldest item at question 6.3, the duplicate and false-merge rate with queue depth and age at 1.3, interface inventory completeness and the date last reviewed at 6.2, the date of the last unannounced downtime drill at 8.2, whether a read-only record exists with its currency and the date last used in anger at 8.3, and the percentage of devices with a known software version, support status and patching owner at 7.1. A seventh, the last successful restore from offline backup, sits inside 8.2's level-5 descriptor. Break-glass use and alert suppression are carried by the principle register in item 19 rather than here, and three of the twelve — laboratory orderable mapping, critical-result acknowledgement and synthetic transaction response — appear in neither.

Eleven domains, not eight phases

A generic instrument shapes its domains like the ADM's phases, which puts this industry's largest exposures inside other people's boxes. Identity, coding, consent, the record and its clock, classification, the seams, devices, resilience, measurement, governance and sourcing are the eleven things that decide whether a provider's architecture survives contact with its estate, and each one here is a domain with its own questions, its own evidence source and its own score.

Domain	What it asks	Where the evidence sits
Identity and matching	Can the estate tell two people apart, and tell one person from themselves twice over?	O43, O37, O20
Coding and terminology	Does anything record what a code means, and which edition it meant it in?	O37, O20, O06, O26, O63
Consent	Is a consent an object with a scope and an expiry, and is a disclosure recorded?	O20, R05, O42, O41, O65
The clinical record and retention	Is there a clock, is retrieval measured, and can the record be reproduced as it stood?	O01, O63, O32, O42, O39
Classification	Do the grades match the data, and do they inherit downward without loosening?	O20, O37, O05 PRN-14, O65
Interoperability and the seams	Does traffic go where the architecture says, and does anybody watch the queue?	R06, O24, O23, RPT04
Devices	Is there a register, and is the exposure of the device estate known?	O20, R05, O27, O32 STD-11
Resilience and downtime	Are the objectives clinical, tested and dated, and is there a path when they fail?	O49, O23, O03, O25
Measurement and definitions	Can a figure be reproduced, and does it say which boundary it used?	O04, O20, O42, O14, O16
Governance and evidence	Does the governance layer reach an obligation, and does it raise what the reports find?	O57, O58, O63, O29, O61
Sourcing	Are the dependencies you do not own named, and is anything testable required of them?	O19, O33, O32 STD-16

Every question names the sheet, the rows or the report that evidences the answer. **You are not eliciting opinions from a room; you are reading 820 records that already exist and most of which are internally consistent and wrong.** That is what makes this instrument worth running at all, and it is why seven of the twenty-nine questions — 1.3, 3.2, 4.2, 4.3, 6.3, 8.3 and 9.2, across six of the eleven domains — are the hardest: they have no repository sheet to read, so they are scored from what is absent.

Eleven domains, twenty-nine questions, and the one hundred weight points

Shaded rows cannot leave level 1 until a sheet that does not exist is created. Together they hold 46 of the 100 weight points.

#	Domain	Weight	Questions	Blocked by an absence
1	Identity and matching	9	3	
2	Coding and terminology	9	3	yes
3	Consent	9	2	yes
4	The clinical record and retention	10	3	yes
5	Classification	9	2	
6	Interoperability and the seams	9	3	
7	Devices	9	2	yes
8	Resilience and downtime	9	3	
9	Measurement and definitions	9	3	yes
10	Governance and evidence	9	3	
11	Sourcing	9	2	
	Totals - the weight check must read 100	**100**	**29**	

Ten domains carry 9 and one carries 10. The odd point sits on the clinical record and retention, where the cost of level 1 is published rather than inferred.

Figure 20.1 The eleven domains with their weights and question counts. Five of the eleven, holding 46 of the 100 weight points, cannot leave level 1 until a sheet that does not exist is created.

Twenty-nine questions: seven domains carry three and four carry two. **Eleven domains do not divide a hundred.** Ten ship at 9 and one at 10, which sums to 100 and makes the weighted mean differ from the plain mean by exactly that single point. The asymmetry is left visible rather than hidden in a decimal, and the reason is stated on the sheet: the extra point sits on *the clinical record and retention*, because it is the one domain where the cost of being at level 1 is published rather than inferred. Provinces pay in the order of R1,5 billion a year settling medico-legal claims [5], the Auditor-General's review of twenty-seven facilities found fifteen with missing or lost records and seventeen not tracking records after issue, and **three of nine paid claims were conceded because the record could not be located** [4]. A records-retrieval capability that prevented one R20 million concession would fund more digital health than the national direct grant budgeted for it.

Everything else is flat on purpose. No evidence available to this volume supports ranking one of these eleven above another, and inventing a ranking to look rigorous is how a maturity score stops being defensible. **Move the weights before you score, write down why, record it at `060 Governance Decision` with the date column item 19 adds, and keep them stable between runs** — a figure whose definition moved is not a trend, which is exactly what `PRN-29` says about every other derived figure.

> **DO NOT WEIGHT A DOMAIN DOWN BECAUSE THE REPOSITORY HAS NO SHEET FOR IT**
>
> That is the same mistake the file made. A domain with no evidence source is still a domain, and giving it a weight is how it gets a sheet. **Five of the eleven cannot leave level 1 until a sheet that does not exist is created** — consent, devices, coding, measurement, and the record and retention — which is forty-six of the hundred weight points stuck at 1.0. That is the finding, not a defect in the instrument.
>
> The opposite temptation is just as bad: a domain nobody can evidence twice running. Question its weight rather than its score. Either it does not matter here, or nobody owns it — and the second is a finding about the practice, which belongs in the governance domain.

Who to ask, in what order, over what period

Four weeks, one interview per scored question, and the sequence matters more than the calendar.

20.1 — Ask for documents in week one, not for decisions

Sheet `R06`, `004`, `019` and `049`, open beside you for comparison · **Input** volume 1 chapter 22's first-day list · **Output** eight artefacts, or eight dated findings that they do not exist · **Signed by** `ROL-01` the Chief Enterprise Architect · **Time** a week

1. Ask for the interface inventory and the date it was last reviewed.
2. Ask for the exception queues, on screen, with counts and the age of the oldest item.
3. Ask for the identifier types in the index with their assigning authorities, including defaulted and placeholder values.
4. Ask for every statutory and programmatic report sent out, with its recipient and its clock.
5. Ask for the three figures the executive argues about, with the query behind each.

6. Ask for the date of the last unannounced downtime drill, the retention schedule with the authority cited per period, and the list of shared services the organisation depends on and does not own.
7. Where an artefact does not exist, write that down with the date you asked and who you asked. **A request for a document that does not exist is a finding with a date on it**, and it is worth more than the document would have been.

Nothing on that list requires a decision, a budget or a steering committee, and nobody has to agree to anything to hand it over. It also tells you, before any scoring, which half of the instrument will be scored from absence.

Then the order of people. Week one is the information officer, the head of health records, the clinical coding manager, whoever runs integration, the chief executive with the information officer in the room, and a ward sister or charge nurse. Week two is the people who work queues and loads: the duplicate queue, the code-update loader, the off-site store, and the chief information officer on recovery objectives and tests. Week three is the technical reconstruction questions — the as-at record, the grade applied at read, clinical and biomedical engineering, the networks engineer, and the end-to-end lineage trace. Week four is the slowest to arrange and the most senior: the compliance lead, the chief enterprise architect on the gap register, and the contracts owner.

Three questions go to a clinician rather than to IT, and those three produce the answers nobody in IT has. What do you read when the record is down, and when did you last use it in anger. What is the recovery objective for the clinical record, and who decided it. Which boundary does your length-of-stay figure use. IT will answer all three confidently and differently.

Two people not to interview. The vendor, who turns an assessment into a pre-sales conversation; and the person who built the repository, who will explain the sheet rather than the estate. The sheet is not what you are scoring.

THREE THINGS TO SETTLE BEFORE THE FIRST INTERVIEW

Agree these with whoever commissioned the assessment, and write the answers here.

Who sees the scores before the findings, and in what forum?

What happens to a question nobody will answer - a date and an owner, or a level?

Which three artefacts, if they do not exist, stop the assessment rather than scoring 1?

How to score honestly

One rule, and it does the work of a page of guidance: **if you cannot name the artefact, the score is the level below.** A score with no artefact behind it is an opinion with a number on it, and it is the first thing a second assessor will overturn.

Ask for the artefact in front of you rather than for an opinion — "show me the last timed retrieval", never "are we compliant". And **write the bad answer down verbatim**, because a bad answer from an interview is usually a true answer about the estate. The hedging is the finding: "it's tested annually" and "there is no record of the last test" are the same fact, and only the first one is quotable, which is why it is the one to record.

What a level 1 answer sounds like, against a level 3

Question	Level 1 sounds like	Level 3 is
Is there a written match rule somebody signed?	"The system does it." That names a product, not a rule.	A written rule with thresholds, a review queue with an owner, and a reversal mechanism.
Which edition of each code set is loaded today?	"The latest." Latest is not an edition.	Every code system in use has a row with its edition and the date it was loaded.
Show me one consent with its scope and its expiry.	"It's a tick on the admission screen."	A consent record with a type, a subject, a scope, a grant date and an expiry.
How long does it take to retrieve a ten-year-old record?	"A few days." An estimate is not a measurement.	A measured retrieval time per record class, with the date it was measured.
Show me this record as it stood before it was amended.	"We'd restore a backup." A backup is a disaster-recovery artefact, not a reconstruction.	Corrections supersede rather than overwrite on the clinical systems in scope.
How many items are in the exception queues right now?	"We monitor them." Watch whether they open the queue or open a dashboard.	Queue depth and the age of the oldest item reported, with a named owner.
What do you read when the record is down?	"We ring the ward." That is the true answer and it is the finding.	A read-only record exists with a stated currency and a named owner.

Level 5 has one shape across all eleven domains: **a number, a date and a named person.** The thing is measured, the measurement has a date on it, and somebody owns the figure. Nothing reaches 5 on a description of intent.

Levels 2 and 4 are interpolated, and the instrument says so on its face. Levels 1, 3 and 5 are written per question, because defining every level in words produces a page of distinctions nobody can apply. Score 2 where the thing exists for part of the estate, or exists and is not used. Score 4 where it is defined, owned and used and the only thing missing is the measurement — which is the step most estates never take, and the step that separates a practice that works from one that can prove it. The two interpolated levels are still *named*, because the score sheet looks the level name up against the score you enter and a lookup needs five keys.

Level 1 is not zero. It means the thing does not exist, or exists as one person's private knowledge. A blank is different and worse: a blank reads as not assessed and loses the finding entirely.

> SCORE IT TWICE AND KEEP BOTH COLUMNS
>
> Once from the repository and once from the interviews. Where the two disagree by more than a level, the disagreement is the finding in its own right — and **the repository is usually the optimistic one**, because it records intentions as statuses: twenty deliverables of which twelve read `Approved` with no artefact in the file, and seven work packages reading `Delivered` before any has reached its start date.
>
> Where they disagree, score the lower and keep the higher visible. Never average them.

The scoring and the roll-up

Four numbers come out, and each answers a different question.

The domain score is the mean of its questions, to one decimal, on the 1 to 5 scale and not a percentage. **The assessment mean** is the mean of the eleven domain scores: the first answer, the one that moves when anything improves, and therefore the right number for a trend. **The weighted mean** is the sum of weight times domain score divided by 100; while the weights ship as they are it differs from the assessment mean by the single point parked on the record and retention, and if it differs by more then somebody has re-weighted and the decision belongs at `O60` with a date. **The flat mean of the twenty-nine questions** differs from the domain mean whenever the domains have unequal question counts, which they do — seven carry three and four carry two — so keep it only to spot a domain whose two questions are carrying more weight than they should.

Three arithmetic checks catch the ways this goes wrong.

- **Questions scored must reach 29** before any domain score means anything. An average that ignores blanks will report a confident mean over the questions you happened to answer.
- **The weight check must read 100.** Eleven domains do not divide a hundred, so if you re-weight, this is the cell that catches it.
- **Questions with no evidence recorded must read zero** by the end, and more than four low-confidence scores means the assessment is not ready to present. Low confidence is honest, and it is also a work item: name what you would have to see.

The domain floor, and why the lowest subdomain governs

The freely published health-system interoperability maturity toolkit sets each domain at its **lowest subdomain** and the overall level at its **lowest domain**, rather than at an average [1][2]. This instrument does the same: the domain floor is the lowest question in the domain, and the overall level is the lowest domain floor in the instrument.

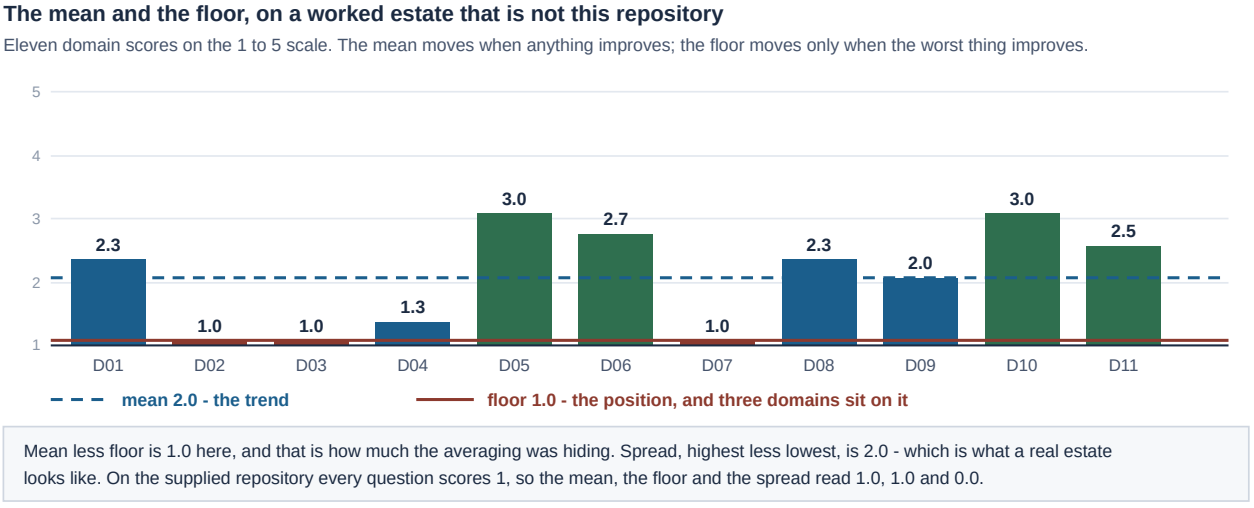


Figure 20.2 Why both numbers are quoted. The mean is the trend and the floor is the position, and the difference between them is how much the averaging was hiding.

The argument for it is not modesty, it is mechanism. **Interoperability maturity genuinely is limited by its weakest component.** A seam that cannot carry an identifier is not half-solved by a good terminology position; a record you cannot reconstruct is not half-defensible because the retention schedule is excellent. Where the capability composes, the minimum is the honest reading, and taking the minimum resists the averaging that makes most maturity scores meaningless.

So quote both, and say which is which. **The mean is the trend. The floor is the position.** The difference between them — mean less floor — is how much the averaging was hiding, and it is the number to put in front of anybody who wants to quote the mean alone: if it reads 1.4, the organisation is a level and a half worse than its own scorecard says.

On the supplied repository the two are identical at 1.0, because **every one of the twenty-nine questions scores 1.** That is the least comfortable thing in this item and it is what the cells say rather than a judgement about the people who filled them. The spread — highest less lowest — is 0.0, and that is the most useful reading on the sheet. A spread of zero does not describe an estate with strong and weak areas; it describes a repository built to one standard throughout, where the structure is good, the referential integrity is perfect across all 430 relationship rows, and the content is uniformly unexamined. The nearest thing to real maturity in the file is structural — a sound capability tree, perfect integrity, and governance bodies with a genuine cadence, membership and mandate — and none of the three is a domain here, because none of them is about the content. On a real estate expect a spread of 1.5 to 2.5, and the domains at the bottom of it are where the next year goes.

How to score the sheets that are missing

This is the part no sibling volume has, and it is the reason the instrument exists. Six of the sheets this volume tells the reader to create do not exist in the repository, and four further absences need no new sheet at all. **All ten are named, all ten score 1, and all ten get a sentence.**

The absence register: ten named absences, and what each one costs the score

Every absence scores 1 and gets a sentence. A blank would read as not assessed and lose the finding entirely.

SIX NEED A NEW SHEET - the numbering is fixed and there is no seventh			
AB-1	Consent scope and expiry	O67 Consent	caps 3.1
AB-2	The medical device register	O68 Medical Device	caps 7.1
AB-3	A clinical coding system	O69 Code System	caps 2.1, 2.2
AB-4	The measure definition	O70 Measure Definition	caps 9.1, 9.2
AB-5	The disclosure record	O71 Disclosure Event	caps 3.2
AB-6	The retention clock	O72 Retention Clock	caps 4.1, 4.2
FOUR NEED NO NEW SHEET AT ALL - repaired inside columns the file already has			
AB-7	Identity structure	Retype MDR-01 on O43; add the identifier set to O37	caps 1.2, 1.3
AB-8	Clinical safety casework	Clinical hazards into O62; the officer into O10 and O11	caps nothing
AB-9	The clinical organisation	Clinical units into O09; clinical roles into O10 and O11	caps 10.1
AB-10	Ward, clinic, theatre and bed	A care setting column on O12; attributes for E11 on O37	caps nothing

Twelve of the twenty-nine questions cannot exceed 1 while the thing behind them is missing.
1.2, 1.3, 2.1, 2.2, 3.1, 3.2, 4.1, 4.2, 7.1, 9.1, 9.2 and 10.1. The other seventeen are capped by content, which is a slower repair.

Figure 20.3 The ten absences, split by whether a new sheet is needed. Two of the ten cap no question in the instrument and are registered anyway, because an absence the instrument cannot score is the one that gets lost.

20.2 — Score an absence

Sheet `O20`, `O37`, `O43`, `O04`, `O63`, and the absence register you are keeping · **Input** the ten named absences · **Output** a score of 1 per capped question with a sentence of evidence · **Signed by** `ROL-01` the Chief Enterprise Architect · **Time** two days

1. Name what the file actually shows *instead*, by cell. For consent that is one attributeless row at `O20` row 17 whose `Lineage` reads `Full`, plus `MTD-07 Consent basis` at coverage `Partial` and referenced by nothing. Never write "nothing exists" where a cell exists and says the wrong thing.
2. Name the sheet to create or the column to fix. Six take a new sheet, `O67` to `O72`, and the numbering is fixed with no seventh. Four are repaired inside columns the file already has.
3. Name what the absence costs, in the terms the organisation already uses. For devices that is one of volume 1 chapter 19's twelve measures being not computable; for retention it is a single corporate period being unlawfully short for the longest class and needlessly expensive for everything else, both at once.
4. Name who signs the repair, and name a person rather than a body. Two of the ten are countersigned, because the treatment side and the information side of consent are different people.
5. Score the questions the absence caps at 1 and write the sentence in the evidence column. Do not leave the cell blank.
6. Register the two absences that cap no question anyway — clinical safety casework, and ward, clinic, theatre and bed. **An absence the instrument cannot score is the one that gets lost**, and the first of those two is the one most likely to matter in a year.

WHY AN ABSENCE CAPS A SCORE RATHER THAN REDUCING IT

A reduction says the thing is weak. A cap says the thing cannot be demonstrated at all, which is a different claim and a different repair.

Score the question 1 and hold it there until the register exists. Then the score moves on its own, without anybody re-arguing the question — nine rows at `O72` with a period, an event, an instrument and an authority takes the record and retention domain off its floor in a fortnight, and nothing else has to change for the number to move. A reduction would have the domain creeping up as the conversation improved, which is how a maturity score comes to measure the quality of the conversation instead of the quality of the estate.

The cap is also what makes the instrument honest about seventeen of the twenty-nine. **Twelve questions are capped by structure and seventeen by content** — a different repair, and usually a slower one.

Three of the ten absences are repaired in a single sitting, with no budget, no product and no decision above the data owner: the retention clock, retyping `MDR-01` on `O43` from `Golden source` to `Match rule` and writing the thresholds down, and adding a care setting column to `O12`. Between them they move three domains off the floor. A fourth, counting the exception queue depth and the age of the oldest item, is an afternoon with read access and gives you the best integration measure there is.

An instrument that skipped the absences would score this estate well above 2 and tell you nothing — which is what every stage model applied to this file would do.

The forms

Form 20-A — Question score sheet (part one: score)

Feeds no sheet in the repository · **Rows for a first pass** 29, one per question · **Signed by** `ROL-01` the Chief Enterprise Architect · **Ratified by** `GB-01`

None of these six columns exists anywhere in the repository, and `O04 Measure` is the only sheet that could hold the output — it has `Ref`, `Measure`, `Unit`, `Domain`, `Baseline`, `Cadence`, so the assessment mean goes in as one `O04` row with its `O70` definition beside it, and the per-question detail stays on this form. Keyed on the question reference.

Ref	Domain	Question	Score 1-5	Target	Gap
4.2	The clinical record and retention	Is the time to retrieve an old record measured? — example row	1	3	2

Form 20-B — Evidence and confidence (part two: assess)

Feeds no sheet in the repository · **Rows for a first pass** 29, keyed on `Ref` to part one · **Signed by** `ROL-01` the Chief Enterprise Architect · **Ratified by** `GB-01`

Write the evidence verbatim, including the bad answer. The last column is the one a second assessor reads first: where you and the repository disagree, score the lower and record both.

Ref	Evidence seen — what you actually looked at	Confidence	Assessed by	Date	Disagreement with the file
4.2	"A few days" — no timed retrieval on record; off-site store manager, 14 Mar — example row	High			None; the file has no measure either

Form 20-C — The absence register

Feeds the six new sheets and four existing-column repairs · **Rows for a first pass** 10, one per named absence · **Signed by** the post named in column five · **Ratified by** `GB-01`

Two of the ten cap no question in the instrument and are registered anyway. Column three is the one that connects this form to the score: the questions each absence holds at 1.

Ref	The absence	Questions it caps	Sheet to create, or column to fix	Who signs it	Score
AB-6	The retention clock	4.1, 4.2	O72 Retention Clock, keyed RTN- — example row	Head of health records, countersigned by the information officer	1

Form 20-D — Candidate gap rows for O29

Feeds `O29 Gap` · **Rows for a first pass** one per domain scoring below 2, plus one per absence · **Signed by** `ROL-01` the Chief Enterprise Architect · **Ratified by** `GB-01`

`O29` has seven columns — `Ref`, `Gap`, `Domain`, `Building block`, `Severity`, `Status`, `Due`. Six are below; **`Due` is typed straight into Excel**, because it is a date you will be held to and it should not be pre-drafted. `Domain` takes the file's own values, which are the four architecture domains and not this instrument's eleven, so a domain score maps to `Data`, `Application`, `Technology` or `Business` on the way across.

Ref	Gap	Domain	Building block	Severity	Status
GAP-03	No retention clock, no measured retrieval time and no way to reproduce the record as it stood — example row	Data		Critical	Open

What to do with the output

The number is the cover slide and the findings are the paper. An audience given a 1.0 will spend the hour on whether it is really a 1.0. **Lead on three findings with the cell reference behind each and let somebody ask for the score.**

Then convert. That conversion is the point of the whole exercise: `O29 Gap` holds two records against everything the rest of the file says is wrong — ten capabilities with a maturity gap, six standards in breach, three absent controls, six obligations without evidence and a component past support — and thirty-five of the thirty-seven building blocks on `O55` have no gap row at all.

20.3 — Turn the scores into gap rows somebody owns

Sheet `O29 Gap`, `O30 Work Package`, `O61 Governance Change Request` · **Input** `Form 20-A`, `Form 20-C` and the eleven domain scores · **Output** six to eight gap rows with an owner and a date · **Signed by** `ROL-01` the Chief Enterprise Architect · **Ratified by** `GB-01` · **Time** two days

1. Draft one candidate row per domain scoring below 2 and one per absence. On this repository that is eleven plus ten, and **you do not paste twenty-one rows into `O29`.**
2. Consolidate where two rows describe one repair. The consent domain row and the consent absence are one repair; so are the retention row and the retention clock. Say on the row which ones were merged.
3. Cut to six to eight. A gap register with twenty-one open rows and no owner is the same failure as one with two rows and no coverage, from the other direction.
4. Set `Severity` yourself. A domain at 1.0 whose repair is nine register rows is not the same risk as a domain at 1.0 whose repair is a procurement, and the register should say so.
5. Fill `Building block` by hand against `O55`. This column is where the register joins the rest of the file, and thirty-five of thirty-seven blocks have nothing pointing at them.
6. Then raise the change request. `CR-G03` was raised against the repository itself to turn a reporting gap into gap rows and is **still `Raised`** — so a gap row with no work package at `O30` and no owner is a complaint with a reference number.

Put nothing on that register you have not seen a cell for. Every candidate row carries rows, columns or counts from the file, and that is why they survive a hostile reading; a finding argued from experience rather than from the evidence in front of the room is the one that gets talked out at eleven o'clock.

FOUR QUESTIONS TO CLOSE THE ASSESSMENT ON

Answer these on the register itself before the pack is written.

Which three findings lead, and what is the cell reference behind each?

Which domain is the floor, and what single repair would move it?

What is the mean, what is the floor, and what is the difference between them?

Which gap rows were consolidated, and who owns each surviving row?

Next quarter, re-run the twenty-nine questions rather than the whole instrument. Keep the weights, keep the level-1 and level-5 anchors, and change a level-3 descriptor only on a recorded decision — moving an anchor between runs makes the trend meaningless. Set a target you can defend as achievable inside the period rather than one that reads well: three domains moving one level each is a credible year, and on this repository the three should be the record and retention, identity, and the seams, because between them nine register rows, one retyped rule and one afternoon of counting move all three.

THE CHAPTER IN TEN LINES

- Run it in month six, not month three: a maturity score in a first-quarter pack displaces the findings that change anything.
- Eleven domains rather than eight phase-shaped ones, because the phase shape puts this industry's largest exposures inside other people's boxes.
- Twenty-nine questions, ten domains weighted 9 and one weighted 10, and the odd point sits on the clinical record and retention for a stated reason.
- Ask for documents in week one and for decisions never; a request for a document that does not exist is a finding with a date on it.
- Three questions go to a clinician rather than to IT, and those three produce the answers nobody in IT has.
- If you cannot name the artefact, the score is the level below, and the bad answer is written down verbatim.
- Levels 1, 3 and 5 are written per question; 2 and 4 are interpolated and marked so, because the score sheet needs five keys to look a name up.
- Every level-5 descriptor reduces to a number, a date and a named person, so nothing reaches 5 on a description of intent.
- The domain floor is the lowest question and the overall level is the lowest domain, because maturity is limited by its weakest component.
- Quote the mean for the trend and the floor for the position, and quote the difference as the amount the averaging was hiding.
- Ten named absences each score 1 and each get a sentence: six need a new sheet and four are repaired inside columns the file already has.
- An absence caps a score rather than reducing it, so the number moves when the register appears rather than when the conversation improves.
- Six to eight gap rows reach the board, consolidated from twenty-one candidates, with the first six fields shaped to paste into O29 Gap.

SOURCES FOR THIS CHAPTER

- **1.** Health Information Systems Interoperability Maturity Toolkit: Users' Guide, MEASURE Evaluation with the Health Data Collaborative (each domain scored at its lowest subdomain and the overall level at the lowest domain rather than an average) https://www.healthdatacollaborative.org/fileadmin/uploads/hdc/Documents/Working_Groups/Digital_Health_Interoperability_Working_Group/HIS_Interoperabiity_Maturity_Toolkit
- **2.** MEASURE Evaluation, Health Information Systems Interoperability Maturity Toolkit: Model (the domain and subdomain structure the lowest-score rule is applied to) https://www.measureevaluation.org/resources/publications/tl-17-03c.html
- **3.** HIMSS, Electronic Medical Record Adoption Model (a stage model measuring adoption and pervasiveness; stage criteria partly self-assessed and no sourced resilience requirement) https://www.himss.org/maturity-models/emram/
- **4.** Auditor-General findings reported via MedicalBrief, 14 October 2026 (audit of 27 facilities across nine provinces; 15 with missing or lost records; 17 not tracking records after issue; three of nine paid claims conceded for want of a record) https://www.medicalbrief.co.za/health-sector-crippled-by-recurring-systemic-weaknesses-ag-warns/
- **5.** National Treasury, 2026 Budget Review, Chapter 6 (provinces spend about R1,5 billion a year settling medico-legal claims; health conditional grant tables) https://archive.treasury.gov.za/documents/National%20Budget/2026/review/Chapter%206.pdf